

Data Communications and Networking Fourth Edition

Forouzan

Chapter 1

Introduction

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1-1 DATA COMMUNICATIONS

The term telecommunication means communication at a distance. The word data refers to information presented in whatever form is agreed upon by the parties creating and using the data. Data communications are the exchange of data between two devices via some form of transmission medium such as a wire cable.

Topics discussed in this section:

Components
Data Representation
Data Flow

Data?

- Data is a collection of raw, unorganized facts, figures, observations, or symbols. On its own, it lacks meaning or context. Once collected and analyzed, it is processed into **information**, which reveals actionable insights, patterns, and trends used for decision-making.

Data Communication

- Transferring data over a transmission medium between two or more devices, systems, or places is known as data communication. Nowadays, computing and telecommunications depend heavily on this data transmission, which makes a variety of applications conceivable, including email, video chatting, the Internet, and many more things.

Why Data Communication

- Data communication is the backbone of the modern, connected world. It is essential because it allows electronic devices to exchange information, enabling instant global connectivity, real-time collaboration, and data-driven decision-making. Without it, the internet, cloud computing, and remote work would be impossible

Figure 1.1 *Five components of data communication*

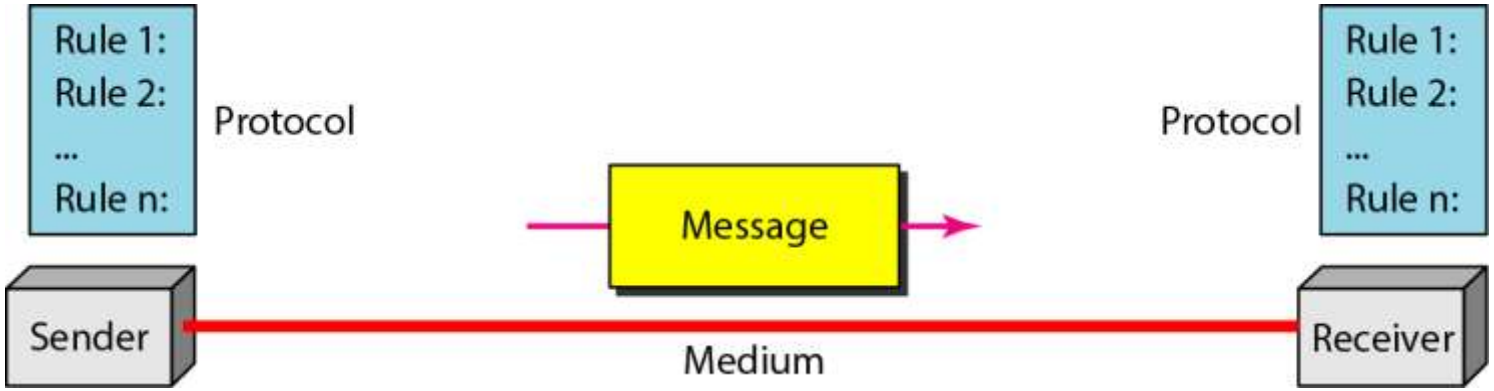
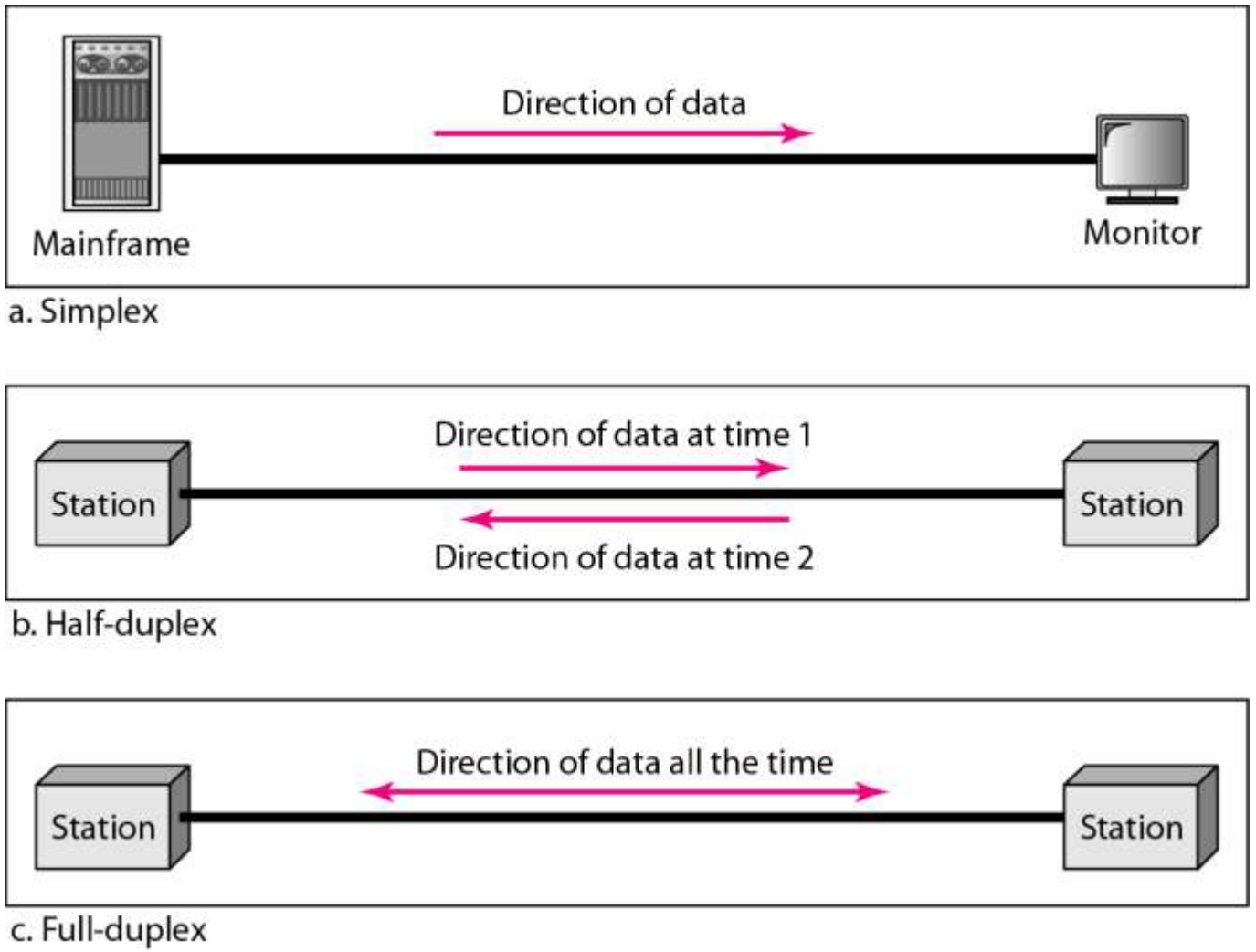


Figure 1.2 *Data flow (simplex, half-duplex, and full-duplex)*



1-2 NETWORKS

A network is a set of devices (often referred to as nodes) connected by communication links. A node can be a computer, printer, or any other device capable of sending and/or receiving data generated by other nodes on the network.

Topics discussed in this section:

Distributed Processing
Network Criteria
Physical Structures
Network Models
Categories of Networks
Interconnection of Networks: Internetwork

Figure 1.3 *Types of connections: point-to-point and multipoint*

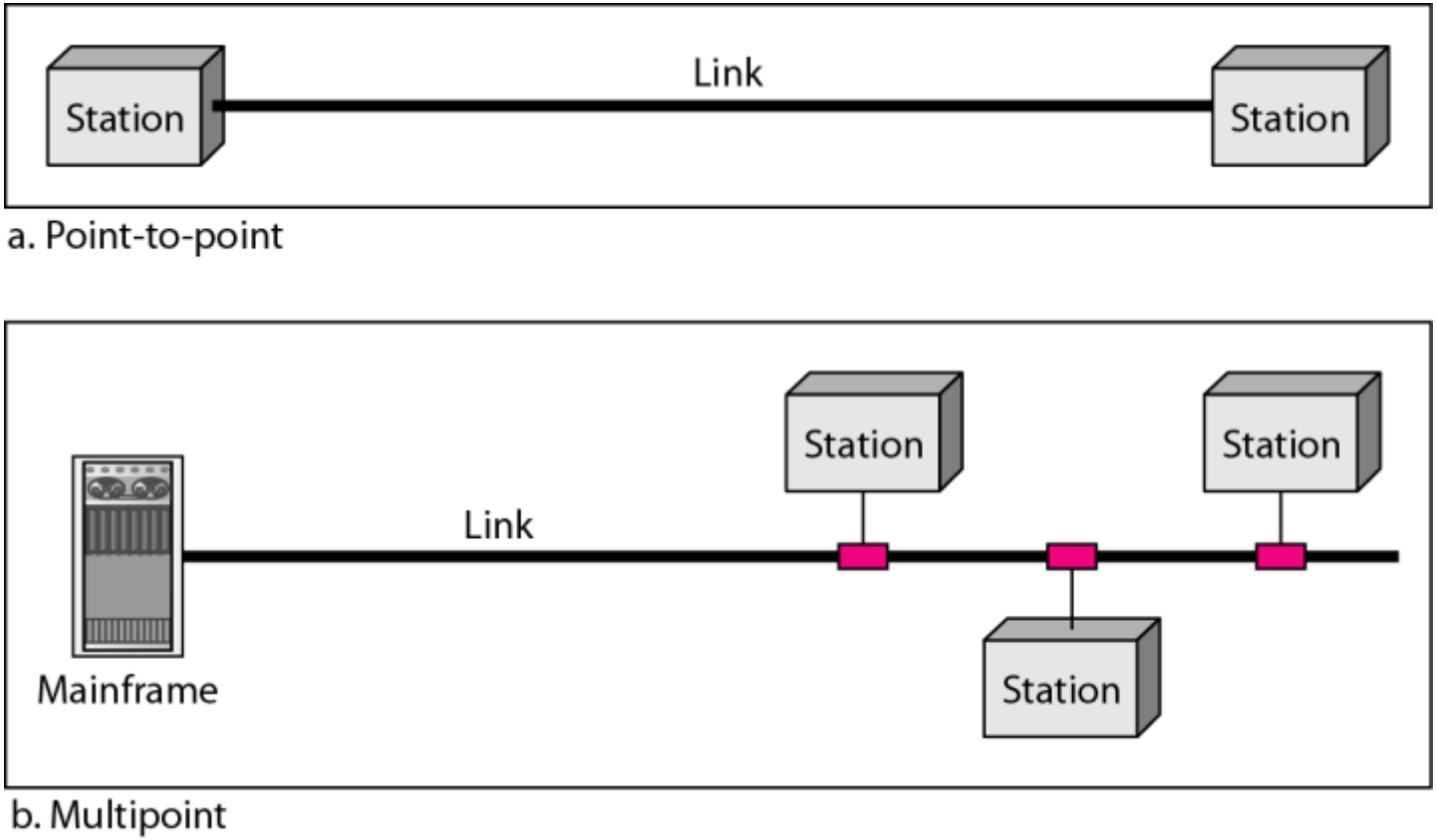


Figure 1.4 *Categories of topology*

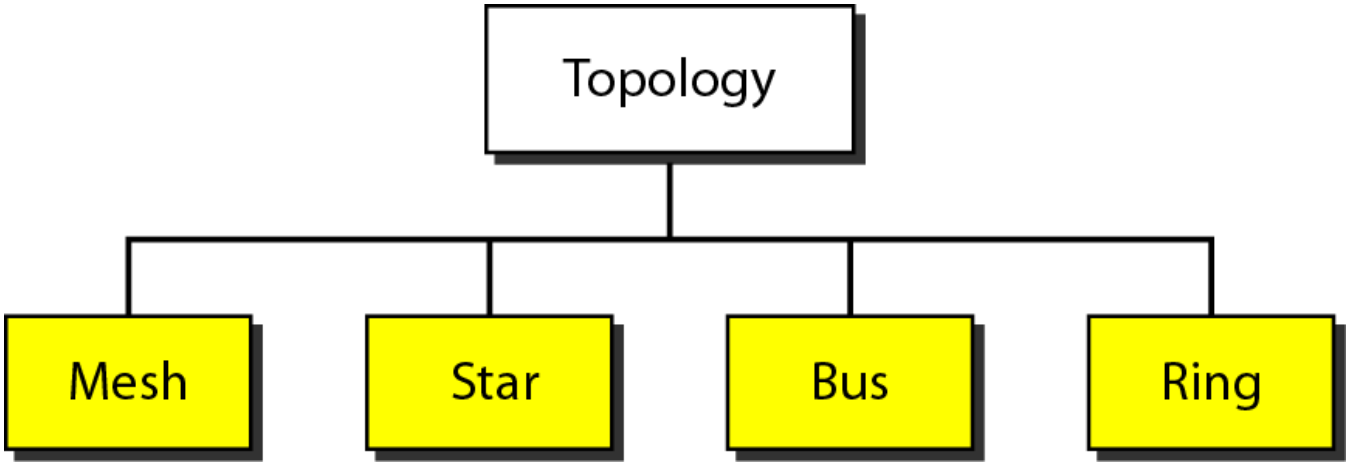


Figure 1.5 *A fully connected mesh topology (five devices)*

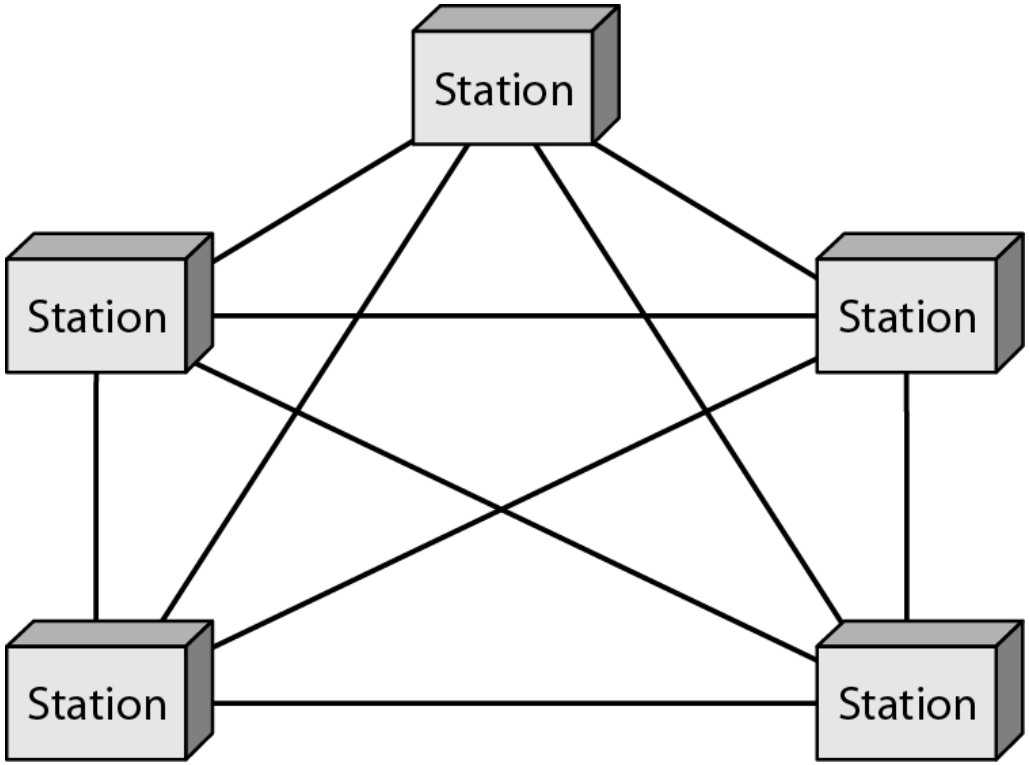


Figure 1.6 *A star topology connecting four stations*

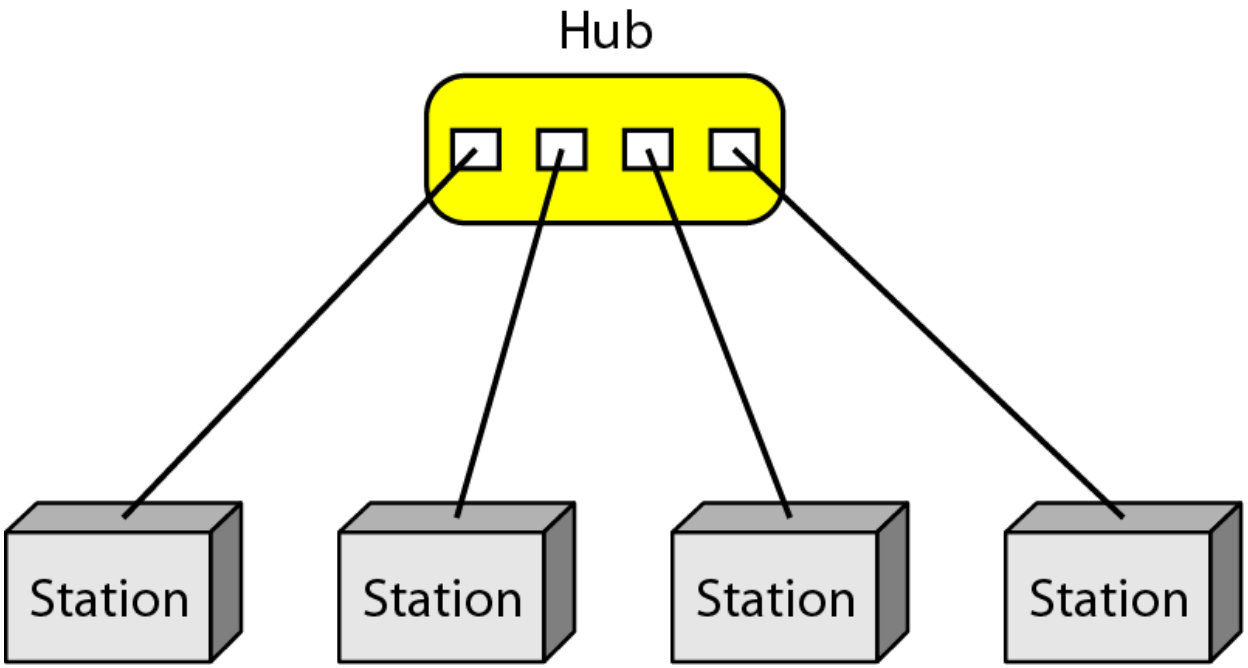


Figure 1.7 *A bus topology connecting three stations*

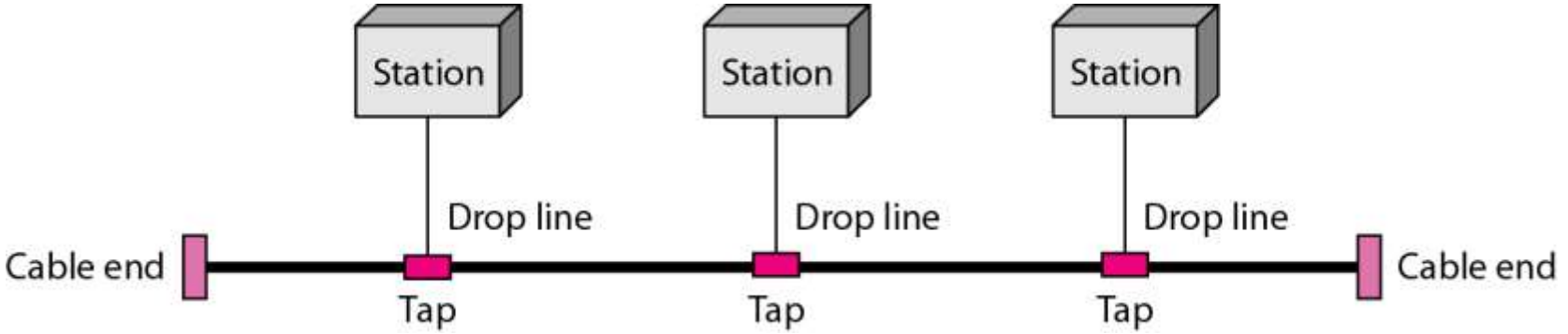


Figure 1.8 *A ring topology connecting six stations*

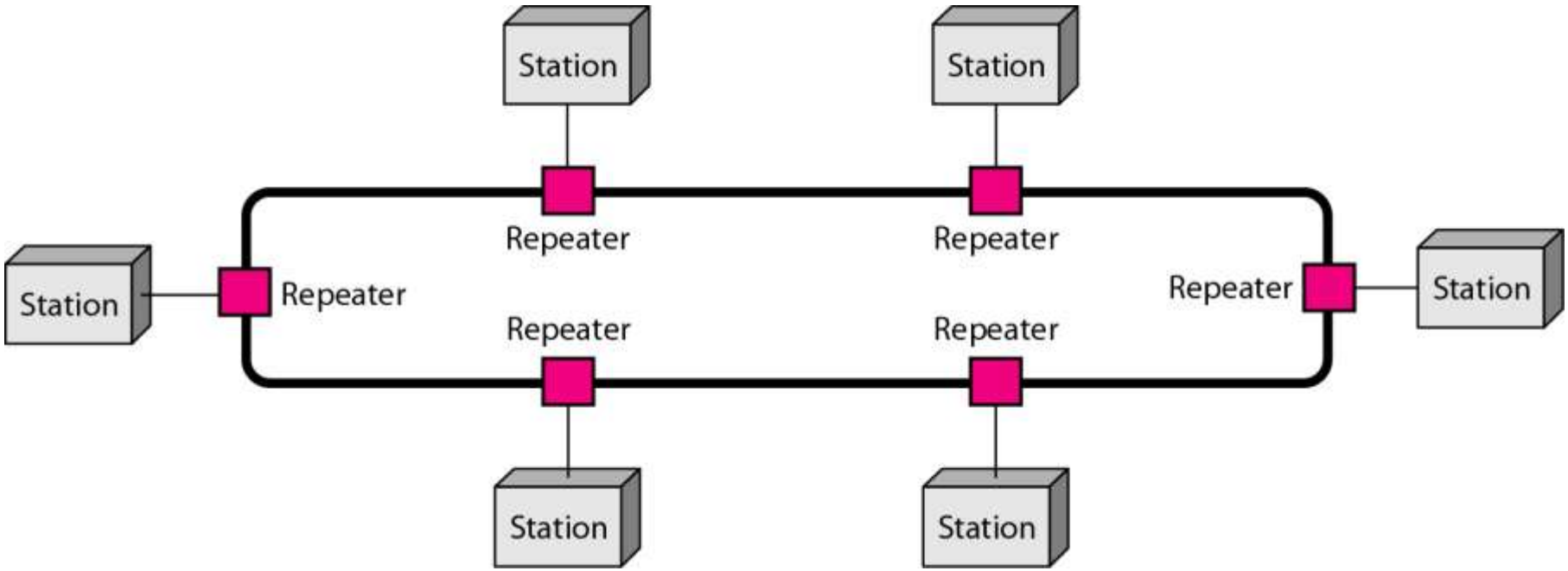


Figure 1.9 *A hybrid topology: a star backbone with three bus networks*

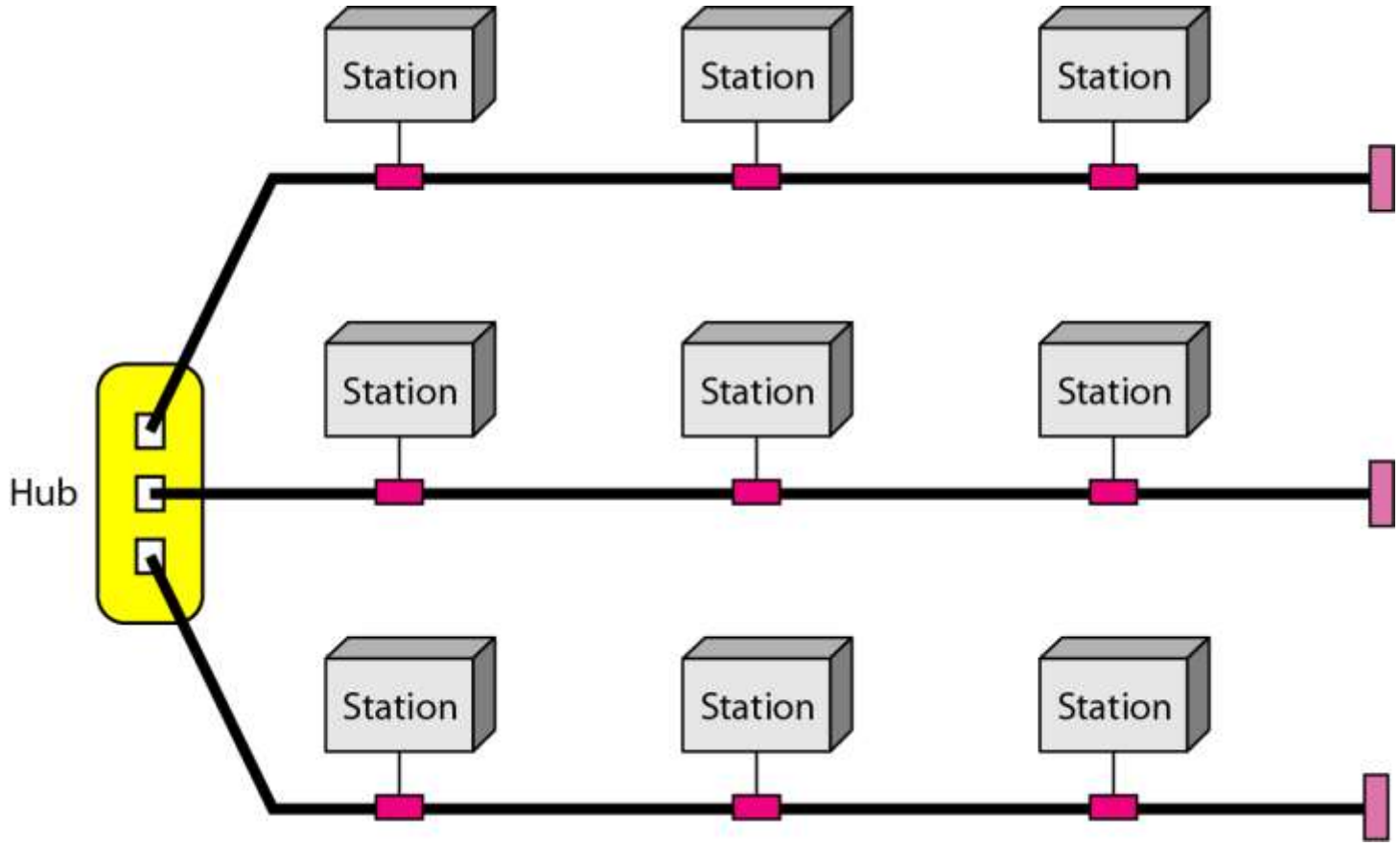


Figure 1.10 *An isolated LAN connecting 12 computers to a hub in a closet*

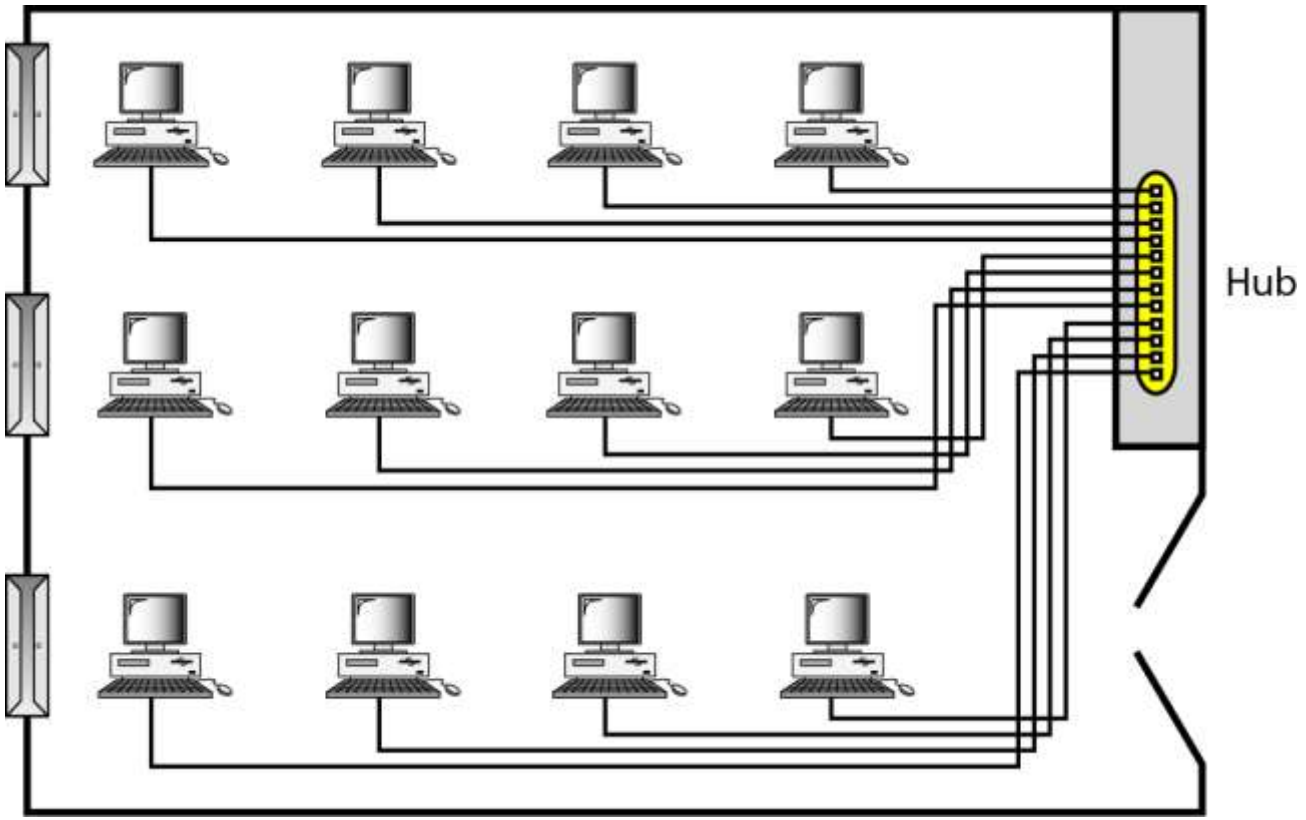
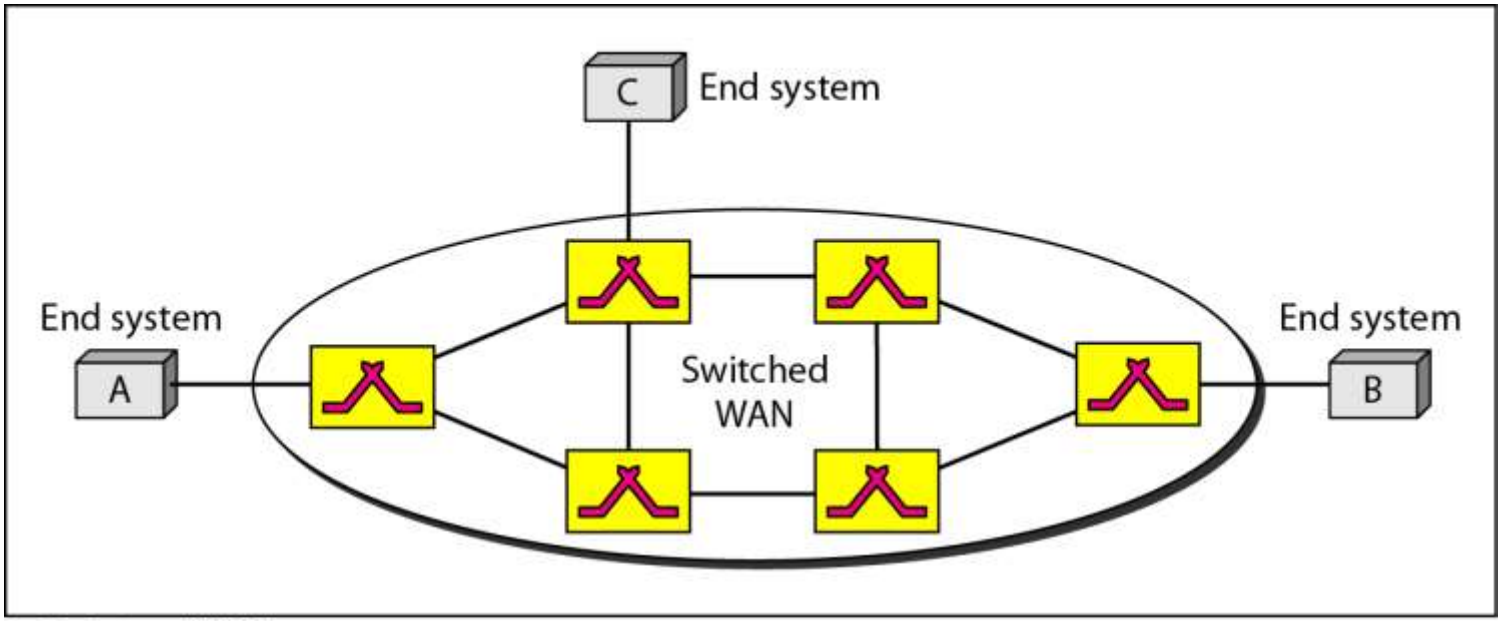
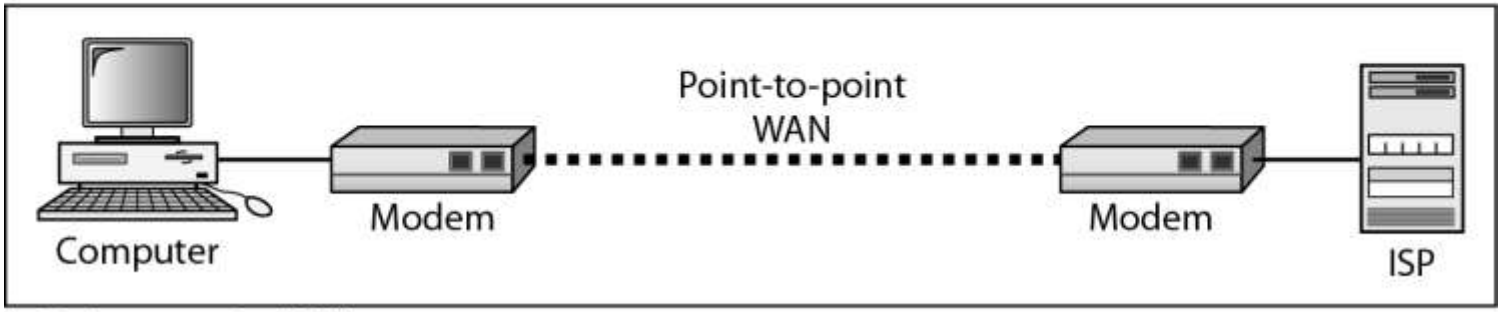


Figure 1.11 *WANs: a switched WAN and a point-to-point WAN*

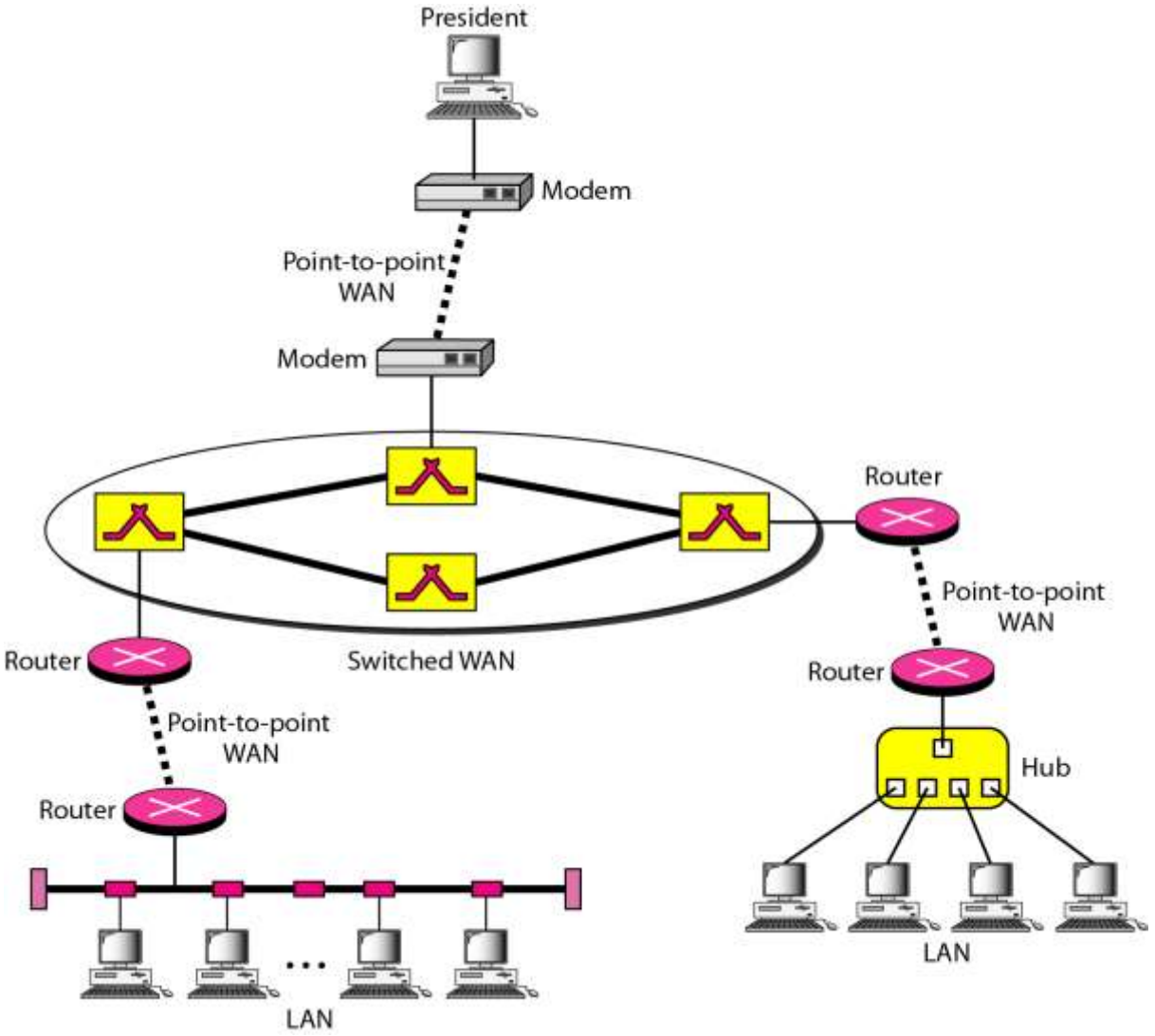


a. Switched WAN



b. Point-to-point WAN

Figure 1.12 *A heterogeneous network made of four WANs and two LANs*



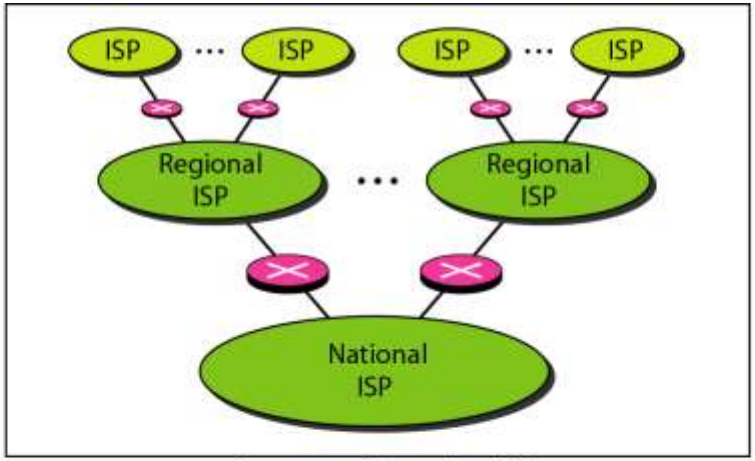
1-3 THE INTERNET

The Internet has revolutionized many aspects of our daily lives. It has affected the way we do business as well as the way we spend our leisure time. The Internet is a communication system that has brought a wealth of information to our fingertips and organized it for our use.

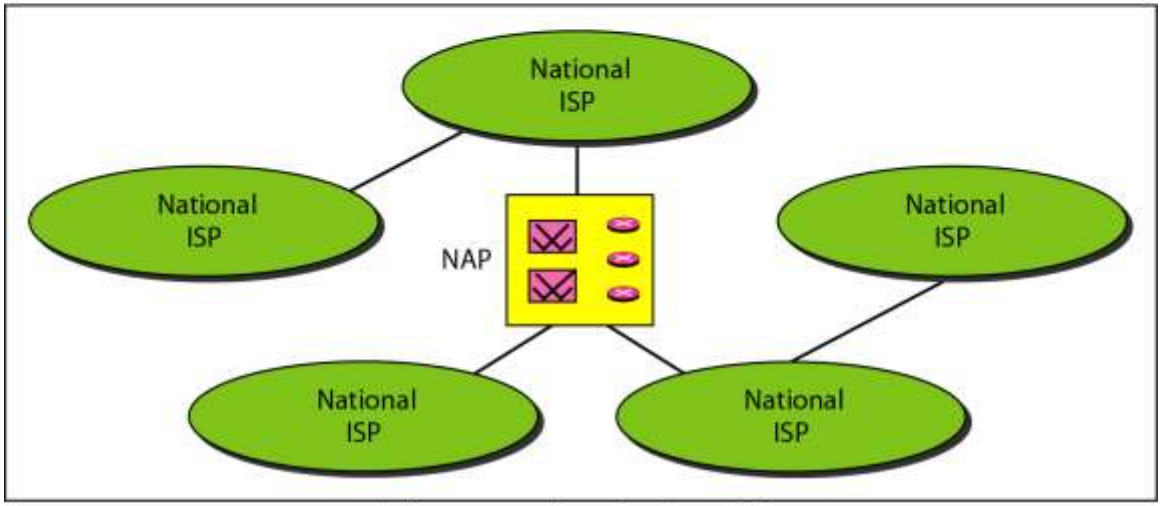
Topics discussed in this section:

A Brief History
The Internet Today (ISPs)

Figure 1.13 *Hierarchical organization of the Internet*



a. Structure of a national ISP



b. Interconnection of national ISPs

1-4 PROTOCOLS AND STANDARDS

In this section, we define two widely used terms: protocols and standards. First, we define protocol, which is synonymous with rule. Then we discuss standards, which are agreed-upon rules.

Topics discussed in this section:

Protocols
Standards
Standards Organizations
Internet Standards



Chapter 3

Data and Signals

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Note

To be transmitted, data must be transformed to electromagnetic signals.

3-1 ANALOG AND DIGITAL

*Data can be **analog** or **digital**. The term **analog data** refers to information that is continuous; **digital data** refers to information that has discrete states. Analog data take on continuous values. Digital data take on discrete values.*

Topics discussed in this section:

Analog and Digital Data
Analog and Digital Signals
Periodic and Nonperiodic Signals

Note

Data can be analog or digital. Analog data are continuous and take continuous values. Digital data have discrete states and take discrete values.

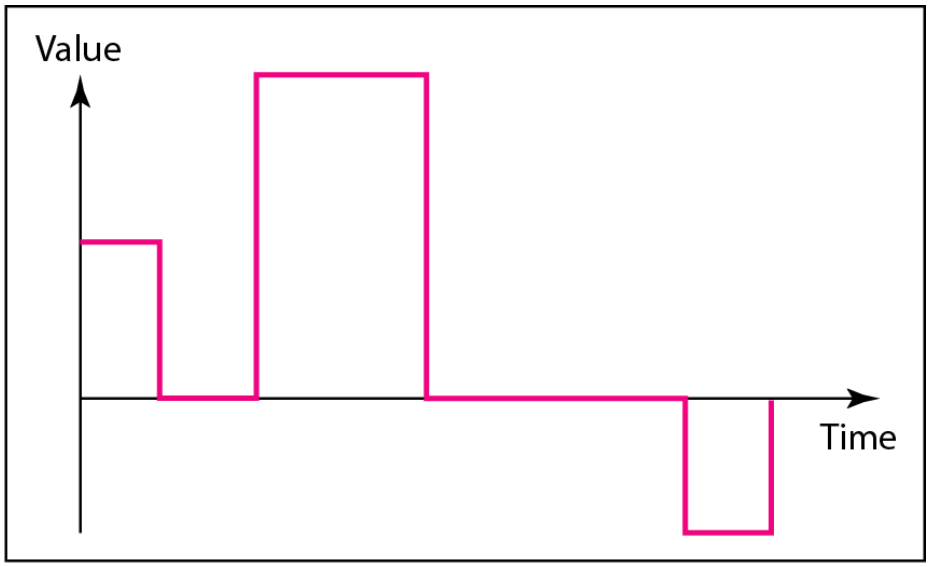
Note

Signals can be analog or digital. Analog signals can have an infinite number of values in a range; digital signals can have only a limited number of values.

Figure 3.1 *Comparison of analog and digital signals*



a. Analog signal



b. Digital signal

Note

In data communications, we commonly use periodic analog signals and nonperiodic digital signals.

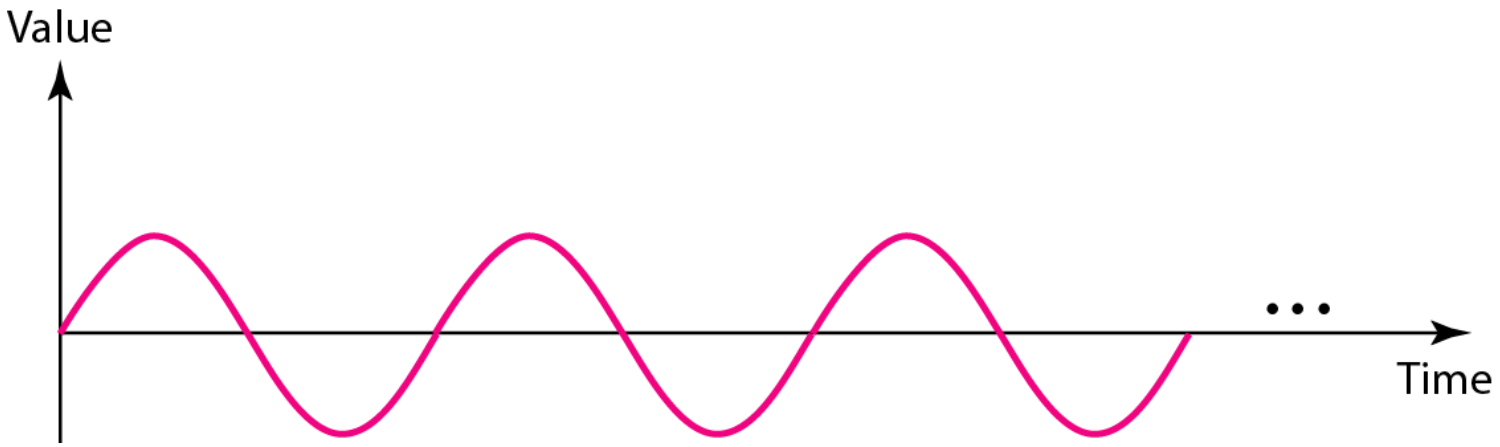
3-2 PERIODIC ANALOG SIGNALS

*Periodic analog signals can be classified as **simple** or **composite**. A simple periodic analog signal, a **sine wave**, cannot be decomposed into simpler signals. A composite periodic analog signal is composed of multiple sine waves.*

Topics discussed in this section:

- Sine Wave
- Wavelength
- Time and Frequency Domain
- Composite Signals
- Bandwidth

Figure 3.2 *A sine wave*



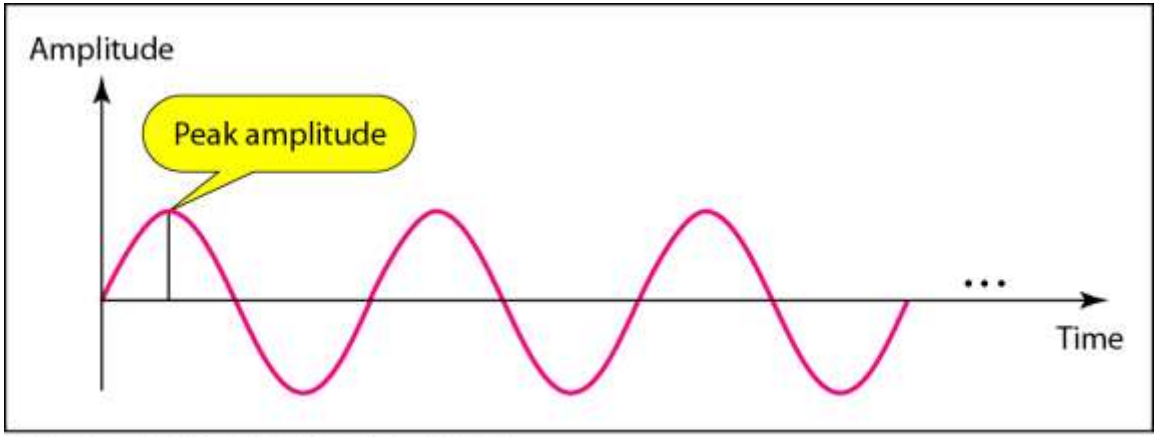
Note

We discuss a mathematical approach to sine waves in Appendix C.

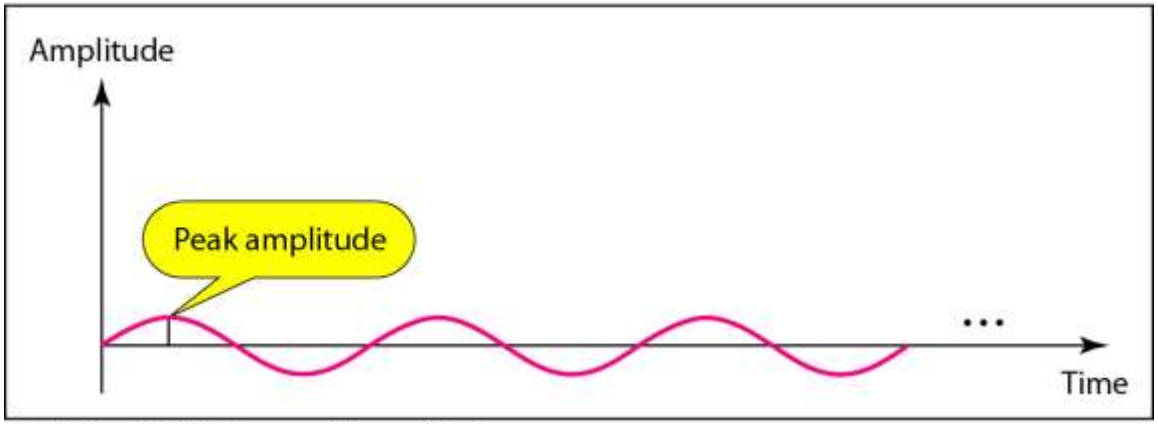
Example 3.1

The power in your house can be represented by a sine wave with a peak amplitude of 155 to 170 V. However, it is common knowledge that the voltage of the power in U.S. homes is 110 to 120 V. This discrepancy is due to the fact that these are root mean square (rms) values. The signal is squared and then the average amplitude is calculated. The peak value is equal to $2^{1/2} \times$ rms value.

Figure 3.3 *Two signals with the same phase and frequency, but different amplitudes*



a. A signal with high peak amplitude



b. A signal with low peak amplitude

Example 3.2

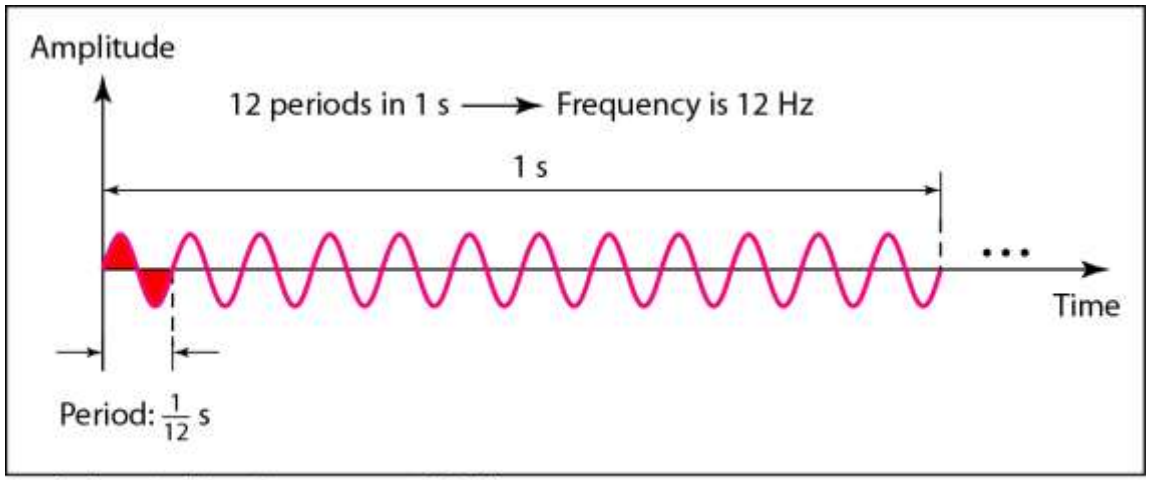
The voltage of a battery is a constant; this constant value can be considered a sine wave, as we will see later. For example, the peak value of an AA battery is normally 1.5 V.

Note

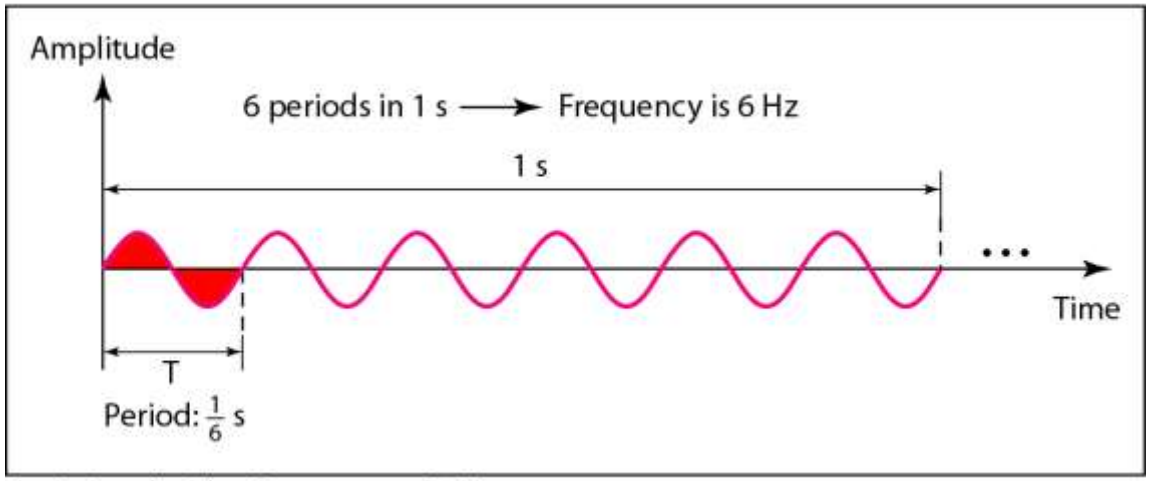
Frequency and period are the inverse of each other.

$$f = \frac{1}{T} \quad \text{and} \quad T = \frac{1}{f}$$

Figure 3.4 *Two signals with the same amplitude and phase, but different frequencies*



a. A signal with a frequency of 12 Hz



b. A signal with a frequency of 6 Hz

Table 3.1 *Units of period and frequency*

Unit	*Equivalent*	*Unit*	*Equivalent*
Seconds (s)	1 s	Hertz (Hz)	1 Hz
Milliseconds (ms)	10^{-3} s	Kilohertz (kHz)	10^{3} Hz
Microseconds (μs)	10^{-6} s	Megahertz (MHz)	10^{6} Hz
Nanoseconds (ns)	10^{-9} s	Gigahertz (GHz)	10^{9} Hz
Picoseconds (ps)	10^{-12} s	Terahertz (THz)	10^{12} Hz

Example 3.3

The power we use at home has a frequency of 60 Hz. The period of this sine wave can be determined as follows:

$$T = \frac{1}{f} = \frac{1}{60} = 0.0166 \text{ s} = 0.0166 \times 10^3 \text{ ms} = 16.6 \text{ ms}$$

Example 3.4

Express a period of 100 ms in microseconds.

Solution

From Table 3.1 we find the equivalents of 1 ms (1 ms is 10^{-3} s) and 1 s (1 s is 10^{6} μs). We make the following substitutions:.

$$100\ \text{ms} = 100 \times 10^{-3}\ \text{s} = 100 \times 10^{-3} \times 10^{6}\ \mu\text{s} = 10^{2} \times 10^{-3} \times 10^{6}\ \mu\text{s} = 10^{5}\ \mu\text{s}$$

Example 3.5

The period of a signal is 100 ms. What is its frequency in kilohertz?

Solution

First we change 100 ms to seconds, and then we calculate the frequency from the period (1 Hz = 10^{-3} kHz).

$$100 \text{ ms} = 100 \times 10^{-3} \text{ s} = 10^{-1} \text{ s}$$

$$f = \frac{1}{T} = \frac{1}{10^{-1}} \text{ Hz} = 10 \text{ Hz} = 10 \times 10^{-3} \text{ kHz} = 10^{-2} \text{ kHz}$$

Note

Frequency is the rate of change with respect to time.

Change in a short span of time means high frequency.

Change over a long span of time means low frequency.

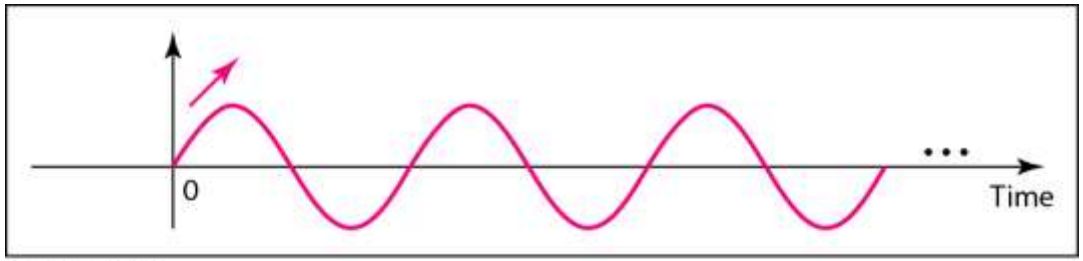
Note

If a signal does not change at all, its frequency is zero.
If a signal changes instantaneously, its frequency is infinite.

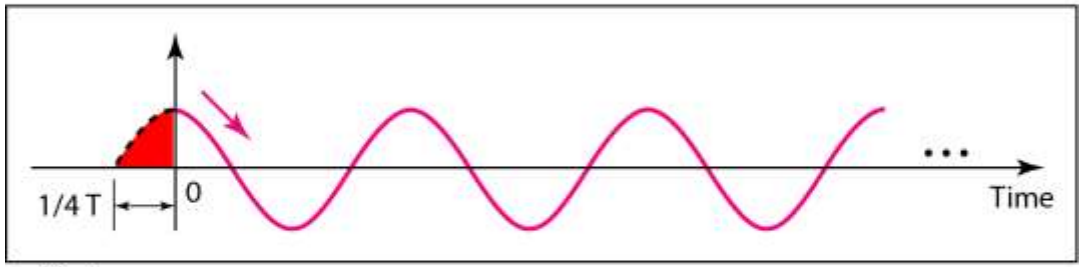
Note

Phase describes the position of the waveform relative to time 0.

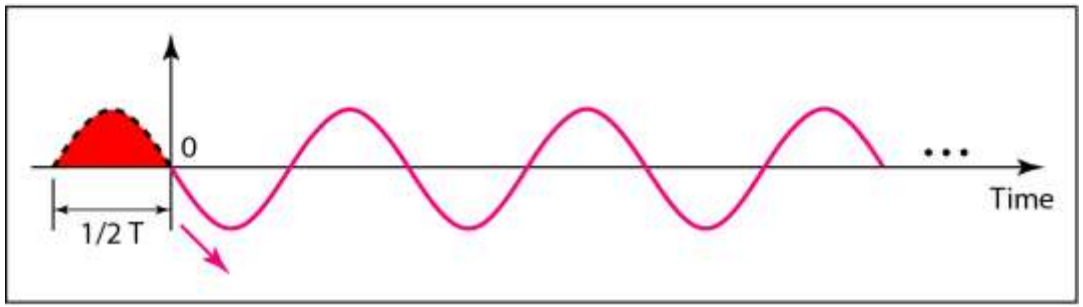
Figure 3.5 *Three sine waves with the same amplitude and frequency, but different phases*



a. 0 degrees



b. 90 degrees



c. 180 degrees

Example 3.6

A sine wave is offset 1/6 cycle with respect to time 0. What is its phase in degrees and radians?

Solution

We know that 1 complete cycle is 360°. Therefore, 1/6 cycle is

$$\frac{1}{6} \times 360 = 60° = 60 \times \frac{2\pi}{360} \text{ rad} = \frac{\pi}{3} \text{ rad} = 1.046 \text{ rad}$$

Figure 3.6 *Wavelength and period*

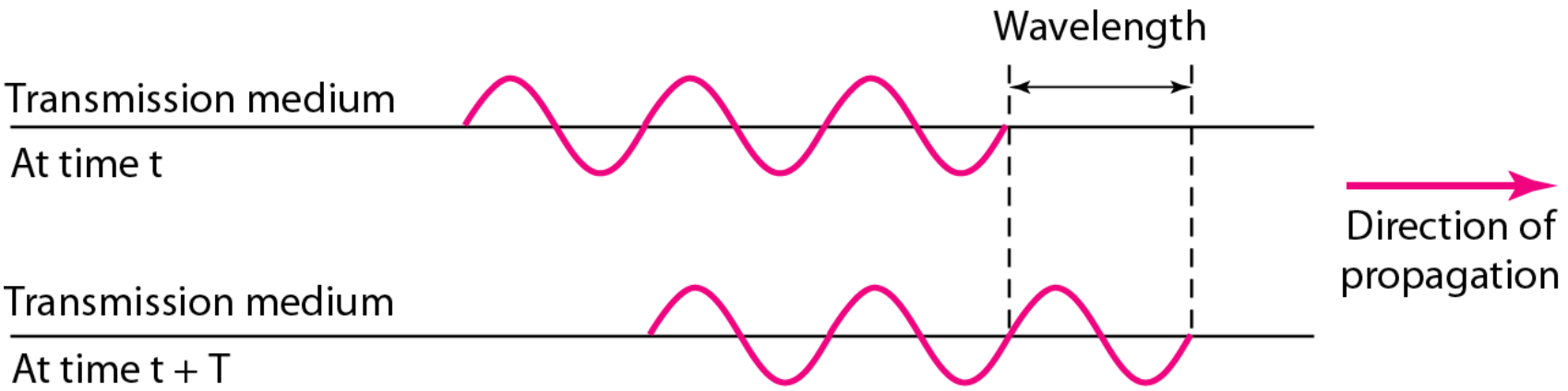
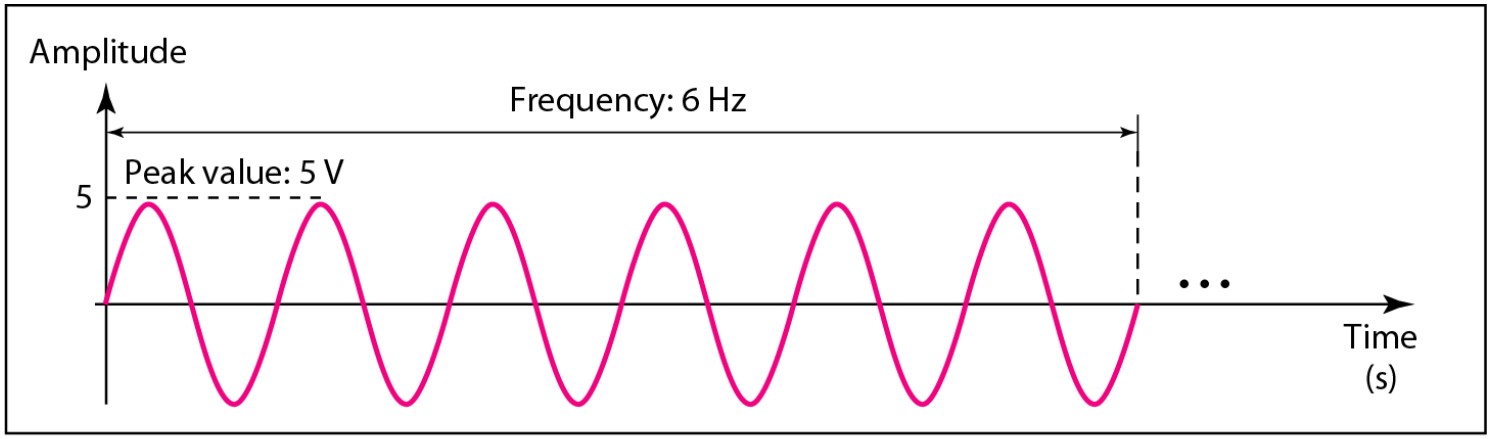
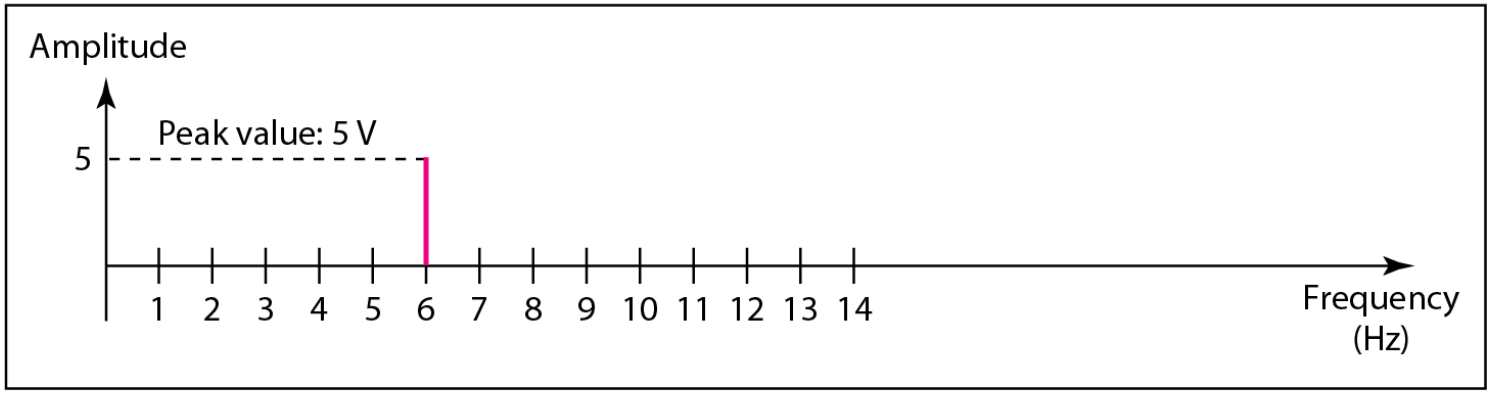


Figure 3.7 *The time-domain and frequency-domain plots of a sine wave*



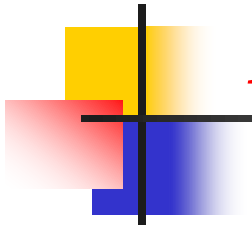
a. A sine wave in the time domain (peak value: 5 V, frequency: 6 Hz)



b. The same sine wave in the frequency domain (peak value: 5 V, frequency: 6 Hz)

Note

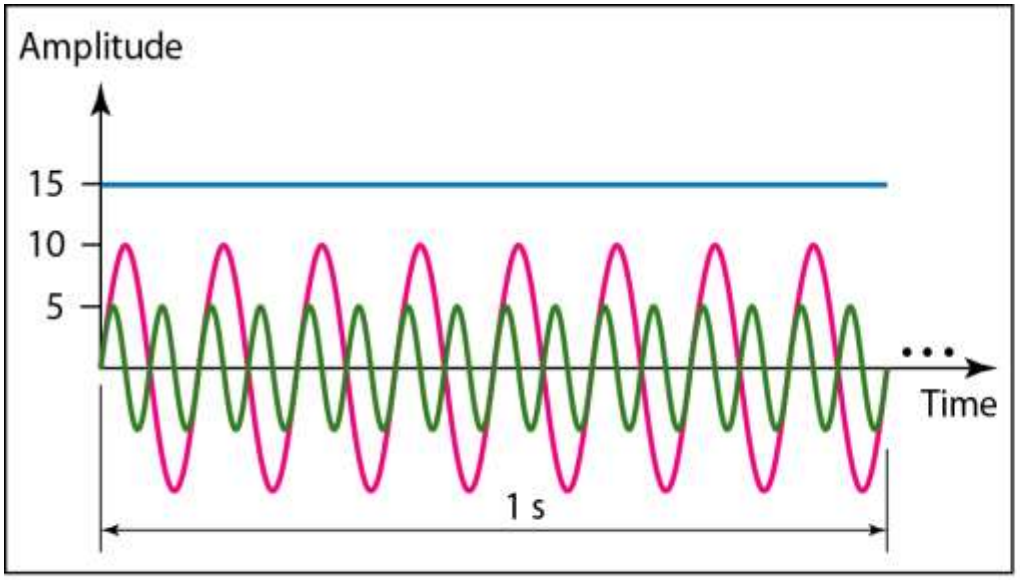
A complete sine wave in the time domain can be represented by one single spike in the frequency domain.



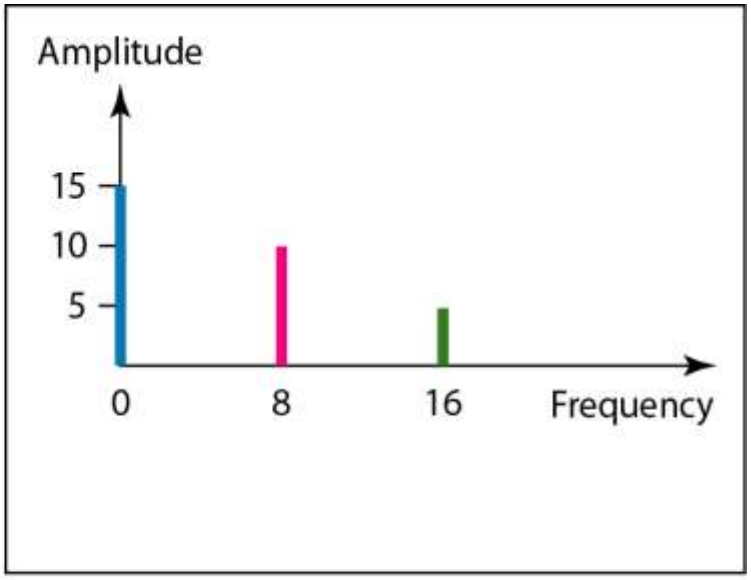
Example 3.7

The frequency domain is more compact and useful when we are dealing with more than one sine wave. For example, Figure 3.8 shows three sine waves, each with different amplitude and frequency. All can be represented by three spikes in the frequency domain.

Figure 3.8 *The time domain and frequency domain of three sine waves*



a. Time-domain representation of three sine waves with frequencies 0, 8, and 16



b. Frequency-domain representation of the same three signals

Note

A single-frequency sine wave is not useful in data communications; we need to send a composite signal, a signal made of many simple sine waves.

Note

According to Fourier analysis, any composite signal is a combination of simple sine waves with different frequencies, amplitudes, and phases. Fourier analysis is discussed in Appendix C.

Note

If the composite signal is periodic, the decomposition gives a series of signals with discrete frequencies;
if the composite signal is nonperiodic, the decomposition gives a combination of sine waves with continuous frequencies.

Example 3.8

Figure 3.9 shows a periodic composite signal with frequency f. This type of signal is not typical of those found in data communications. We can consider it to be three alarm systems, each with a different frequency. The analysis of this signal can give us a good understanding of how to decompose signals.

Figure 3.9 *A composite periodic signal*

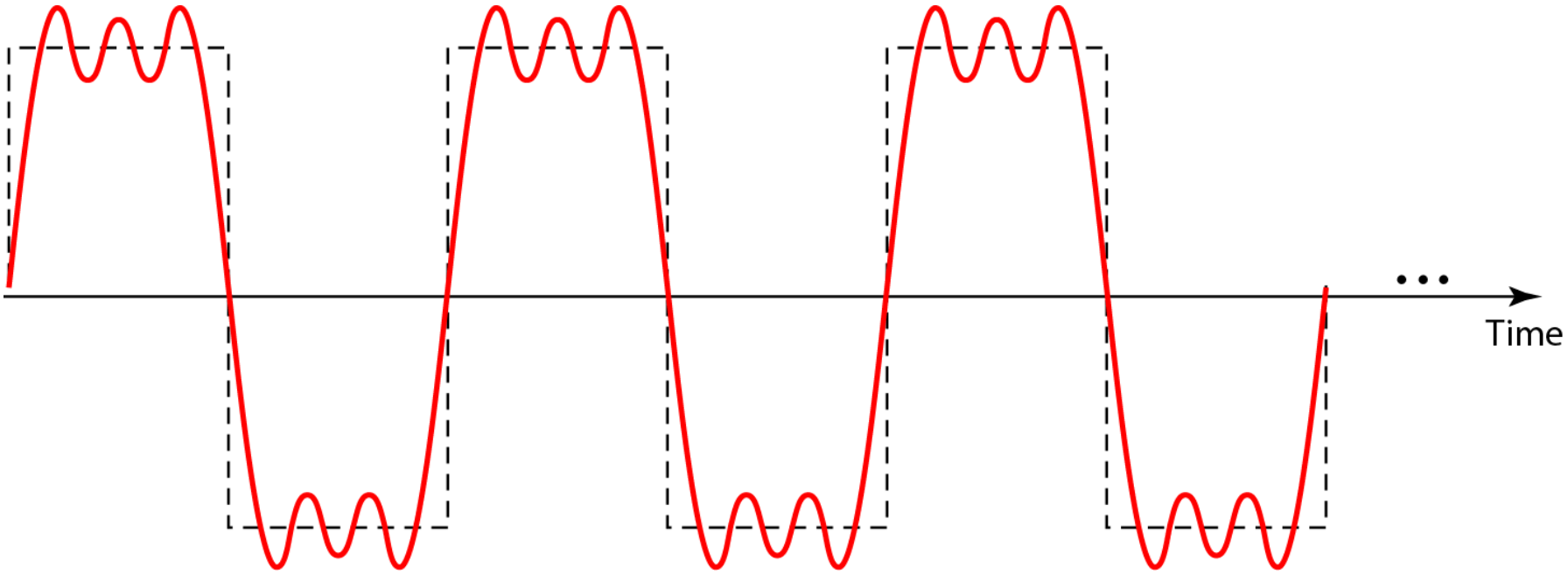
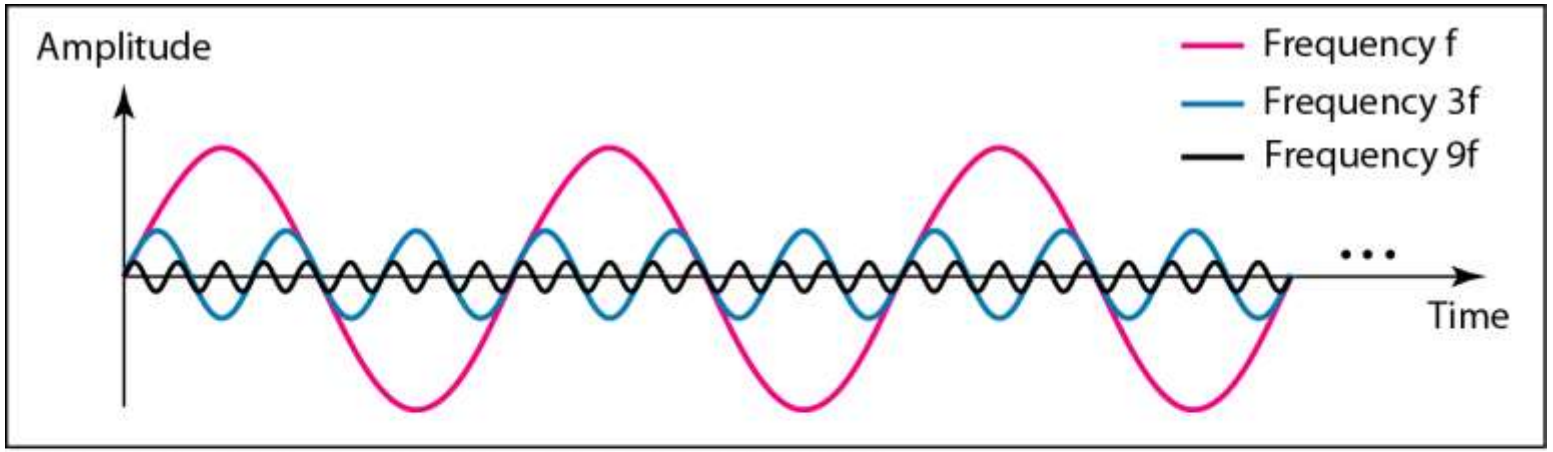
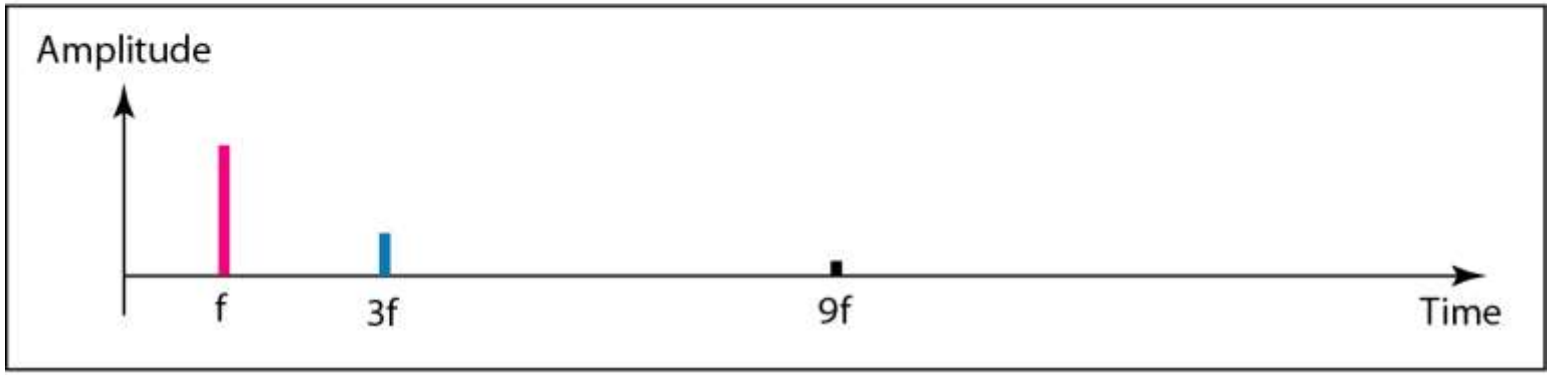


Figure 3.10 *Decomposition of a composite periodic signal in the time and frequency domains*



a. Time-domain decomposition of a composite signal

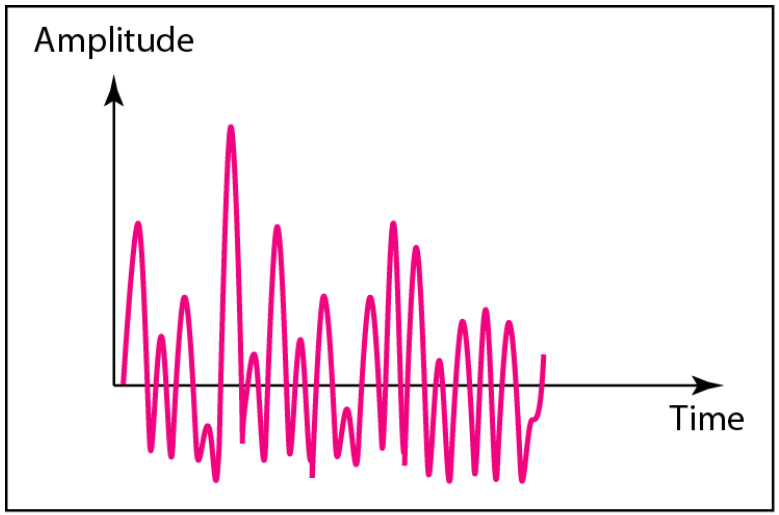


b. Frequency-domain decomposition of the composite signal

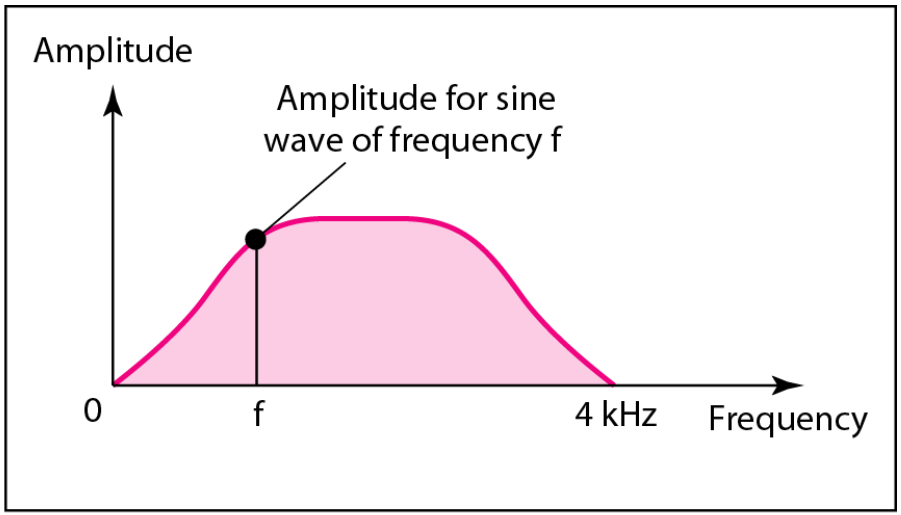
Example 3.9

Figure 3.11 shows a nonperiodic composite signal. It can be the signal created by a microphone or a telephone set when a word or two is pronounced. In this case, the composite signal cannot be periodic, because that implies that we are repeating the same word or words with exactly the same tone.

Figure 3.11 *The time and frequency domains of a nonperiodic signal*



a. Time domain

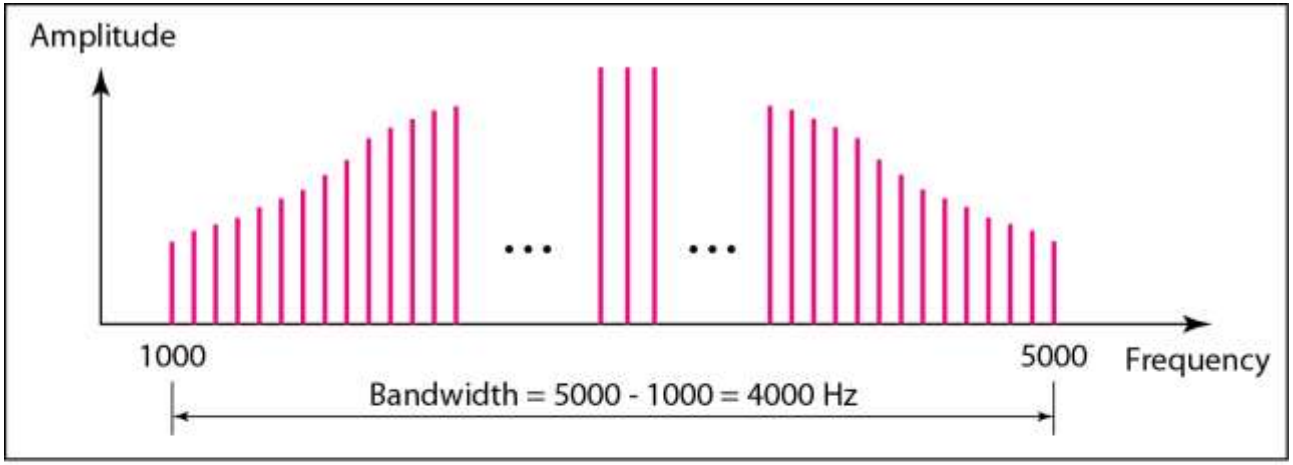


b. Frequency domain

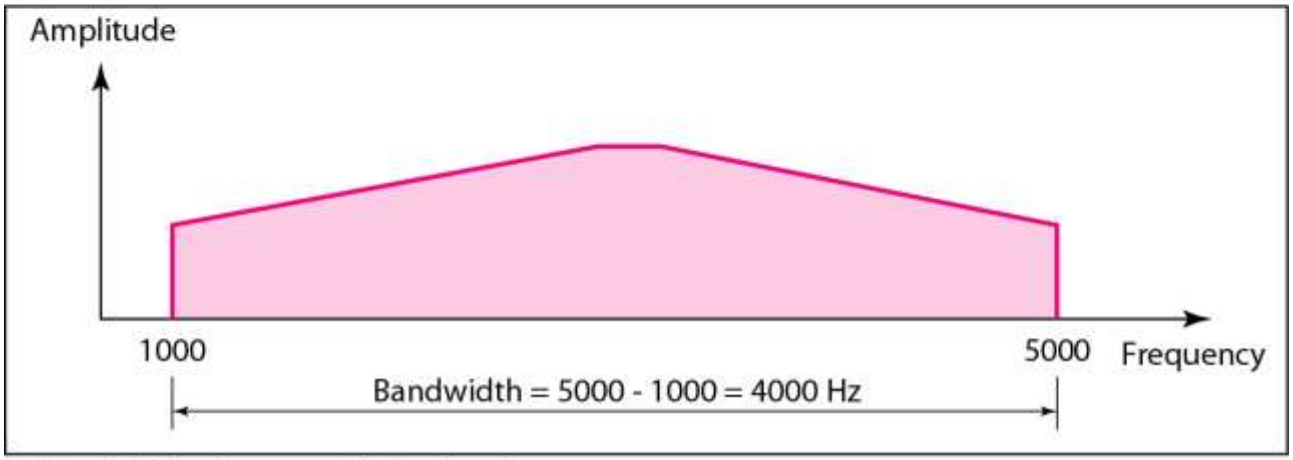
Note

The bandwidth of a composite signal is the difference between the highest and the lowest frequencies contained in that signal.

Figure 3.12 *The bandwidth of periodic and nonperiodic composite signals*



a. Bandwidth of a periodic signal



b. Bandwidth of a nonperiodic signal

Example 3.10

If a periodic signal is decomposed into five sine waves with frequencies of 100, 300, 500, 700, and 900 Hz, what is its bandwidth? Draw the spectrum, assuming all components have a maximum amplitude of 10 V.

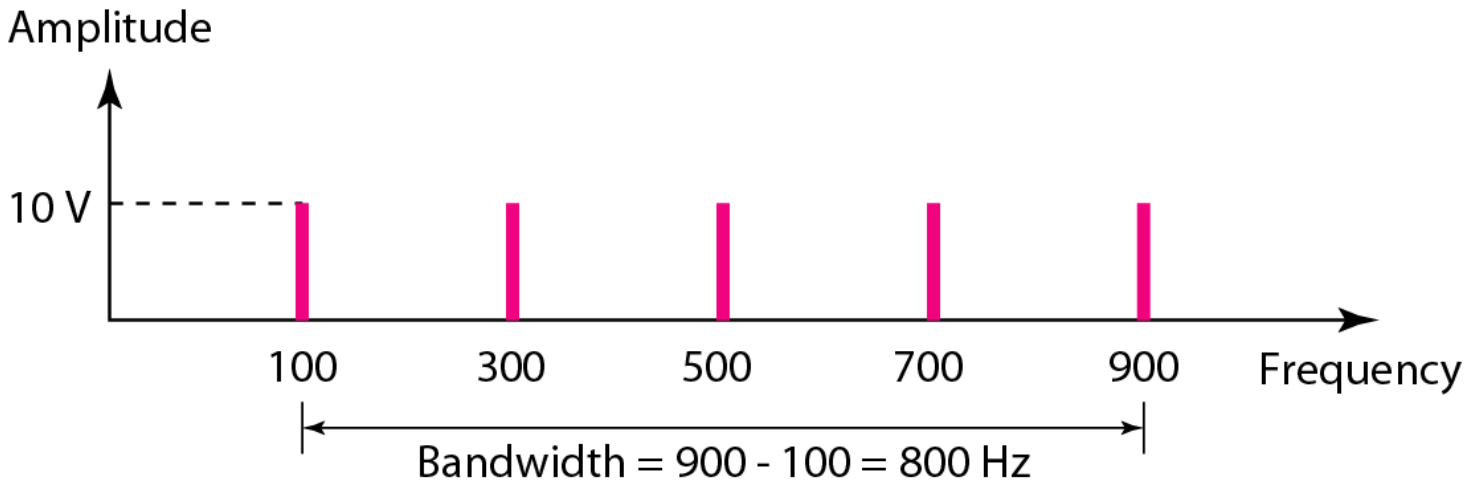
Solution

Let f_h be the highest frequency, f_l the lowest frequency, and B the bandwidth. Then

$$B = f_h - f_l = 900 - 100 = 800 \text{ Hz}$$

The spectrum has only five spikes, at 100, 300, 500, 700, and 900 Hz (see Figure 3.13).

Figure 3.13 *The bandwidth for Example 3.10*



Example 3.11

A periodic signal has a bandwidth of 20 Hz. The highest frequency is 60 Hz. What is the lowest frequency? Draw the spectrum if the signal contains all frequencies of the same amplitude.

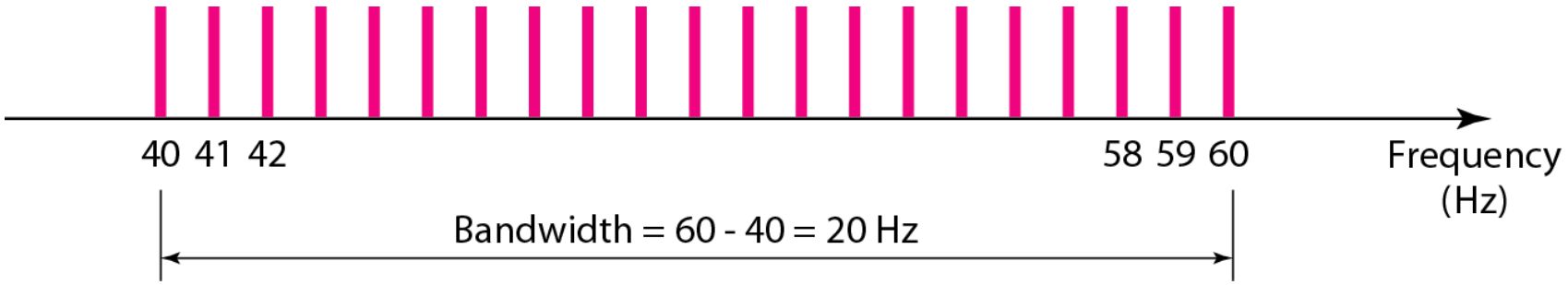
Solution

Let f_h be the highest frequency, f_l the lowest frequency, and B the bandwidth. Then

$$B = f_h - f_l \Rightarrow 20 = 60 - f_l \Rightarrow f_l = 60 - 20 = 40 \text{ Hz}$$

The spectrum contains all integer frequencies. We show this by a series of spikes (see Figure 3.14).

Figure 3.14 *The bandwidth for Example 3.11*



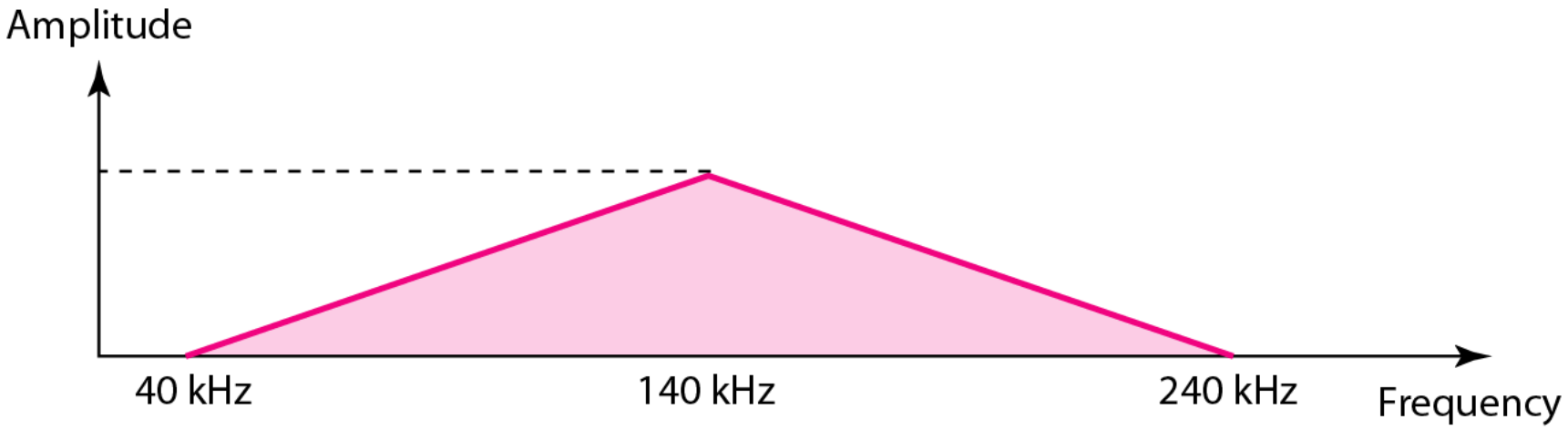
Example 3.12

A nonperiodic composite signal has a bandwidth of 200 kHz, with a middle frequency of 140 kHz and peak amplitude of 20 V. The two extreme frequencies have an amplitude of 0. Draw the frequency domain of the signal.

Solution

The lowest frequency must be at 40 kHz and the highest at 240 kHz. Figure 3.15 shows the frequency domain and the bandwidth.

Figure 3.15 *The bandwidth for Example 3.12*



Example 3.13

An example of a nonperiodic composite signal is the signal propagated by an AM radio station. In the United States, each AM radio station is assigned a 10-kHz bandwidth. The total bandwidth dedicated to AM radio ranges from 530 to 1700 kHz. We will show the rationale behind this 10-kHz bandwidth in Chapter 5.

Example 3.14

Another example of a nonperiodic composite signal is the signal propagated by an FM radio station. In the United States, each FM radio station is assigned a 200-kHz bandwidth. The total bandwidth dedicated to FM radio ranges from 88 to 108 MHz. We will show the rationale behind this 200-kHz bandwidth in Chapter 5.

Example 3.15

Another example of a nonperiodic composite signal is the signal received by an old-fashioned analog black-and-white TV. A TV screen is made up of pixels. If we assume a resolution of 525 × 700, we have 367,500 pixels per screen. If we scan the screen 30 times per second, this is 367,500 × 30 = 11,025,000 pixels per second. The worst-case scenario is alternating black and white pixels. We can send 2 pixels per cycle. Therefore, we need 11,025,000 / 2 = 5,512,500 cycles per second, or Hz. The bandwidth needed is 5.5125 MHz.

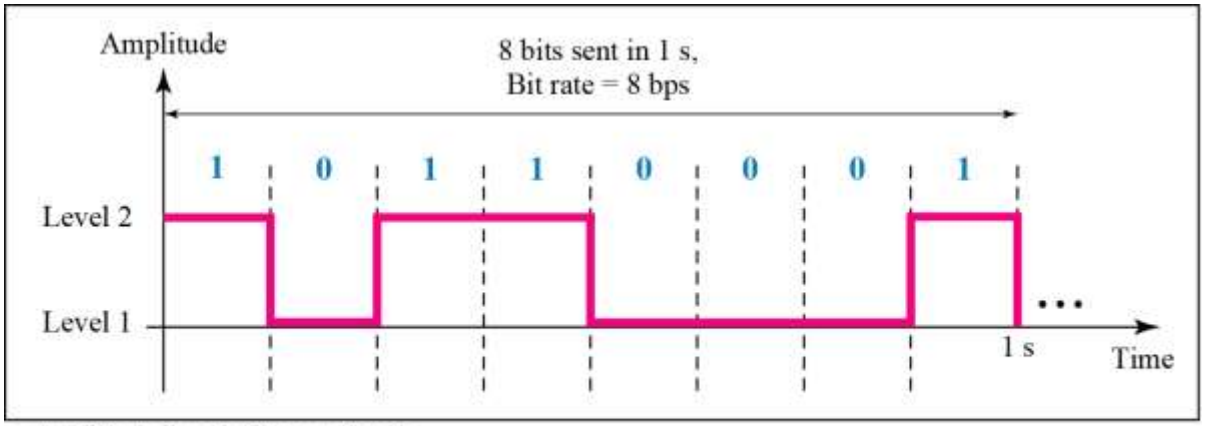
3-3 DIGITAL SIGNALS

*In addition to being represented by an analog signal, information can also be represented by a **digital signal**. For example, a 1 can be encoded as a positive voltage and a 0 as zero voltage. A digital signal can have more than two levels. In this case, we can send more than 1 bit for each level.*

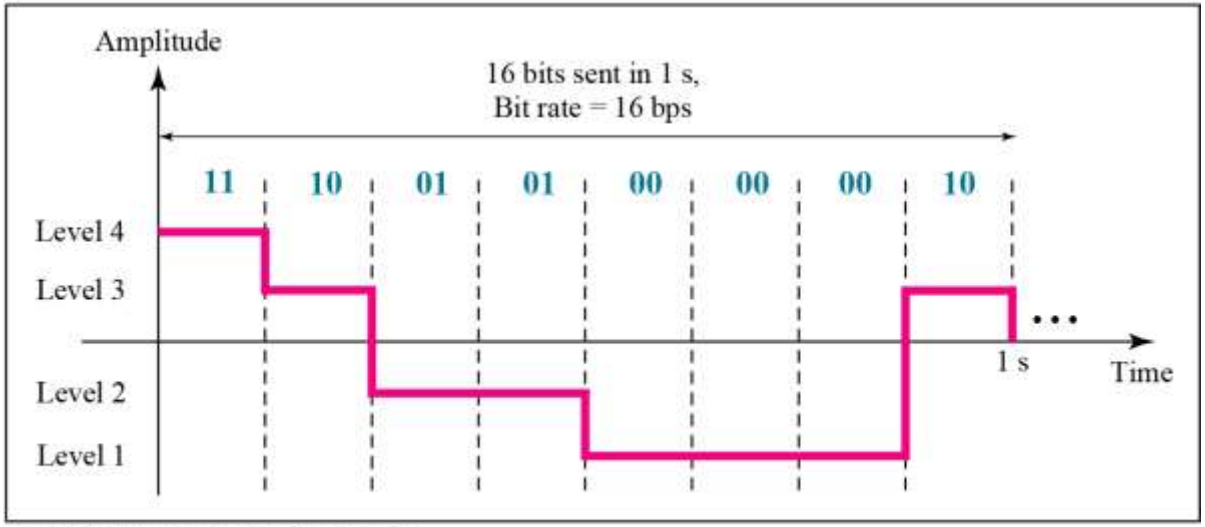
Topics discussed in this section:

- Bit Rate
- Bit Length
- Digital Signal as a Composite Analog Signal
- Application Layer

Figure 3.16 *Two digital signals: one with two signal levels and the other with four signal levels*



a. A digital signal with two levels



b. A digital signal with four levels

Note

Appendix C reviews information about exponential and logarithmic functions.

Example 3.16

A digital signal has eight levels. How many bits are needed per level? We calculate the number of bits from the formula

$$\text{Number of bits per level} = \log_2 8 = 3$$

Each signal level is represented by 3 bits.

Example 3.17

A digital signal has nine levels. How many bits are needed per level? We calculate the number of bits by using the formula. Each signal level is represented by 3.17 bits. However, this answer is not realistic. The number of bits sent per level needs to be an integer as well as a power of 2. For this example, 4 bits can represent one level.

Example 3.18

Assume we need to download text documents at the rate of 100 pages per minute. What is the required bit rate of the channel?

Solution

A page is an average of 24 lines with 80 characters in each line. If we assume that one character requires 8 bits, the bit rate is

$$100 \times 24 \times 80 \times 8 = 1{,}636{,}000 \text{ bps} = 1.636 \text{ Mbps}$$

Example 3.19

A digitized voice channel, as we will see in Chapter 4, is made by digitizing a 4-kHz bandwidth analog voice signal. We need to sample the signal at twice the highest frequency (two samples per hertz). We assume that each sample requires 8 bits. What is the required bit rate?

Solution

The bit rate can be calculated as

$$2 \times 4000 \times 8 = 64{,}000 \text{ bps} = 64 \text{ kbps}$$

Example 3.20

What is the bit rate for high-definition TV (HDTV)?

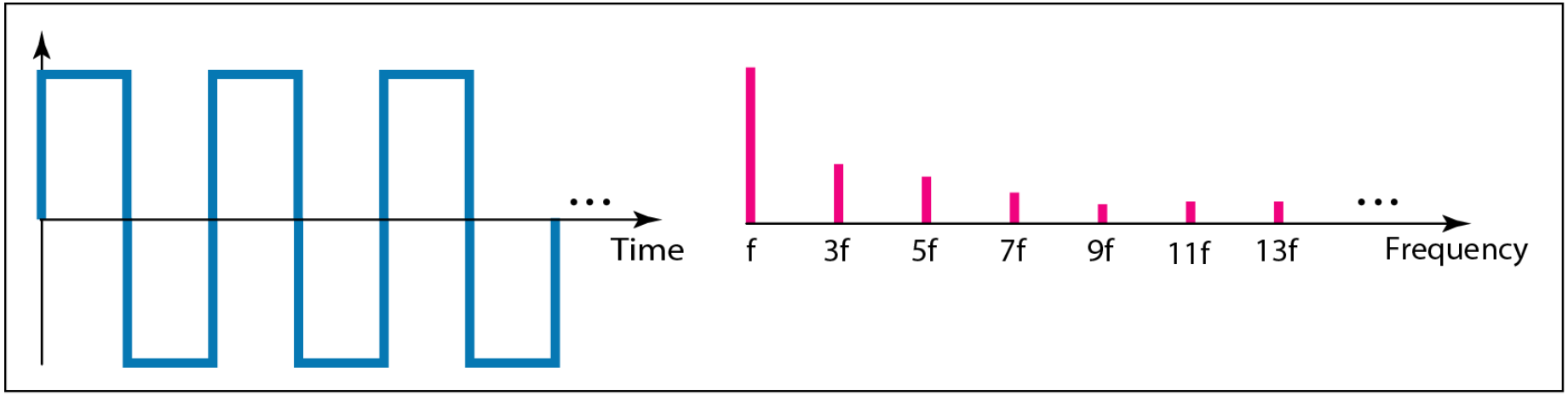
Solution

HDTV uses digital signals to broadcast high quality video signals. The HDTV screen is normally a ratio of 16 : 9. There are 1920 by 1080 pixels per screen, and the screen is renewed 30 times per second. Twenty-four bits represents one color pixel.

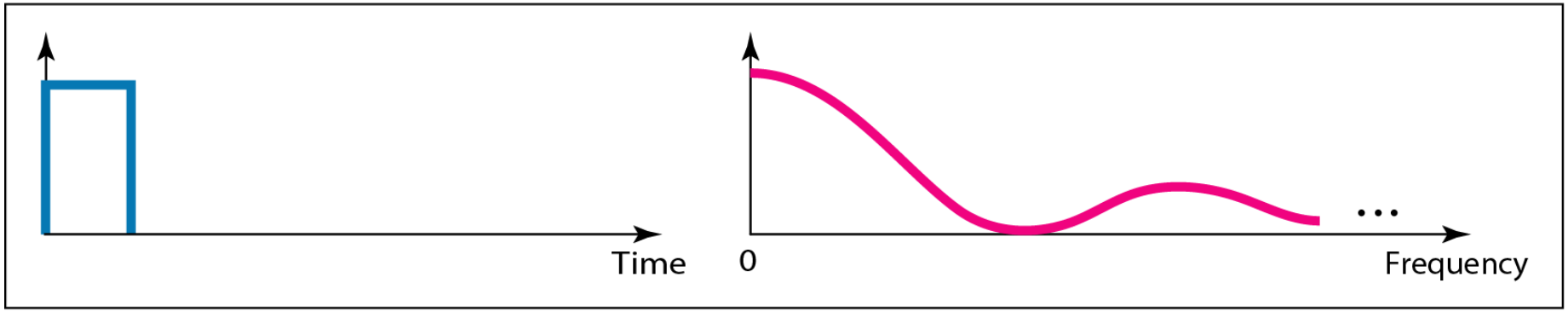
$$1920 \times 1080 \times 30 \times 24 = 1{,}492{,}992{,}000 \text{ or } 1.5 \text{ Gbps}$$

The TV stations reduce this rate to 20 to 40 Mbps through compression.

Figure 3.17 *The time and frequency domains of periodic and nonperiodic digital signals*

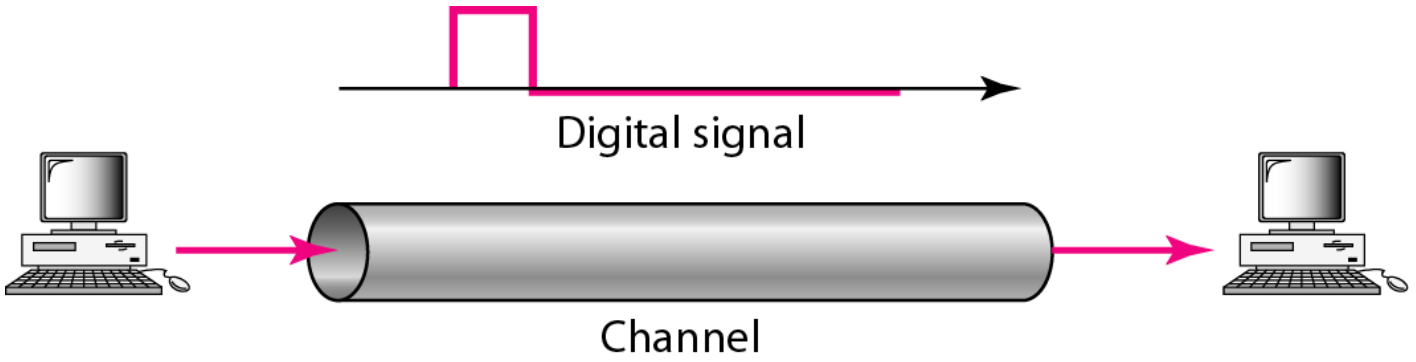


a. Time and frequency domains of periodic digital signal



b. Time and frequency domains of nonperiodic digital signal

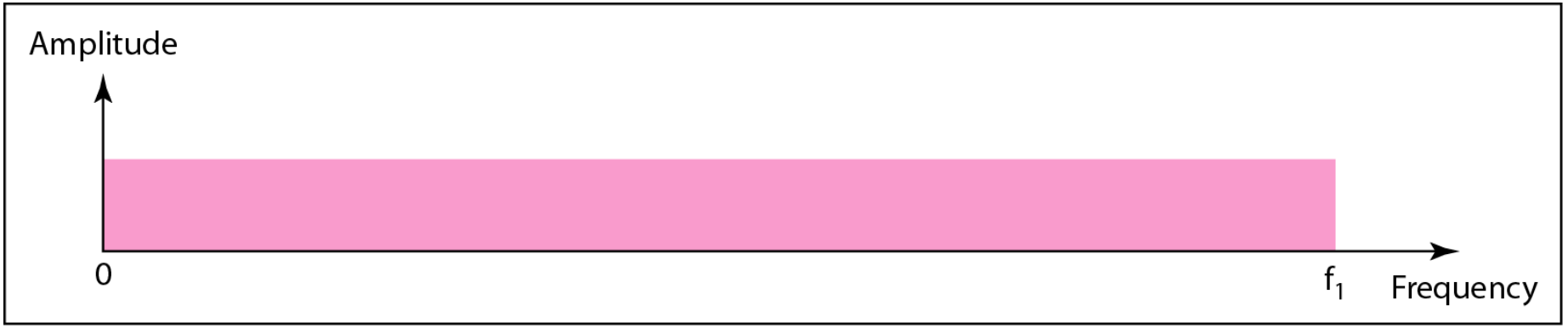
Figure 3.18 *Baseband transmission*



Note

A digital signal is a composite analog signal with an infinite bandwidth.

Figure 3.19 *Bandwidths of two low-pass channels*

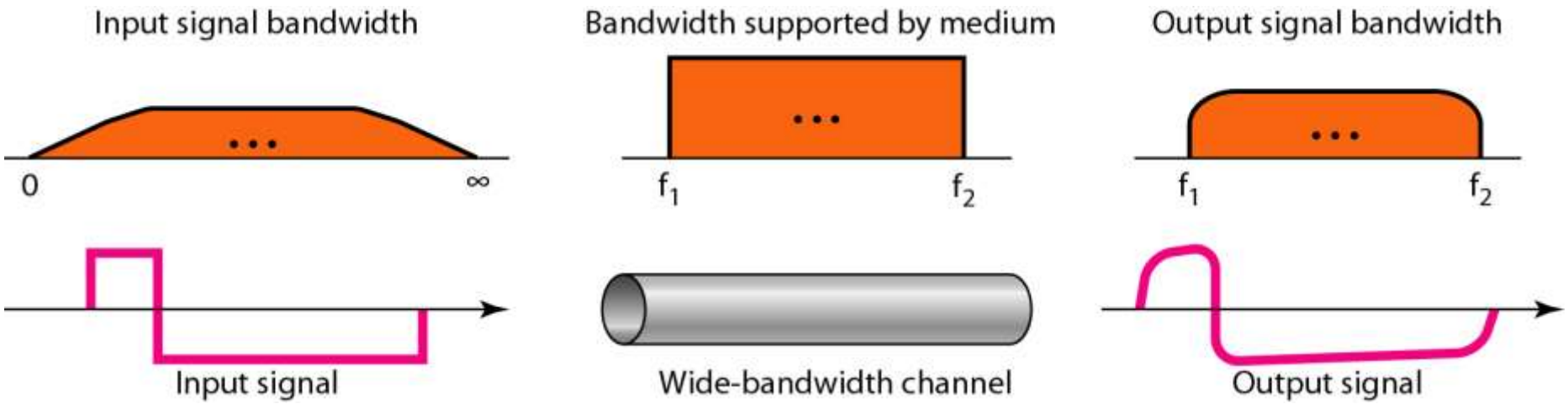


a. Low-pass channel, wide bandwidth



b. Low-pass channel, narrow bandwidth

Figure 3.20 *Baseband transmission using a dedicated medium*



Note

Baseband transmission of a digital signal that preserves the shape of the digital signal is possible only if we have a low-pass channel with an infinite or very wide bandwidth.

Example 3.21

An example of a dedicated channel where the entire bandwidth of the medium is used as one single channel is a LAN. Almost every wired LAN today uses a dedicated channel for two stations communicating with each other. In a bus topology LAN with multipoint connections, only two stations can communicate with each other at each moment in time (timesharing); the other stations need to refrain from sending data. In a star topology LAN, the entire channel between each station and the hub is used for communication between these two entities. We study LANs in Chapter 14.

Figure 3.21 *Rough approximation of a digital signal using the first harmonic for worst case*

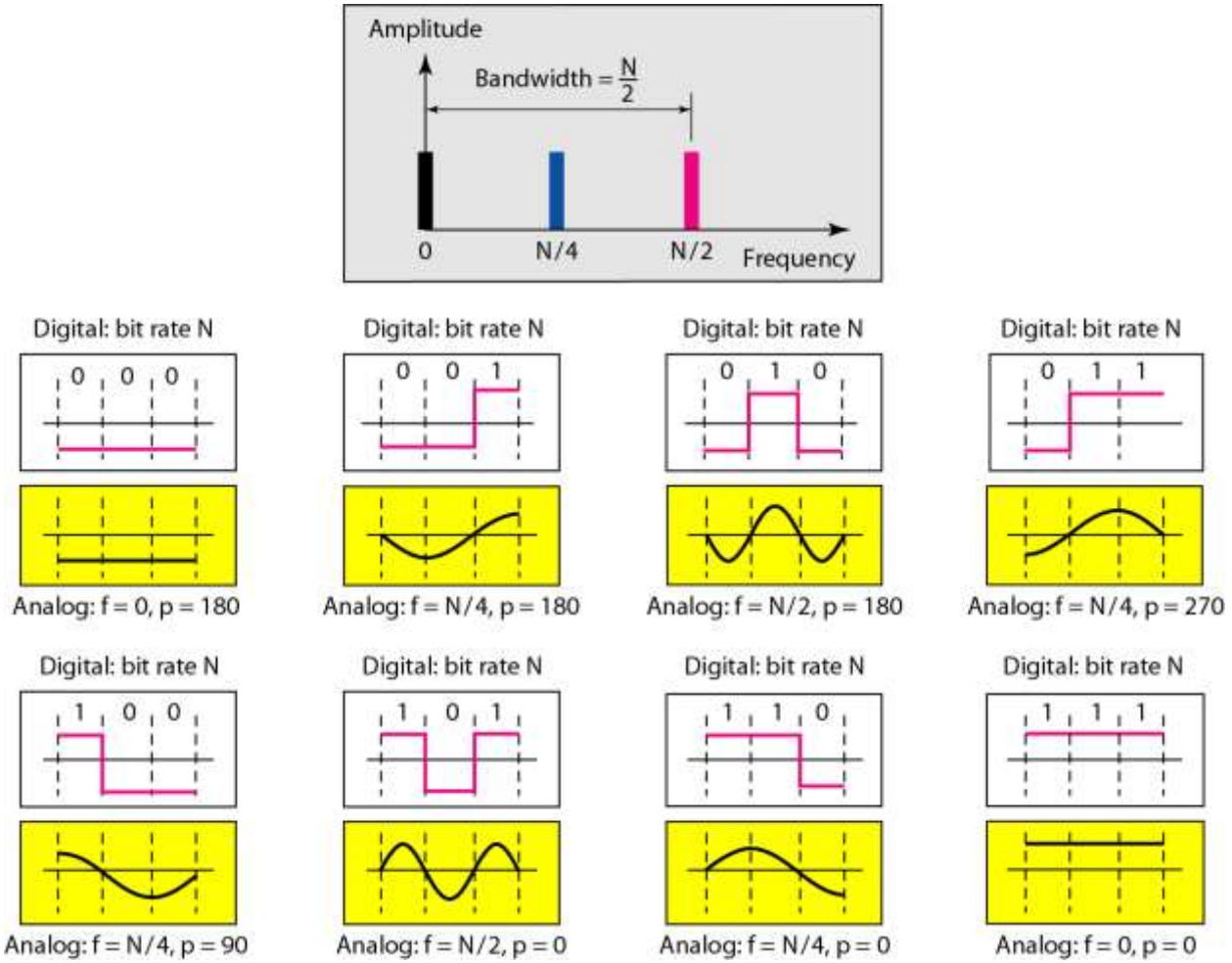
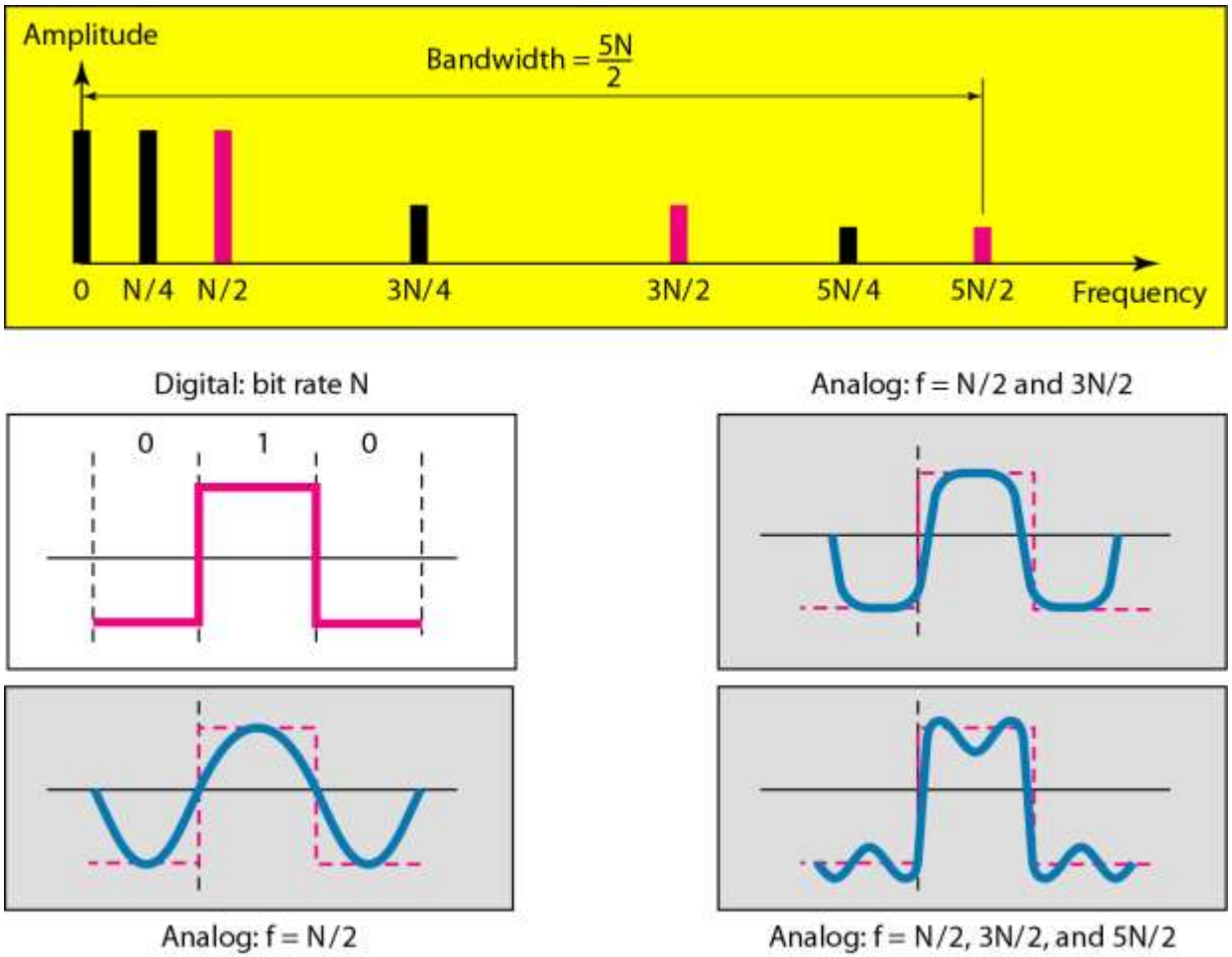


Figure 3.22 *Simulating a digital signal with first three harmonics*



Note

In baseband transmission, the required bandwidth is proportional to the bit rate; if we need to send bits faster, we need more bandwidth.

Table 3.2 *Bandwidth requirements*

Bit Rate	*Harmonic 1*	*Harmonics 1, 3*	*Harmonics 1, 3, 5*
$n = 1$ kbps	$B = 500$ Hz	$B = 1.5$ kHz	$B = 2.5$ kHz
$n = 10$ kbps	$B = 5$ kHz	$B = 15$ kHz	$B = 25$ kHz
$n = 100$ kbps	$B = 50$ kHz	$B = 150$ kHz	$B = 250$ kHz

Example 3.22

What is the required bandwidth of a low-pass channel if we need to send 1 Mbps by using baseband transmission?

Solution

The answer depends on the accuracy desired.

a. The minimum bandwidth, is B = bit rate /2, or 500 kHz.

b. A better solution is to use the first and the third harmonics with B = 3 × 500 kHz = 1.5 MHz.

c. Still a better solution is to use the first, third, and fifth harmonics with B = 5 × 500 kHz = 2.5 MHz.

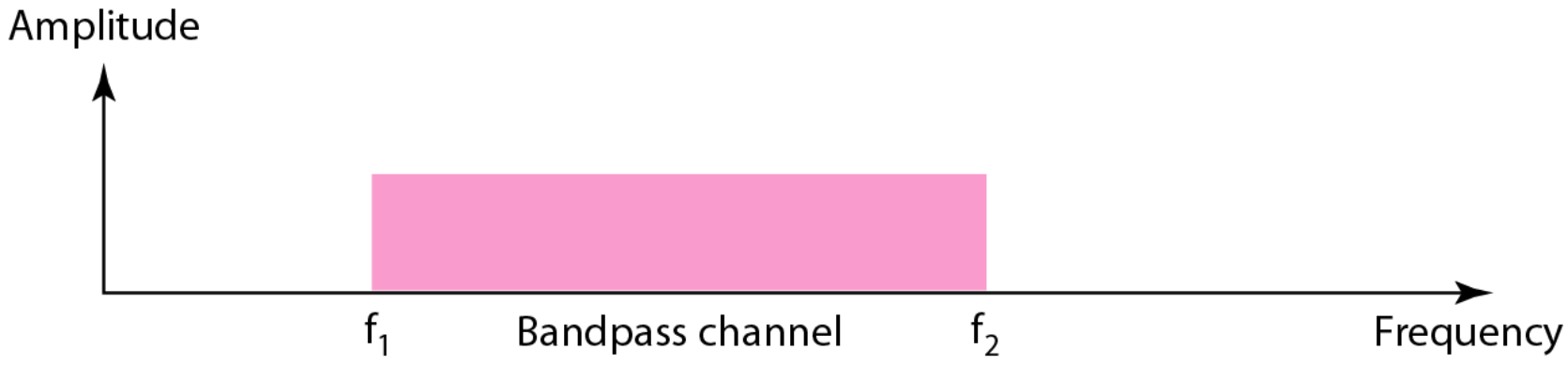
Example 3.22

We have a low-pass channel with bandwidth 100 kHz. What is the maximum bit rate of this channel?

Solution

The maximum bit rate can be achieved if we use the first harmonic. The bit rate is 2 times the available bandwidth, or 200 kbps.

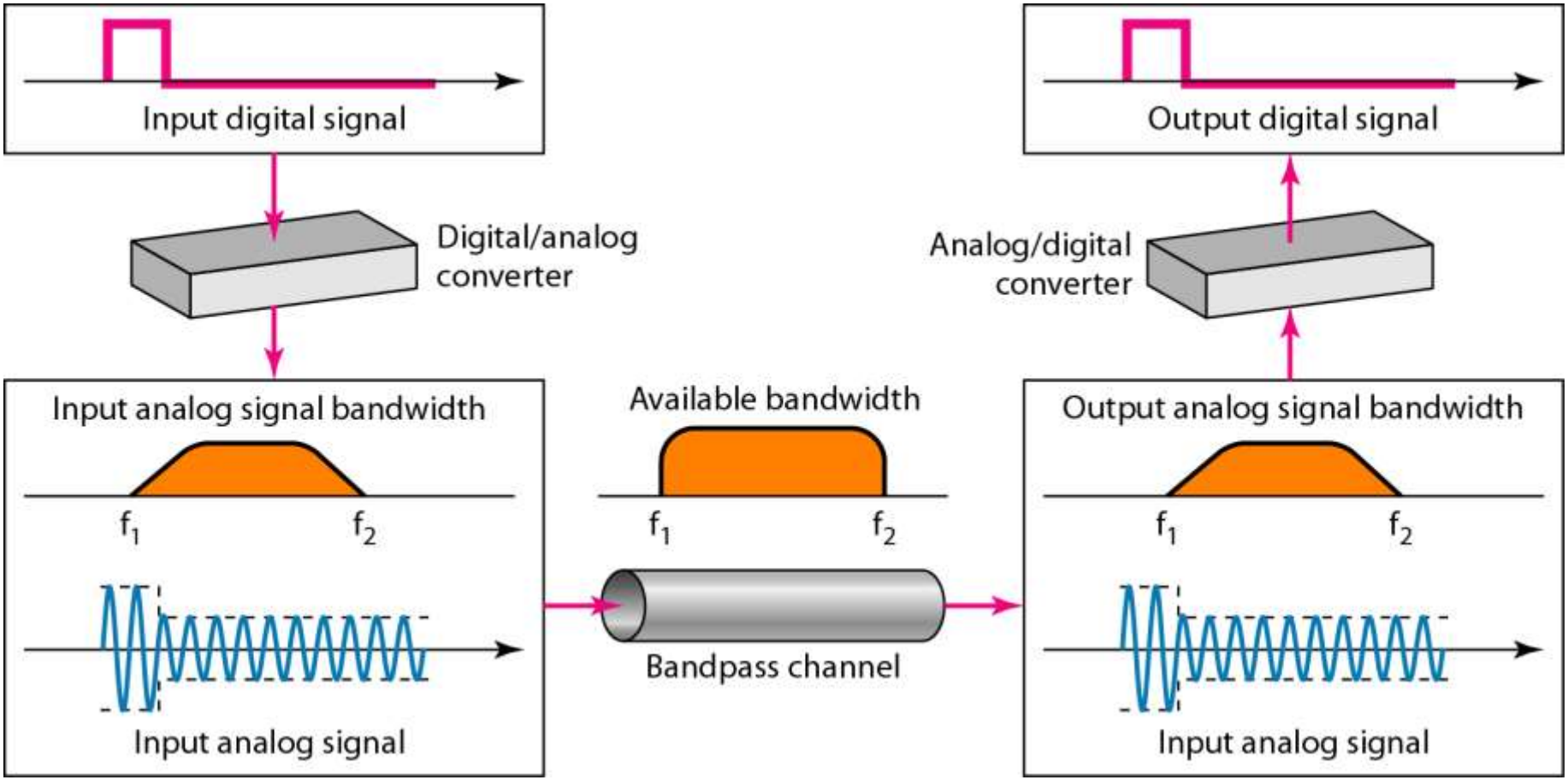
Figure 3.23 *Bandwidth of a bandpass channel*



Note

If the available channel is a bandpass channel, we cannot send the digital signal directly to the channel; we need to convert the digital signal to an analog signal before transmission.

Figure 3.24 *Modulation of a digital signal for transmission on a bandpass channel*



Example 3.24

An example of broadband transmission using modulation is the sending of computer data through a telephone subscriber line, the line connecting a resident to the central telephone office. These lines are designed to carry voice with a limited bandwidth. The channel is considered a bandpass channel. We convert the digital signal from the computer to an analog signal, and send the analog signal. We can install two converters to change the digital signal to analog and vice versa at the receiving end. The converter, in this case, is called a modem which we discuss in detail in Chapter 5.

Example 3.25

A second example is the digital cellular telephone. For better reception, digital cellular phones convert the analog voice signal to a digital signal (see Chapter 16). Although the bandwidth allocated to a company providing digital cellular phone service is very wide, we still cannot send the digital signal without conversion. The reason is that we only have a bandpass channel available between caller and callee. We need to convert the digitized voice to a composite analog signal before sending.

3-4 TRANSMISSION IMPAIRMENT

Signals travel through transmission media, which are not perfect. The imperfection causes signal impairment. This means that the signal at the beginning of the medium is not the same as the signal at the end of the medium. What is sent is not what is received. Three causes of impairment are ***attenuation****,* ***distortion****, and* ***noise****.*

Topics discussed in this section:

Attenuation
Distortion
Noise

Figure 3.25 *Causes of impairment*

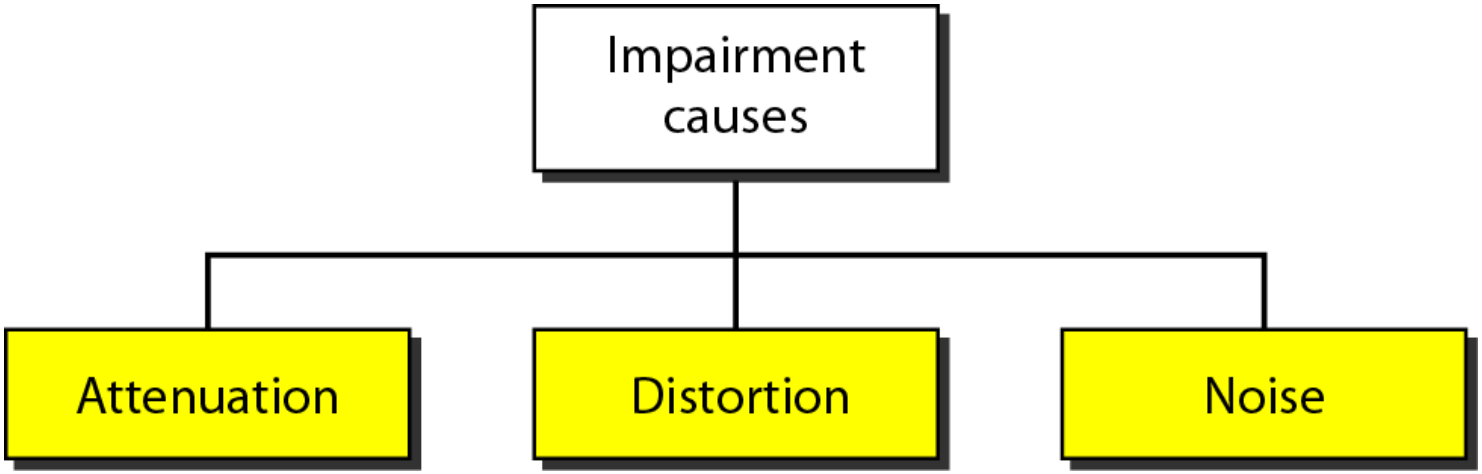
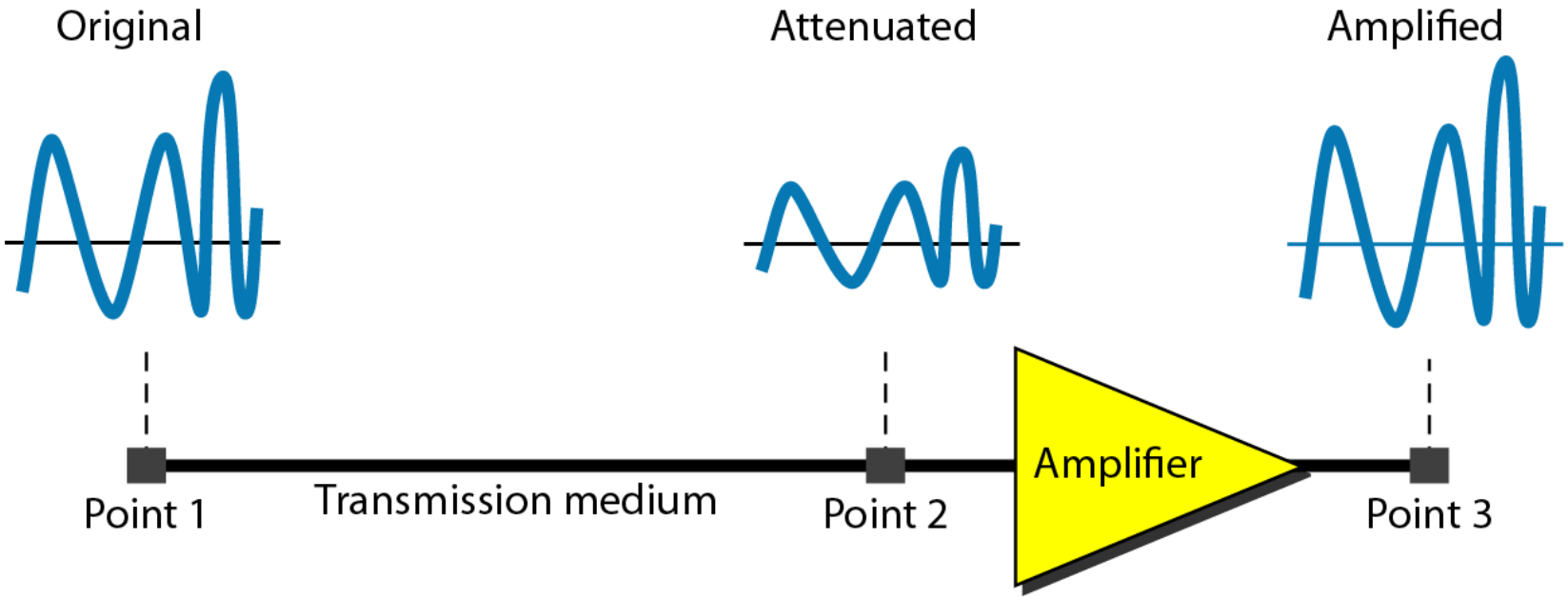


Figure 3.26 *Attenuation*



Example 3.26

Suppose a signal travels through a transmission medium and its power is reduced to one-half. This means that P_2 is $(1/2)P_1$. In this case, the attenuation (loss of power) can be calculated as

$$10 \log_{10} \frac{P_2}{P_1} = 10 \log_{10} \frac{0.5 P_1}{P_1} = 10 \log_{10} 0.5 = 10(-0.3) = -3 \text{ dB}$$

A loss of 3 dB (–3 dB) is equivalent to losing one-half the power.

Example 3.27

A signal travels through an amplifier, and its power is increased 10 times. This means that $P_2 = 10P_1$. In this case, the amplification (gain of power) can be calculated as

$$10 \log_{10} \frac{P_2}{P_1} = 10 \log_{10} \frac{10P_1}{P_1}$$

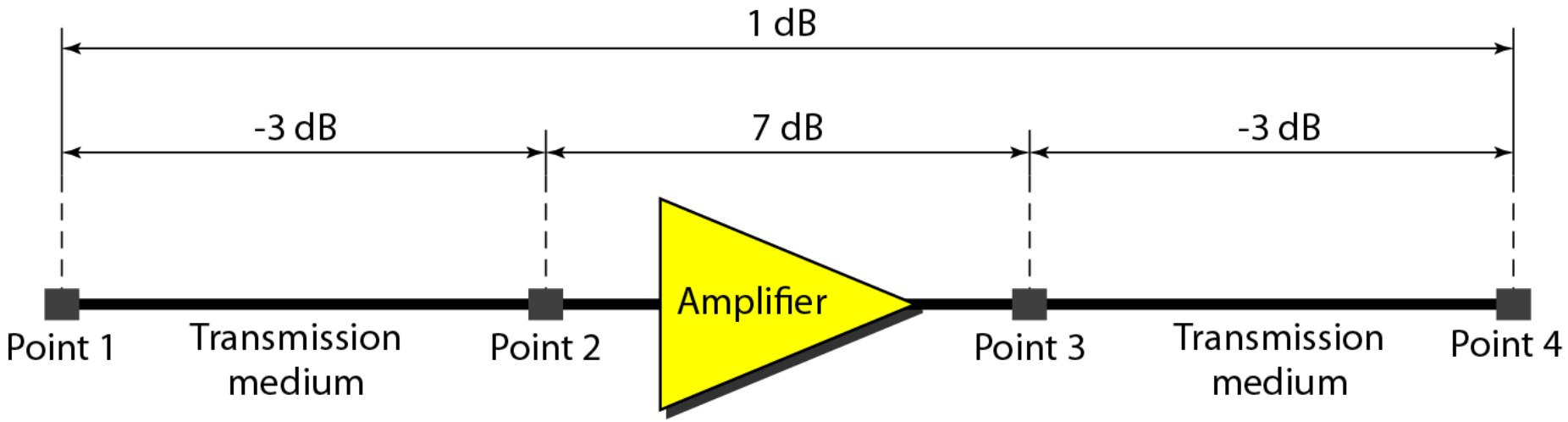
$$= 10 \log_{10} 10 = 10(1) = 10 \text{ dB}$$

Example 3.28

One reason that engineers use the decibel to measure the changes in the strength of a signal is that decibel numbers can be added (or subtracted) when we are measuring several points (cascading) instead of just two. In Figure 3.27 a signal travels from point 1 to point 4. In this case, the decibel value can be calculated as

$$\text{dB} = -3 + 7 - 3 = +1$$

Figure 3.27 *Decibels for Example 3.28*



Example 3.29

Sometimes the decibel is used to measure signal power in milliwatts. In this case, it is referred to as dB_m and is calculated as $dB_m = 10 \log10 P_m$, where P_m is the power in milliwatts. Calculate the power of a signal with $dB_m = -30$.

Solution

We can calculate the power in the signal as

$$\mathrm{dB_m} = 10 \log_{10} P_m = -30$$

$$\log_{10} P_m = -3 \qquad P_m = 10^{-3}\ \mathrm{mW}$$

Example 3.30

The loss in a cable is usually defined in decibels per kilometer (dB/km). If the signal at the beginning of a cable with −0.3 dB/km has a power of 2 mW, what is the power of the signal at 5 km?

Solution

The loss in the cable in decibels is 5 × (−0.3) = −1.5 dB. We can calculate the power as

$$\text{dB} = 10 \log_{10} \frac{P_2}{P_1} = -1.5$$

$$\frac{P_2}{P_1} = 10^{-0.15} = 0.71$$

$$P_2 = 0.71 P_1 = 0.7 \times 2 = 1.4 \text{ mW}$$

Figure 3.28 *Distortion*

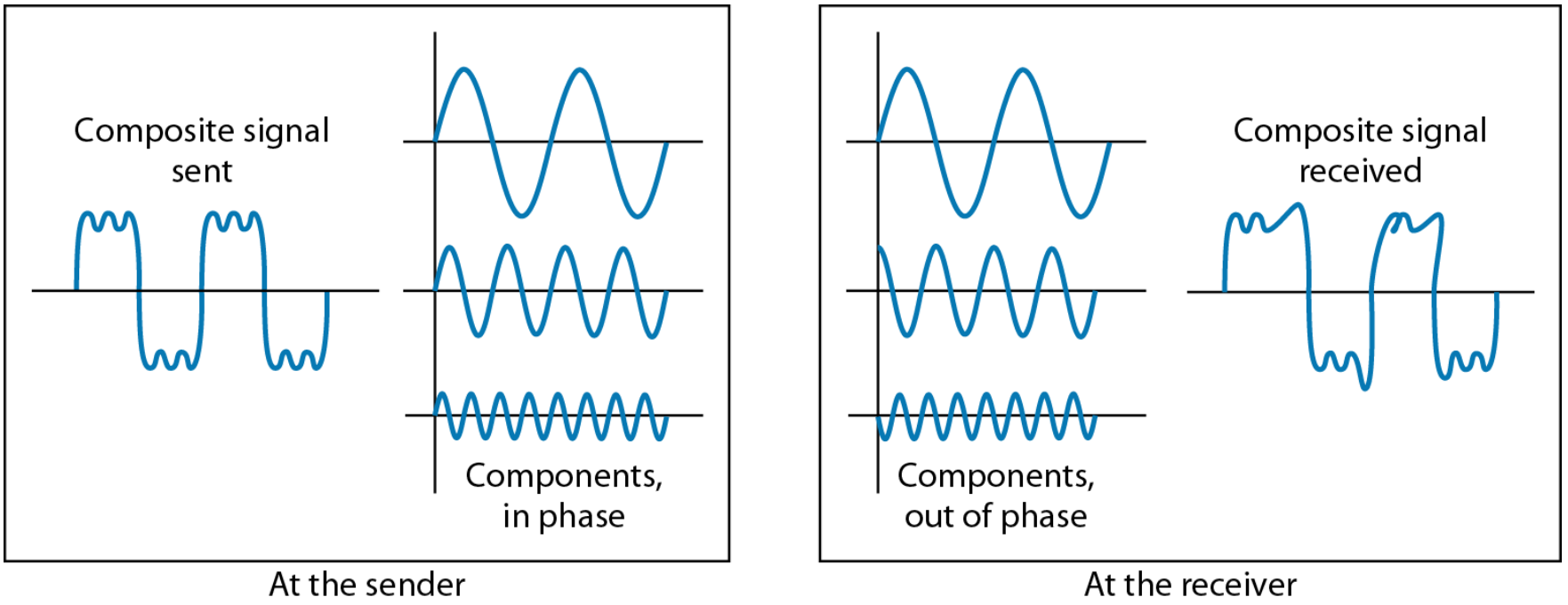
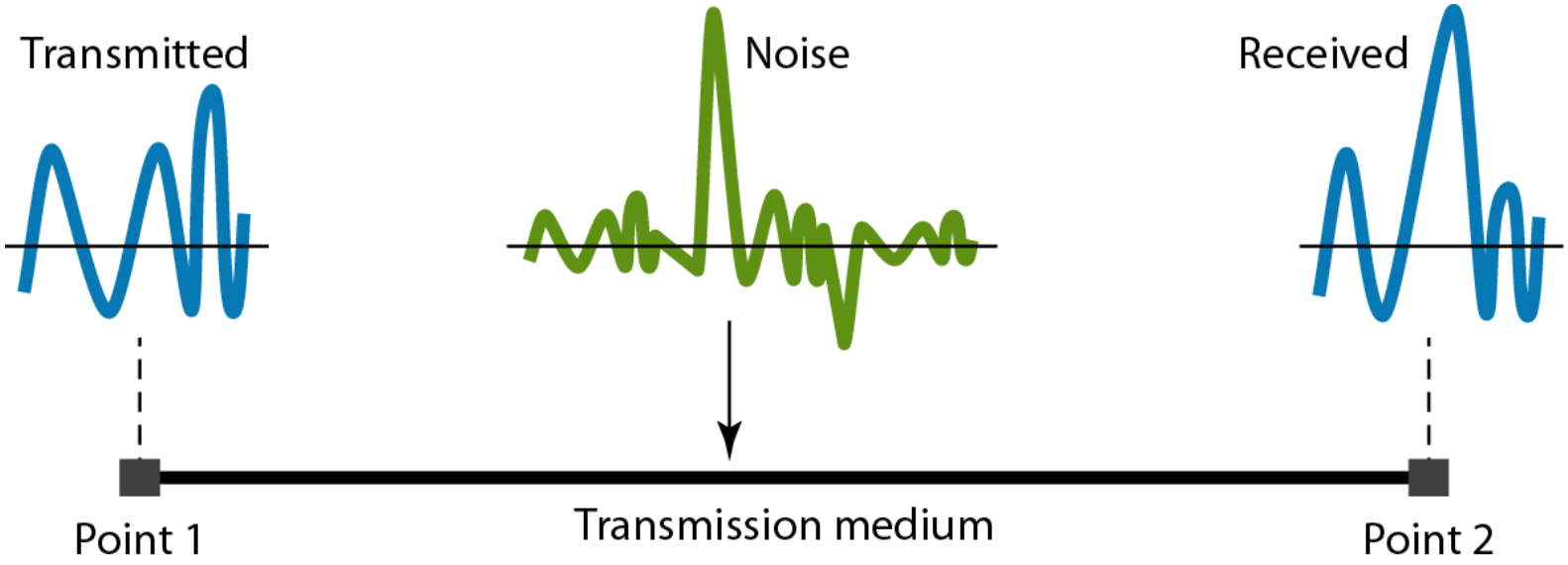


Figure 3.29 *Noise*



Example 3.31

The power of a signal is 10 mW and the power of the noise is 1 μW; what are the values of SNR and SNR_{dB} ?

Solution

The values of SNR and SNR_{dB} can be calculated as follows:

$$\text{SNR} = \frac{10{,}000\ \mu\text{W}}{1\ \text{mW}} = 10{,}000$$

$$\text{SNR}_{\text{dB}} = 10 \log_{10} 10{,}000 = 10 \log_{10} 10^4 = 40$$

Example 3.32

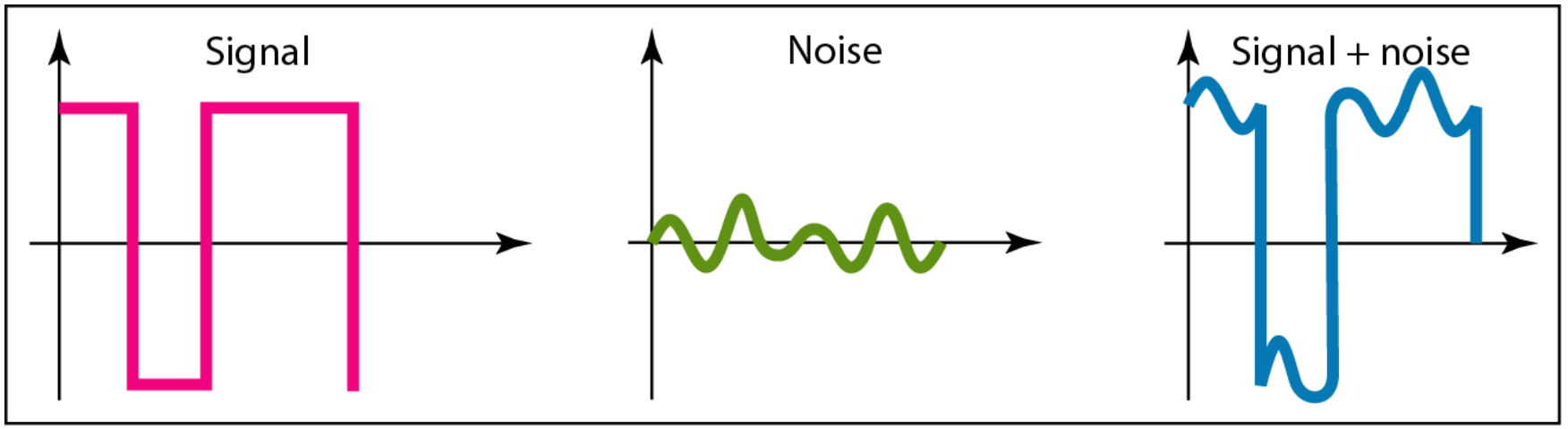
The values of SNR and SNR_{dB} for a noiseless channel are

$$\text{SNR} = \frac{\text{signal power}}{0} = \infty$$

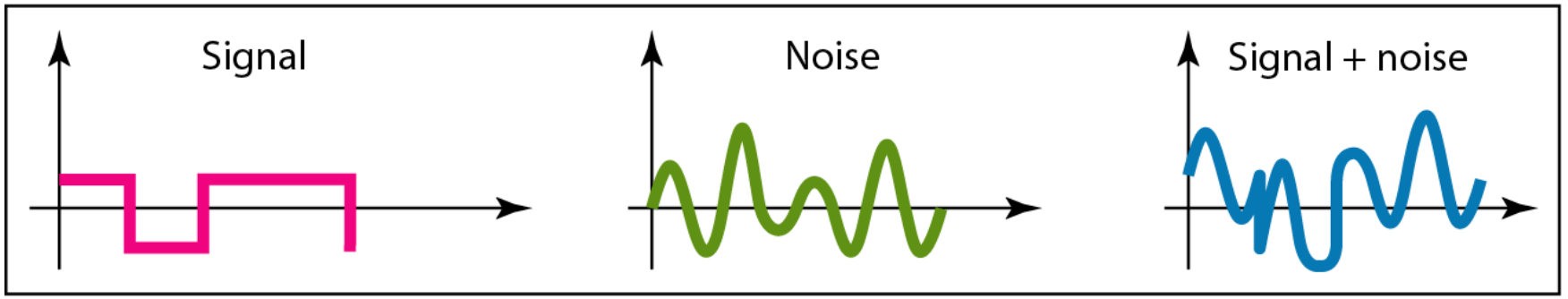
$$\text{SNR}_{\text{dB}} = 10 \log_{10} \infty = \infty$$

We can never achieve this ratio in real life; it is an ideal.

Figure 3.30 *Two cases of SNR: a high SNR and a low SNR*



a. Large SNR



b. Small SNR

3-5 DATA RATE LIMITS

A very important consideration in data communications is how fast we can send data, in bits per second, over a channel. Data rate depends on three factors:

1. *The bandwidth available*
2. *The level of the signals we use*
3. *The quality of the channel (the level of noise)*

Topics discussed in this section:

Noiseless Channel: Nyquist Bit Rate
Noisy Channel: Shannon Capacity
Using Both Limits

Note

Increasing the levels of a signal may reduce the reliability of the system.

Example 3.33

Does the Nyquist theorem bit rate agree with the intuitive bit rate described in baseband transmission?

Solution

They match when we have only two levels. We said, in baseband transmission, the bit rate is 2 times the bandwidth if we use only the first harmonic in the worst case. However, the Nyquist formula is more general than what we derived intuitively; it can be applied to baseband transmission and modulation. Also, it can be applied when we have two or more levels of signals.

Example 3.34

Consider a noiseless channel with a bandwidth of 3000 Hz transmitting a signal with two signal levels. The maximum bit rate can be calculated as

$$\text{BitRate} = 2 \times 3000 \times \log_2 2 = 6000 \text{ bps}$$

Example 3.35

Consider the same noiseless channel transmitting a signal with four signal levels (for each level, we send 2 bits). The maximum bit rate can be calculated as

$$\text{BitRate} = 2 \times 3000 \times \log_2 4 = 12{,}000 \text{ bps}$$

Example 3.36

We need to send 265 kbps over a noiseless channel with a bandwidth of 20 kHz. How many signal levels do we need?

Solution

We can use the Nyquist formula as shown:

$$265{,}000 = 2 \times 20{,}000 \times \log_2 L$$

$$\log_2 L = 6.625 \qquad L = 2^{6.625} = 98.7 \text{ levels}$$

Since this result is not a power of 2, we need to either increase the number of levels or reduce the bit rate. If we have 128 levels, the bit rate is 280 kbps. If we have 64 levels, the bit rate is 240 kbps.

Example 3.37

Consider an extremely noisy channel in which the value of the signal-to-noise ratio is almost zero. In other words, the noise is so strong that the signal is faint. For this channel the capacity C is calculated as

$$C = B \log_2 (1 + \text{SNR}) = B \log_2 (1 + 0) = B \log_2 1 = B \times 0 = 0$$

This means that the capacity of this channel is zero regardless of the bandwidth. In other words, we cannot receive any data through this channel.

Example 3.38

We can calculate the theoretical highest bit rate of a regular telephone line. A telephone line normally has a bandwidth of 3000. The signal-to-noise ratio is usually 3162. For this channel the capacity is calculated as

$$C = B \log_2 (1 + \text{SNR}) = 3000 \log_2 (1 + 3162) = 3000 \log_2 3163$$
$$= 3000 \times 11.62 = 34{,}860 \text{ bps}$$

This means that the highest bit rate for a telephone line is 34.860 kbps. If we want to send data faster than this, we can either increase the bandwidth of the line or improve the signal-to-noise ratio.

Example 3.39

The signal-to-noise ratio is often given in decibels. Assume that SNR_{dB} = 36 and the channel bandwidth is 2 MHz. The theoretical channel capacity can be calculated as

$$\text{SNR}_{\text{dB}} = 10 \log_{10} \text{SNR} \rightarrow \text{SNR} = 10^{\text{SNR}_{\text{dB}}/10} \rightarrow \text{SNR} = 10^{3.6} = 3981$$

$$C = B \log_2 (1 + \text{SNR}) = 2 \times 10^6 \times \log_2 3982 = 24 \text{ Mbps}$$

Example 3.40

For practical purposes, when the SNR is very high, we can assume that SNR + 1 is almost the same as SNR. In these cases, the theoretical channel capacity can be simplified to

$$C = B \times \frac{\text{SNR}_{\text{dB}}}{3}$$

For example, we can calculate the theoretical capacity of the previous example as

$$C = 2 \text{ MHz} \times \frac{36}{3} = 24 \text{ Mbps}$$

Example 3.41

We have a channel with a 1-MHz bandwidth. The SNR for this channel is 63. What are the appropriate bit rate and signal level?

Solution

First, we use the Shannon formula to find the upper limit.

$$C = B \log_2 (1 + \text{SNR}) = 10^6 \log_2 (1 + 63) = 10^6 \log_2 64 = 6 \text{ Mbps}$$

Example 3.41 (continued)

The Shannon formula gives us 6 Mbps, the upper limit. For better performance we choose something lower, 4 Mbps, for example. Then we use the Nyquist formula to find the number of signal levels.

$$4 \text{ Mbps} = 2 \times 1 \text{ MHz} \times \log_2 L \quad \rightarrow \quad L = 4$$

Note

The Shannon capacity gives us the upper limit; the Nyquist formula tells us how many signal levels we need.

3-6 PERFORMANCE

*One important issue in networking is the **performance** of the network—how good is it? We discuss quality of service, an overall measurement of network performance, in greater detail in Chapter 24. In this section, we introduce terms that we need for future chapters.*

Topics discussed in this section:

Bandwidth
Throughput
Latency (Delay)
Bandwidth-Delay Product

Note

In networking, we use the term bandwidth in two contexts.

- **The first, bandwidth in hertz, refers to the range of frequencies in a composite signal or the range of frequencies that a channel can pass.**

- **The second, bandwidth in bits per second, refers to the speed of bit transmission in a channel or link.**

Example 3.42

The bandwidth of a subscriber line is 4 kHz for voice or data. The bandwidth of this line for data transmission can be up to 56,000 bps using a sophisticated modem to change the digital signal to analog.

Example 3.43

If the telephone company improves the quality of the line and increases the bandwidth to 8 kHz, we can send 112,000 bps by using the same technology as mentioned in Example 3.42.

Example 3.44

A network with bandwidth of 10 Mbps can pass only an average of 12,000 frames per minute with each frame carrying an average of 10,000 bits. What is the throughput of this network?

Solution

We can calculate the throughput as

$$\text{Throughput} = \frac{12{,}000 \times 10{,}000}{60} = 2 \text{ Mbps}$$

The throughput is almost one-fifth of the bandwidth in this case.

Example 3.45

What is the propagation time if the distance between the two points is 12,000 km? Assume the propagation speed to be 2.4 × 108 m/s in cable.

Solution

We can calculate the propagation time as

$$\text{Propagation time} = \frac{12{,}000 \times 1000}{2.4 \times 10^8} = 50 \text{ ms}$$

The example shows that a bit can go over the Atlantic Ocean in only 50 ms if there is a direct cable between the source and the destination.

Example 3.46

What are the propagation time and the transmission time for a 2.5-kbyte message (an e-mail) if the bandwidth of the network is 1 Gbps? Assume that the distance between the sender and the receiver is 12,000 km and that light travels at 2.4 × 108 m/s.

Solution

We can calculate the propagation and transmission time as shown on the next slide:

Example 3.46 (continued)

$$\text{Propagation time} = \frac{12{,}000 \times 1000}{2.4 \times 10^8} = 50 \text{ ms}$$

$$\text{Transmission time} = \frac{2500 \times 8}{10^9} = 0.020 \text{ ms}$$

Note that in this case, because the message is short and the bandwidth is high, the dominant factor is the propagation time, not the transmission time. The transmission time can be ignored.

Example 3.47

What are the propagation time and the transmission time for a 5-Mbyte message (an image) if the bandwidth of the network is 1 Mbps? Assume that the distance between the sender and the receiver is 12,000 km and that light travels at 2.4×10^8 m/s.

Solution

We can calculate the propagation and transmission times as shown on the next slide.

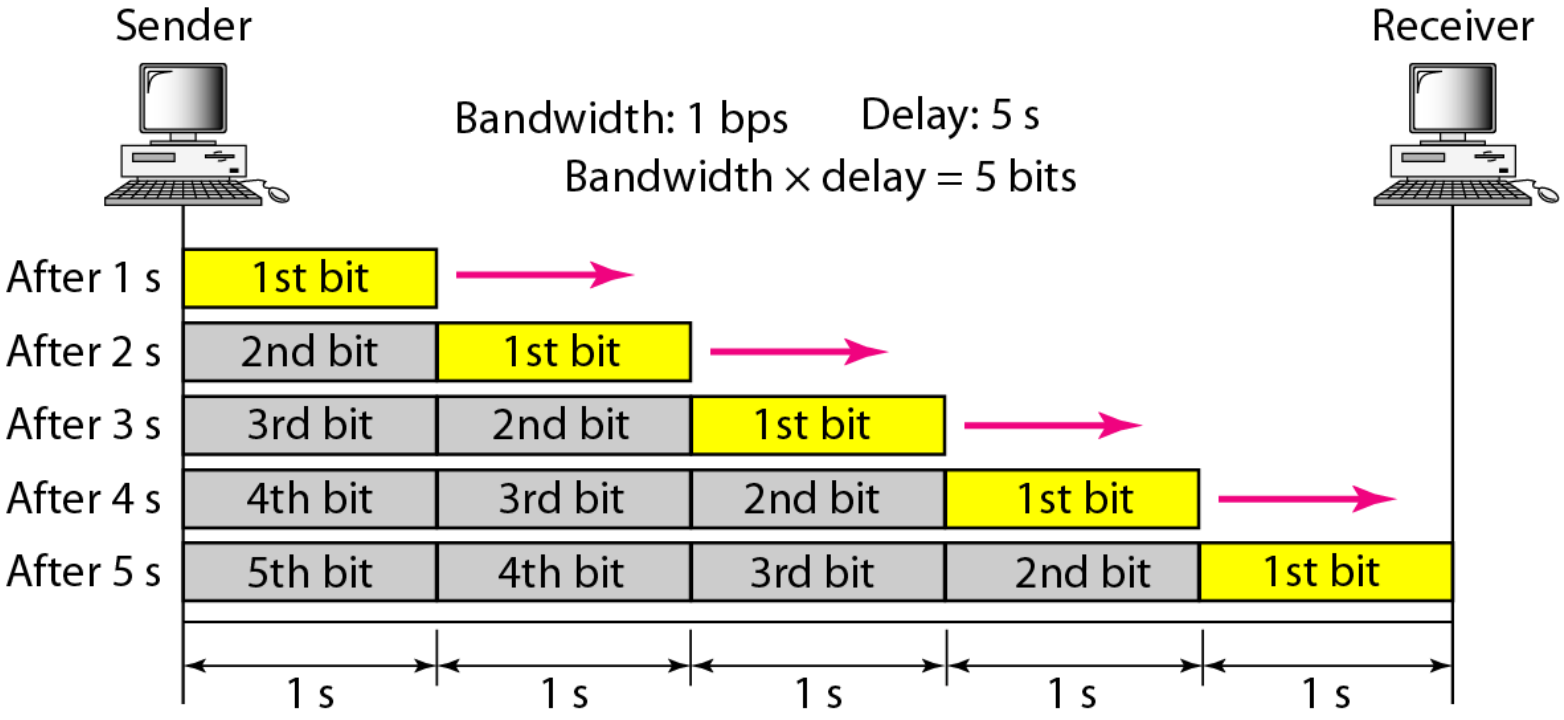
Example 3.47 (continued)

$$\text{Propagation time} = \frac{12{,}000 \times 1000}{2.4 \times 10^8} = 50 \text{ ms}$$

$$\text{Transmission time} = \frac{5{,}000{,}000 \times 8}{10^6} = 40 \text{ s}$$

Note that in this case, because the message is very long and the bandwidth is not very high, the dominant factor is the transmission time, not the propagation time. The propagation time can be ignored.

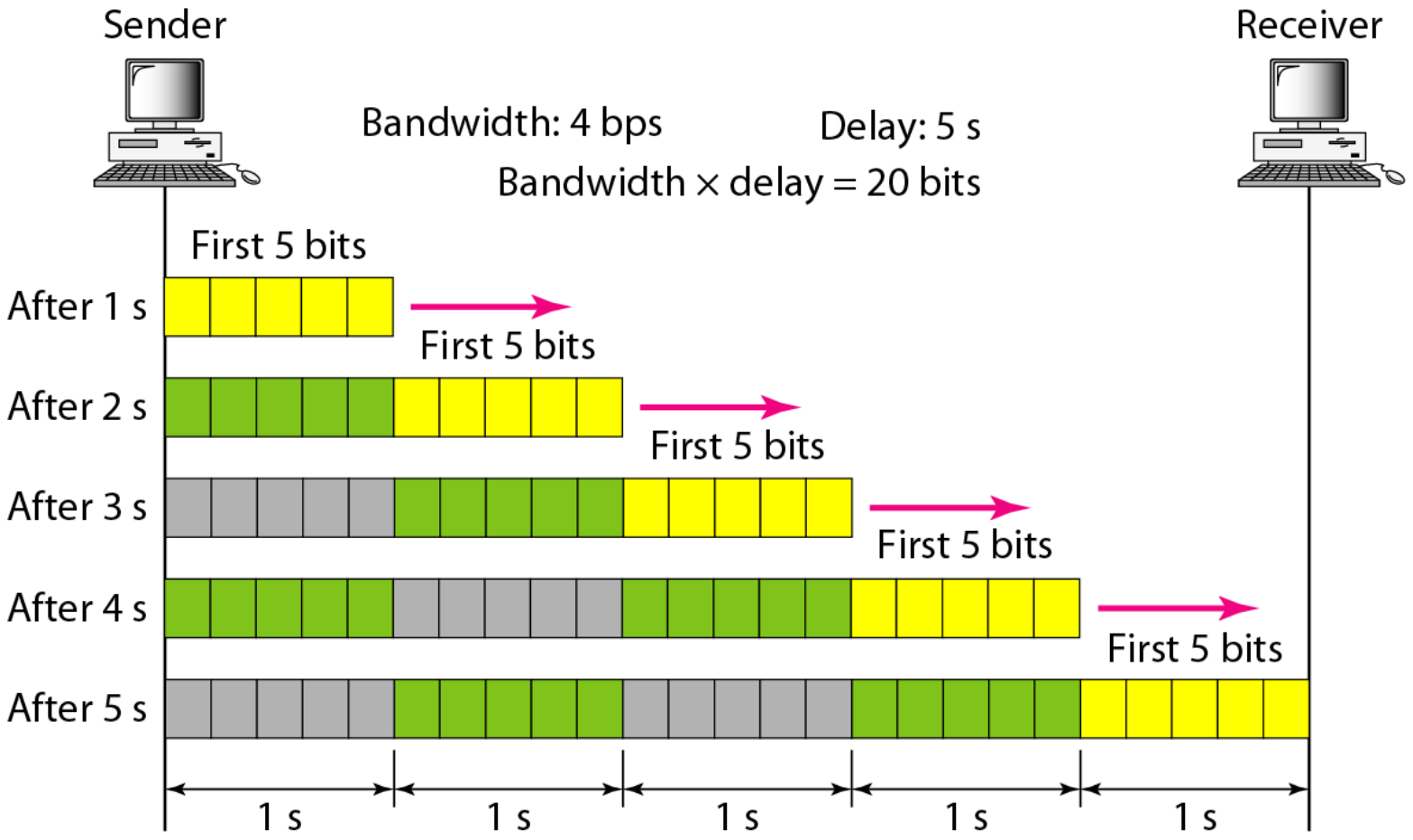
Figure 3.31 *Filling the link with bits for case 1*



Example 3.48

We can think about the link between two points as a pipe. The cross section of the pipe represents the bandwidth, and the length of the pipe represents the delay. We can say the volume of the pipe defines the bandwidth-delay product, as shown in Figure 3.33.

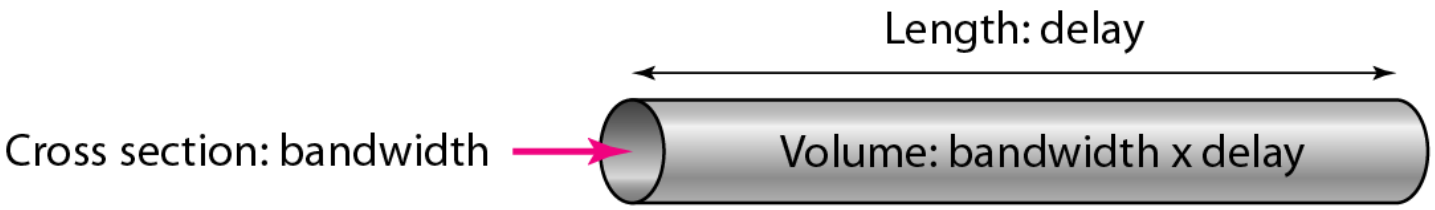
Figure 3.32 *Filling the link with bits in case 2*



Note

The bandwidth-delay product defines the number of bits that can fill the link.

Figure 3.33 *Concept of bandwidth-delay product*



PULSE CODE MODULATION

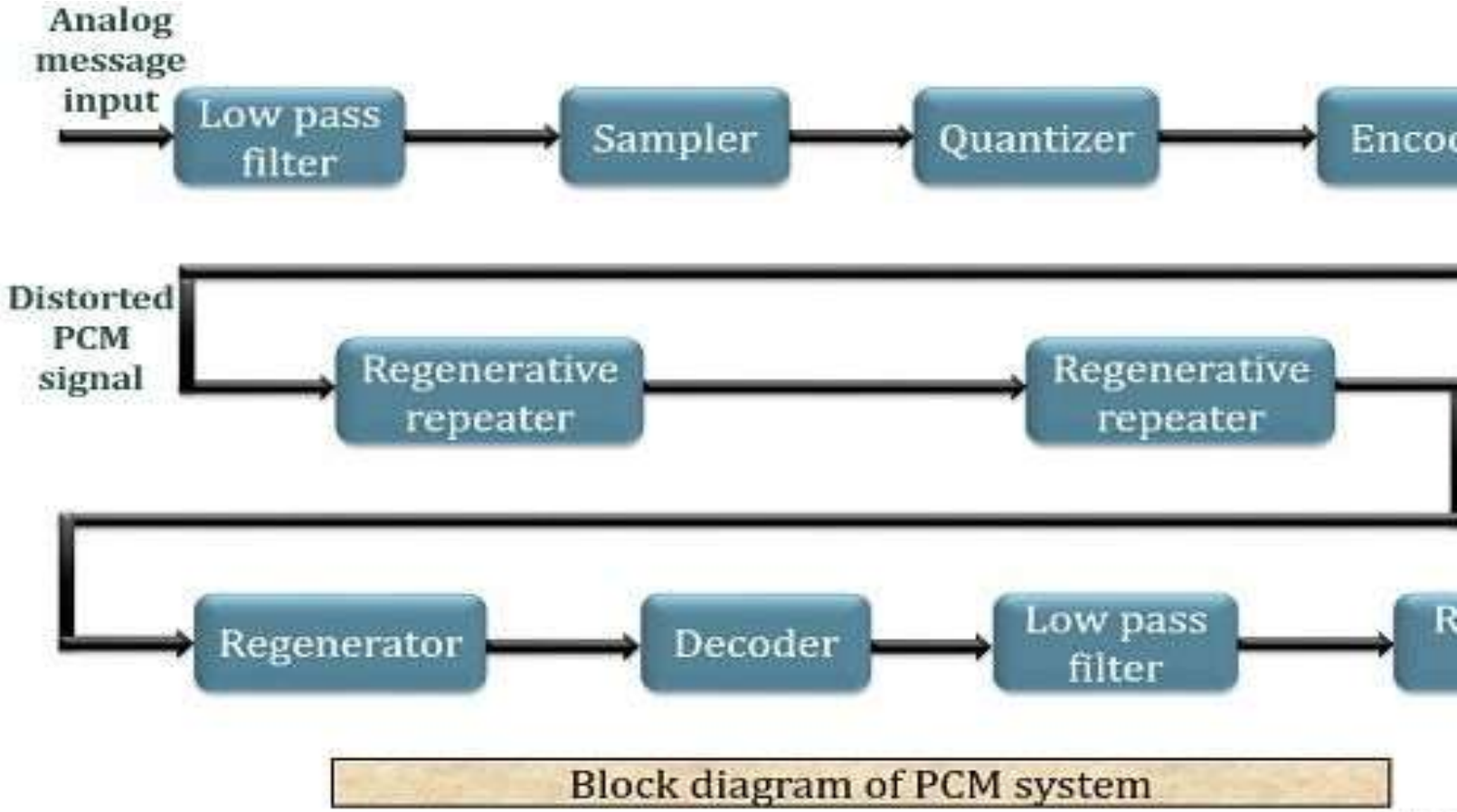
1. PCM quantization Techniques
2. PCM Transmission Bandwidth
3. PCM Coding Techniques
4. PCM Integrated Circuits
5. Advantages of PCM
6. Delta Modulation
7. Adaptive Delta Modulation
8. Differential PCM

Pulse Code Modulation (PCM)

Definition: A technique by which **analog signal get**
digital form in order to have signal transmission thr
network is known as Pulse Code Modulation. It is ab
as **PCM**.

PCM systems are basically signal coders also known as **waveform coders**. PCM allows th
continuous time message signal as a sequence of binary coded pulses. The binary form pe
i.e., **0 and 1**.
The major steps involved in PCM is **sampling, quantizing and encoding** which will be dis
sections.

Block diagram of Pulse Code Modulatio



Block diagram of PCM system

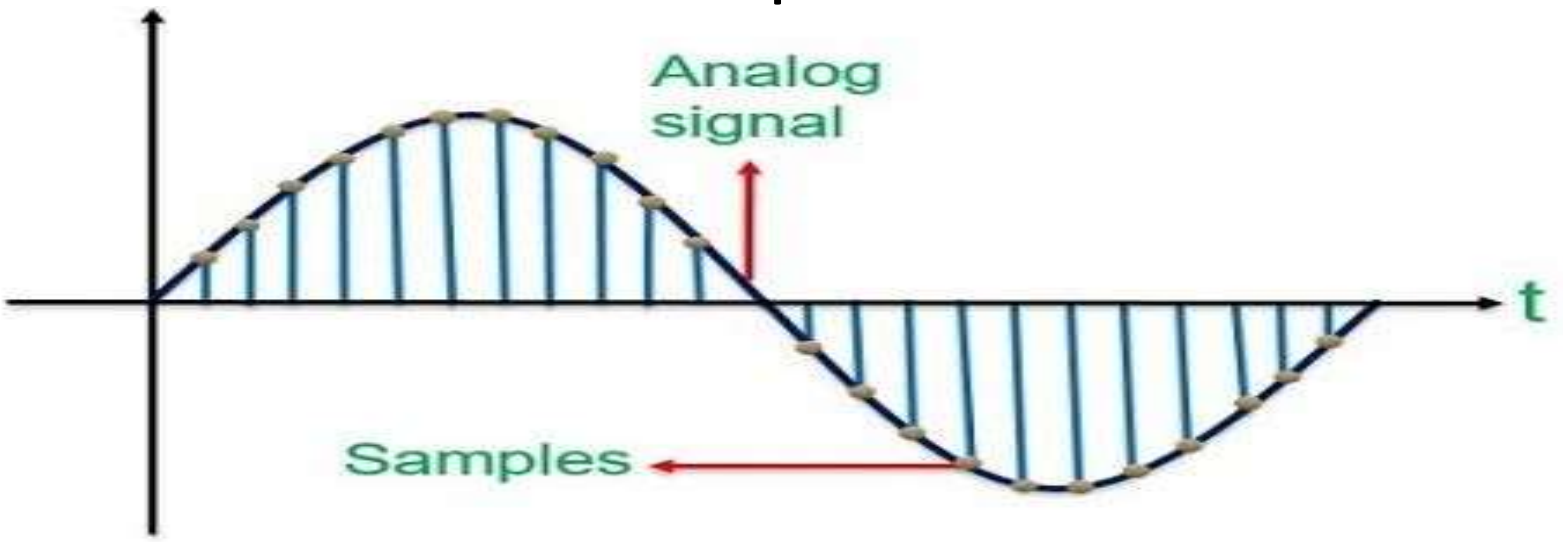
Quantizer

A quantizer is a unit that rounds off each sample to the n
discrete level. The sampler provides a continuous range s
hence still an analog one. The quantizer performs the app
of each sample thus assigning it a particular discrete leve

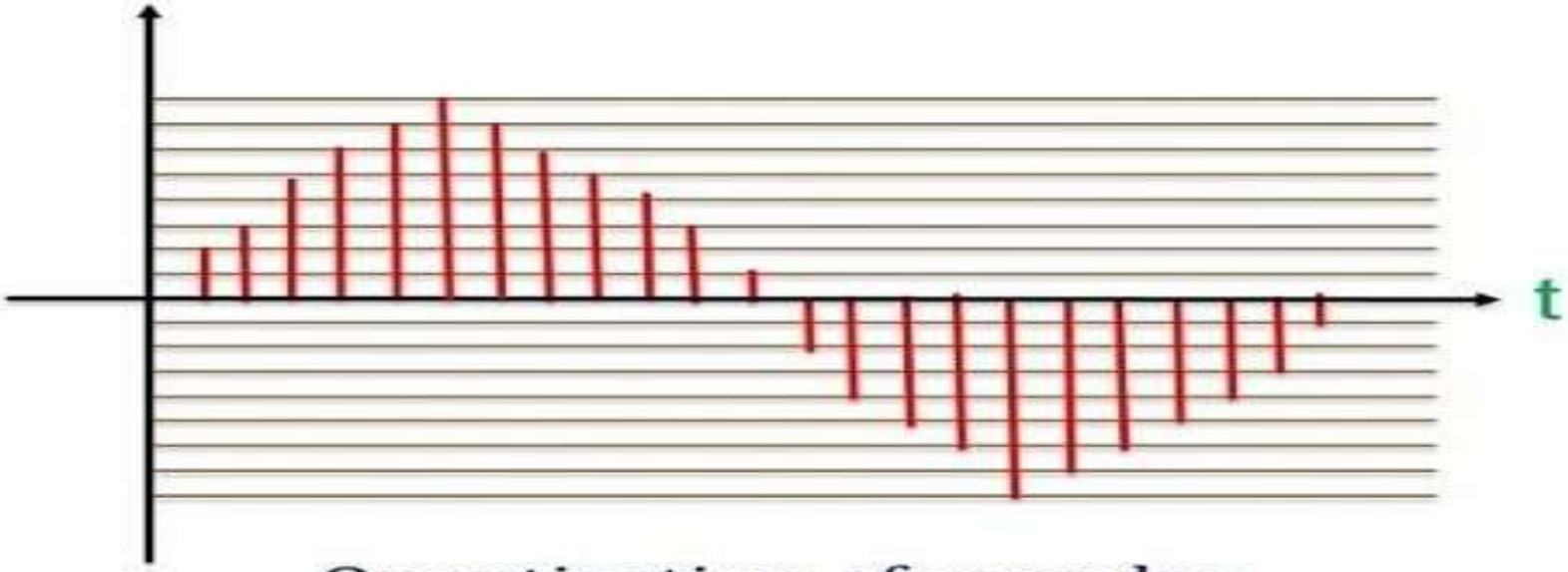
As it basically rounds off the value to a certain level this s
variation by the actual amount. Thus we can say, quantizi
introduces some distortion or noise into it. This is known
as **quantization error**.

This noise factor is somewhat better than the channel no
controllable.

The figure below shows the sampling of analog sign
quantization of the samples



Sampling of analog signal



Quantization of samples

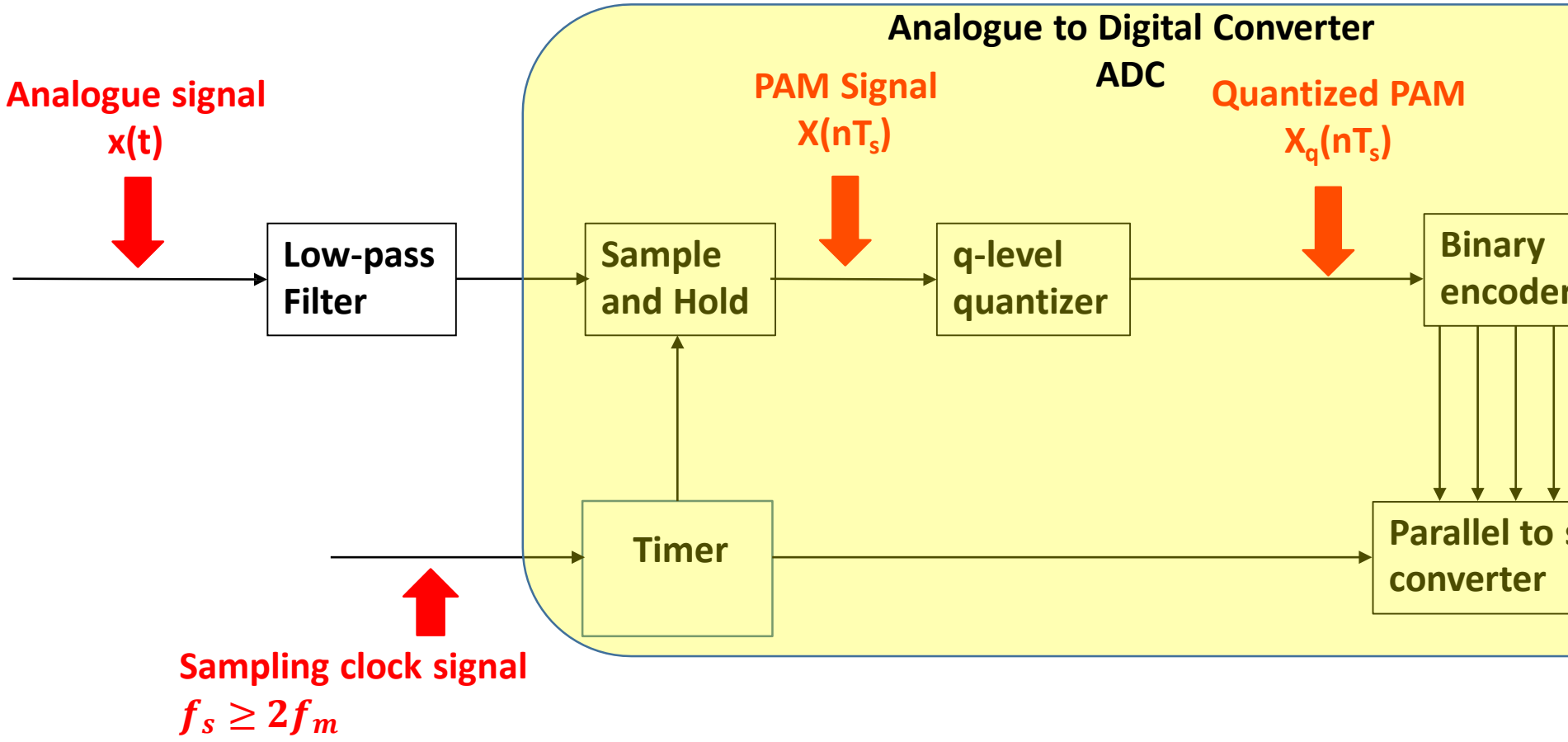
Encoder

An encoder performs the conversion of the quantized sig
binary codes. This unit generates a digitally encoded sign
a sequence of binary pulses that acts as the modulated o

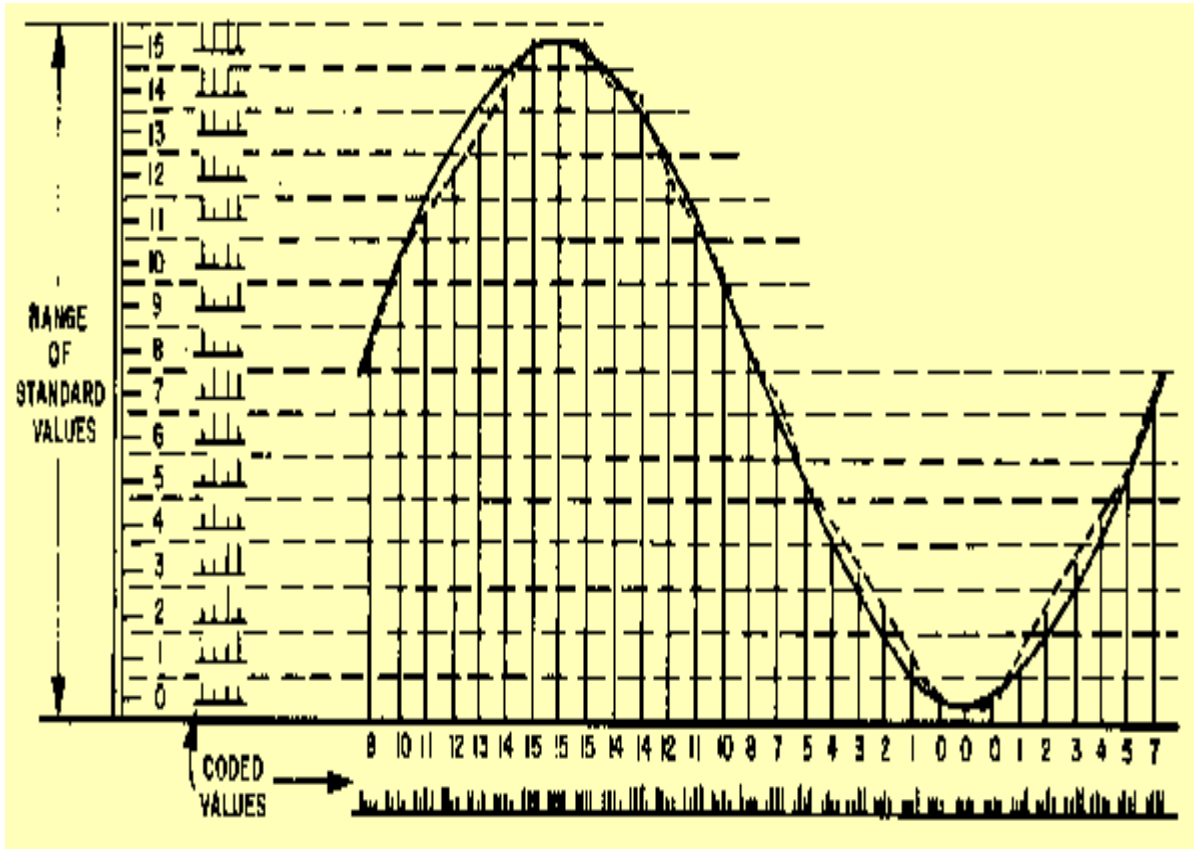
POINTS ABOUT PC

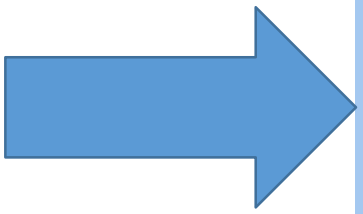
1. While PCM is a pulse modulation technique much like
   or PPM.
2. PCM is digital while the others are either analogue in
   amplitude, i.e PCM pulses are discrete in time and
   unlike PAM, PWM or PPM.
3. Essential aspects of a PCM transmitter are sampling,
   and encoding.
4. PCM is not a modulation in the conventional sense be
   not rely on varying a characteristic of a carrier (amplit
   frequency or phase).

PCM TRANSMIT



BINARY EQUIVALENTS AND PULSE CODE W

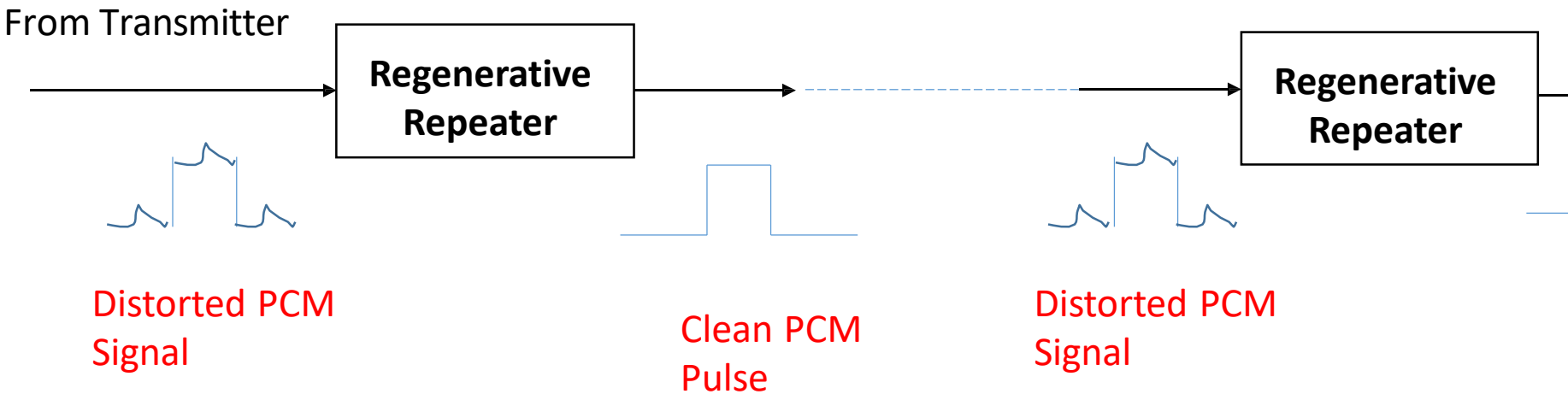




DECIMAL NUMBER	BINARY EQUIV		
	2^3	2^2	2
0	0	0	0
1	0	0	0
2	0	0	1
3	0	0	1
4	0	1	0
5	0	1	0
6	0	1	1
7	0	1	1
8	1	0	0
9	1	0	0
10	1	0	1
11	1	0	1
12	1	1	0
13	1	1	0
14	1	1	1
15	1	1	1

PCM TRANSMISSION PATH

- **PCM transmission path** refers to the path the the signal travels between the transmitter and the receiver.



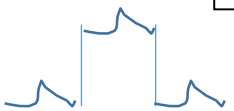
PCM REPEAT

Compensates for the effects of amplitude and phase distortions

Makes a de
whether the
PCM wave i
one

Amplitude and Phase Equalizer

**Decisio
Ci**

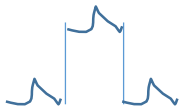


Distorted PCM Signal

Timing Circuit

The timing clock is extracted from the PCM pulse-stream

PCM RECEIV



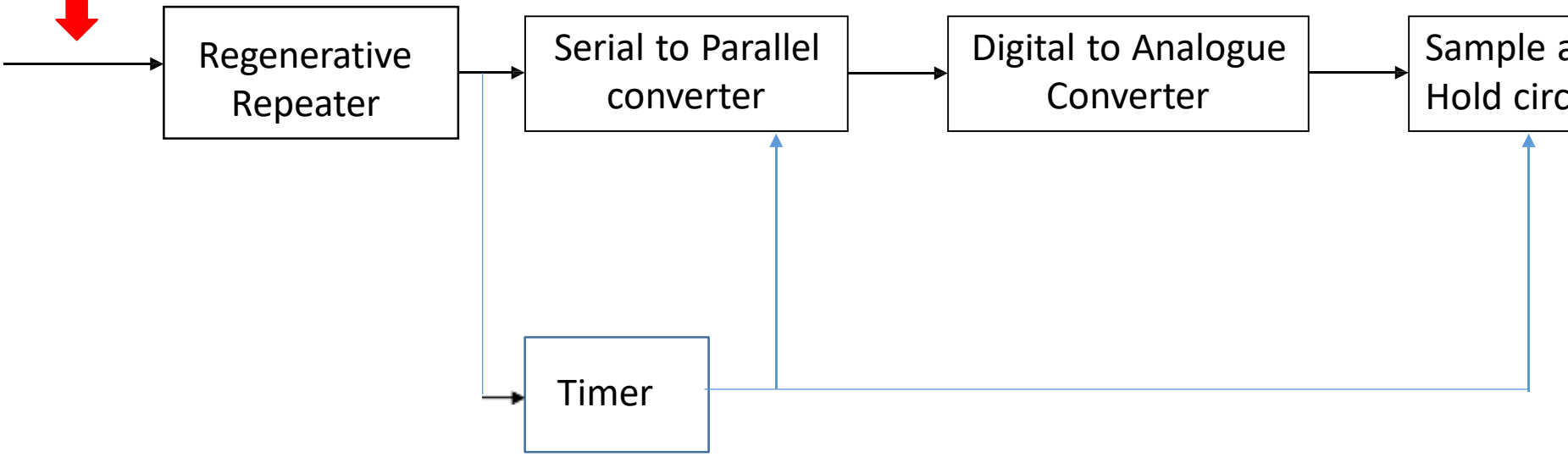


Distorted PCM
Pulse Trainl

Clean PCM
Pulse train







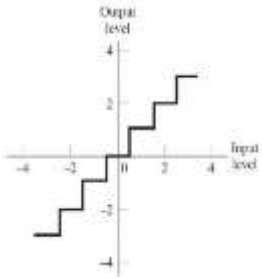
TYPES OF QUANTIZERS

Quantization

Uniform Quantization

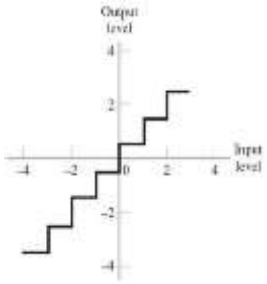
Non-

Midtread Quantization



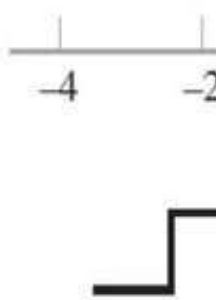
Step size is the same throughout the input signal range

Midrise Quantization



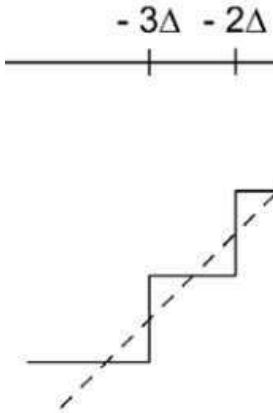
MIDTREAD QUANT

1. **A midtread quantizer** assumes values of the form $\Delta H_i \cdot$where Δ is the step size and $H_i = 0, \pm1, \pm2, \pm3, \ldots$
2. It is called **mid-tread** because the origin lies in the middle of a tread of a staircase- like graph.



MIDRISER QUANTI

1. A **mid-riser quantizer** has output levels are given by $\frac{\Delta}{2}H_i$, where Δ is the step size and H_i
2. = ±1, ±2, ±3, ...
3. The origin lies in the middle of the rising part of the staircase-like characteristic graph.



PCM TRANSMISSION BAND

1. Assume the a PCM encoder has q levels which are encod
   v
2. bits.
3. We can infer

$$4.\ \ q = \quad 2^{v}$$

5. The number of bits per second can be expressed as:

$$f_{pcm} = vf_s \quad \text{where } f_s \geq 2f_m \text{ (Nyquist criterion}$$

4. It therefore follows that the bandwidth, BW of a PCM ch
   is bounded by:

$$BW_{pcm} \geq \ 2vf_m$$

EXAMPLE 1

1. A TV signal with a bandwidth of 4.2 MHz is transmitted
   binary PCM system using 512 quantization levels. Deter
   1. Code word-length
   2. The PCM bandwidth/bit rate

2. **SOLUTION**
3. $f_m = 4.2\ MHz$
4. $q = \quad 2^v = 512$
5. $v = \quad log_2 512 \quad$ = 9 bits
6. Bandwidth, BW $=$ $2vf_m$ = 2 x9 × 4.2 = 75.6 M

WHY IT IS NECESSARY TO HAVE NON-UN QUANTIZATION?

1. Using linear quantization, the quantization error is given by:

$$\epsilon = \frac{\Delta}{2}$$

2. If q quantization levels of a bipolar signal are used, we can write:

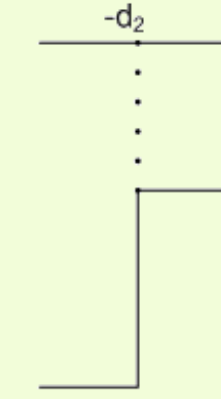
$$\Delta = \frac{2x_{max}}{q}$$

3. Consider a PCM system with $v = 4$ bits and $x_{max} = 16$ Volt, then:

$$q = 2^4 = 16 \qquad \Delta = \frac{2}{q} = \frac{2}{16} = \frac{1}{8}$$

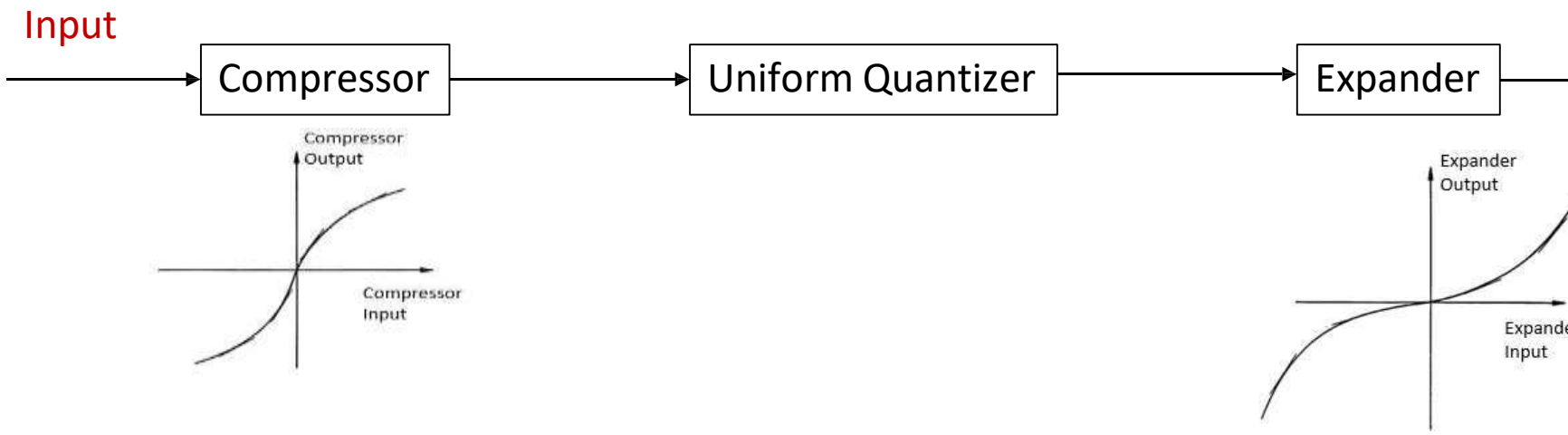
The maximum quantization error is therefore $\varepsilon_{max} = \left|\frac{\Delta}{2}\right| = \frac{1}{16}$

4. At maximum, the relative error is 1 volt out of 16 volts or 6.25%
5. At lower levels, e.g. 2 volts, the relative error is 1 volt out of 2 volts or 50%.
6. To reduce this high relative error at low levels, PCM systems use non- uniform quantization .



COMPANDIN

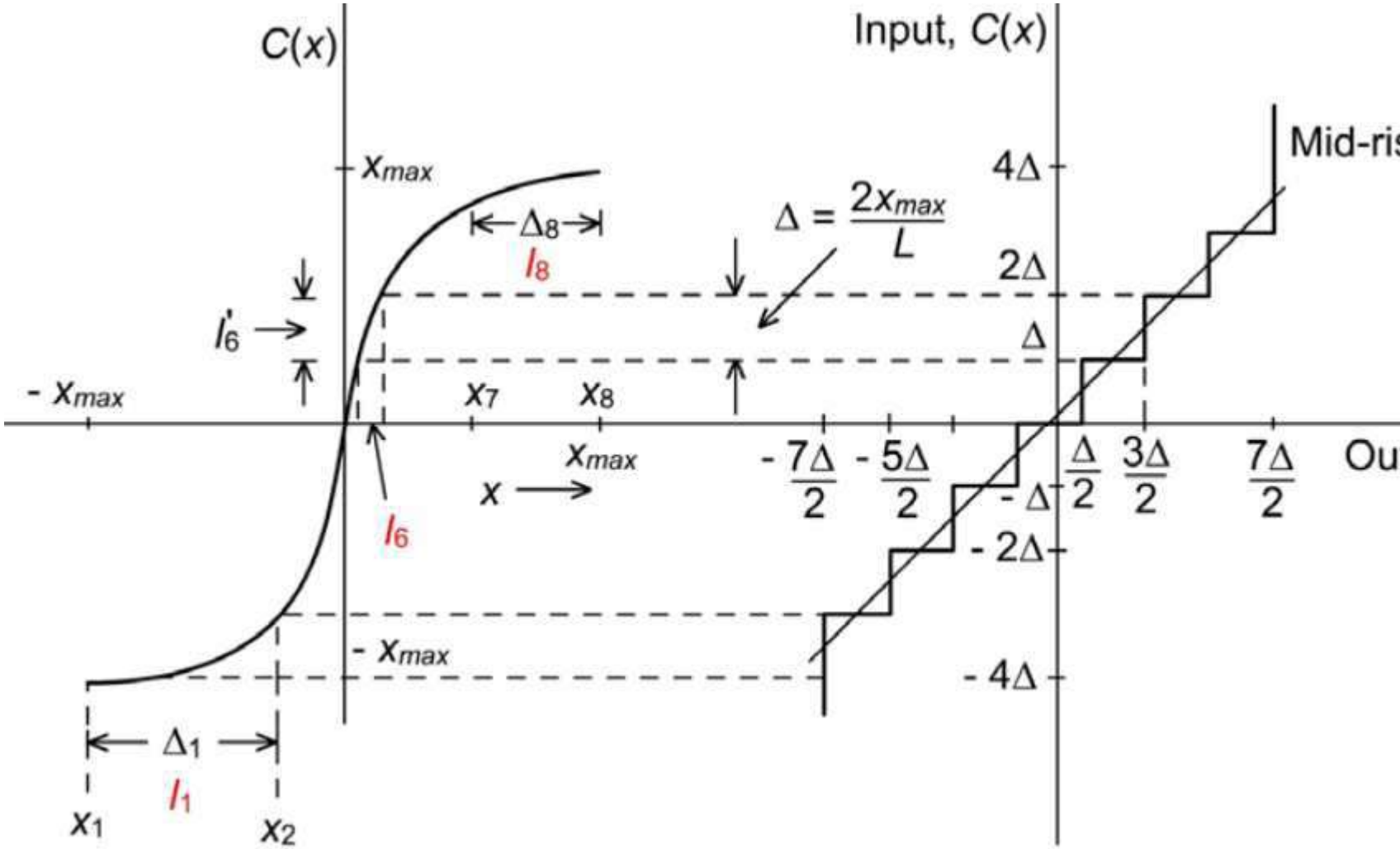
1. With uniform sampling, the quantization step is fixed thus resulti
2. quantization noise power.
3. However signal power is not constant, it is proportional to the sq signal amplitude. This means Quantization Noise is very significar amplitudes.
4. To reduce quantization noise at lower amplitudes, we use comma
5. **Companding** = **Comp**ressing + Ex**panding**



Provides High Gain to Weak Signals and Low Gain to strong Signals

Provides Low Gain t Weak Signals and Hi Gain to strong Signa

COMPRESSING WITH MIDRIZER QU



COMPANDING IN COMMUNICATION

1. The loudest sound that can be tolerated (120 dB SPL) is about one-million times the amplitude of the weakest sound that can be detected (0 dB SPL).

2. If the quantization levels are equally spaced (uniform quantization), 12 bits must be used to obtain telephone quality speech.

3. However, only 8 bits are required if the quantization levels are made unequal (companding) to match the characteristics of human hearing.

THREE METHODS OF REALIZING COMPA
IN COMMUNICATION SYST

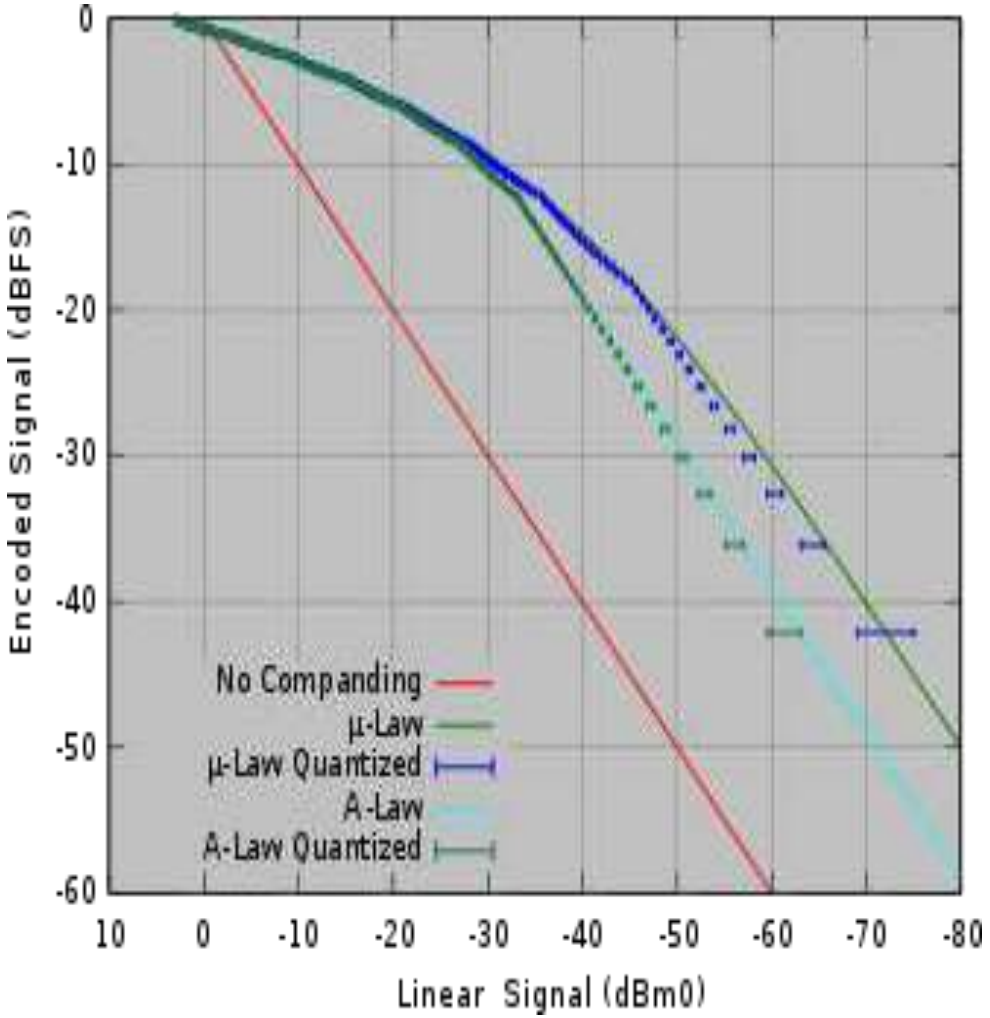
1. Run the analog signal through a nonlinear circuit before reaching a linear 8 bit ADC,
2. Use an 8 bit ADC that internally has unequally spaced steps, or
3. Use a linear 12 bit ADC followed by a digital look-up table (12 bits in, 8 bits out).

1. Each of these three options requires the same nonlinearity, just in a different place: at analog circuit, at the ADC, or a digital circuit after the ADC.

COMPANDING STAND

(1)μ255 law used in North America

(2)"A" law, used in Europe.



"A" LAW COMPAND

$$F(x) = \operatorname{sgn}(x) \begin{cases} \frac{A|x|}{1+\ln(A)}, & |x| < \frac{1}{A} \\ \frac{1+\ln(A|x|)}{1+\ln(A)}, & \frac{1}{A} \leq |x| \leq 1, \end{cases}$$

Where A is the compression parameter

$$F^{-1}(y) = \operatorname{sgn}(y) \begin{cases} \frac{|y|(1+\ln(A))}{A}, & |y| < \frac{1}{1+\ln(A)} \\ \frac{\exp(|y|(1+\ln(A))-1)}{A}, & \frac{1}{1+\ln(A)} \leq |y| < 1. \end{cases}$$

μ-LAW COMPANDING

$$F(x) = \operatorname{sgn}(x)\frac{\ln(1+\mu|x|)}{\ln(1+\mu)} \qquad -1 \le x \le 1,$$

where μ is 255 for 8 bits.

$$F^{-1}(y) = \operatorname{sgn}(y)(1/\mu)((1+\mu)^{|y|}-1) \qquad -1 \le y \le 1$$

BINARY ENCODI

1. Encoding converts the quantized samples into a form that is more convenient for the purpose of transmission.

2. It is a one-to-one mapping of the quantized samples by using code elements or symbols of the required length per sample.

1	2				3				
Decimal Level No.	Natural Binary code				Folded Binary code				
	b_4	b_3	b_2	b_1					
15	1	1	1	1	1	1	1	1	1
14	1	1	1	0	1	1	1	0	1
13	1	1	0	1	1	1	0	1	1
12	1	1	0	0	1	1	0	0	1
11	1	0	1	1	1	0	1	1	1
10	1	0	1	0	1	0	1	0	1
9	1	0	0	1	1	0	0	1	1
8	1	0	0	0	1	0	0	0	1
7	0	1	1	1	0	0	0	0	0
6	0	1	1	0	0	0	0	1	0
5	0	1	0	1	0	0	1	0	0
4	0	1	0	0	0	0	1	1	0
3	0	0	1	1	0	1	0	0	0
2	0	0	1	0	0	1	0	1	0
1	0	0	0	1	0	1	1	0	0
0	0	0	0	0	0	1	1	1	0

FOLDED BINARY C

- The **folded binary code** (also called the sign-magnitude representation) assigns the first (left most) digit to the sign and the remaining digits are used to code the magnitude.
- This code is superior to the natural code in masking transmission errors when encoding speech.

1	2				3				
Decimal Level No.	Natural Binary code				Folded Binary code				
	b_4	b_3	b_2	b_1					
15	1	1	1	1	1	1	1	1	1
14	1	1	1	0	1	1	1	0	1
13	1	1	0	1	1	1	0	1	1
12	1	1	0	0	1	1	0	0	1
11	1	0	1	1	1	0	1	1	1
10	1	0	1	0	1	0	1	0	1
9	1	0	0	1	1	0	0	1	1
8	1	0	0	0	1	0	0	0	1
7	0	1	1	1	0	0	0	0	0
6	0	1	1	0	0	0	0	1	0
5	0	1	0	1	0	0	1	0	0
4	0	1	0	0	0	0	1	1	0
3	0	0	1	1	0	1	0	0	0
2	0	0	1	0	0	1	0	1	0
1	0	0	0	1	0	1	1	0	0
0	0	0	0	0	0	1	1	1	0

INVERTED FOLDED BINARY

1. If only the amplitude digits of a folded binary code are complemented (1's changed to 0's and 0's to 1's), an inverted folded binary code results.
2. This code has the advantage of higher density of 1's for small amplitude signals, which are most probable for voice messages.
3. The higher density of 1's relieves some system timing errors.

1	2				3				
Decimal Level No.	Natural Binary code				Folded Binary code				
	b_4	b_3	b_2	b_1					
15	1	1	1	1	1	1	1	1	1
14	1	1	1	0	1	1	1	0	1
13	1	1	0	1	1	1	0	1	1
12	1	1	0	0	1	1	0	0	1
11	1	0	1	1	1	0	1	1	1
10	1	0	1	0	1	0	1	0	1
9	1	0	0	1	1	0	0	1	1
8	1	0	0	0	1	0	0	0	1
7	0	1	1	1	0	0	0	0	0
6	0	1	1	0	0	0	0	1	0
5	0	1	0	1	0	0	1	0	0
4	0	1	0	0	0	0	1	1	0
3	0	0	1	1	0	1	0	0	0
2	0	0	1	0	0	1	0	1	0
1	0	0	0	1	0	1	1	0	0
0	0	0	0	0	0	1	1	1	0

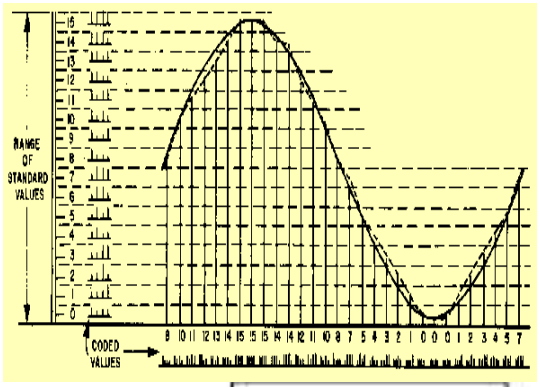
GRAY COD

1. With natural binary encoding, a number of codeword digits can change even when a change of only one quantization level occurs. For example, a change from level 7 to 8 entails every bit changing in the 4-bit code illustrated.

2. In some applications, this behavior is undesirable and a code is desired for which only one digit changes when any transition occurs between adjacent levels.

1	2				3				
Decimal Level No.	Natural Binary code				Folded Binary code				
	b_4	b_3	b_2	b_1					
15	1	1	1	1	1	1	1	1	1
14	1	1	1	0	1	1	1	0	1
13	1	1	0	1	1	1	0	1	1
12	1	1	0	0	1	1	0	0	1
11	1	0	1	1	1	0	1	1	1
10	1	0	1	0	1	0	1	0	1
9	1	0	0	1	1	0	0	1	1
8	1	0	0	0	1	0	0	0	1
7	0	1	1	1	0	0	0	0	0
6	0	1	1	0	0	0	0	1	0
5	0	1	0	1	0	0	1	0	0
4	0	1	0	0	0	0	1	1	0
3	0	0	1	1	0	1	0	0	0
2	0	0	1	0	0	1	0	1	0
1	0	0	0	1	0	1	1	0	0
0	0	0	0	0	0	1	1	1	0

4-BIT PCM TRANSMITTER - CIRCUIT S



$$\begin{cases} V_A > V_B \Rightarrow 1 \\ V_A < V_B \Rightarrow 0 \end{cases}$$

Comparing

x(t)

Analog signal

Sample&Holding

V_A

V_B

Sampling Pulses

Gate

Parallel/Series Converter

Q_A Q_B Q_C Q_D

Binary Counter

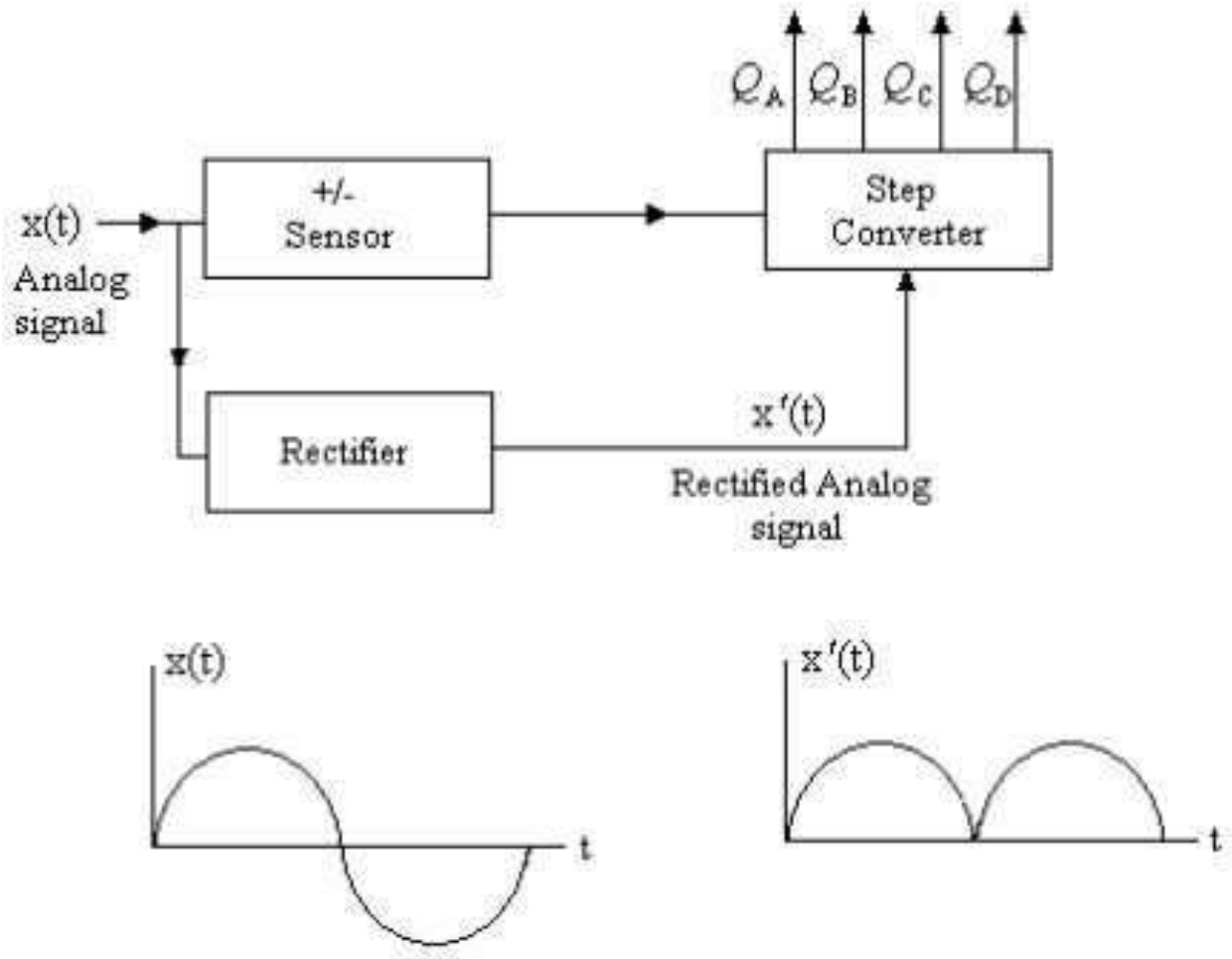
Q_A Q_B Q_C Q_D

D/A Converter

Reference Clock Input

Sampling Pulses

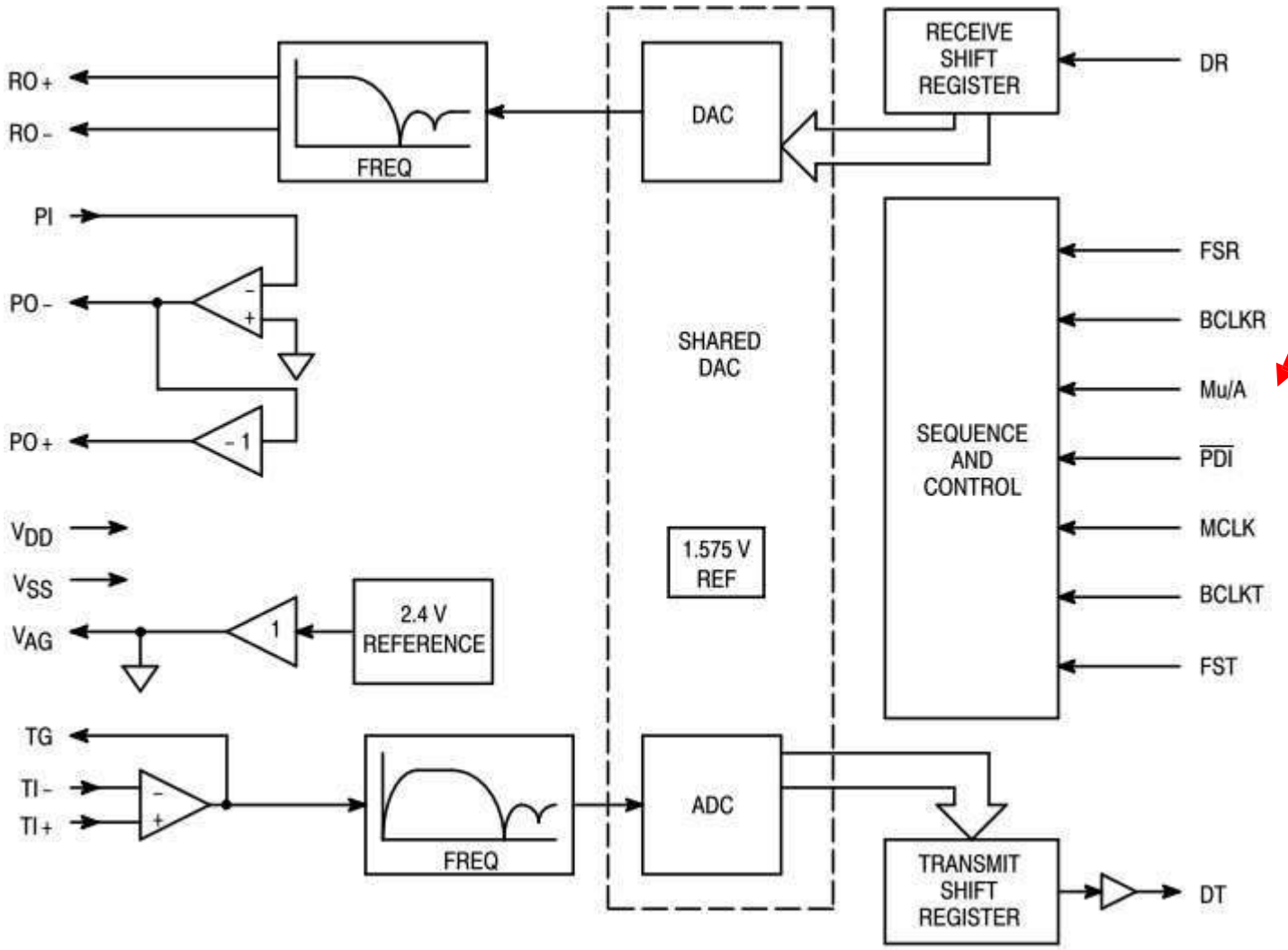
PCM FOR BI-POLAR SIG



PCM INTEGRATED CIRCUITS - MC1

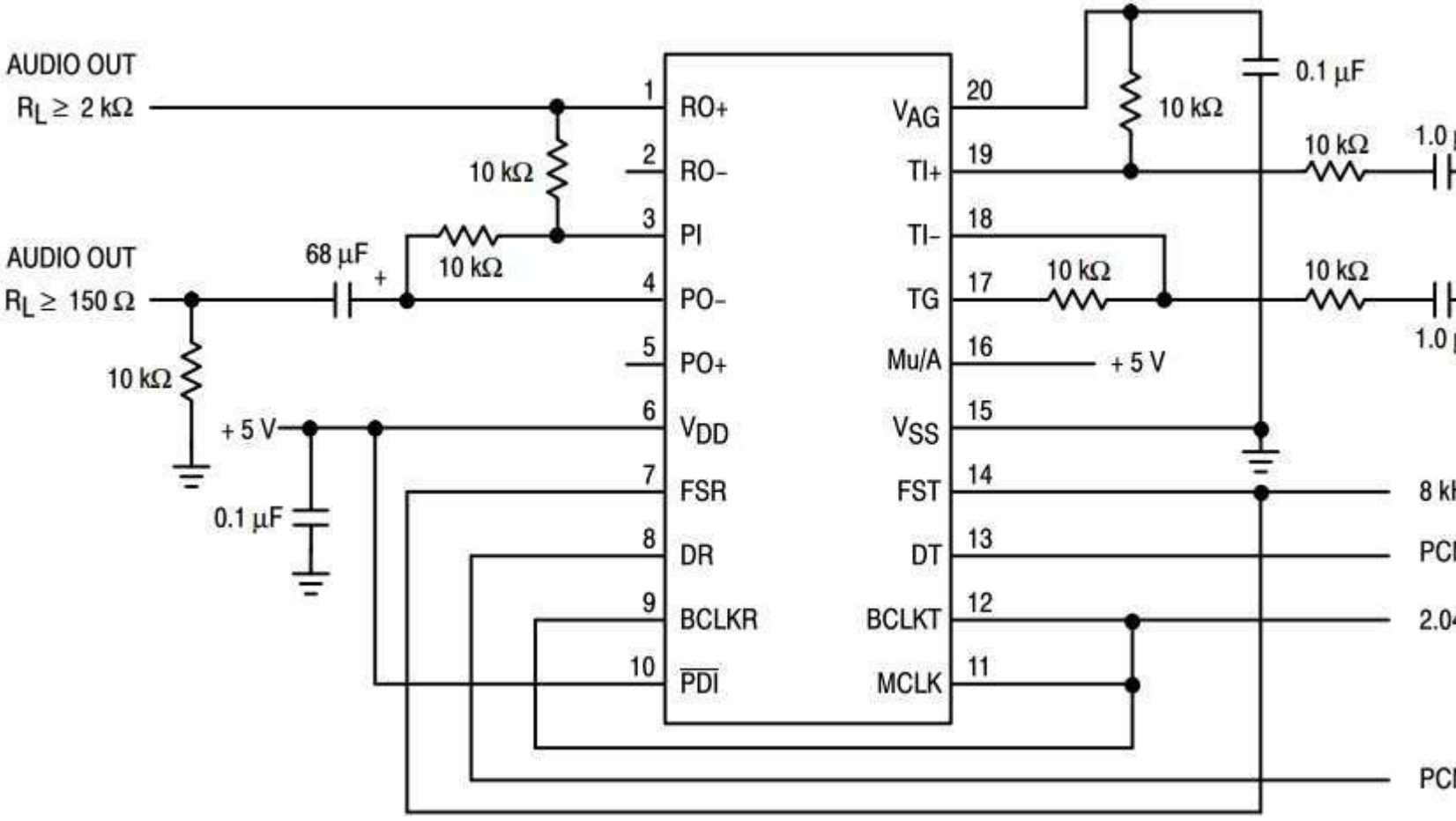
1. The MC14LC5480 is a general purpose per channel PCM Codec–Filter with pin selectable μ–Law or A–Law companding, and is offered in 20–pin DIP, SOG, and SSOP packages.
2. MC14LC5480 performs voice digitization and reconstruction as well as the band limiting and smoothing required for PCM systems.
3. MC14LC5480 designed to operate in both synchronous and asynchronous applications and contains an on–chip precision reference voltage.

MC14LC5480 - BLOCK DIA



µ/A Law
This pin
encoder
decoder
Mu–Law
this pin
A–Law c
pin is co

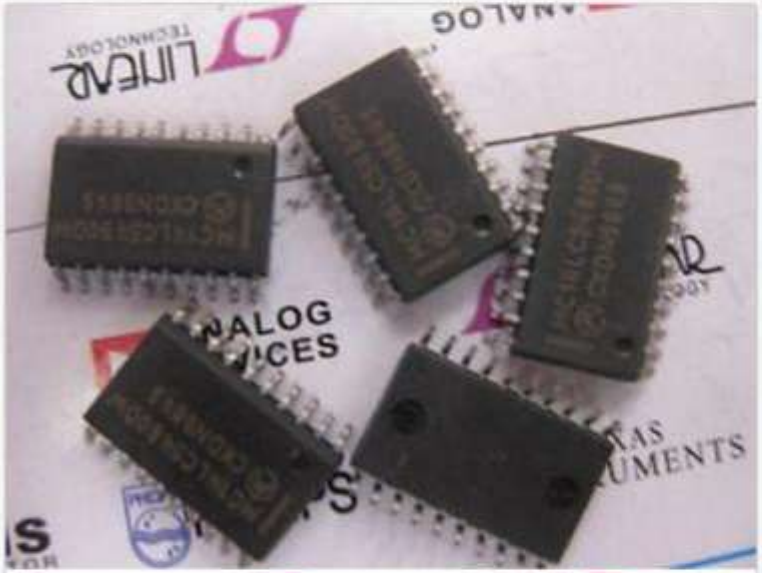
MC14LC5480 - TYPICAL CONN



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2x MC14LC5480DW 5V PCM Codec-filter MC14LC548

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Item location: **Shenzhen, China**
Ships to: **Worldwide** See exclusions

ADVANTAGES OF P

1. PCM provides high noise immunity
2. Allows regeneration of clean signal by using repeaters placed between the transmitter and the receiver.
3. PCM signals can be stored for later use or retransmissionwith high fidelity
4. PCM signals can be encrypted more easily and to very high standards.

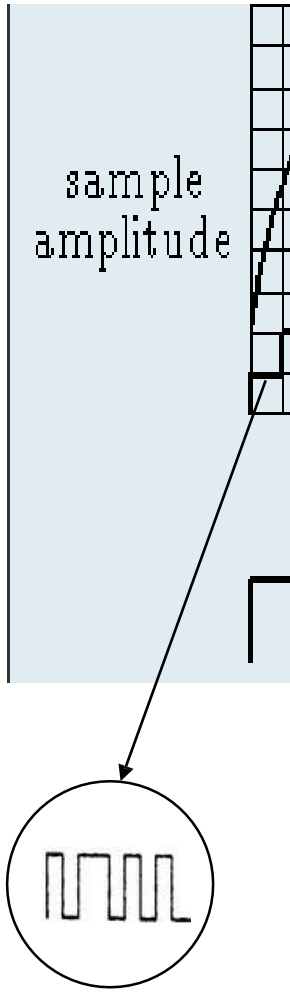
DISADVANTAGES OF

1. PCM requires complex circuitry to sample, quantize, code and decode.
2. PCM requires large bandwidth compared with that of the original analog signal.

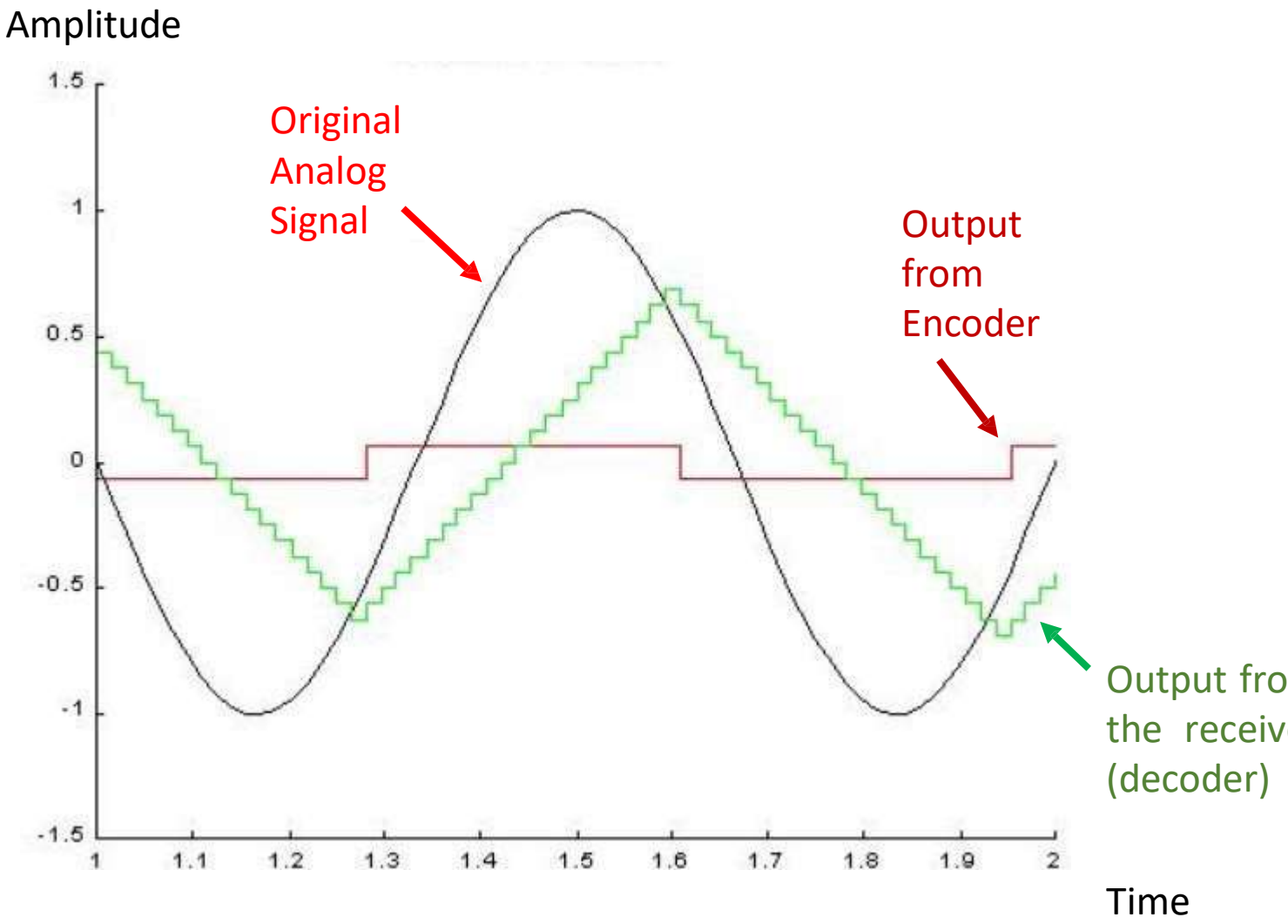
DELTA MODULATION

DELTA MODULAT

1. **Delta modulation** seeks to overcome the problem of high bandwidth requirement in conventional PCM.
2. Instead of generating and transmitting many bits per sample, only one bit is transmitted.
3. During coding, the present sample is compared with the previous and a 0 or 1 transmitted depending on whether the sample is higher or lower than the previous.

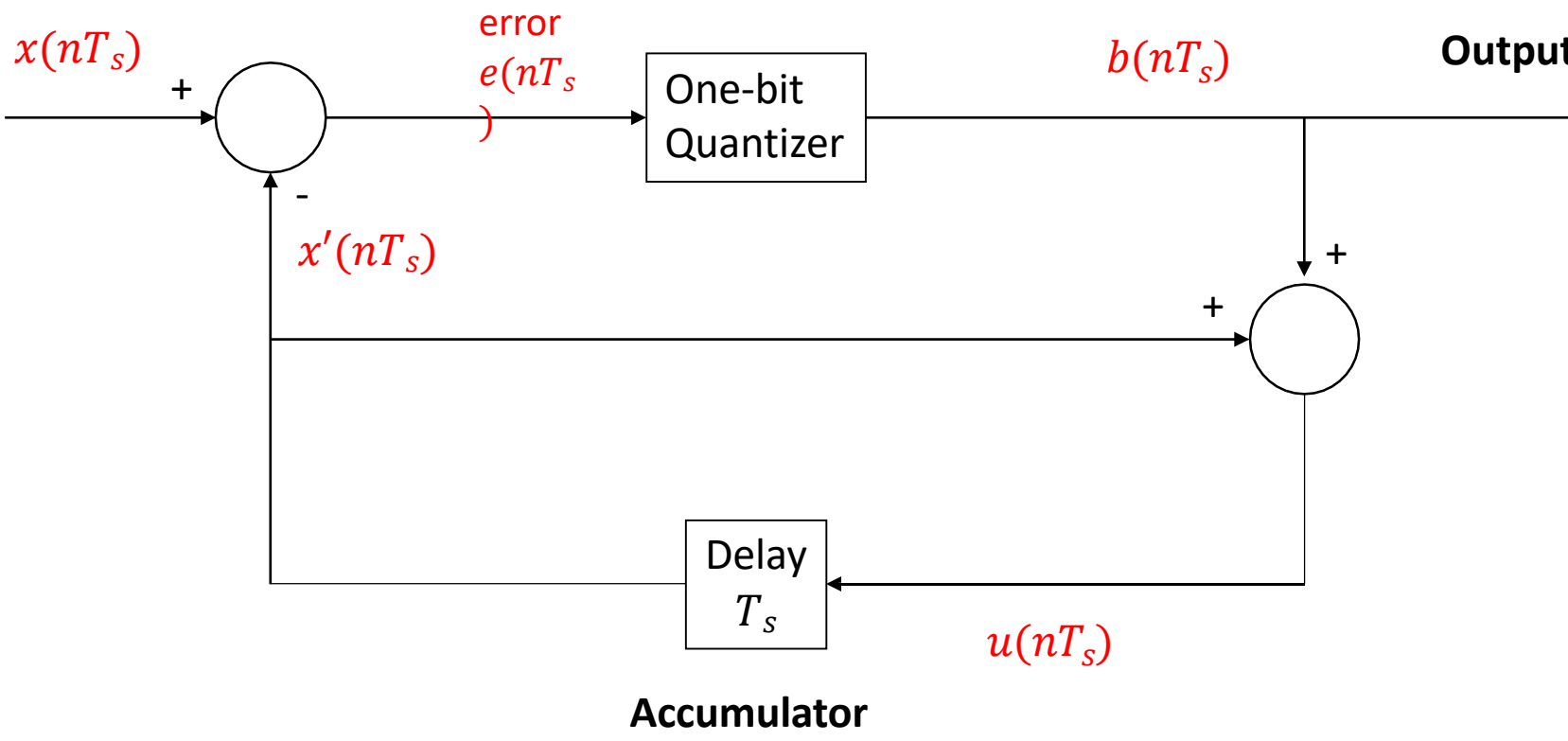


SIGNALS IN A DELTA MODULATION

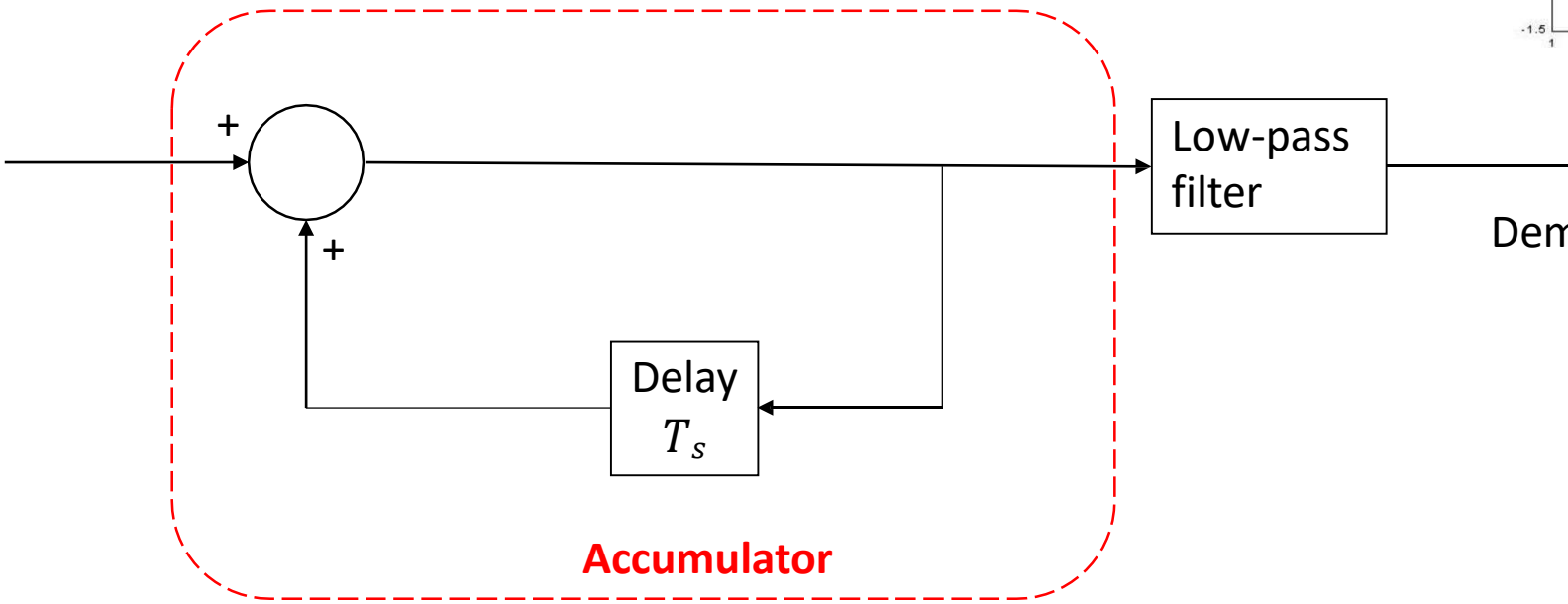


DELTA MODULATION TRANS





DELTA MODULATION REC



ADVANTAGES OF DELTA MODU

Delta Modulation:

1.Requires very small bandwidth since it transmits only one bit per sample

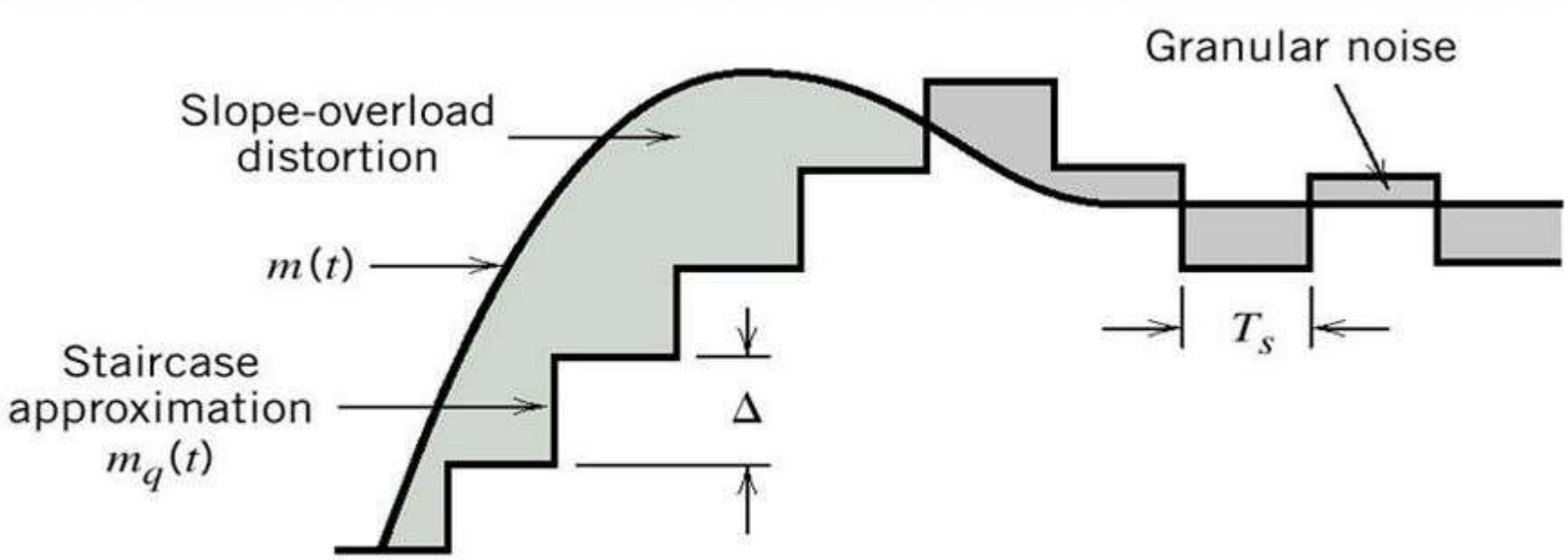
2.Has very simple transmitter and receiver circuitry .

DISADVANTAGES OF DELTA MOD

Delta modulation has:

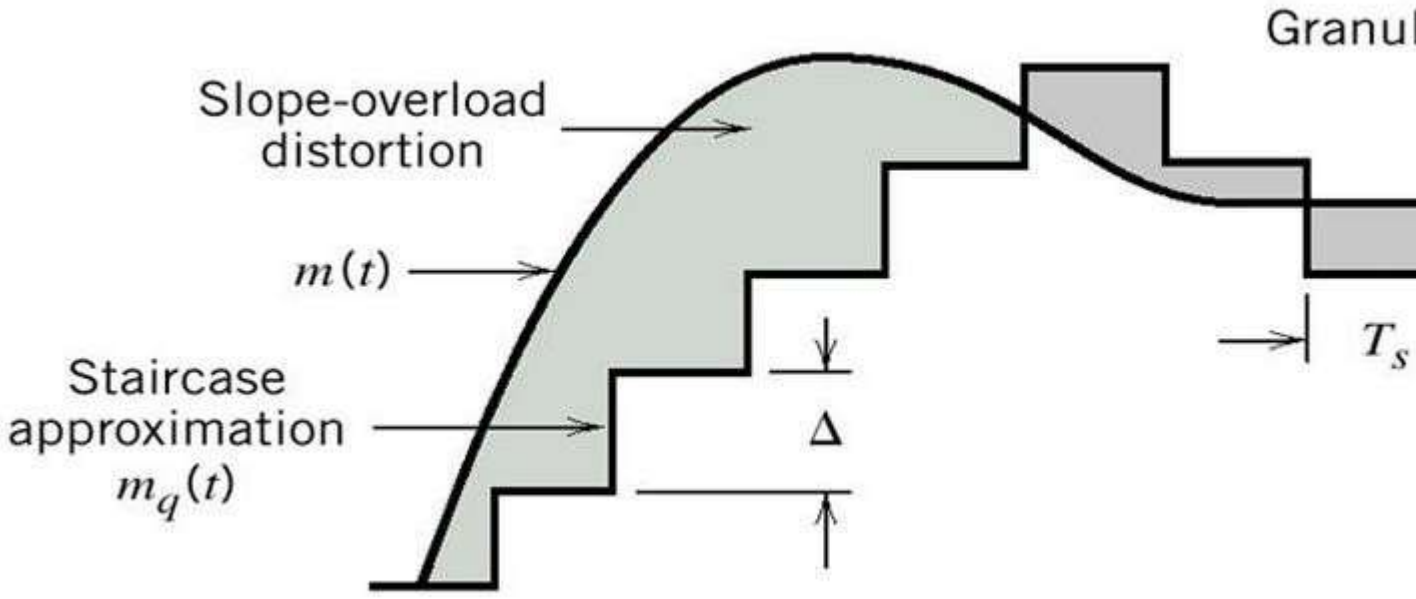
a) Slope and overload distortion

b) Granular and Idle noise



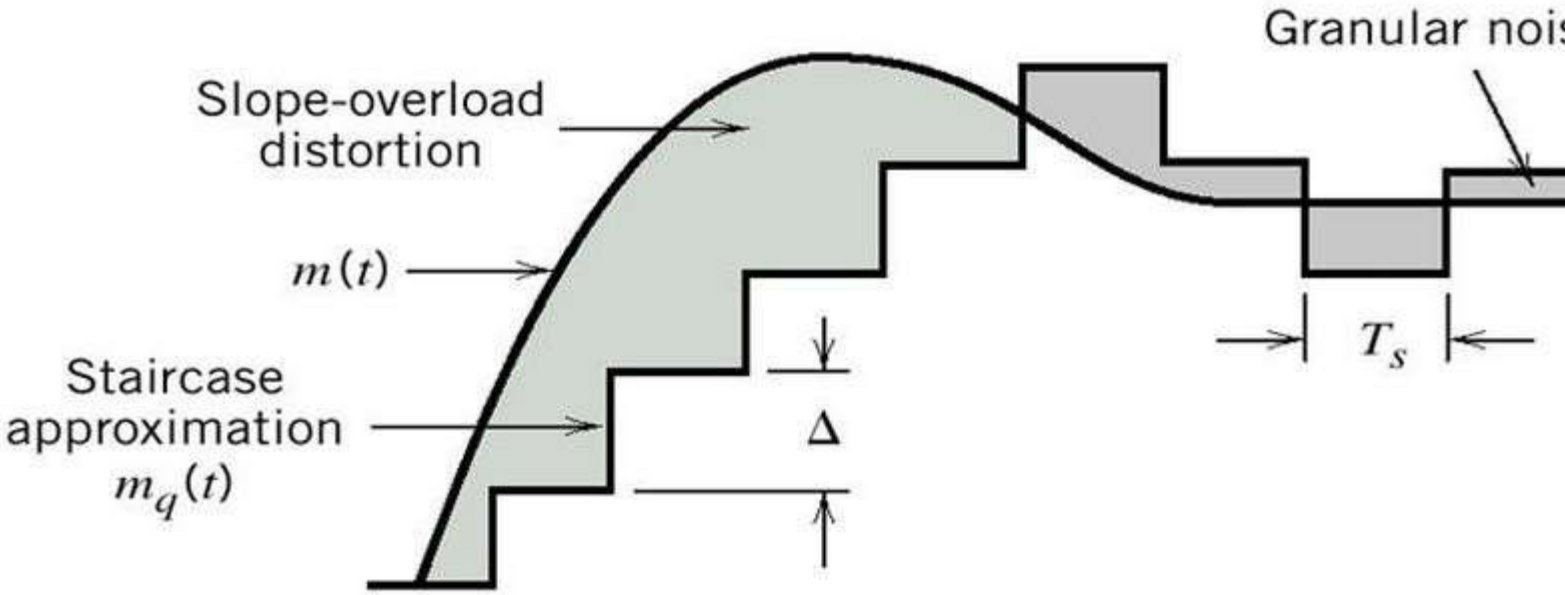
SLOPE OVERLO

- **Slope-overload** occurs when the step size is too small to follow a steep segment of the input waveform x(t).



GRANULARIT

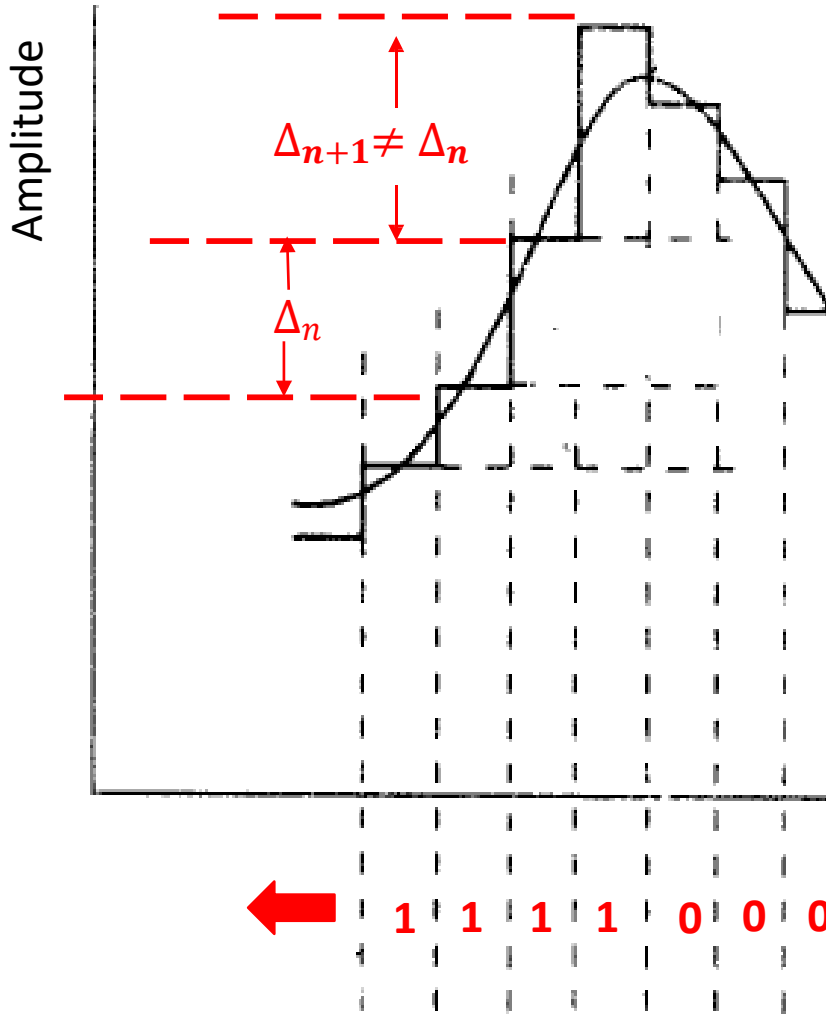
- **Granularity** refers to a situation where the staircase fu
  hunts around a relatively flat segment of the input func
  a step size that is too large relative to the local slope
  characteristic of the input.



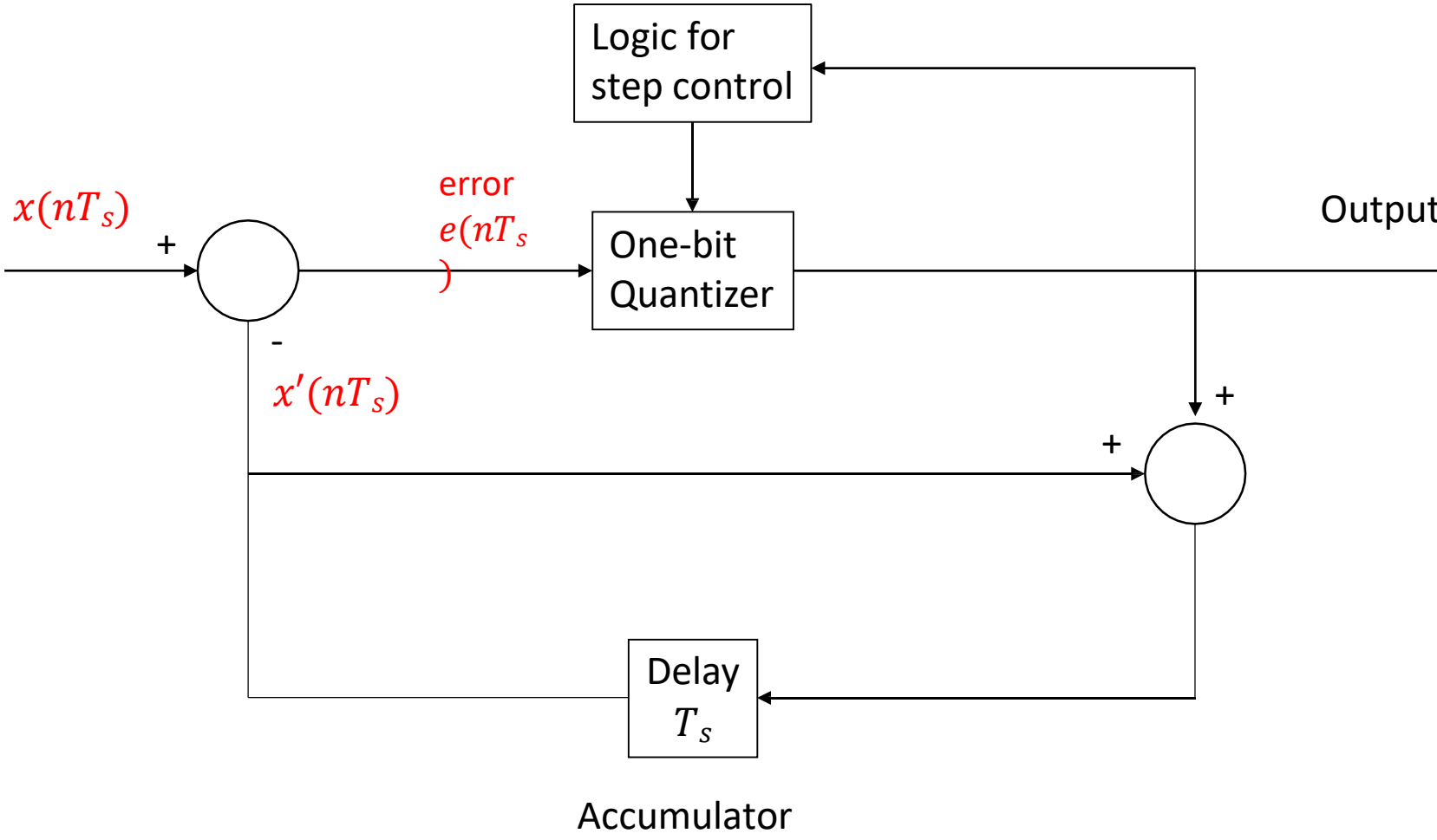
ADAPTIVE DELTA MODULATION

THE PRINCIPLE OF ADAPATIVE DELTA MO

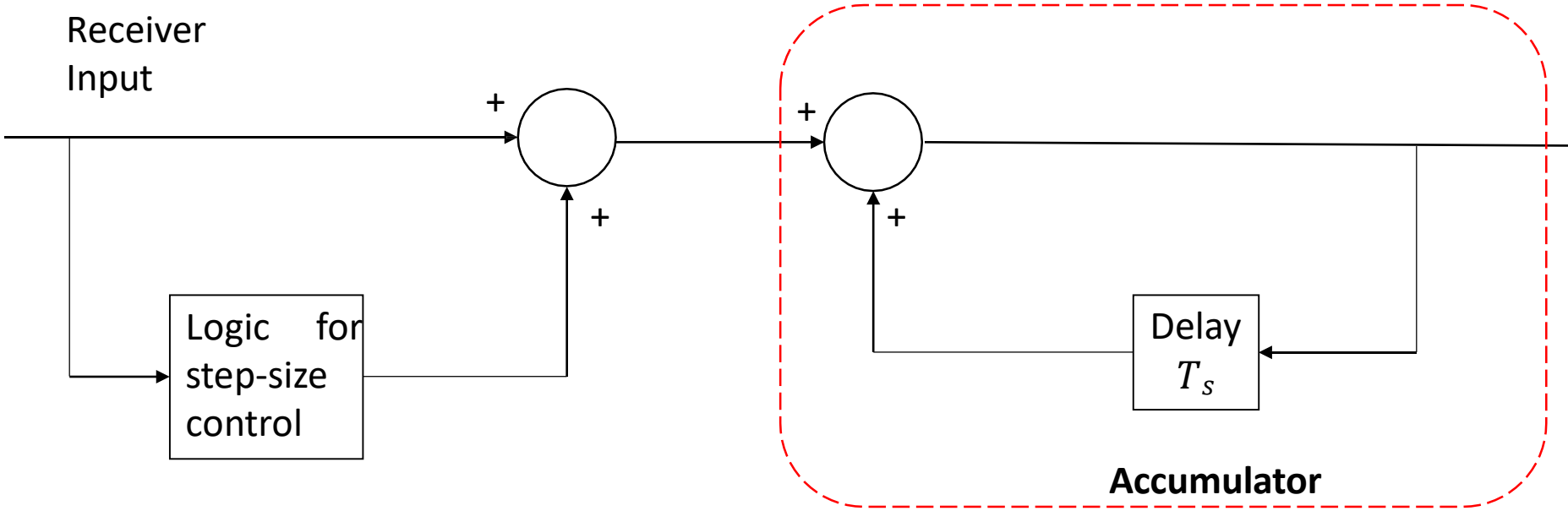
- **Adaptive Delta Modulation** seeks to overcome quantization errors arising from slope overload and granular noise by varying the step size in accordance to the signal amplitude.



ADAPTIVE DELTA MODULATION TRA



ADAPTIVE DELTA MODULATION

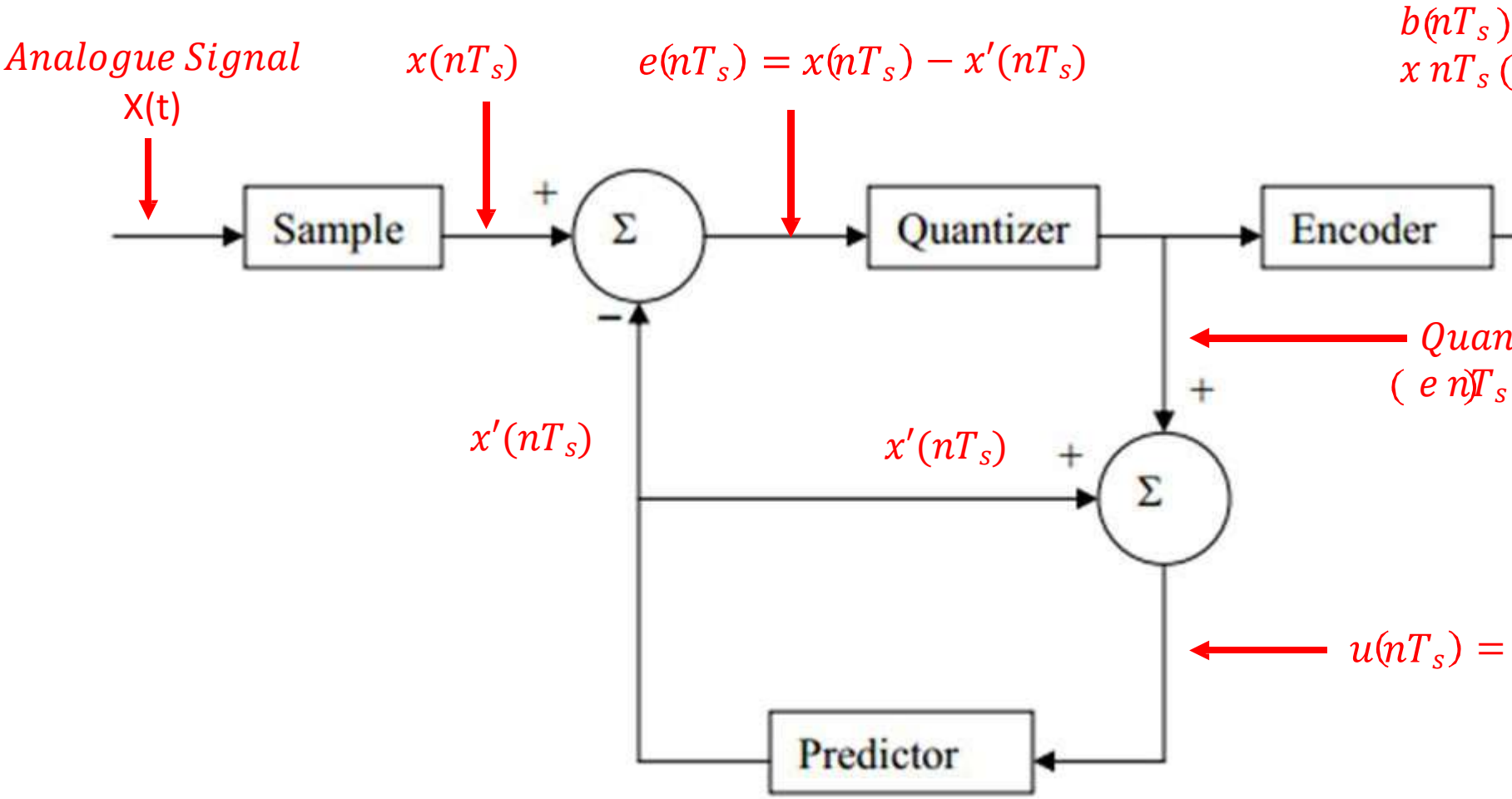


DIFFERENTIAL PULSE CODE MODULATION (DPCM)

DIFFERENTIAL PULSE CODE MOD

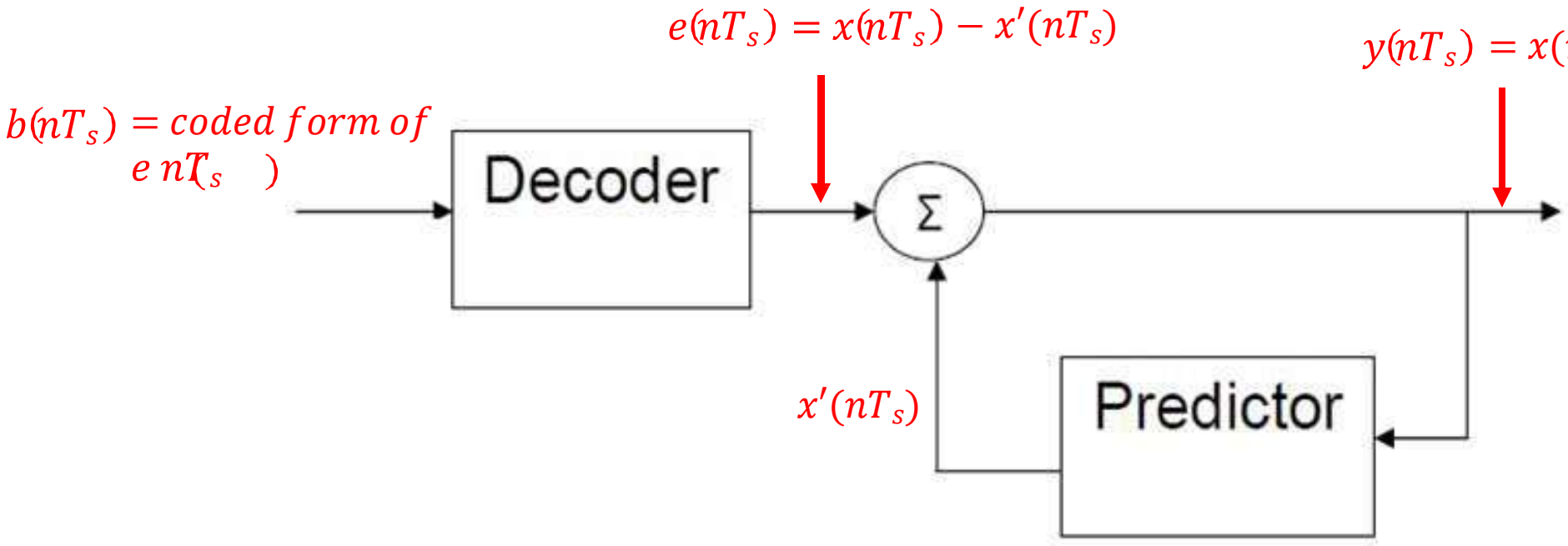
1. Some signals such as speech have high correlation betwe adjacent samples.
2. When such highly correlated samples are encoded using PCM, the resulting code contains a lot of redundant information.
3. In such cases, basic PCM scheme is not the preferred co method.
4. By removing this redundancy before encoding an efficien coded signal can be obtained.
5. One method of removing redundancy is by using the Differential PCM (DPCM) method.
6. **DPCM** is based on the principle that by knowing the pas behaviour of a signal up to a certain point in time, it is possible to predict future values.

DPCM TRANSMIT



Predictor uses $\hat{x}(nT_s)$ and previous values to predict $x'(nT_s)$

DPCM RECEIV



Predictor uses $x^{\wedge}(nT_s)$ and previous values to predict $x'(nT_s)$

COMPARISON OF PCM, DELTA MODULATION AN PULSE CODE MODUL

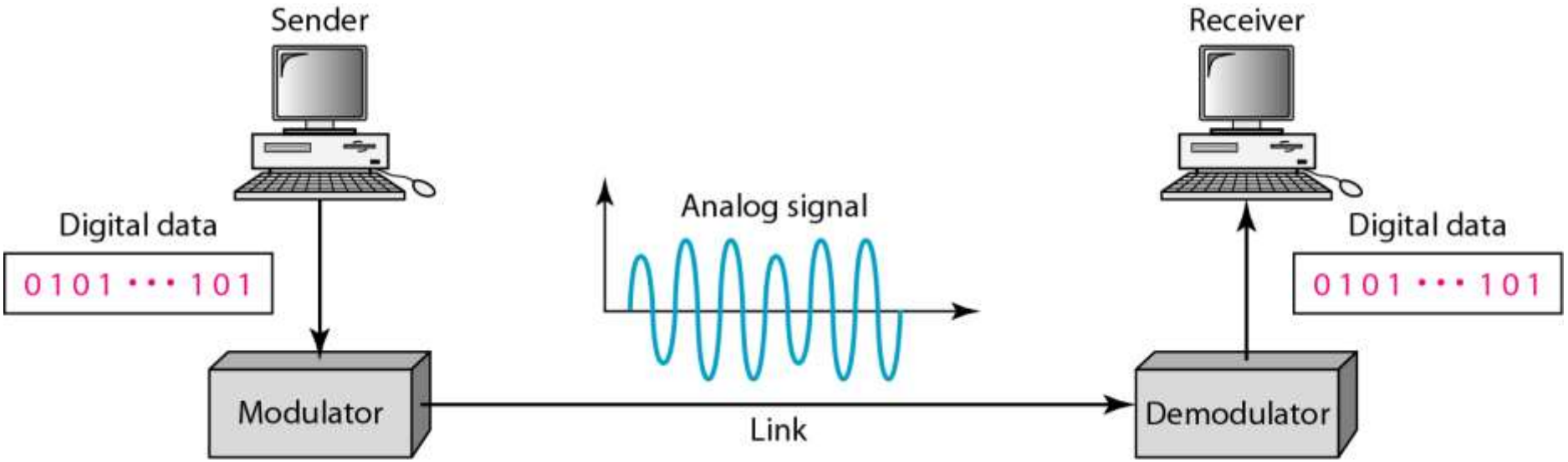
N0	PARAMETER	PULSE CODE MODULATION (PCM)	DELTA MODULATION
1	Levels and Step Size	Number of levels depend on number of bits Level size is fixed	Step size is fixed
2	Number of Bits	Can take 4, 8 or 16 bits per sample	One bit per sample
3	Quantization errors and distortion	Quantization noise is present	Has lope overload and granular noise
4	Bandwidth	Highest bandwidth	Low bandwidth required
5	Feedback in transmitter or receiver	No feedback	Feedback in transmitter
6	Complexity in implementation	High	Simple



Area 9
Analog Transmission

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Figure 5.1 *Digital-to-analog conversion*



5-2 ANALOG AND DIGITAL

Analog-to-analog conversion is the representation of analog information by an analog signal. One may ask why we need to modulate an analog signal; it is already analog. Modulation is needed if the medium is bandpass in nature or if only a bandpass channel is available to us.

Topics discussed in this section:

Amplitude Modulation
Frequency Modulation
Phase Modulation

Figure 5.15 *Types of analog-to-analog modulation*

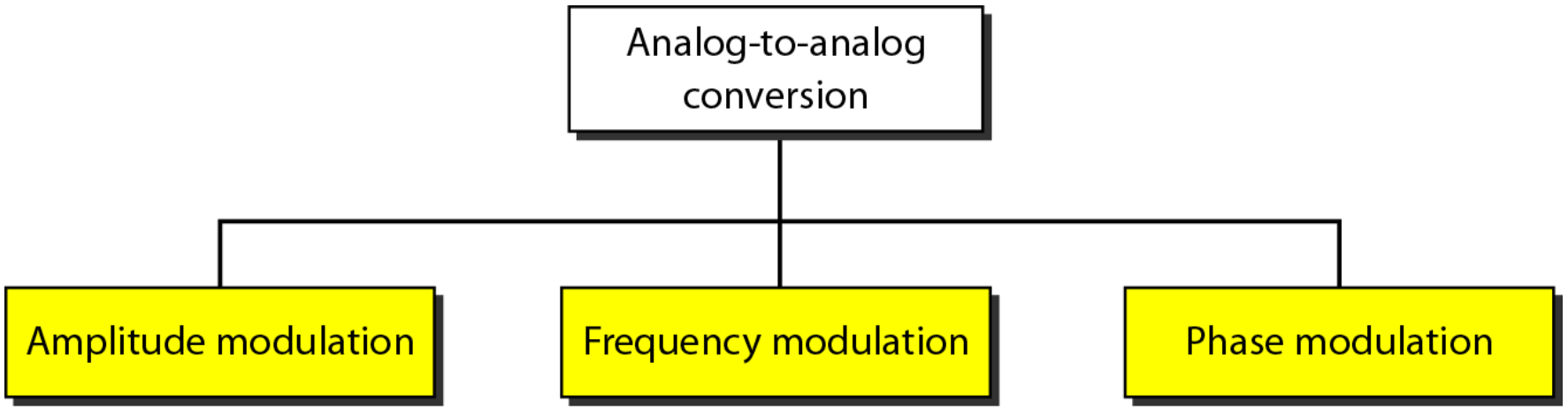
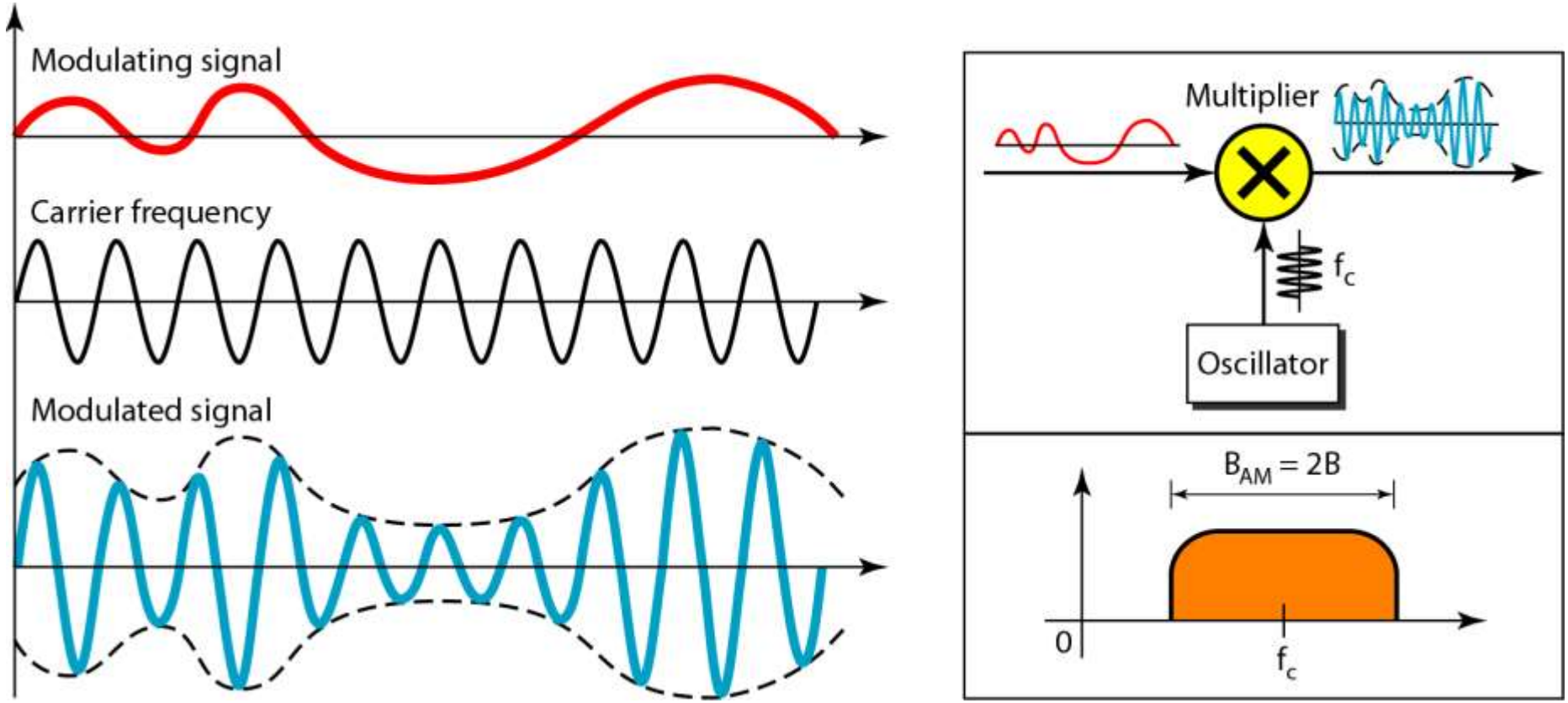


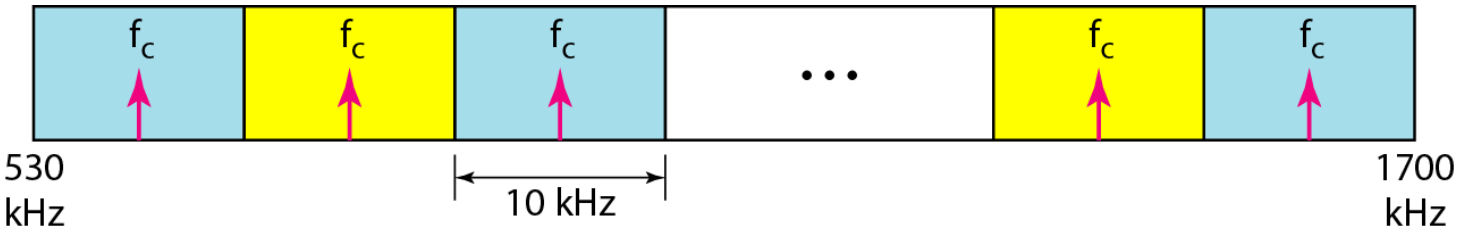
Figure 5.16 *Amplitude modulation*



Note

The total bandwidth required for AM can be determined from the bandwidth of the audio signal: $B_{AM} = 2B$.

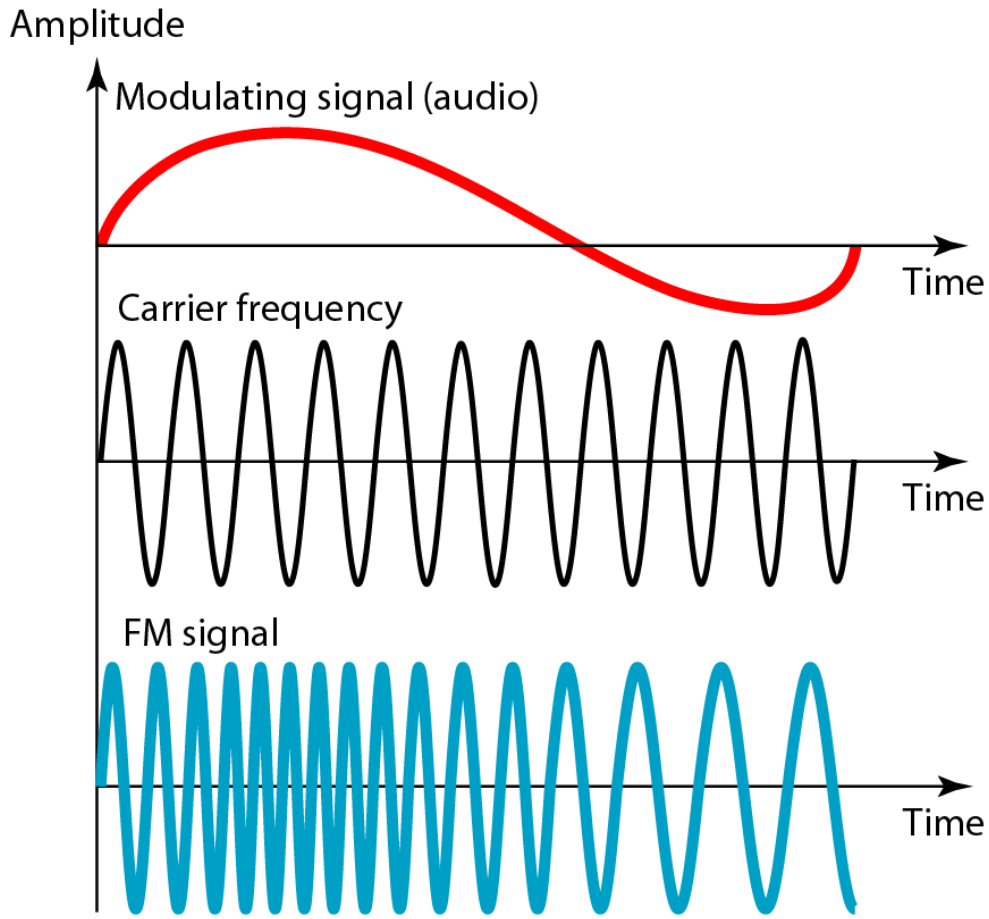
Figure 5.17 *AM band allocation*



Note

The total bandwidth required for FM can be determined from the bandwidth of the audio signal: $B_{FM} = 2(1 + \beta)B$.

Figure 5.18 *Frequency modulation*



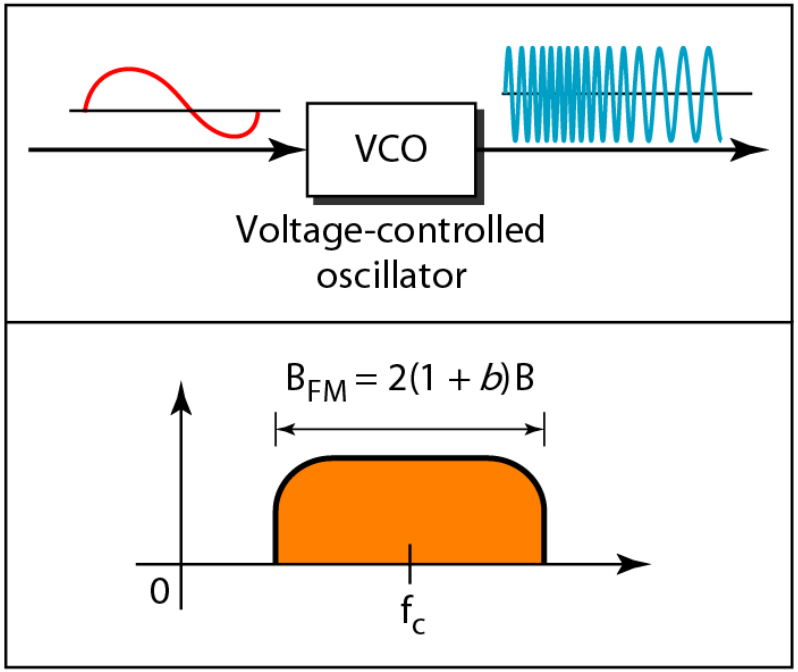


Figure 5.19 *FM band allocation*

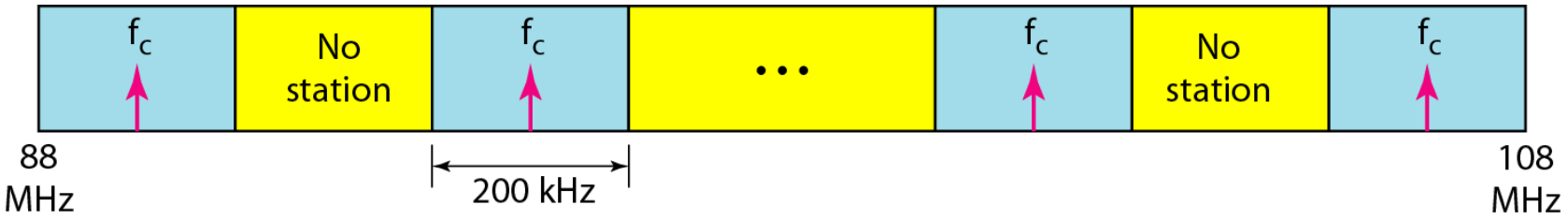
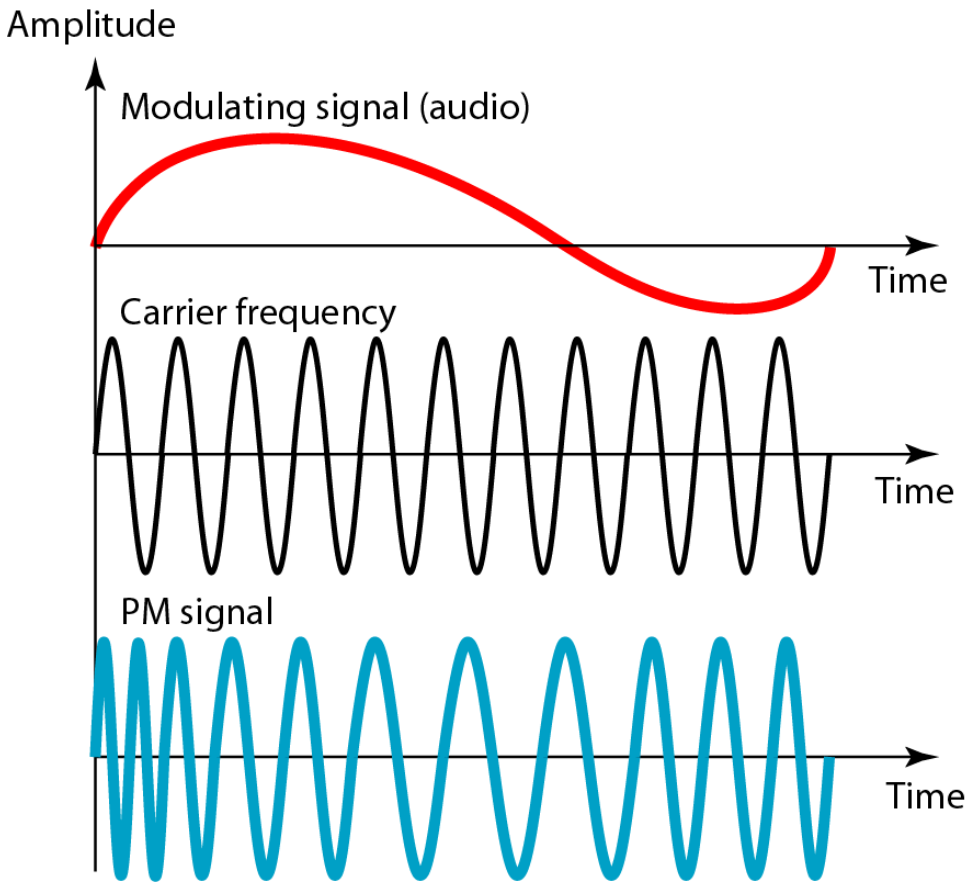
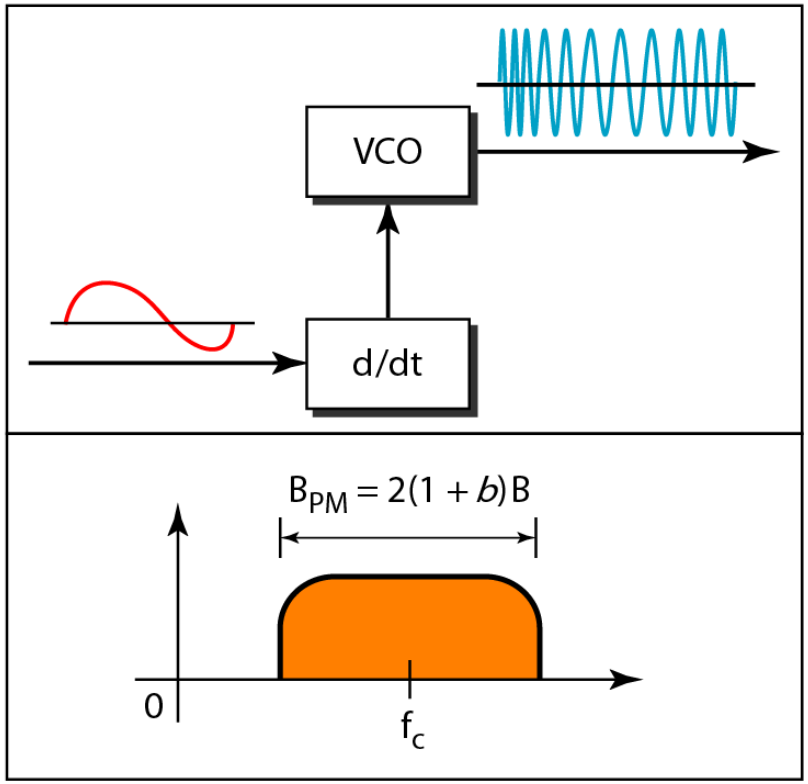


Figure 5.20 *Phase modulation*





Note

The total bandwidth required for PM can be determined from the bandwidth and maximum amplitude of the modulating signal:

$$B_{PM} = 2(1 + \beta)B.$$



Modulation

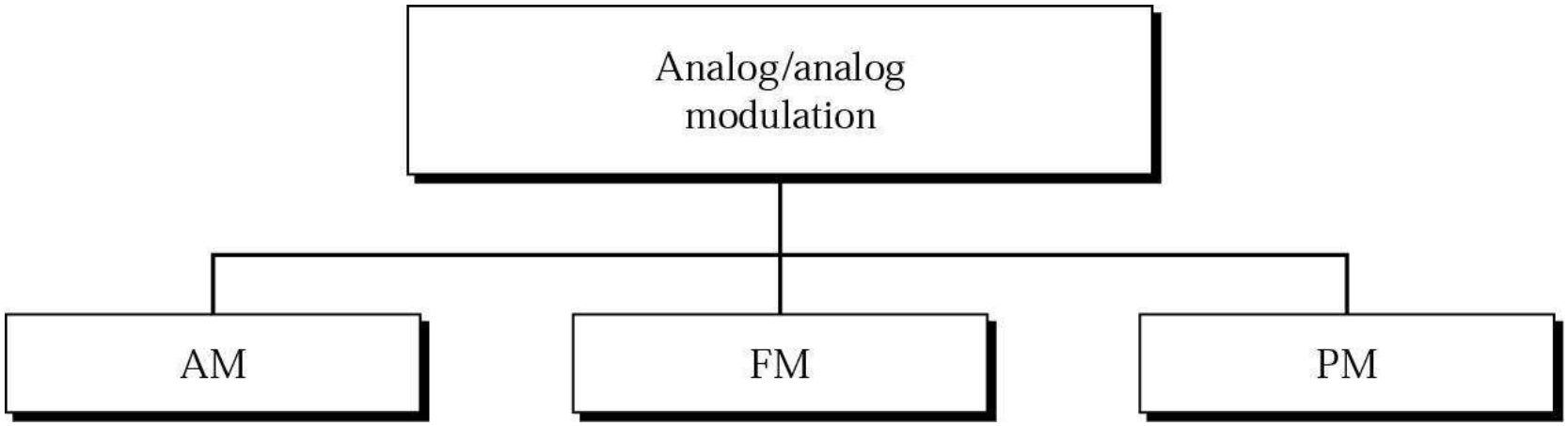
Modulation

Modulation is the process of changing a property of a high-frequency carrier wave (like its amplitude, frequency, or phase) to match a low-frequency message signal. This allows data to travel long distances through the air or cables without losing strength or mixing up with other signals

Analog-to-analog modulation



Main Types of Modulation



Main Types of Modulation Con....

Amplitude Modulation (AM): Changes the height (strength) of th

Frequency Modulation (FM): Changes how fast the wave cycles
second.

Phase Modulation (PM): Shifts the timing or startin
point of the wave's cycle.

Digital variants (ASK, FSK, PSK): Used to send com
bits (0s and 1s) instead of smooth audio waves.

Other Meanings of Modulation

- Music: Moving from one musical key to another.
- Voice: Adjusting the pitch or volume of how you speak to show emotion.

Amplitude Modulation (AM)

- Amplitude Modulation (AM)Amplitude Modulation (AM): Changes the height (strength) of the wave.

Amplitude Modulation

1.3Amplitude Modulation

- The amplitude of high-carrier signal is varied according to the instantaneous amplitude of the modulating message signal m(t).

Ca

- *Amplitude Modulation*
  - The AM signal is generated using a multiplier.
  - All info is carried in the amplitude of the carrier, AM carrier signal has time-varying envelope.
  - In frequency domain the AM waveform are the lower-side frequency/band ($f_c - f_m$), the carrier frequency f_c, the upper-side frequency/band ($f_c + f_m$).

1.3.2 AM Modulation – Example

- The information signal is usually not a single frequency but a range of frequencies (band). For example, frequencies from 20Hz to 15KHz. If we use a carrier of 1.4MHz, what will be the AM spectrum?

In frequency domain the AM waveform are the lower-side frequency/band ($f_c - f_m$), the carrier frequency f_c, the upper-side frequency/band ($f_c + f_m$). Bandwidth: 2x(25K-20)Hz.

1.3.3Modulation Index of AM Signal

- Modulation index k is a measure of the extent to which a carrier voltage is varied by the modulating signal. When k=0 no modulation, when k=1 100% modulation, when k>1 over modulation.

1.3.4 Demodulation of AM Signals

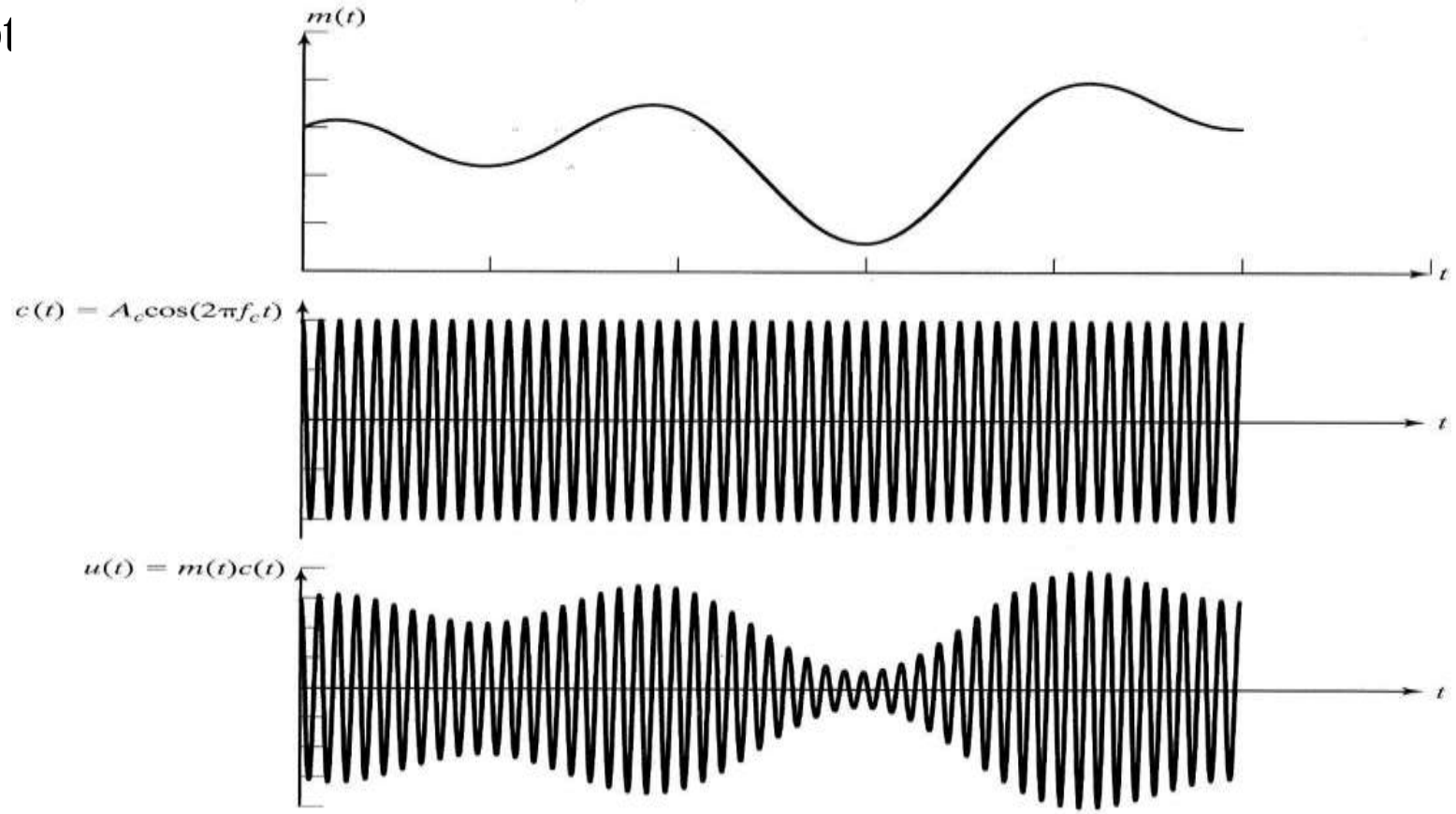
Demodulation extracting the baseband message from the carrier.

There are 2 main methods of AM Demodulation:

1.Envelope or non-coherent detection or demodulation 2.Synchronised or coherent demodulation.

Double-Sideband Suppressed-Carrier AM

A double-sideband, suppressed-carrier (DSB-SC) AM signal is obt



Frequency Modulation (FM): Changes how fast the wave cycles per second.

Phase Modulation (PM)

Phase Modulation (PM): Shifts the timing or starting point of the wave's cycle.



Data Communications and Networking Fourth Edition

Forouzan

Chapter 6

Bandwidth Utilization: Multiplexing and Spreading

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Note

Bandwidth utilization is the wise use of available bandwidth to achieve specific goals.

Efficiency can be achieved by multiplexing; privacy and anti-jamming can be achieved by spreading.

6-1 MULTIPLEXING

Whenever the bandwidth of a medium linking two devices is greater than the bandwidth needs of the devices, the link can be shared. Multiplexing is the set of techniques that allows the simultaneous transmission of multiple signals across a single data link. As data and telecommunications use increases, so does traffic.

Topics discussed in this section:

Frequency-Division Multiplexing
Wavelength-Division Multiplexing
Synchronous Time-Division Multiplexing
Statistical Time-Division Multiplexing

Multiplexing and Demultiplexing

- Multiplexing is the process of combining multiple data signals into one single stream to send over a shared communication channel.
- Demultiplexing is the reverse process of separating that combined stream back into the original, individual signals at the receiving end

Multiplexing and Demultiplexing con.....

- Multiplexing (Sender Side): Takes data from different sources or app programs, adds address headers (like port numbers), and merges them onto one path.
- Demultiplexing (Receiver Side): Examines the header data in the received mixed signal and routes each piece of data to the correct target app or destination

Why Multiplexing is Needed

- Resource Sharing: It combines data from multiple application processes (like web browsers, email clients, and chat apps) into a single stream so they can share one network cable or Wi-Fi channel.
- Cost and Efficiency: It saves money and bandwidth by preventing the need to build a separate physical wire or network path for every single app running on your computer.
- Adding Headers: At the sending side (primarily managed by the Transport Layer via protocols like TCP or UDP), data is bundled with source and destination port numbers to tag which app sent it.

Why Demultiplexing is Needed

- Correct Delivery: It is the reverse process that takes the incoming shared data stream at the receiving device and sorts it out.

- Targeting Apps: It uses the port numbers inside the packet headers to deliver the right data chunks to the correct application process, ensuring a WhatsApp message does not accidentally open inside your web browser.

Figure 6.1 *Dividing a link into channels*

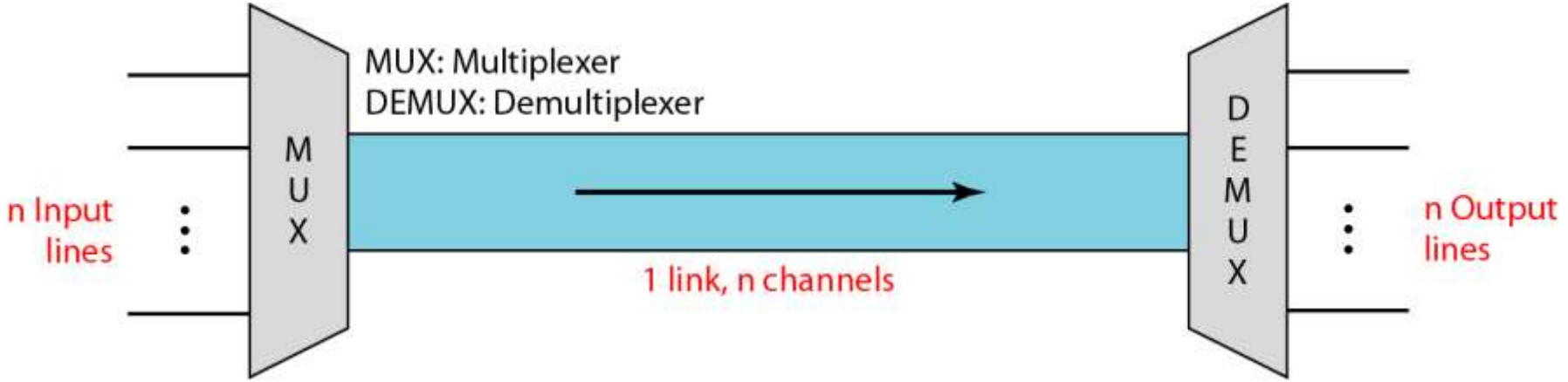


Figure 6.2 *Categories of multiplexing*

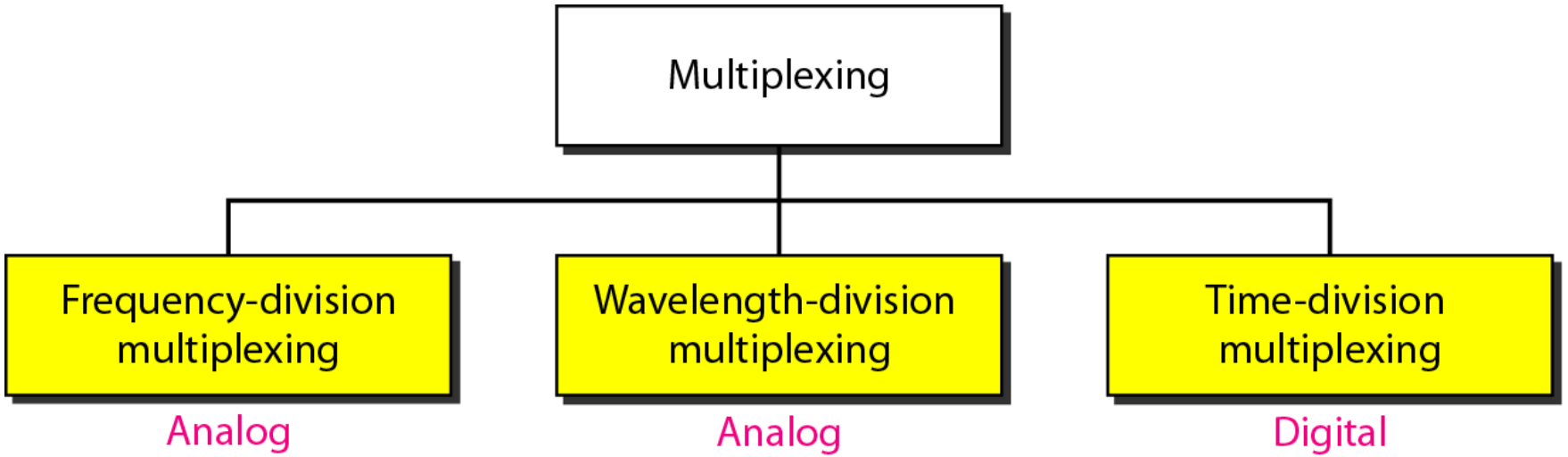


Figure 6.3 *Frequency-division multiplexing*



Note

FDM is an analog multiplexing technique that combines analog signals.

Figure 6.4 *FDM process*

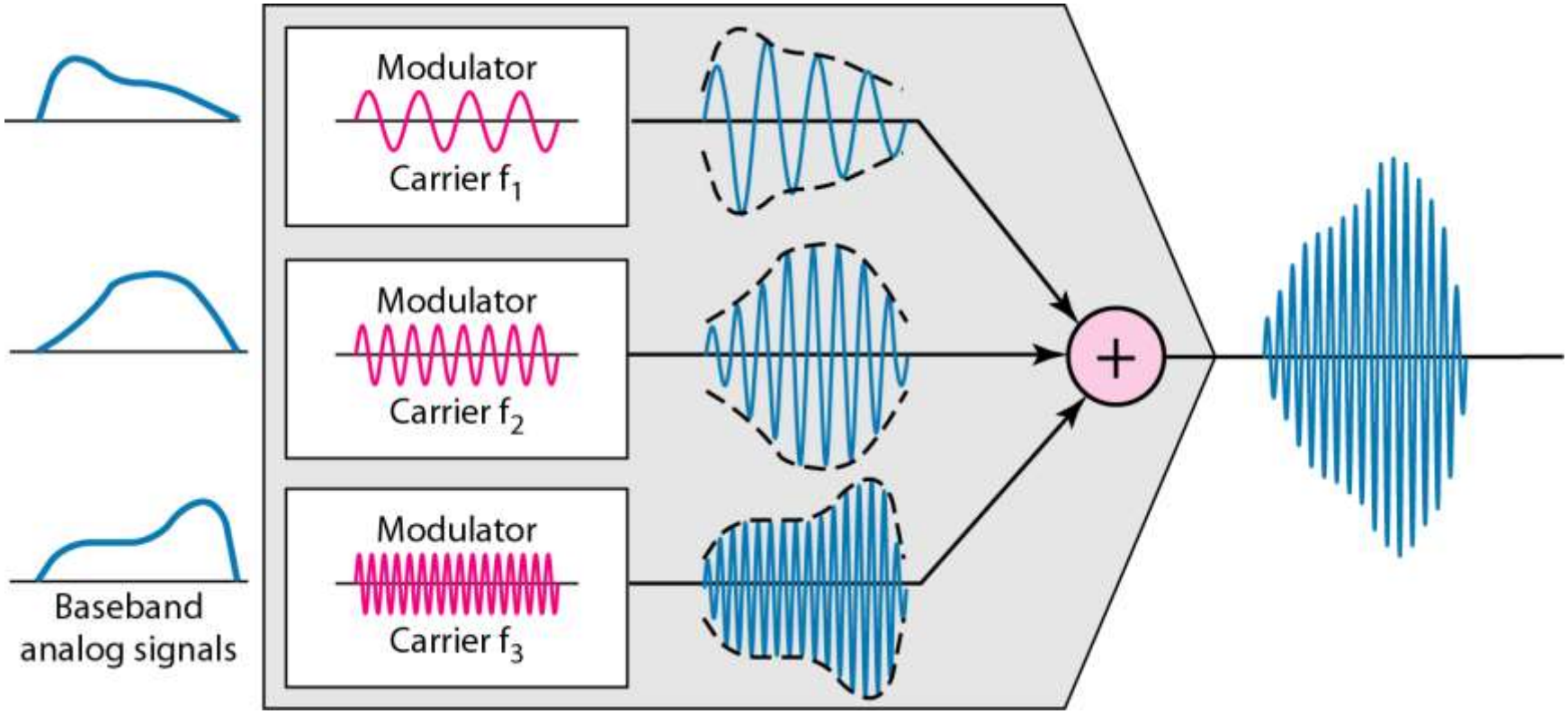
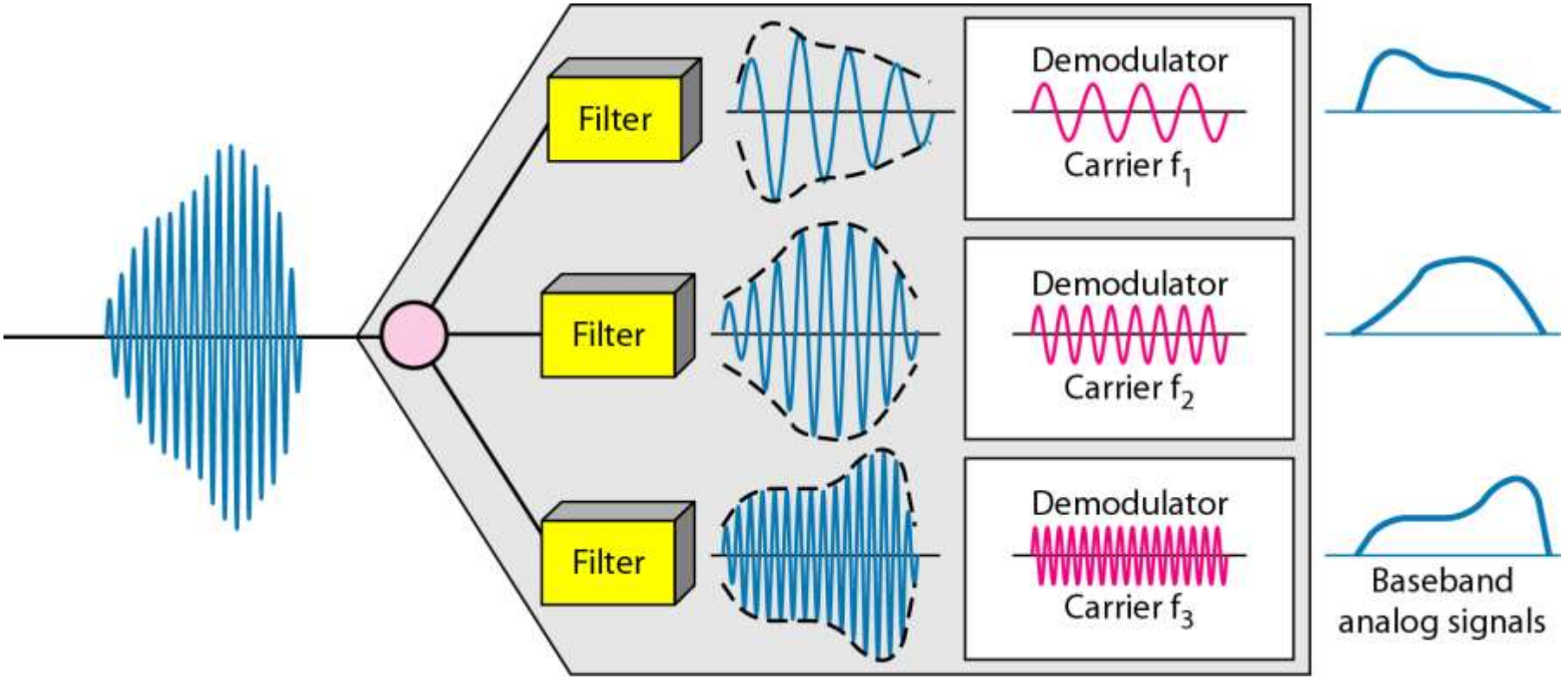


Figure 6.5 *FDM demultiplexing example*



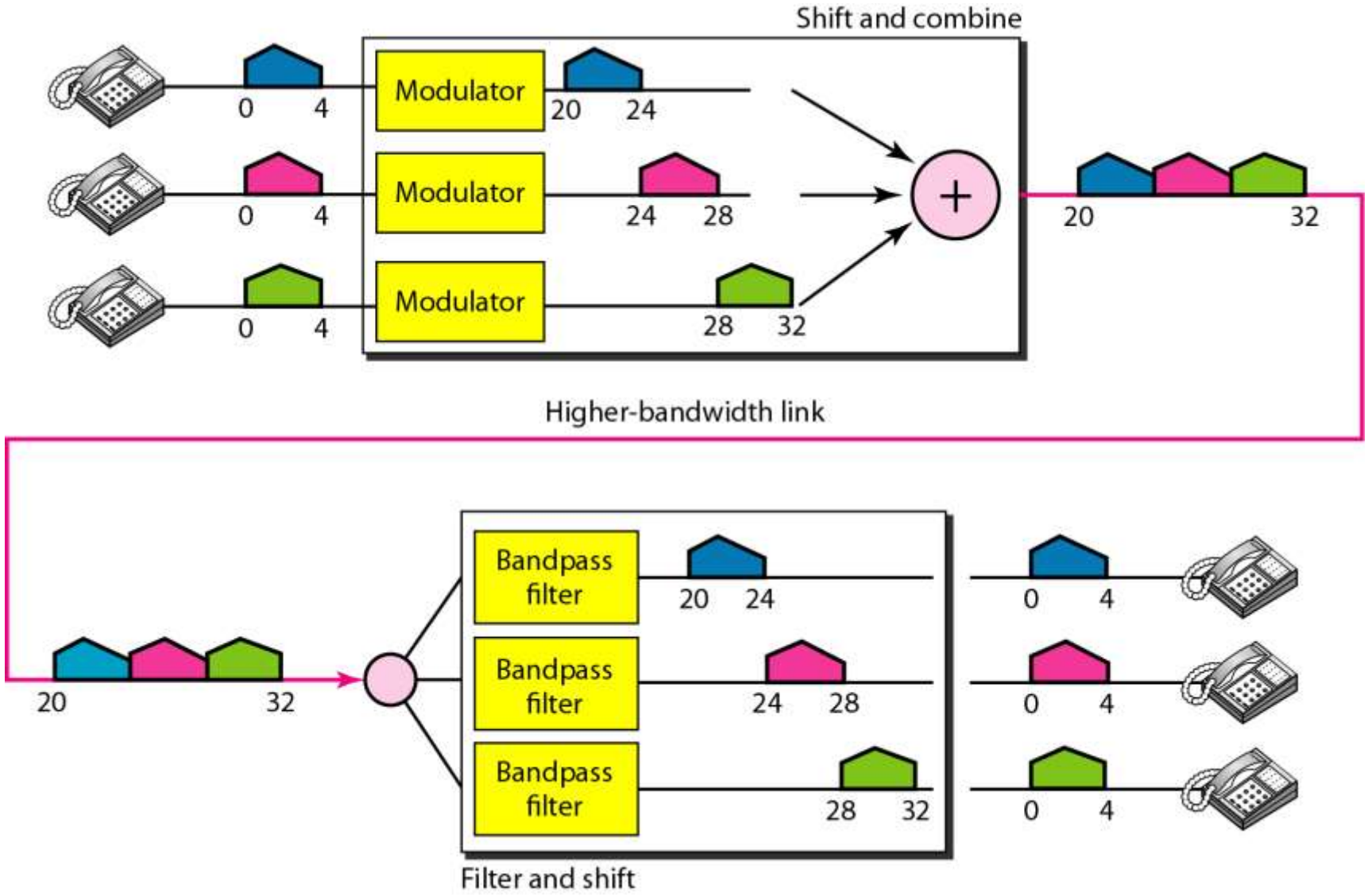
Example 6.1

Assume that a voice channel occupies a bandwidth of 4 kHz. We need to combine three voice channels into a link with a bandwidth of 12 kHz, from 20 to 32 kHz. Show the configuration, using the frequency domain. Assume there are no guard bands.

Solution

We shift (modulate) each of the three voice channels to a different bandwidth, as shown in Figure 6.6. We use the 20- to 24-kHz bandwidth for the first channel, the 24- to 28-kHz bandwidth for the second channel, and the 28- to 32-kHz bandwidth for the third one. Then we combine them as shown in Figure 6.6.

Figure 6.6 *Example 6.1*



Example 6.2

Five channels, each with a 100-kHz bandwidth, are to be multiplexed together. What is the minimum bandwidth of the link if there is a need for a guard band of 10 kHz between the channels to prevent interference?

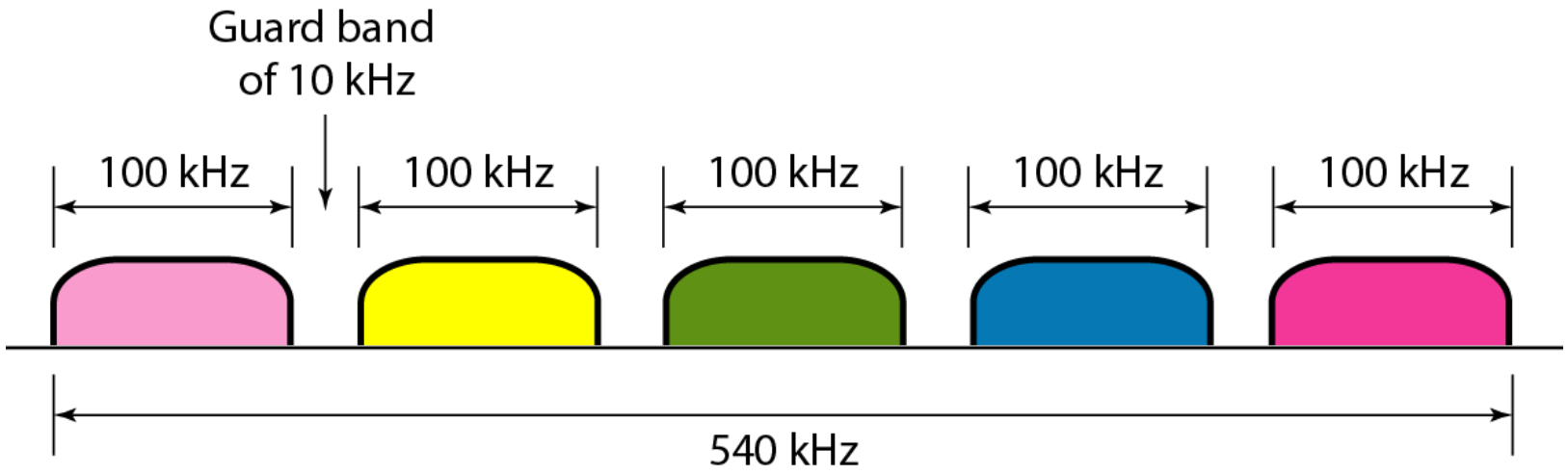
Solution

For five channels, we need at least four guard bands. This means that the required bandwidth is at least

$$5 \times 100 + 4 \times 10 = 540 \text{ kHz},$$

as shown in Figure 6.7.

Figure 6.7 *Example 6.2*



Example 6.3

Four data channels (digital), each transmitting at 1 Mbps, use a satellite channel of 1 MHz. Design an appropriate configuration, using FDM.

Solution

The satellite channel is analog. We divide it into four channels, each channel having a 250-kHz bandwidth. Each digital channel of 1 Mbps is modulated such that each 4 bits is modulated to 1 Hz. One solution is 16-QAM modulation. Figure 6.8 shows one possible configuration.

Figure 6.8 *Example 6.3*

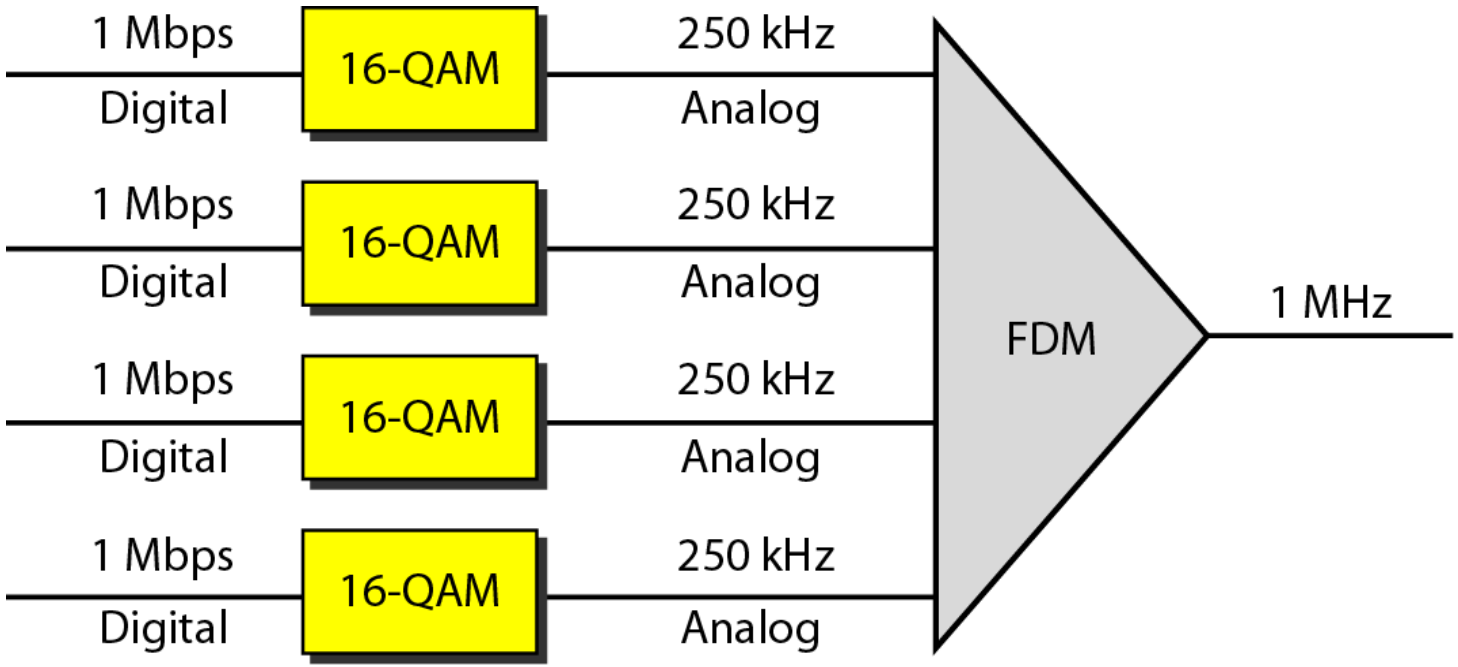
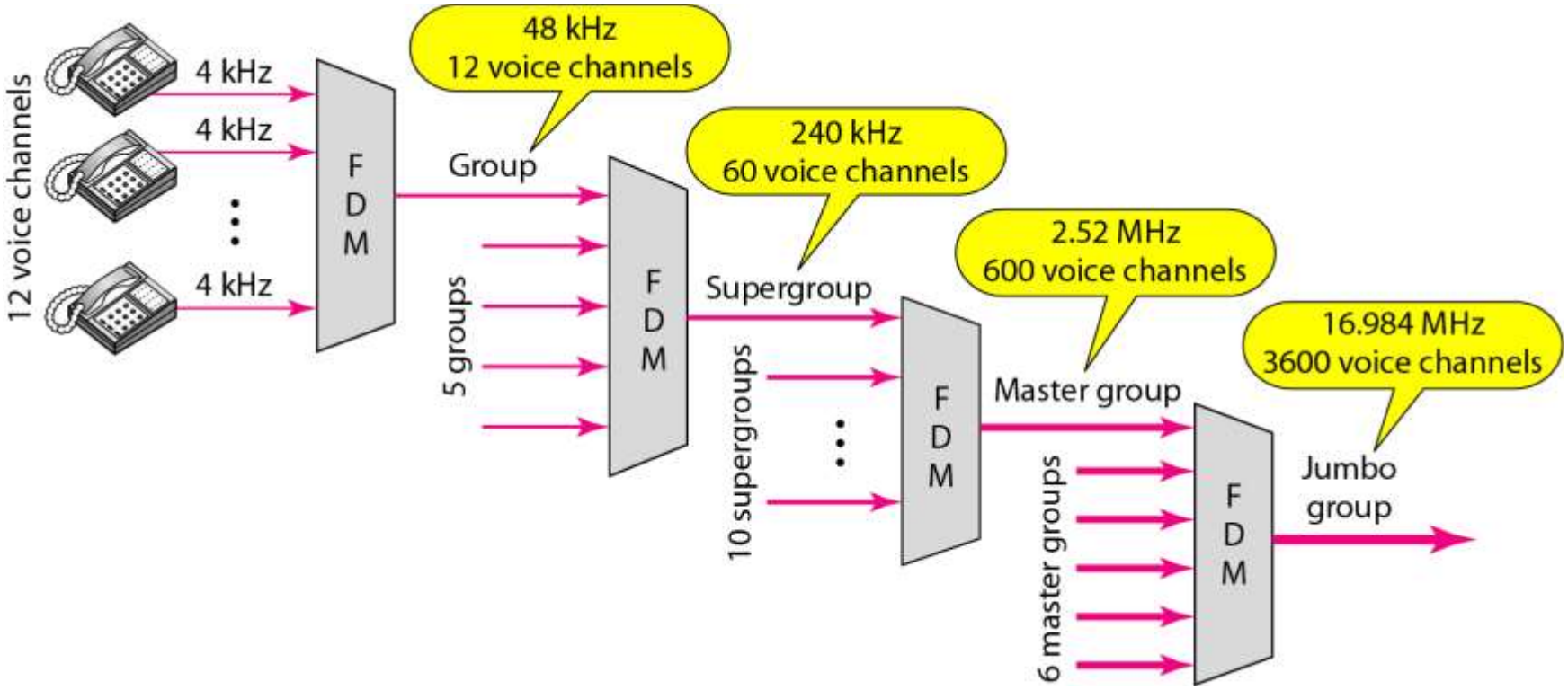


Figure 6.9 *Analog hierarchy*



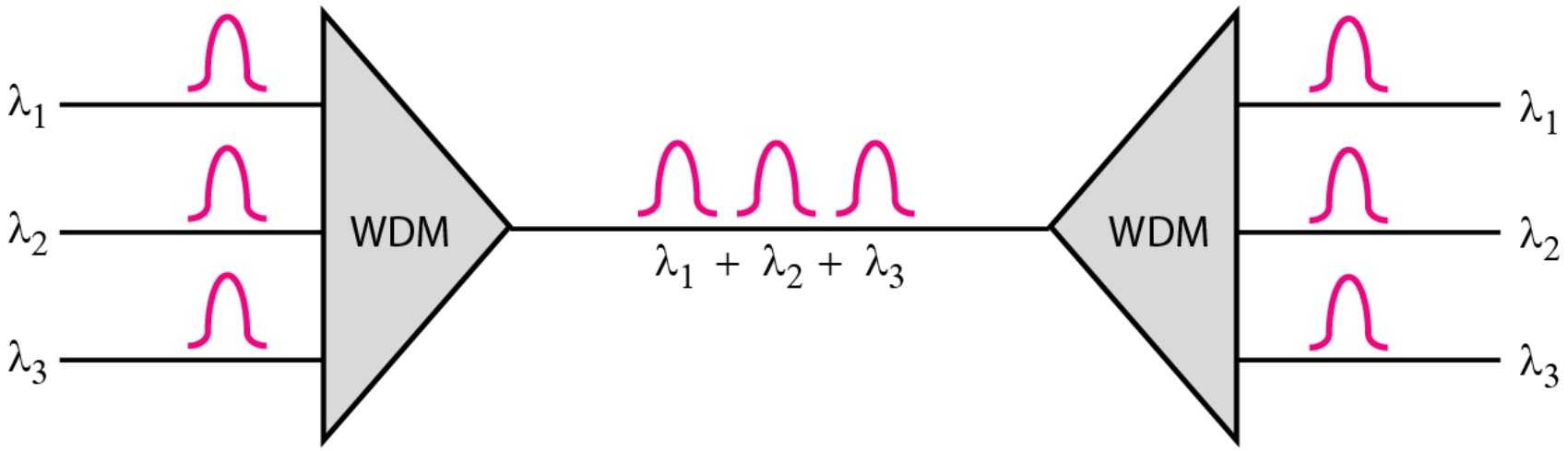
Example 6.4

The Advanced Mobile Phone System (AMPS) uses two bands. The first band of 824 to 849 MHz is used for sending, and 869 to 894 MHz is used for receiving. Each user has a bandwidth of 30 kHz in each direction. How many people can use their cellular phones simultaneously?

Solution

Each band is 25 MHz. If we divide 25 MHz by 30 kHz, we get 833.33. In reality, the band is divided into 832 channels. Of these, 42 channels are used for control, which means only 790 channels are available for cellular phone users.

Figure 6.10 *Wavelength-division multiplexing*



Note

WDM is an analog multiplexing technique to combine optical signals.

Figure 6.11 *Prisms in wavelength-division multiplexing and demultiplexing*

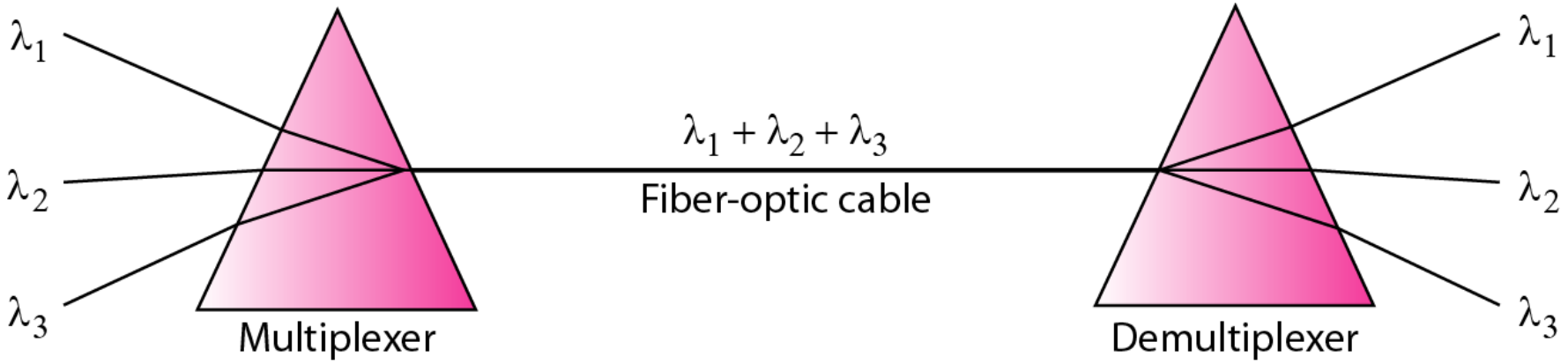
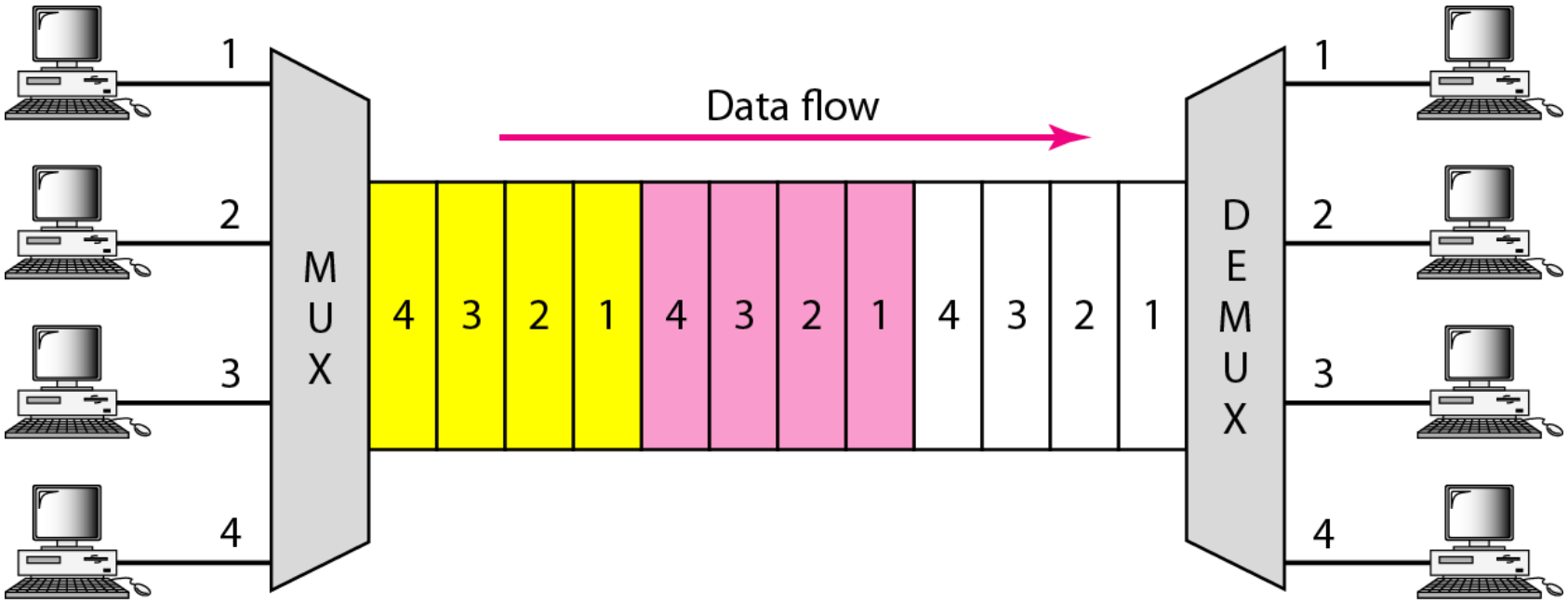


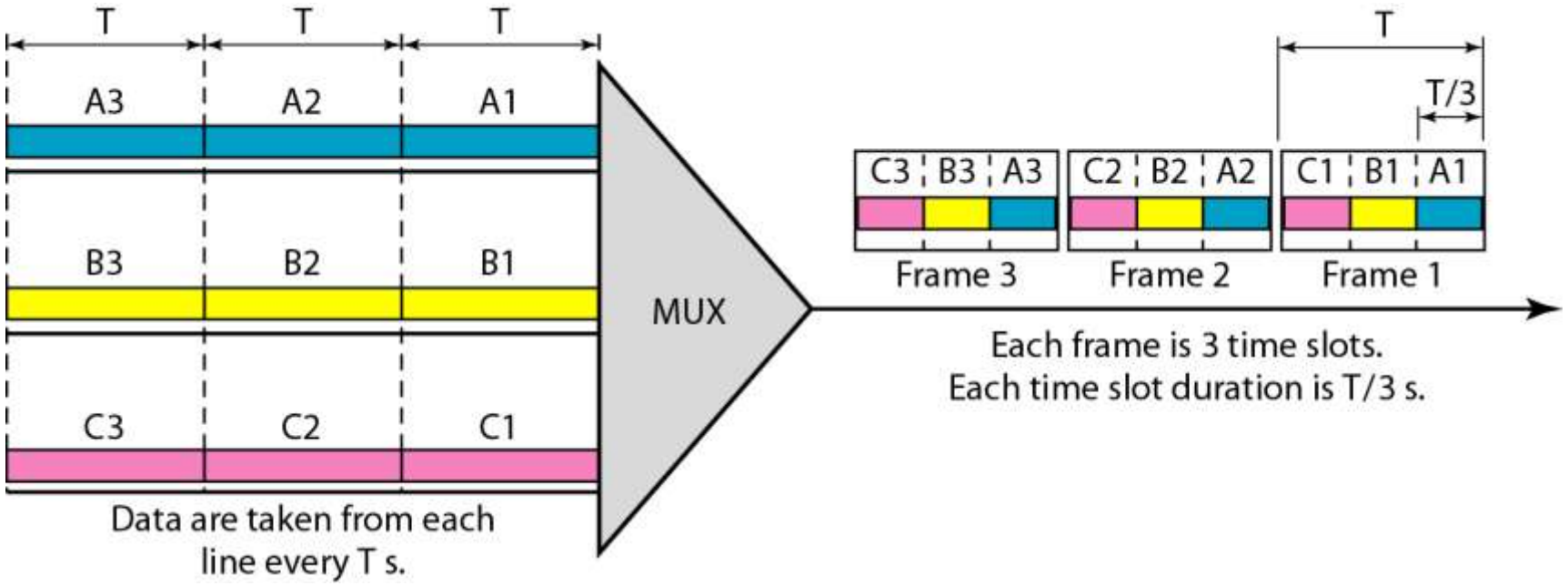
Figure 6.12 *TDM*



Note

TDM is a digital multiplexing technique for combining several low-rate channels into one high-rate one.

Figure 6.13 *Synchronous time-division multiplexing*



Note

In synchronous TDM, the data rate of the link is *n* times faster, and the unit duration is *n* times shorter.

Example 6.5

In Figure 6.13, the data rate for each input connection is 3 kbps. If 1 bit at a time is multiplexed (a unit is 1 bit), what is the duration of (a) each input slot, (b) each output slot, and (c) each frame?

Solution

We can answer the questions as follows:

a. The data rate of each input connection is 1 kbps. This means that the bit duration is 1/1000 s or 1 ms. The duration of the input time slot is 1 ms (same as bit duration).

Example 6.5 (continued)

b. ***The duration of each output time slot is one-third of the input time slot. This means that the duration of the output time slot is 1/3 ms.***

c. ***Each frame carries three output time slots. So the duration of a frame is 3 × 1/3 ms, or 1 ms. The duration of a frame is the same as the duration of an input unit.***

Example 6.6

Figure 6.14 shows synchronous TDM with a data stream for each input and one data stream for the output. The unit of data is 1 bit. Find (a) the input bit duration, (b) the output bit duration, (c) the output bit rate, and (d) the output frame rate.

Solution

We can answer the questions as follows:

a. The input bit duration is the inverse of the bit rate: 1/1 Mbps = 1 µs.

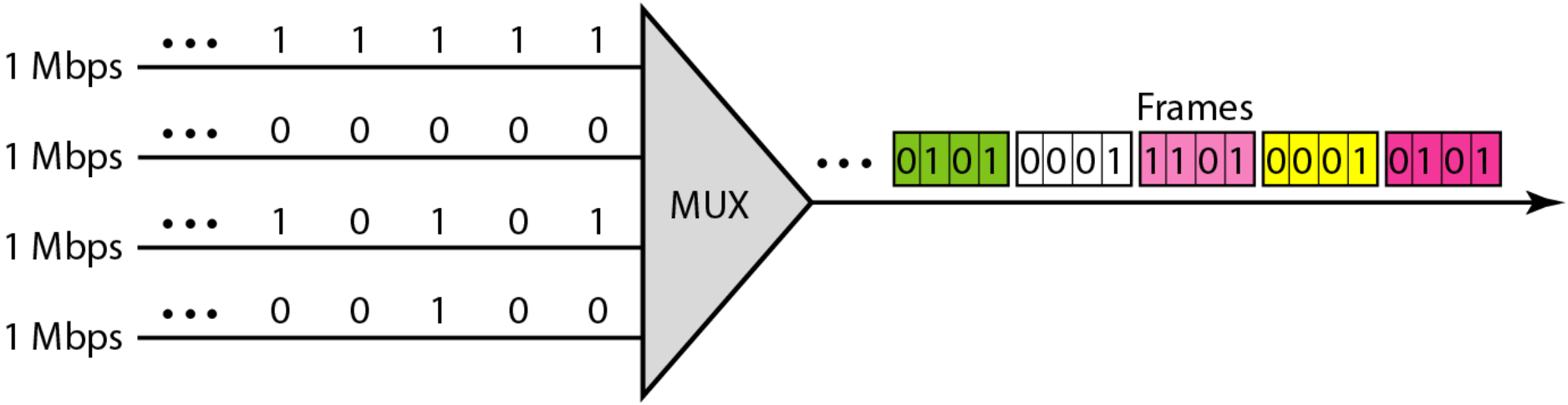
b. The output bit duration is one-fourth of the input bit duration, or ¼ µs.

Example 6.6 (continued)

c. ***The output bit rate is the inverse of the output bit duration or 1/(4μs) or 4 Mbps. This can also be deduced from the fact that the output rate is 4 times as fast as any input rate; so the output rate = 4 × 1 Mbps = 4 Mbps.***

d. ***The frame rate is always the same as any input rate. So the frame rate is 1,000,000 frames per second. Because we are sending 4 bits in each frame, we can verify the result of the previous question by multiplying the frame rate by the number of bits per frame.***

Figure 6.14 *Example 6.6*



Example 6.7

Four 1-kbps connections are multiplexed together. A unit is 1 bit. Find (a) the duration of 1 bit before multiplexing, (b) the transmission rate of the link, (c) the duration of a time slot, and (d) the duration of a frame.

Solution

We can answer the questions as follows:

a. The duration of 1 bit before multiplexing is 1 / 1 kbps, or 0.001 s (1 ms).

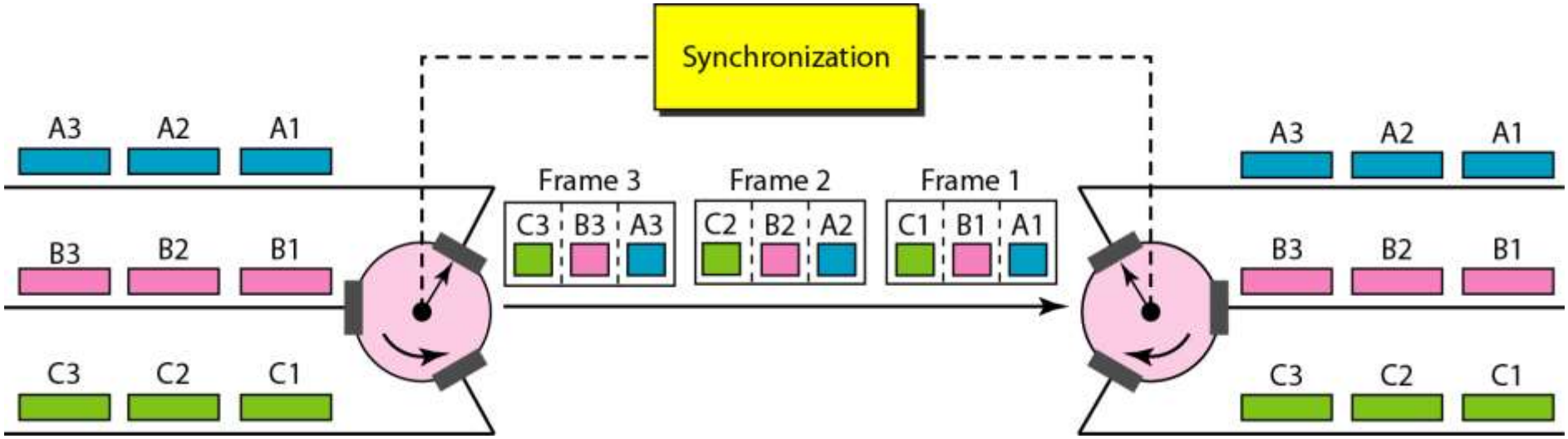
b. The rate of the link is 4 times the rate of a connection, or 4 kbps.

Example 6.7 (continued)

c. ***The duration of each time slot is one-fourth of the duration of each bit before multiplexing, or 1/4 ms or 250 μs. Note that we can also calculate this from the data rate of the link, 4 kbps. The bit duration is the inverse of the data rate, or 1/4 kbps or 250 μs.***

d. ***The duration of a frame is always the same as the duration of a unit before multiplexing, or 1 ms. We can also calculate this in another way. Each frame in this case has four time slots. So the duration of a frame is 4 times 250 μs, or 1 ms.***

Figure 6.15 *Interleaving*



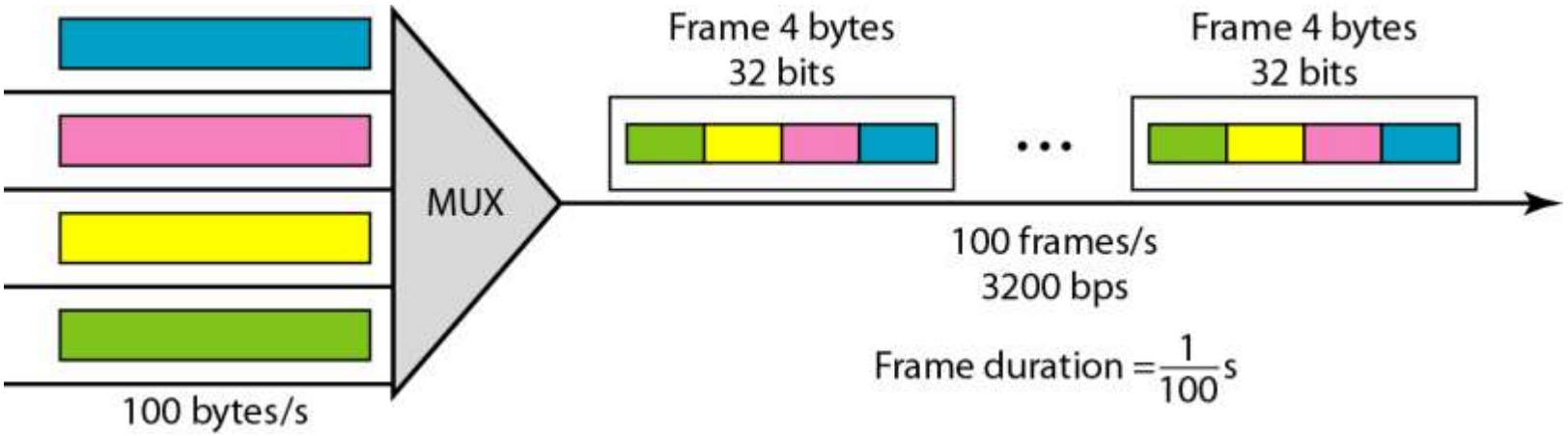
Example 6.8

Four channels are multiplexed using TDM. If each channel sends 100 bytes /s and we multiplex 1 byte per channel, show the frame traveling on the link, the size of the frame, the duration of a frame, the frame rate, and the bit rate for the link.

Solution

The multiplexer is shown in Figure 6.16. Each frame carries 1 byte from each channel; the size of each frame, therefore, is 4 bytes, or 32 bits. Because each channel is sending 100 bytes/s and a frame carries 1 byte from each channel, the frame rate must be 100 frames per second. The bit rate is 100 × 32, or 3200 bps.

Figure 6.16 *Example 6.8*



Example 6.9

A multiplexer combines four 100-kbps channels using a time slot of 2 bits. Show the output with four arbitrary inputs. What is the frame rate? What is the frame duration? What is the bit rate? What is the bit duration?

Solution

Figure 6.17 shows the output for four arbitrary inputs. The link carries 50,000 frames per second. The frame duration is therefore 1/50,000 s or 20 µs. The frame rate is 50,000 frames per second, and each frame carries 8 bits; the bit rate is 50,000 × 8 = 400,000 bits or 400 kbps. The bit duration is 1/400,000 s, or 2.5 µs.

Figure 6.17 *Example 6.9*

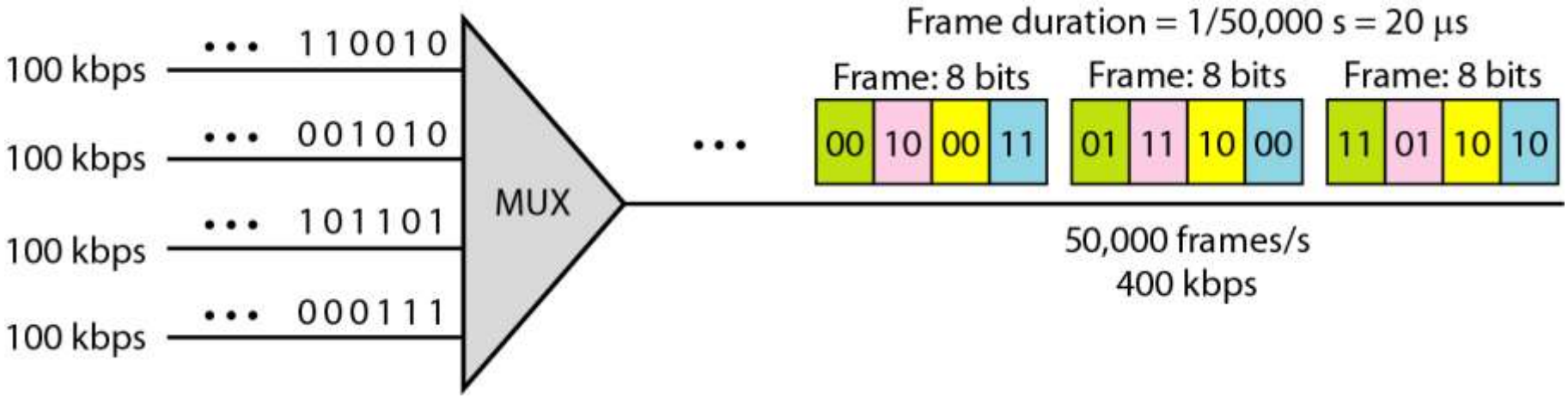


Figure 6.18 *Empty slots*

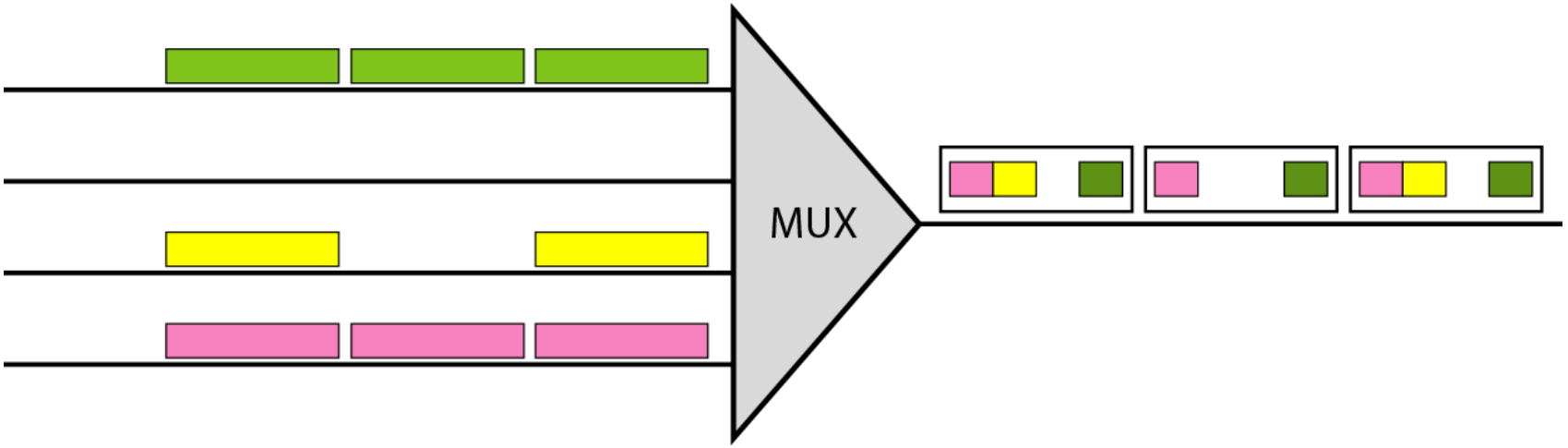


Figure 6.19 *Multilevel multiplexing*

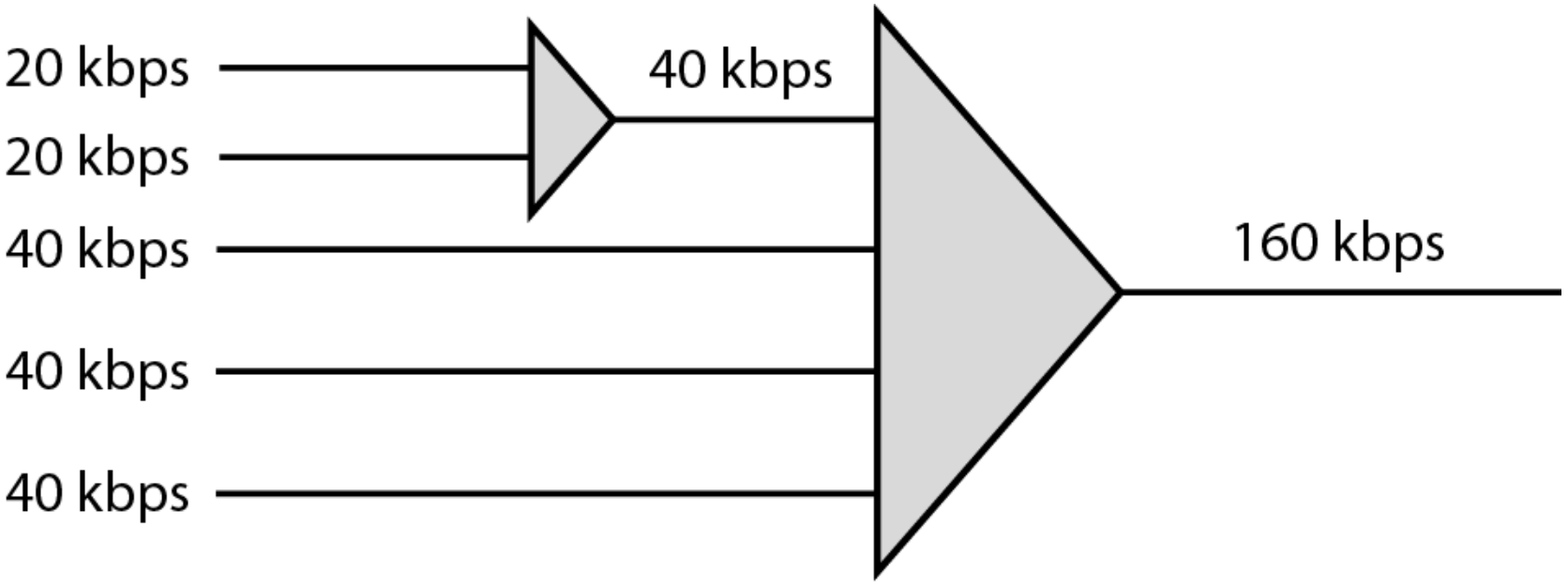


Figure 6.20 *Multiple-slot multiplexing*

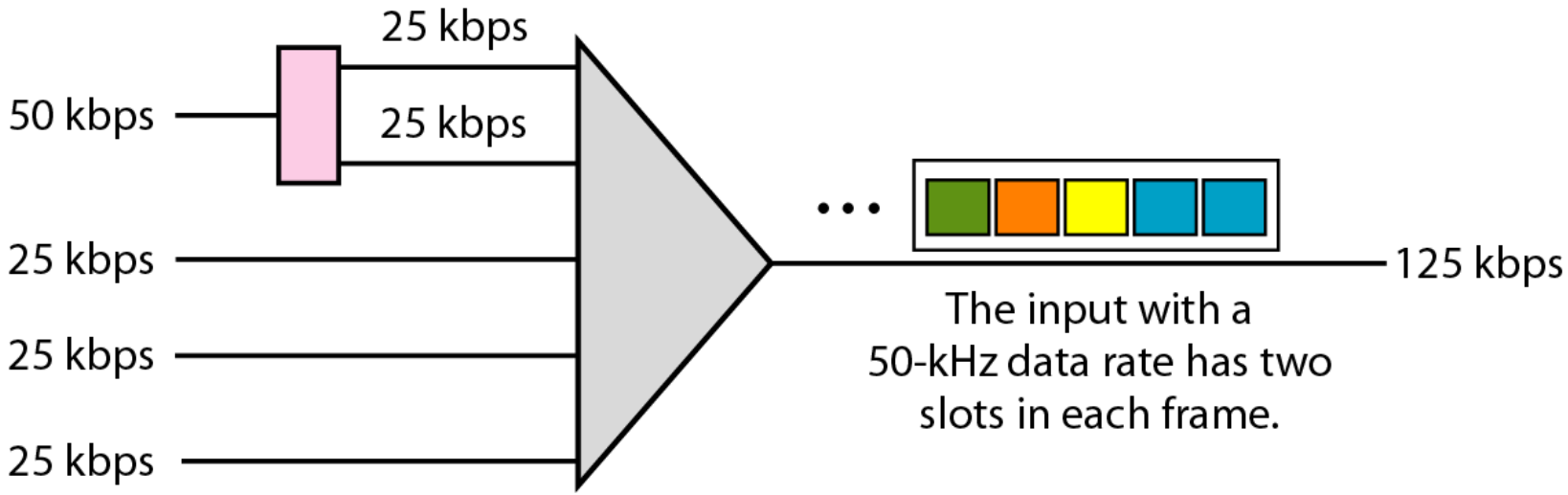


Figure 6.21 *Pulse stuffing*

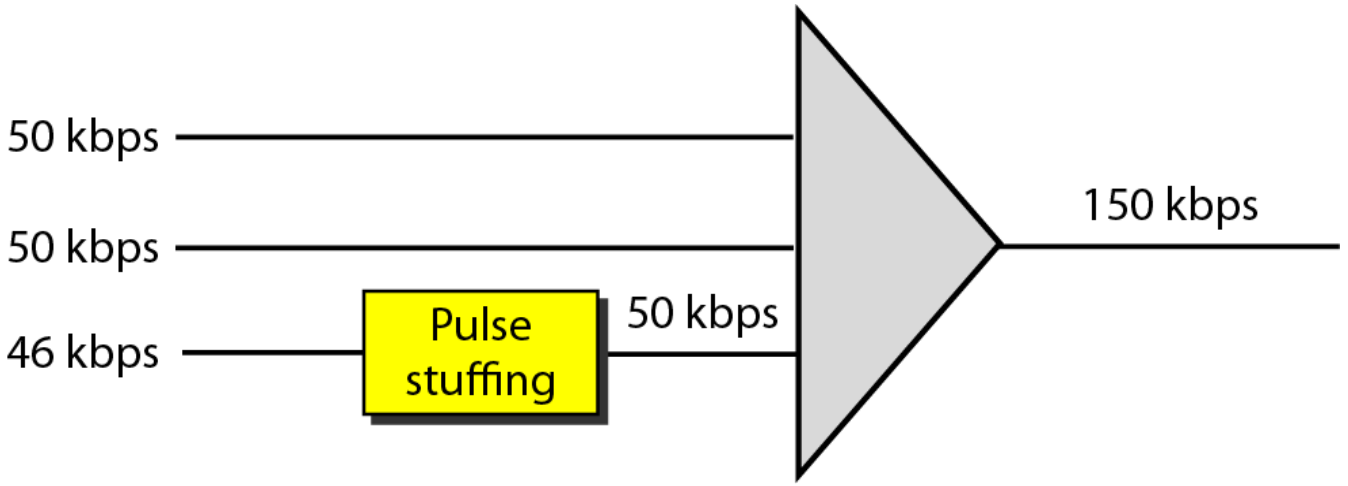
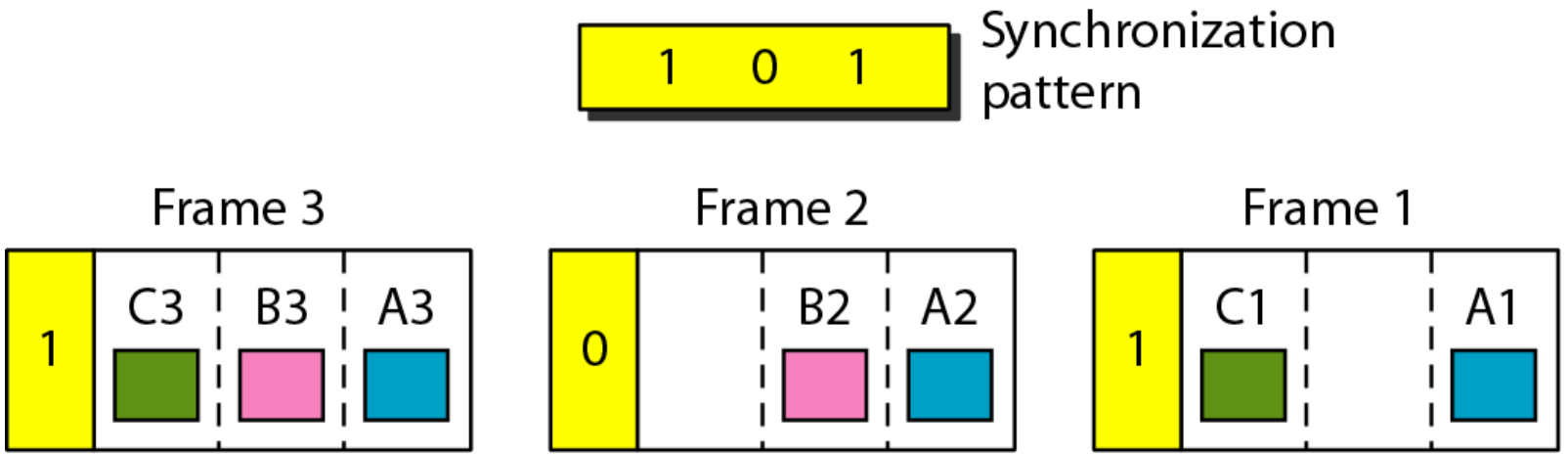


Figure 6.22 *Framing bits*



Example 6.10

We have four sources, each creating 250 characters per second. If the interleaved unit is a character and 1 synchronizing bit is added to each frame, find (a) the data rate of each source, (b) the duration of each character in each source, (c) the frame rate, (d) the duration of each frame, (e) the number of bits in each frame, and (f) the data rate of the link.

Solution

We can answer the questions as follows:

a. The data rate of each source is 250 × 8 = 2000 bps = 2 kbps.

Example 6.10 (continued)

b. ***Each source sends 250 characters per second; therefore, the duration of a character is 1/250 s, or 4 ms.***

c. ***Each frame has one character from each source, which means the link needs to send 250 frames per second to keep the transmission rate of each source.***

d. ***The duration of each frame is 1/250 s, or 4 ms. Note that the duration of each frame is the same as the duration of each character coming from each source.***

e. ***Each frame carries 4 characters and 1 extra synchronizing bit. This means that each frame is*** $4 \times 8 + 1 = 33$ ***bits.***

Example 6.11

Two channels, one with a bit rate of 100 kbps and another with a bit rate of 200 kbps, are to be multiplexed. How this can be achieved? What is the frame rate? What is the frame duration? What is the bit rate of the link?

Solution

We can allocate one slot to the first channel and two slots to the second channel. Each frame carries 3 bits. The frame rate is 100,000 frames per second because it carries 1 bit from the first channel. The bit rate is 100,000 frames/s × 3 bits per frame, or 300 kbps.

Figure 6.23 *Digital hierarchy*

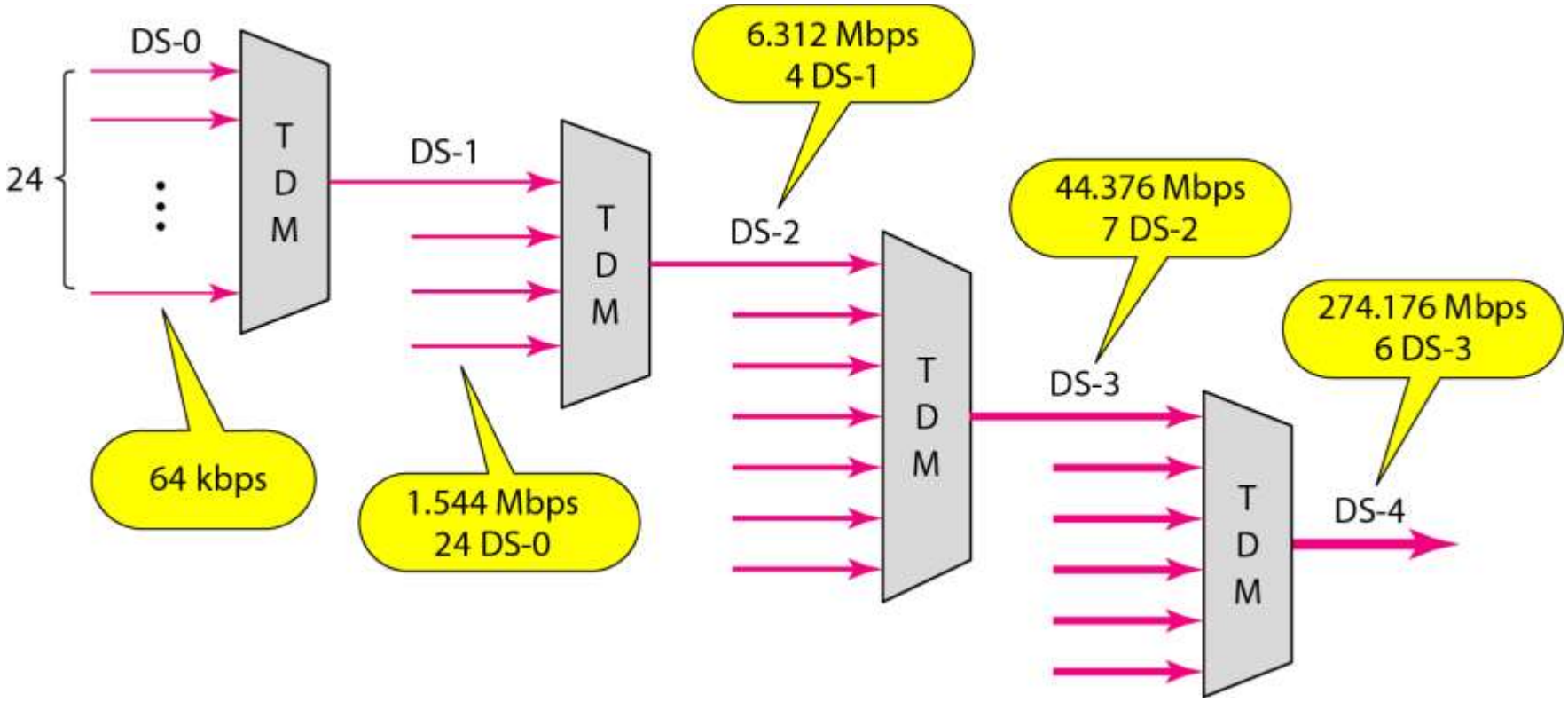


Table 6.1 *DS and T line rates*

Service	*Line*	*Rate (Mbps)*	*Voice Channels*
DS-1	T-1	1.544	24
DS-2	T-2	6.312	96
DS-3	T-3	44.736	672
DS-4	T-4	274.176	4032

Figure 6.24 *T-1 line for multiplexing telephone lines*

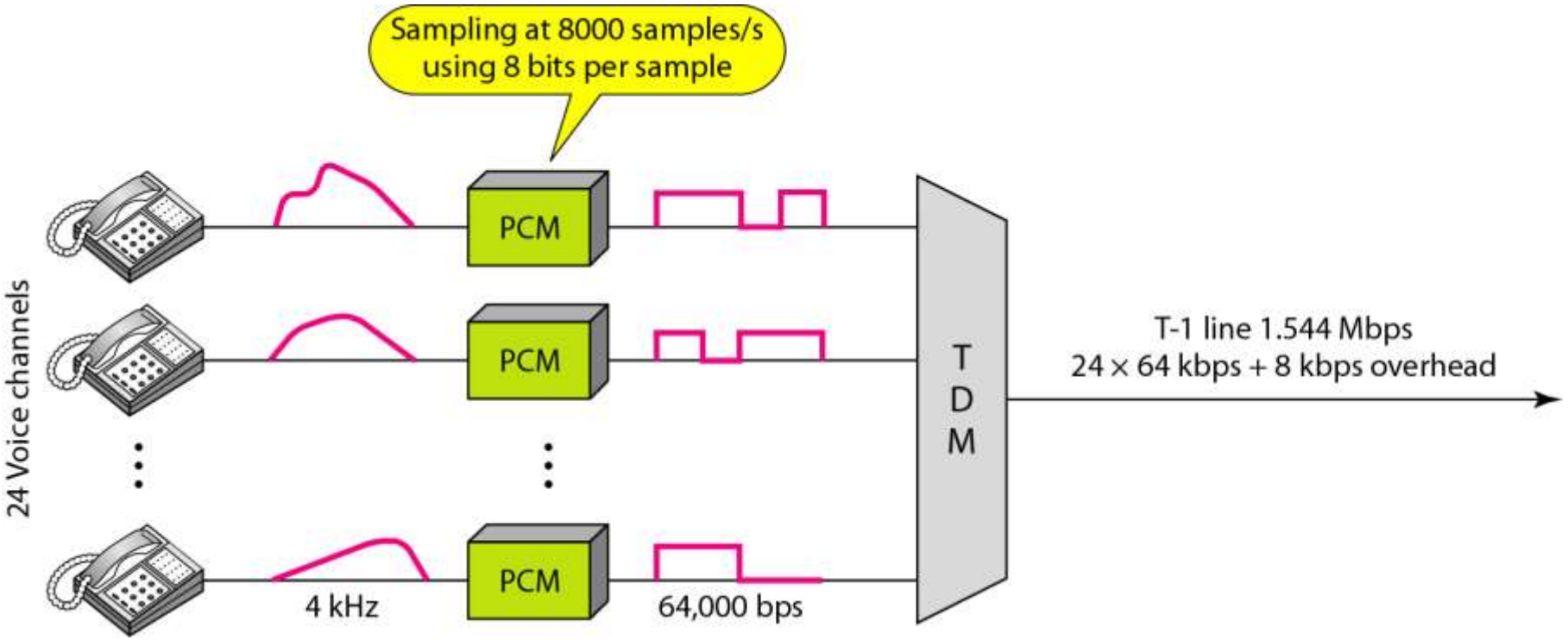


Figure 6.25 *T-1 frame structure*

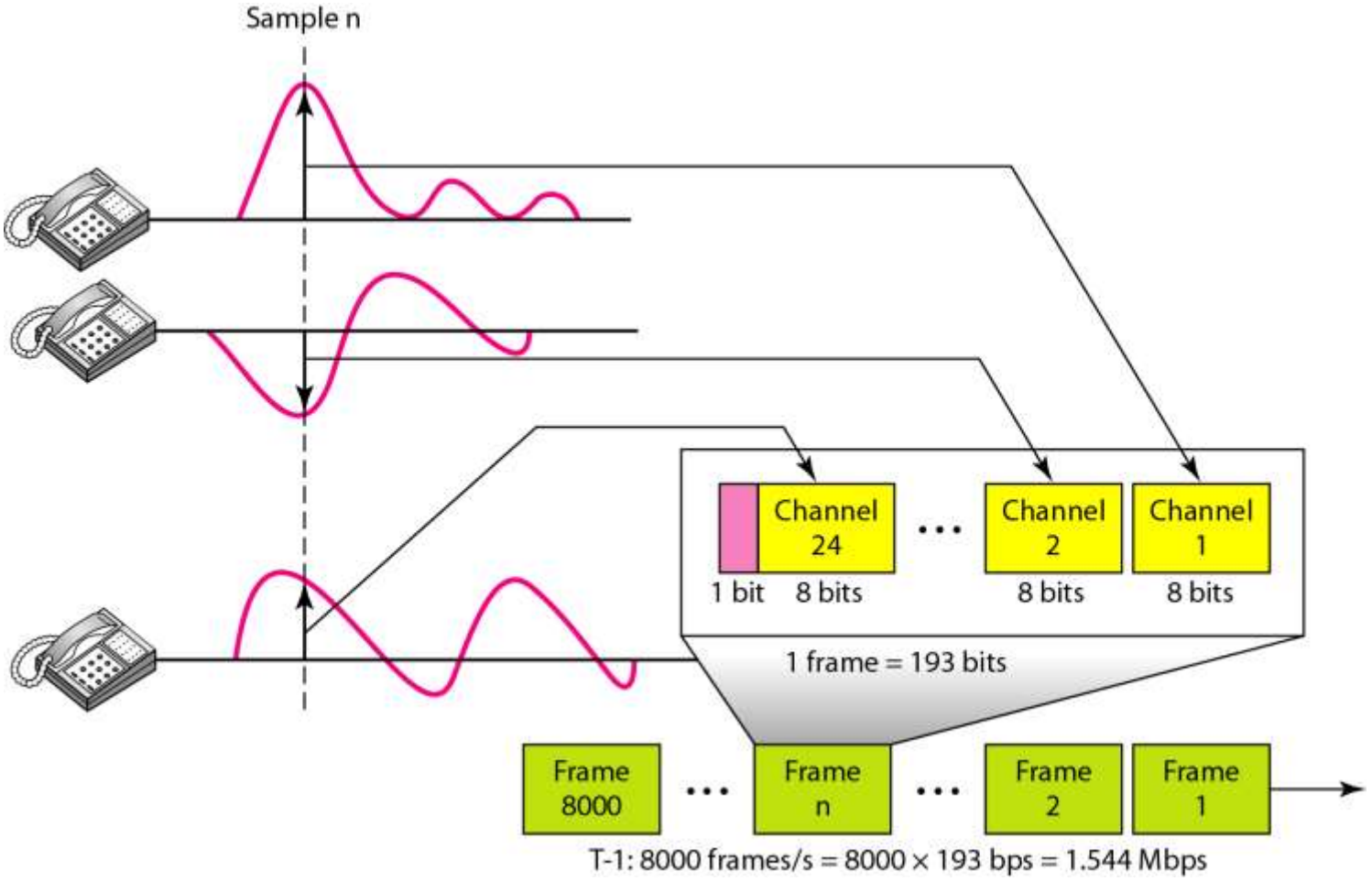
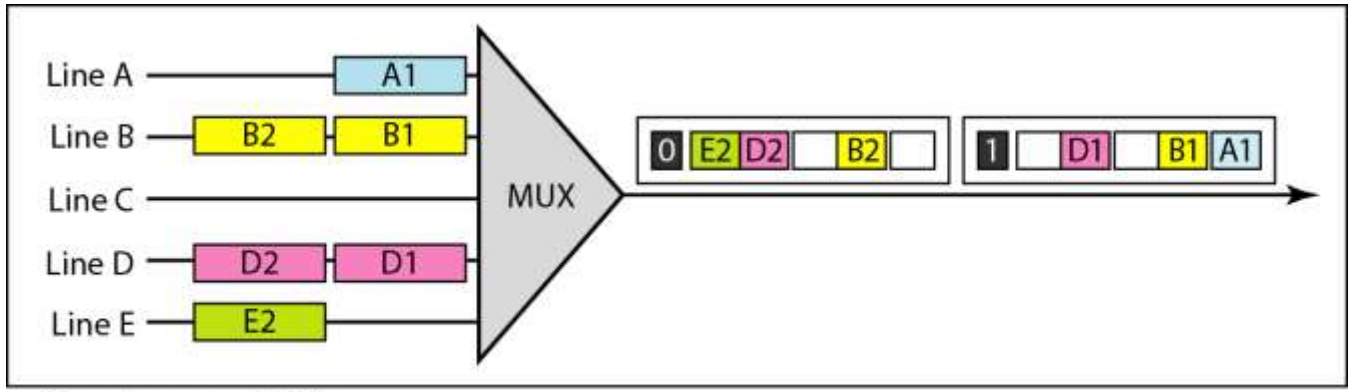


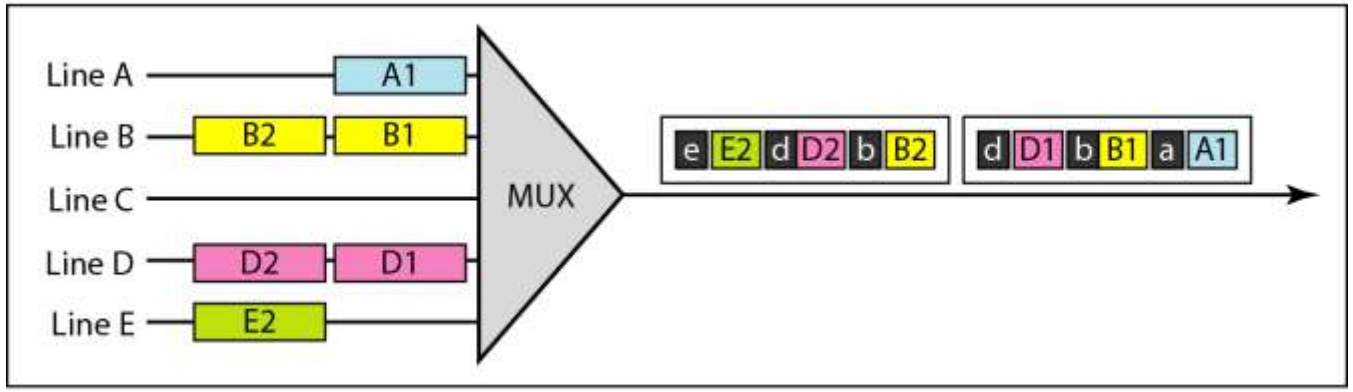
Table 6.2 *E line rates*

Line	*Rate (Mbps)*	*Voice Channels*
E-1	2.048	30
E-2	8.448	120
E-3	34.368	480
E-4	139.264	1920

Figure 6.26 *TDM slot comparison*



a. Synchronous TDM



b. Statistical TDM

6-1 SPREAD SPECTRUM

In spread spectrum (SS), we combine signals from different sources to fit into a larger bandwidth, but our goals are to prevent eavesdropping and jamming. To achieve these goals, spread spectrum techniques add redundancy.

Topics discussed in this section:

Frequency Hopping Spread Spectrum (FHSS)
Direct Sequence Spread Spectrum Synchronous (DSSS)

Figure 6.27 *Spread spectrum*

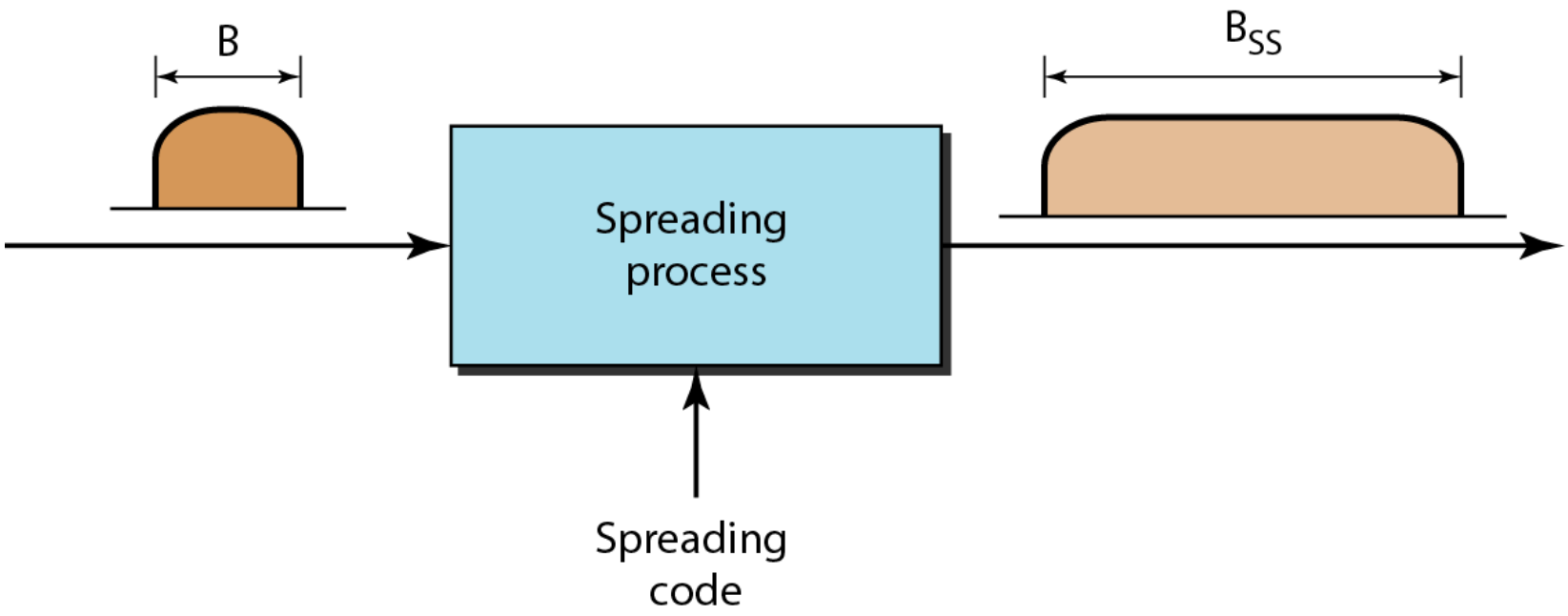


Figure 6.28 *Frequency hopping spread spectrum (FHSS)*

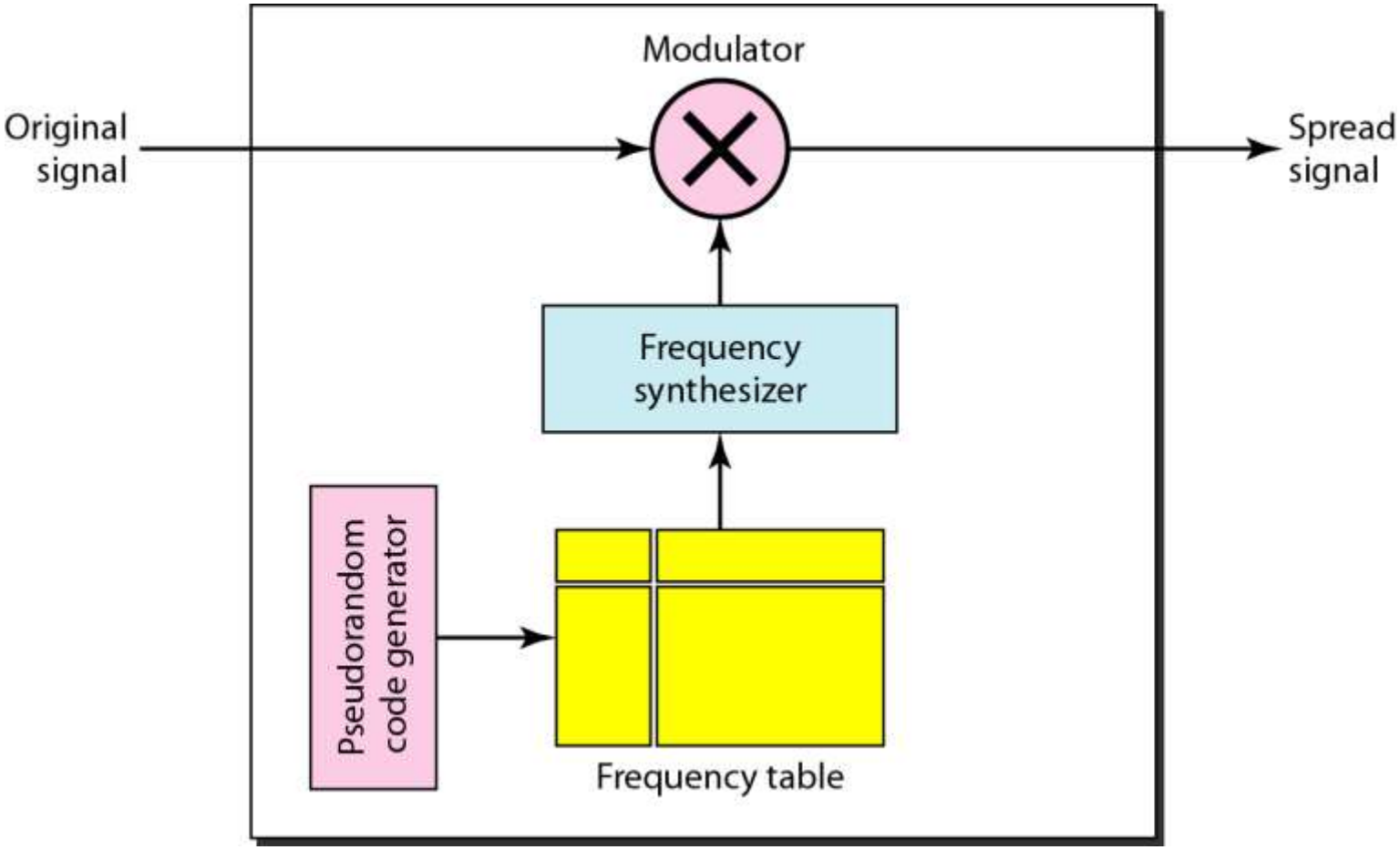


Figure 6.29 *Frequency selection in FHSS*

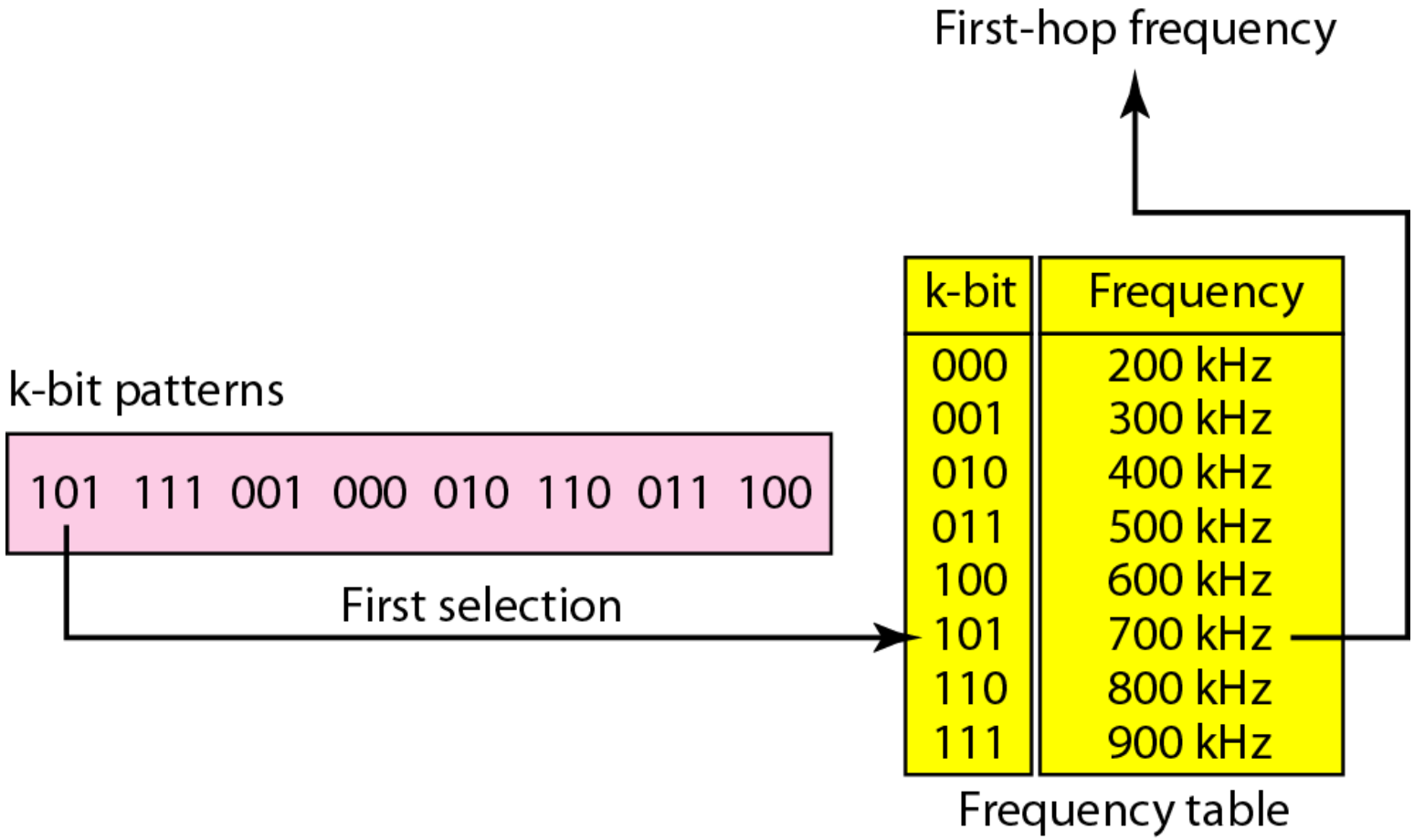


Figure 6.30 *FHSS cycles*

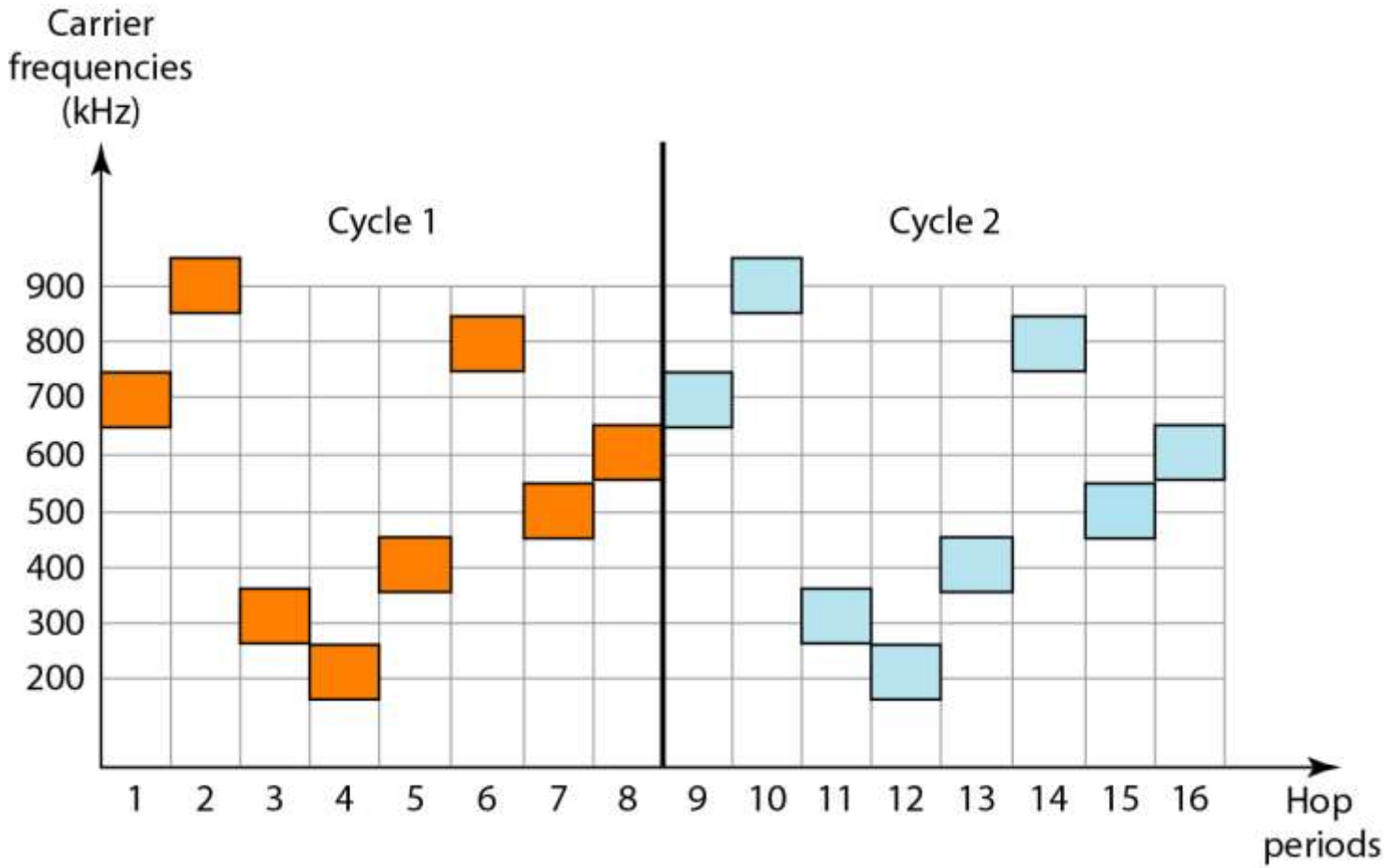
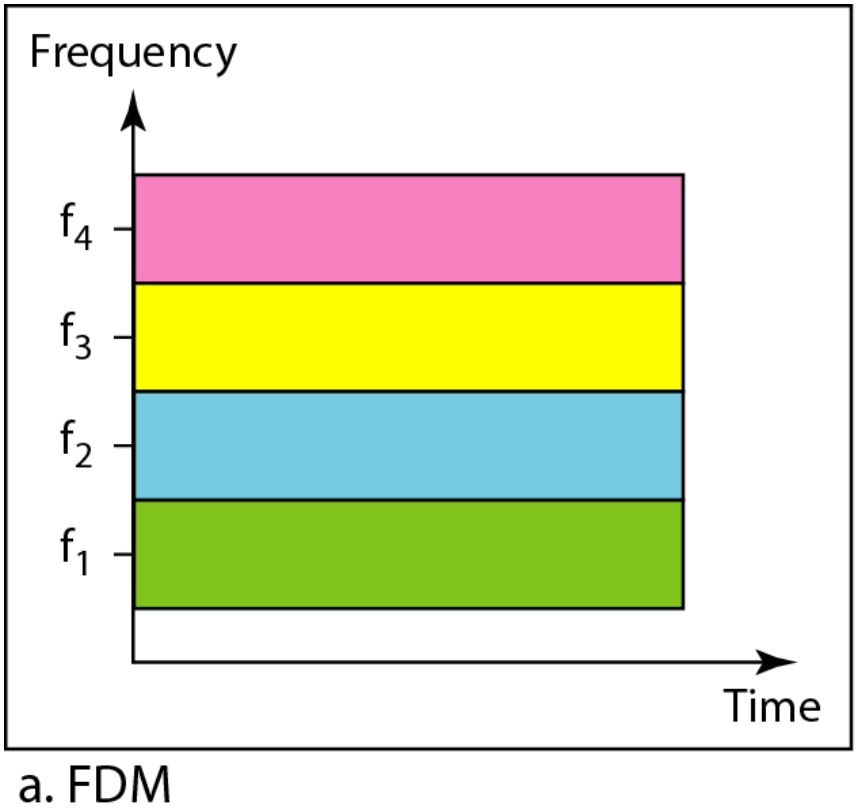
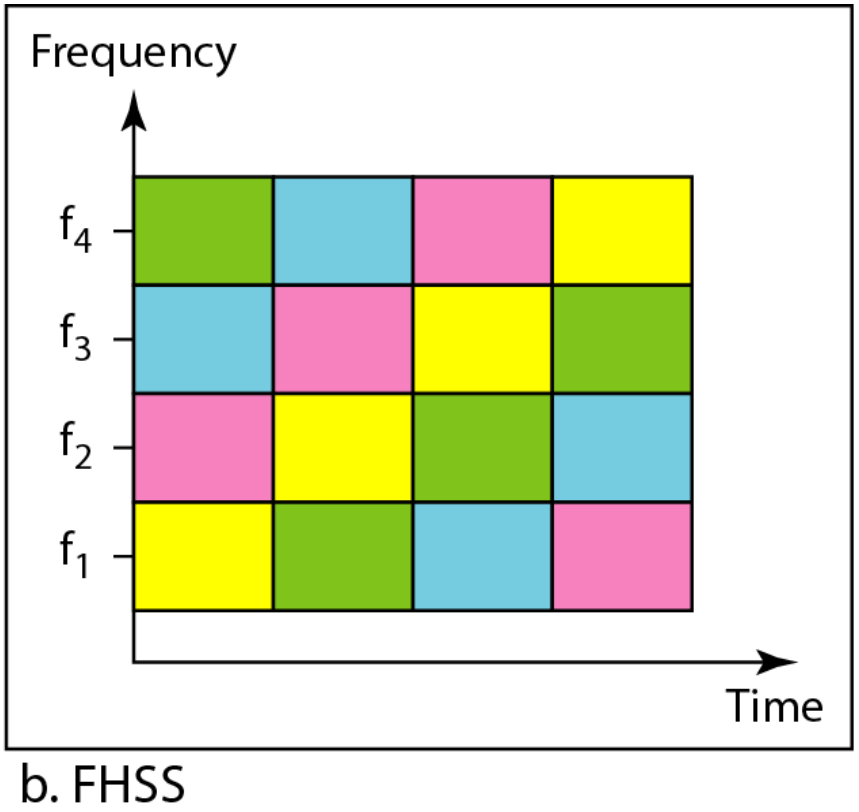


Figure 6.31 *Bandwidth sharing*



a. FDM



b. FHSS

Figure 6.32 *DSSS*

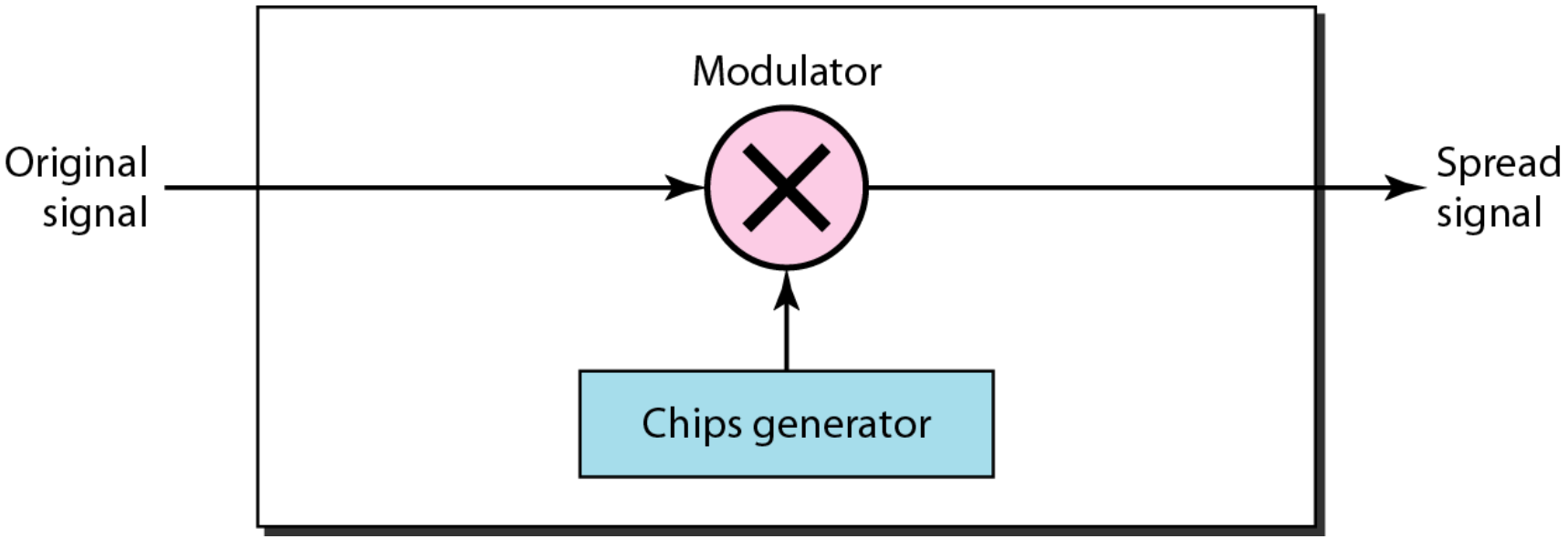
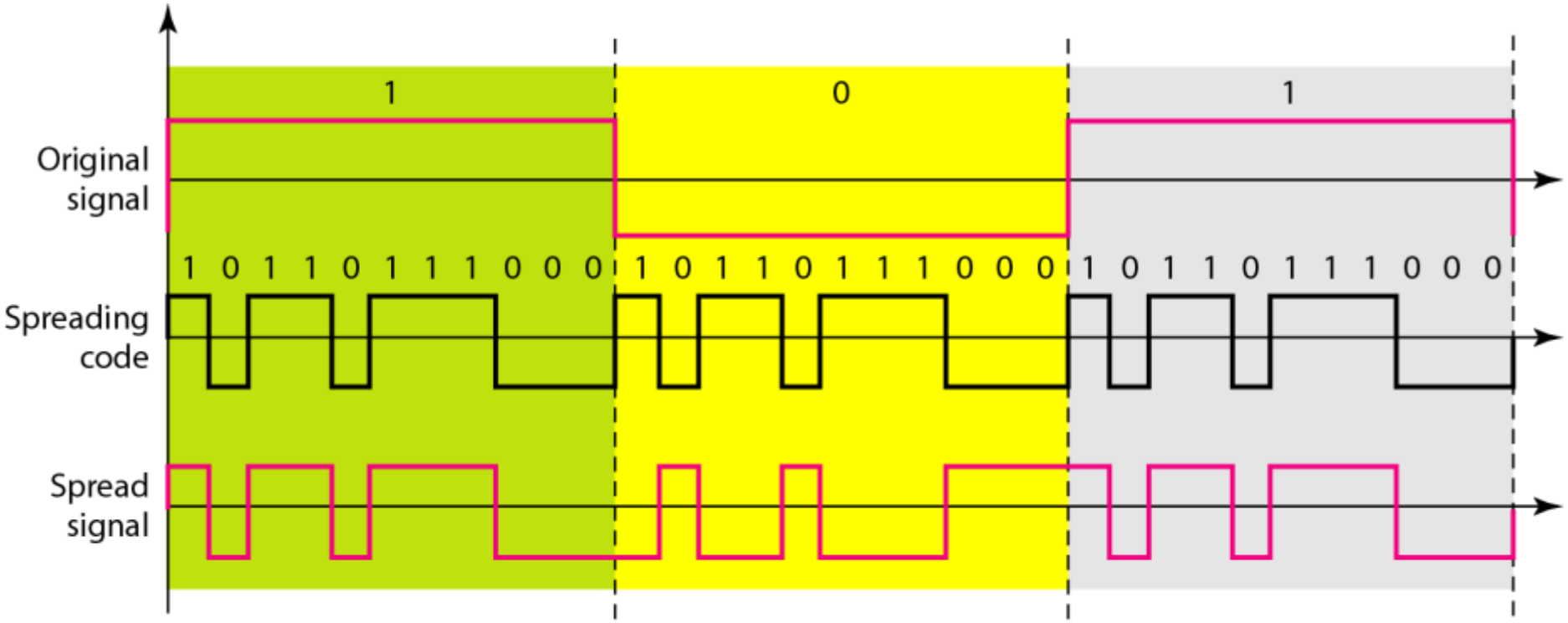


Figure 6.33 *DSSS example*





Data Communications and Networking Fourth Edition

Forouzan

Chapter 8

Switching

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Switching

- Switching is the process of channeling for transferring data from one device to another within the same network or across different networks using switches. It mainly occurs at the Data Link Layer (Layer 2) of the OSI model, where data is forwarded in the form of frames based on MAC addresses.

Switching Con…

- A switch connects multiple devices within a network.
- It reduces interference by sending data only to the intended device.
- It forwards data frames to the correct port using MAC addresses.
- It delivers data directly to connected devices (computers, phones).
- For communication outside the local network, it forwards da to a router.
- **Example:** A website request passes through multiple switches and routers before reaching the server.

Figure 8.1 *Switched network*

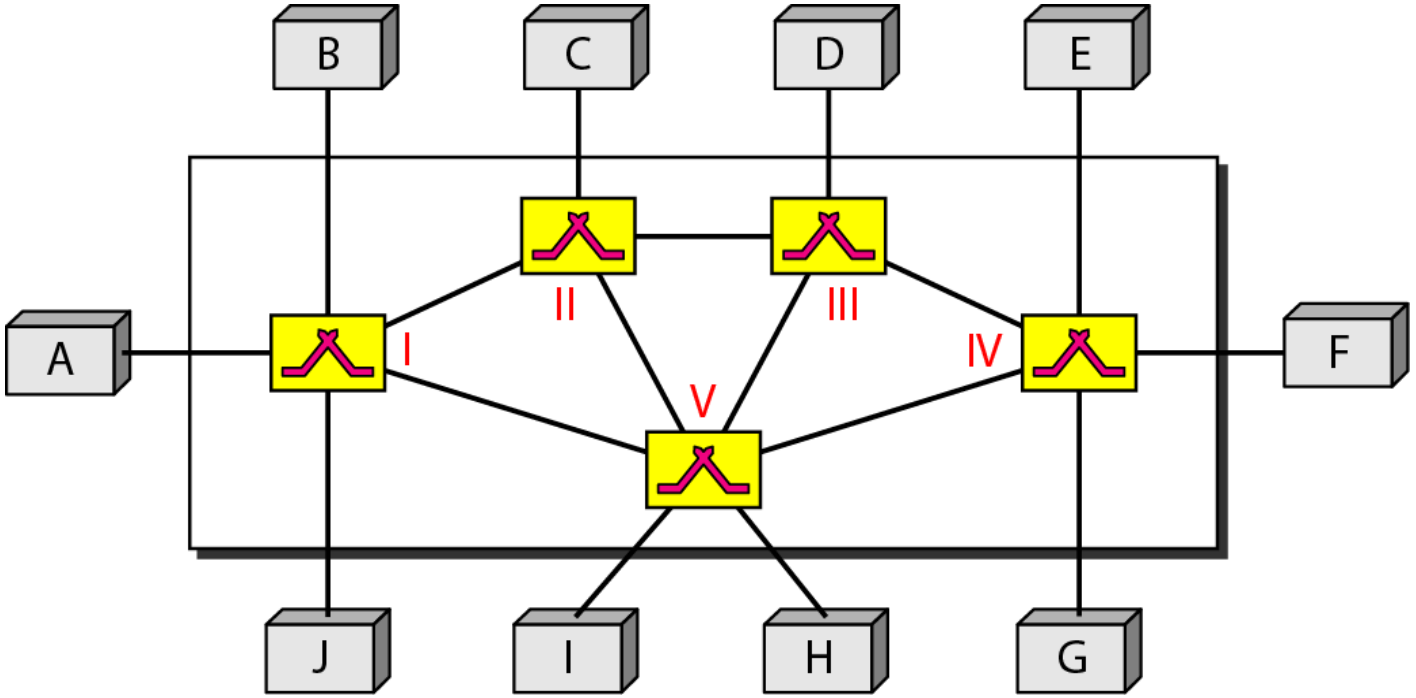
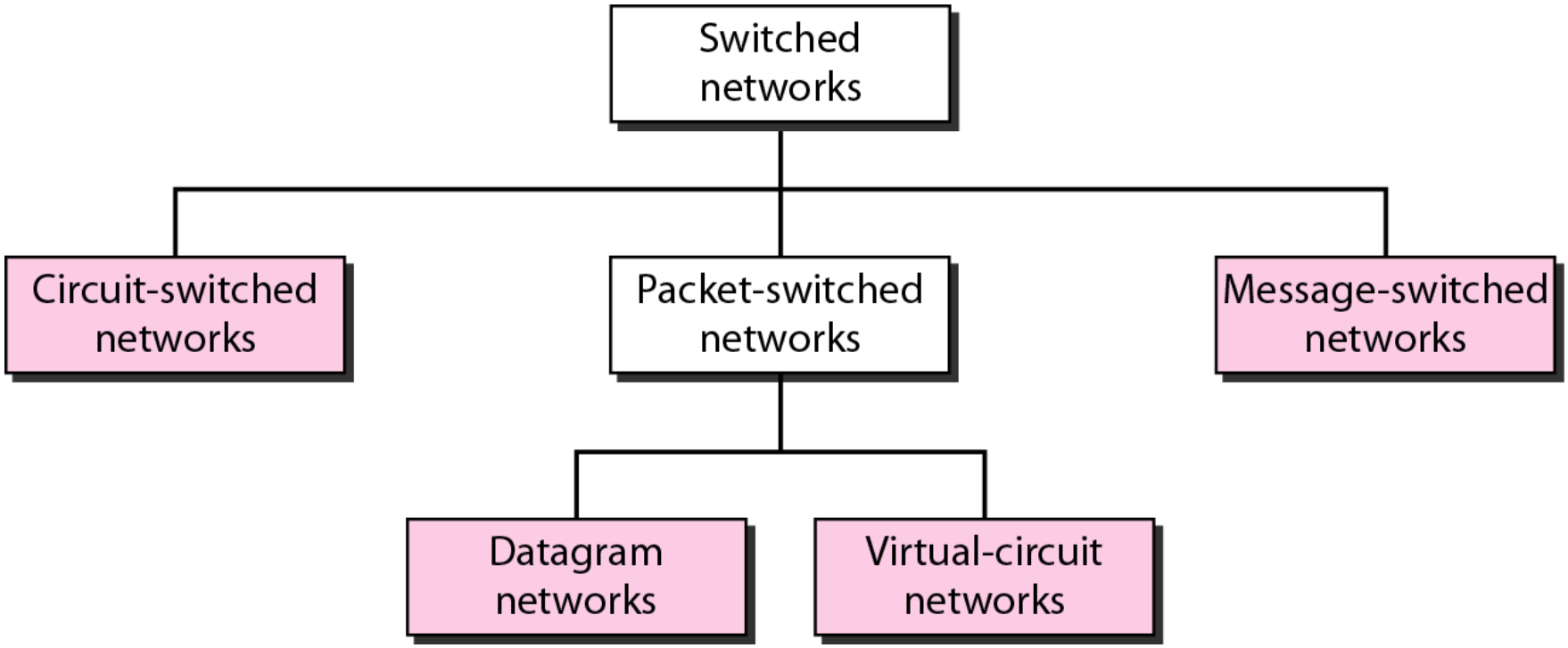


Figure 8.2 *Taxonomy of switched networks*



Circuit Switching:

- Circuit switching is a connection-oriented method widely used in traditional telephone networks. In this method:
- a dedicated path is established between the sender and receiver before any data transmission begins
- The connection remains reserved for the entire duration of the communication session
- Both ends enjoy a fixed bandwidth and low latency.
- **Analogy:** Like reserving a private road for the whole trip, even if no cars are on it for some time.

Packet Switching:

- Packet switching is the foundation of modern computer networks, including the Internet. In this method:
- Data is divided into small packets.
- Each packet carries a portion of the message and the destination address.
- Packets may travel through different paths in the network.
- At the destination, packets are reassembled to reconstruct the original message.
- **Analogy:** Like sending a book in separate chapters through different courier routes, then reassembling them at the destination.

Message Switching

- Message switching is one of the earliest switching techniques. In this method:
- The entire message is sent to an intermediate node.
- The message is stored temporarily at that node before being forwarded to the next node.
- No dedicated path is established between the sender and receiver.
- **Analogy:** Like sending a physical letter that is first stored at a post office before moving to the next branch.

Comparison of Switching Types

Message Switching	Circuit Switching	Packet Switching
No dedicated path; the entire message is stored and forwarded at each node.	A fixed, dedicated path is established before transmission.	Data is split into packets that follow dynamic or virtual paths.
Very inefficient due to full message storage and forwarding.	Inefficient because bandwidth remains reserved even when idle.	Highly efficient due to optimal bandwidth utilization.
High transmission delay.	Low delay after initial setup.	Delay varies based on network conditions.
Less reliable and obsolete.	Reliable with guaranteed connection.	Reliable with error control and retransmission.
Used in telegraph systems.	Used in traditional telephone networks.	Used in the Internet.

8-1 CIRCUIT-SWITCHED NETWORKS

A circuit-switched network consists of a set of switches connected by physical links. A connection between two stations is a dedicated path made of one or more links. However, each connection uses only one dedicated channel on each link. Each link is normally divided into n channels by using FDM or TDM.

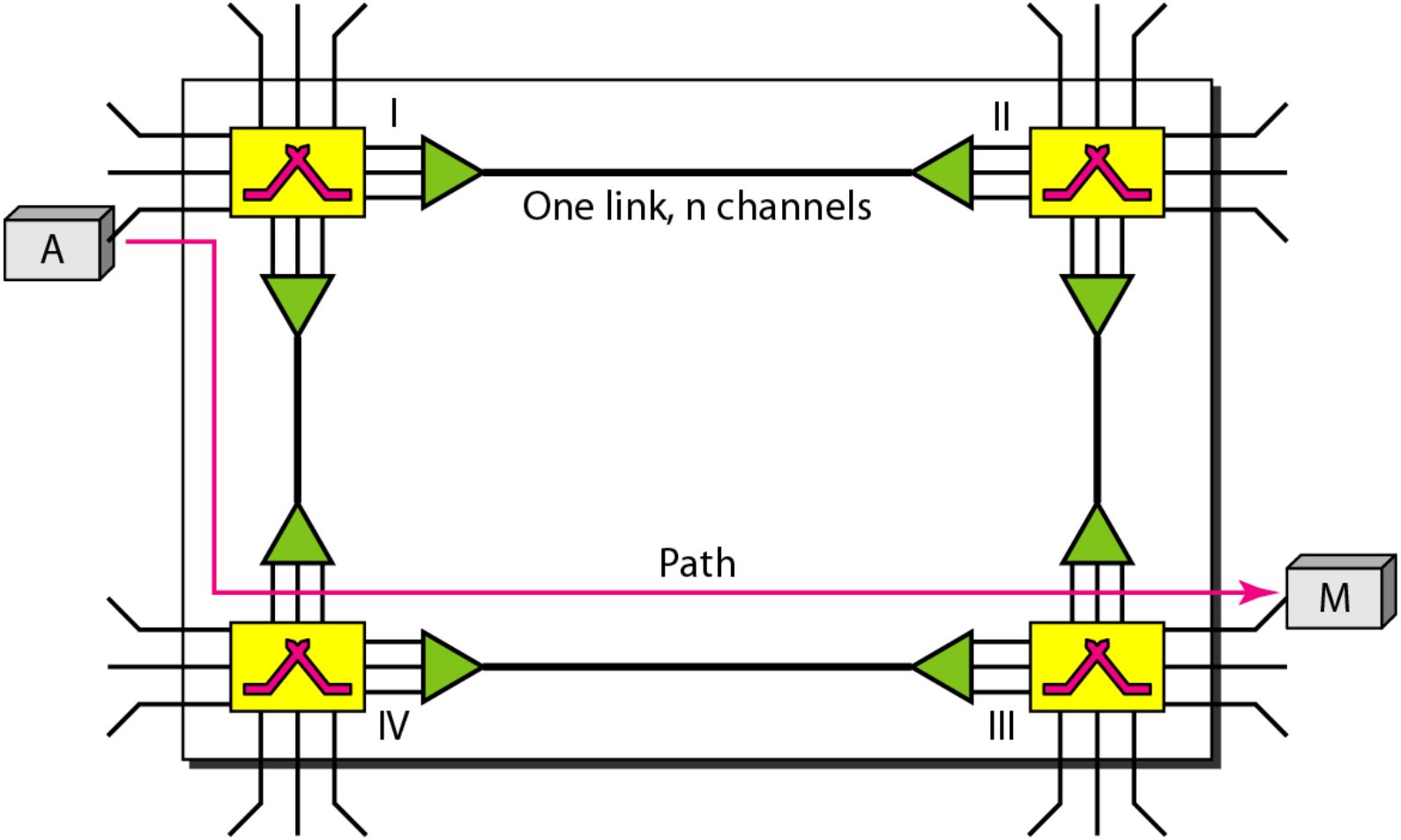
Topics discussed in this section:

Three Phases
Efficiency
Delay
Circuit-Switched Technology in Telephone Networks

Note

A circuit-switched network is made of a set of switches connected by physical links, in which each link is divided into *n* channels.

Figure 8.3 *A trivial circuit-switched network*



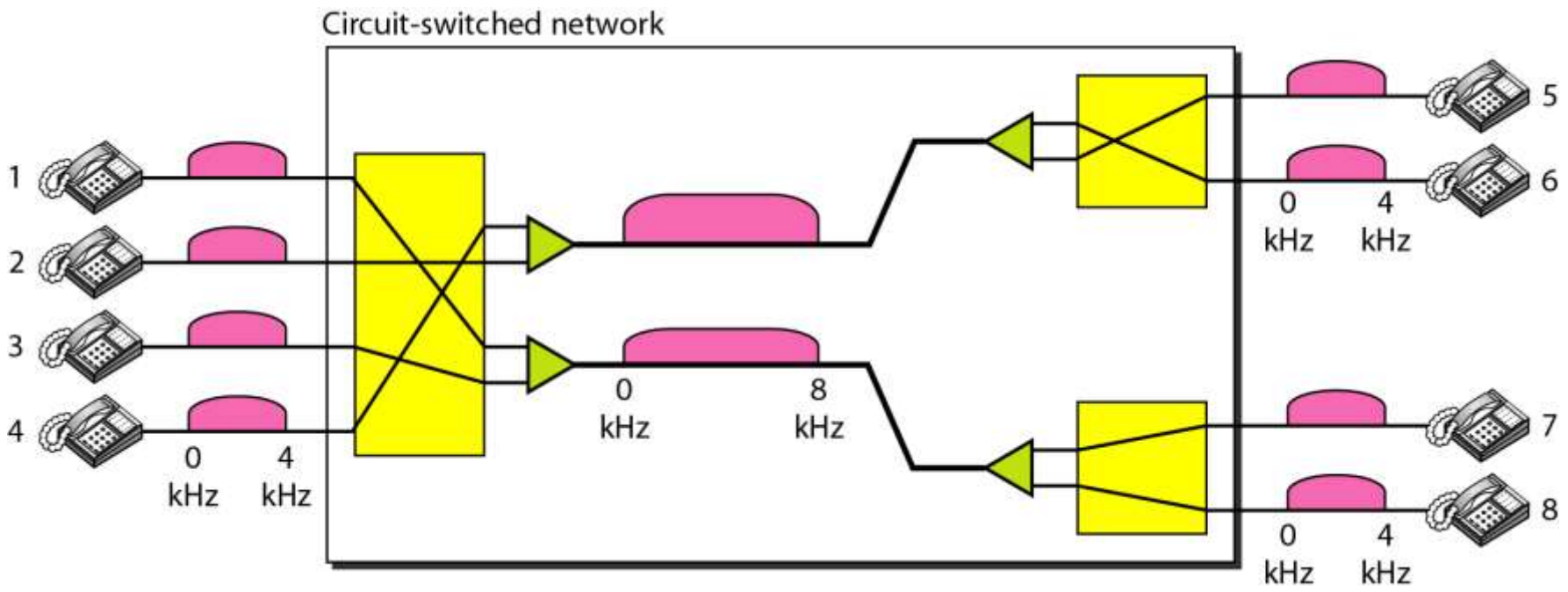
Note

In circuit switching, the resources need to be reserved during the setup phase; the resources remain dedicated for the entire duration of data transfer until the teardown phase.

Example 8.1

As a trivial example, let us use a circuit-switched network to connect eight telephones in a small area. Communication is through 4-kHz voice channels. We assume that each link uses FDM to connect a maximum of two voice channels. The bandwidth of each link is then 8 kHz. Figure 8.4 shows the situation. Telephone 1 is connected to telephone 7; 2 to 5; 3 to 8; and 4 to 6. Of course the situation may change when new connections are made. The switch controls the connections.

Figure 8.4 *Circuit-switched network used in Example 8.1*



Example 8.2

As another example, consider a circuit-switched network that connects computers in two remote offices of a private company. The offices are connected using a T-1 line leased from a communication service provider. There are two 4 × 8 (4 inputs and 8 outputs) switches in this network. For each switch, four output ports are folded into the input ports to allow communication between computers in the same office. Four other output ports allow communication between the two offices. Figure 8.5 shows the situation.

Figure 8.5 *Circuit-switched network used in Example 8.2*

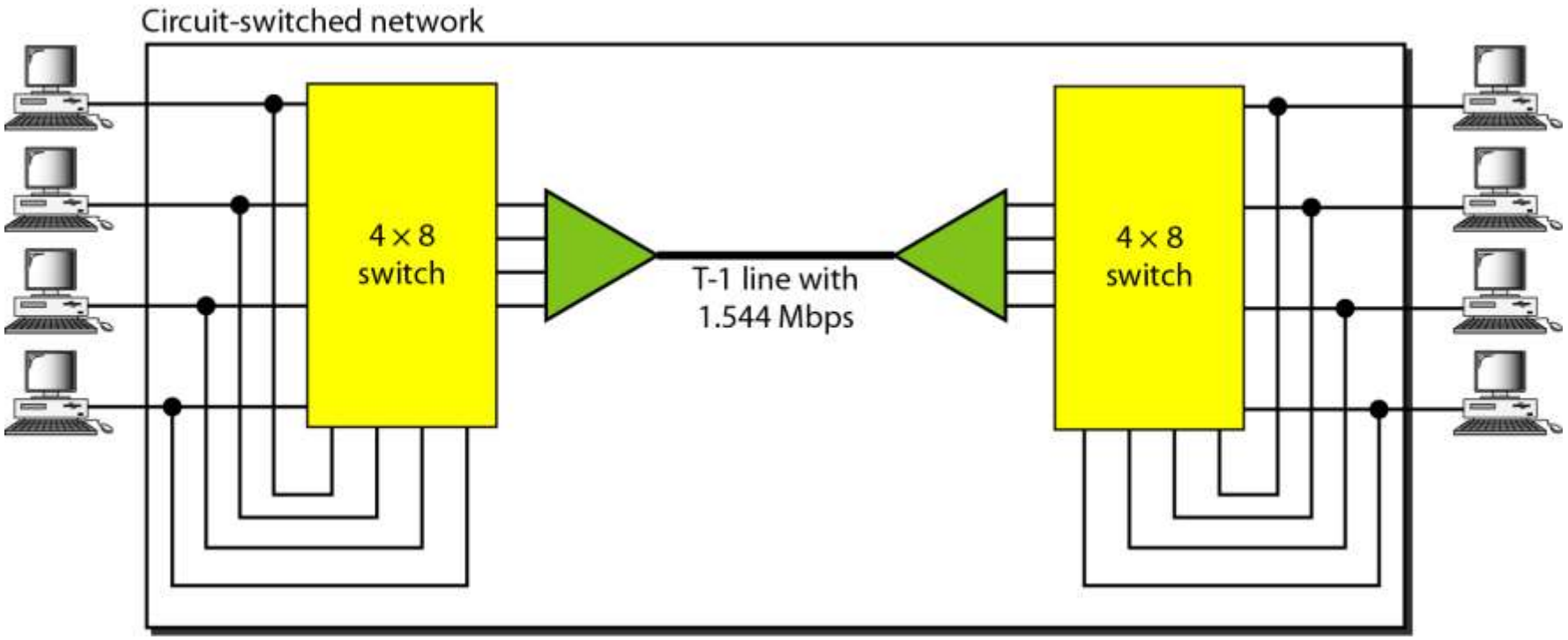
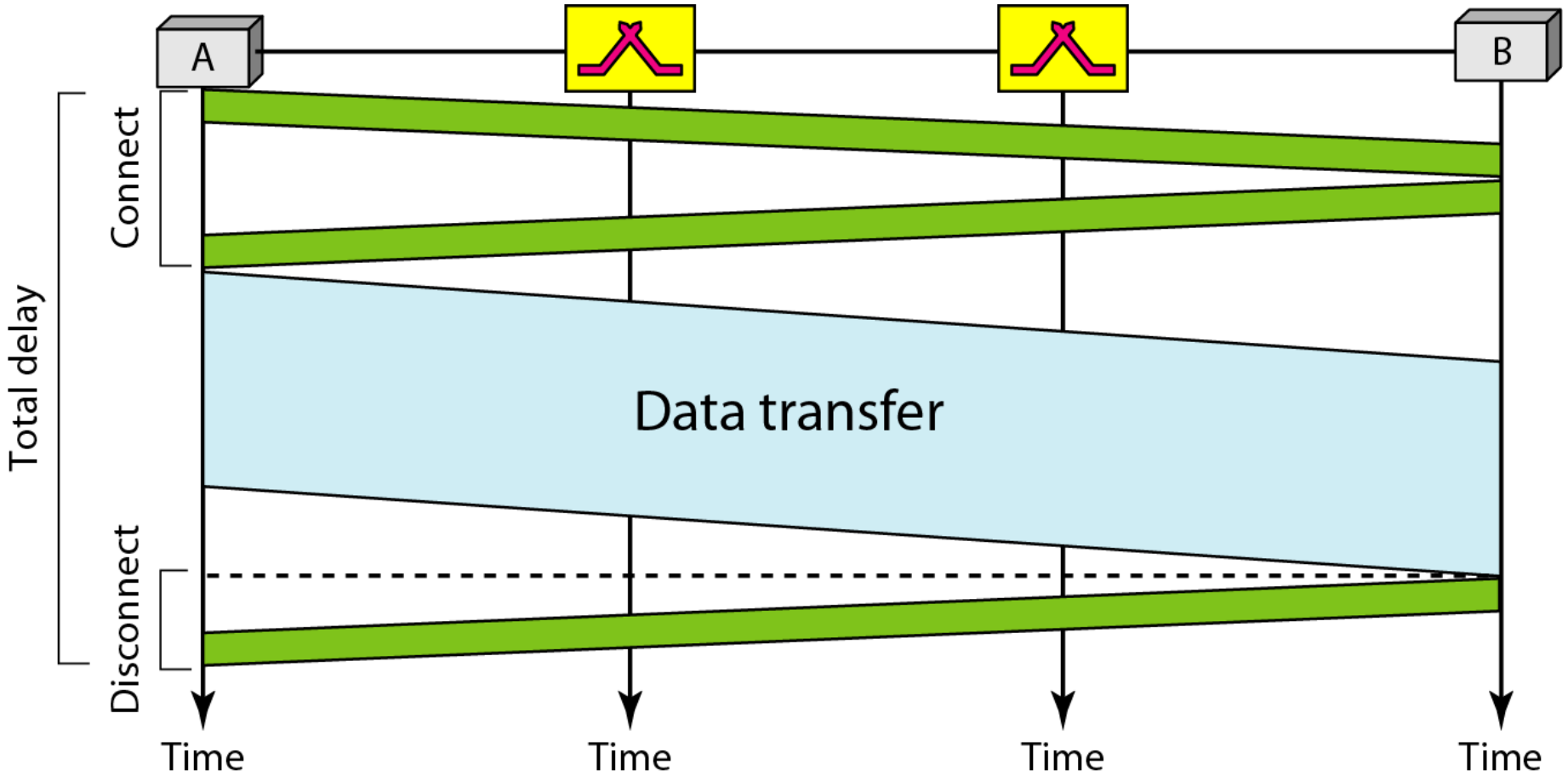


Figure 8.6 *Delay in a circuit-switched network*



Note

Switching at the physical layer in the traditional telephone network uses the circuit-switching approach.

8-2 DATAGRAM NETWORKS

In data communications, we need to send messages from one end system to another. If the message is going to pass through a packet-switched network, it needs to be divided into packets of fixed or variable size. The size of the packet is determined by the network and the governing protocol.

Topics discussed in this section:

Routing Table
Efficiency
Delay
Datagram Networks in the Internet

Note

In a packet-switched network, there is no resource reservation; resources are allocated on demand.

Figure 8.7 *A datagram network with four switches (routers)*

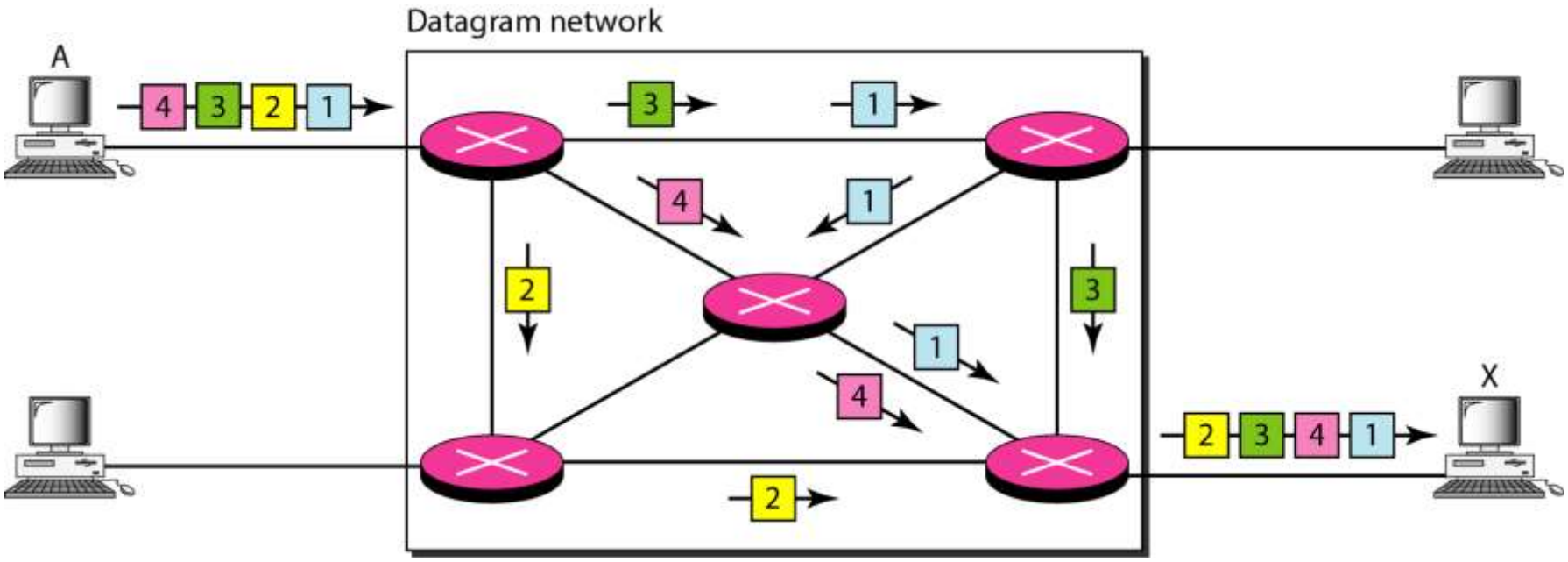
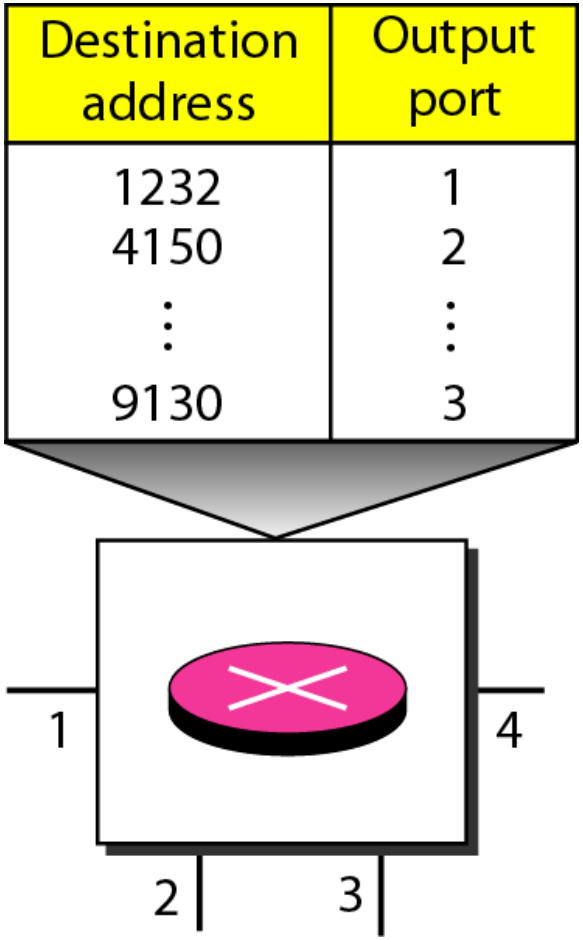


Figure 8.8 *Routing table in a datagram network*



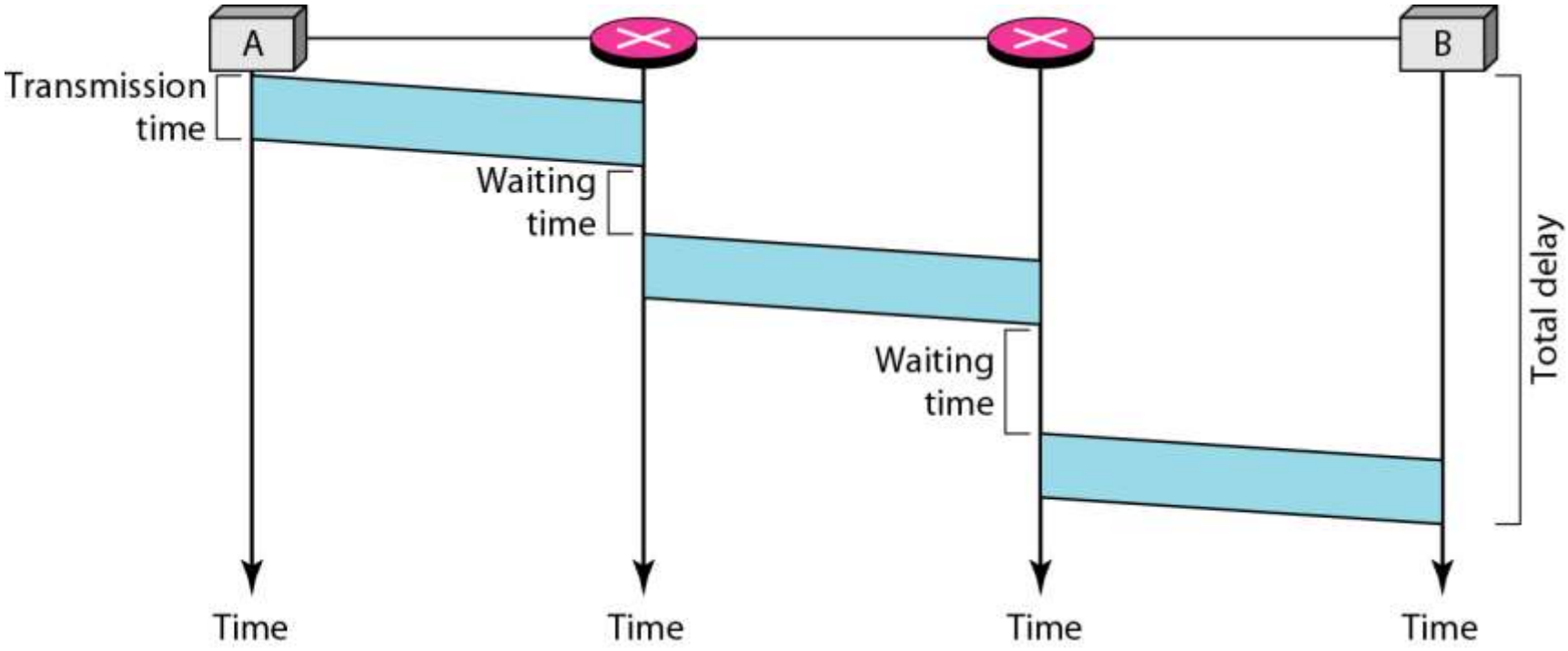
Note

A switch in a datagram network uses a routing table that is based on the destination address.

Note

The destination address in the header of a packet in a datagram network remains the same during the entire journey of the packet.

Figure 8.9 *Delay in a datagram network*



Note

Switching in the Internet is done by using the datagram approach to packet switching at the network layer.

8-3 VIRTUAL-CIRCUIT NETWORKS

A virtual-circuit network is a cross between a circuit-switched network and a datagram network. It has some characteristics of both.

Topics discussed in this section:

Addressing
Three Phases
Efficiency
Delay
Circuit-Switched Technology in WANs

Figure 8.10 *Virtual-circuit network*

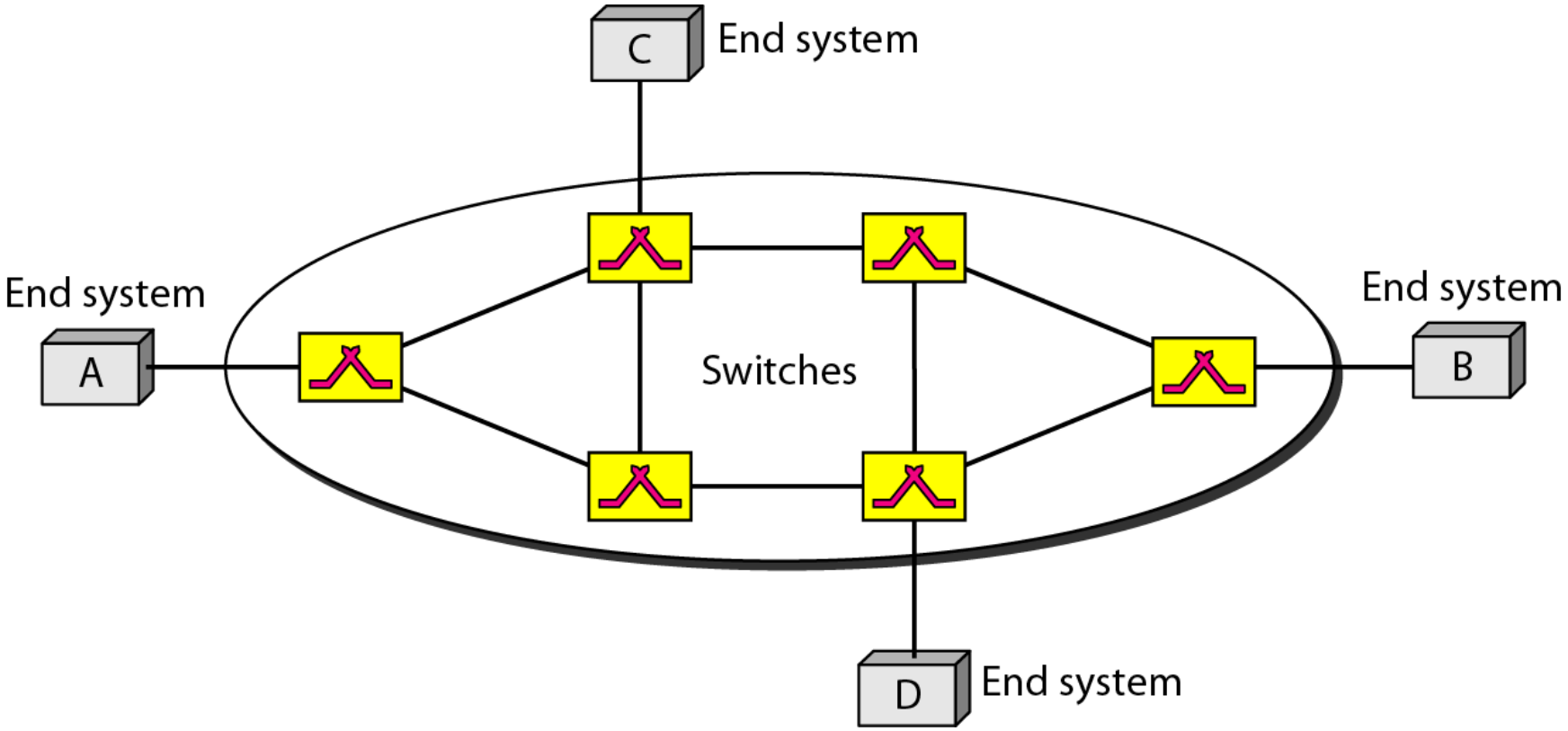


Figure 8.11 *Virtual-circuit identifier*

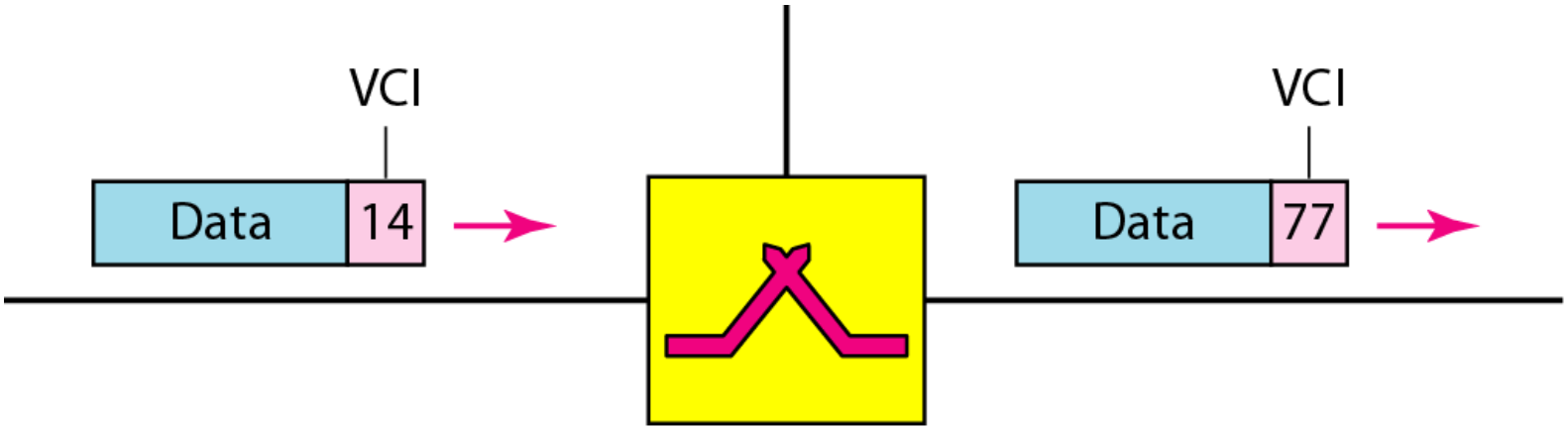


Figure 8.12 *Switch and tables in a virtual-circuit network*

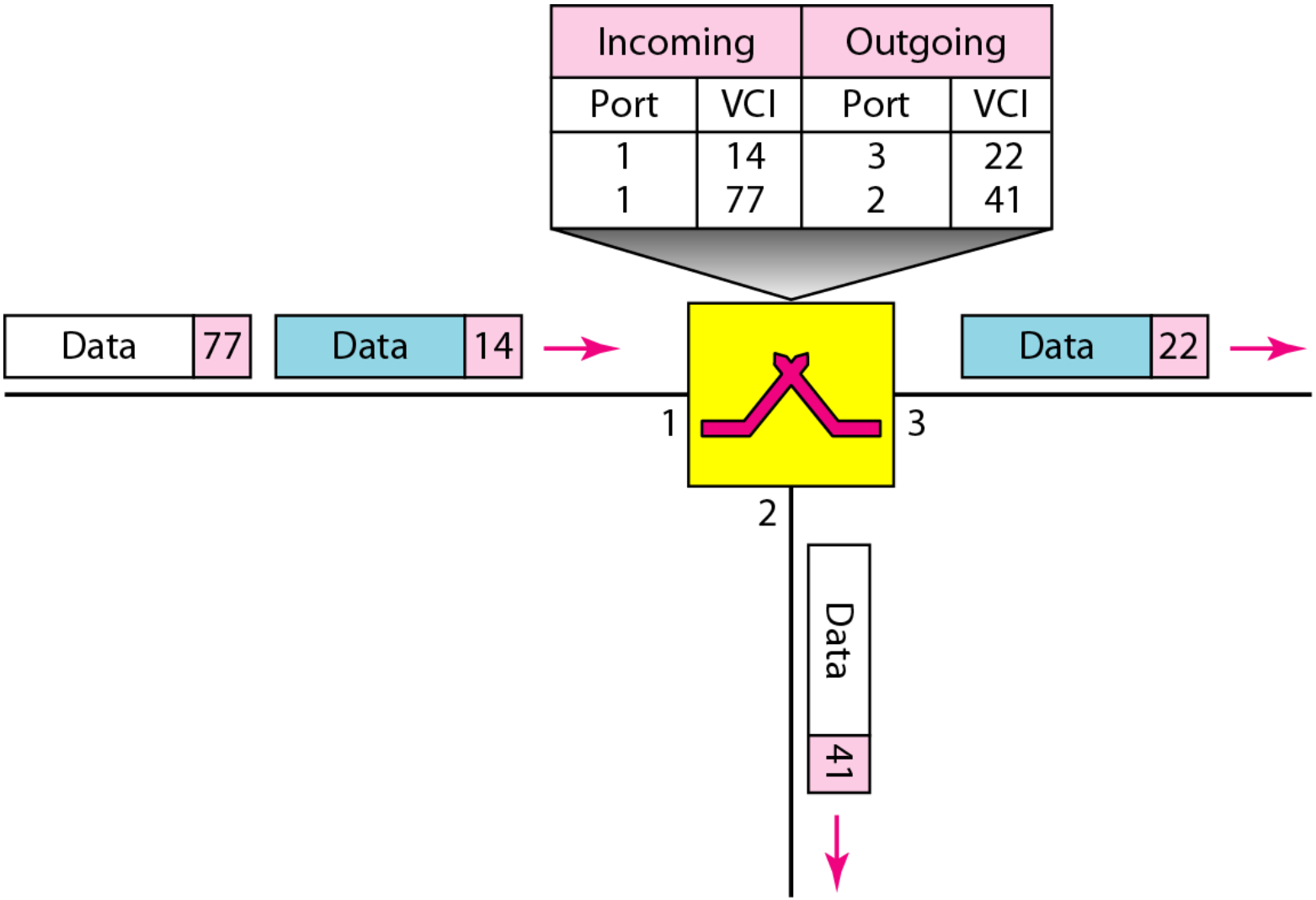


Figure 8.13 *Source-to-destination data transfer in a virtual-circuit network*

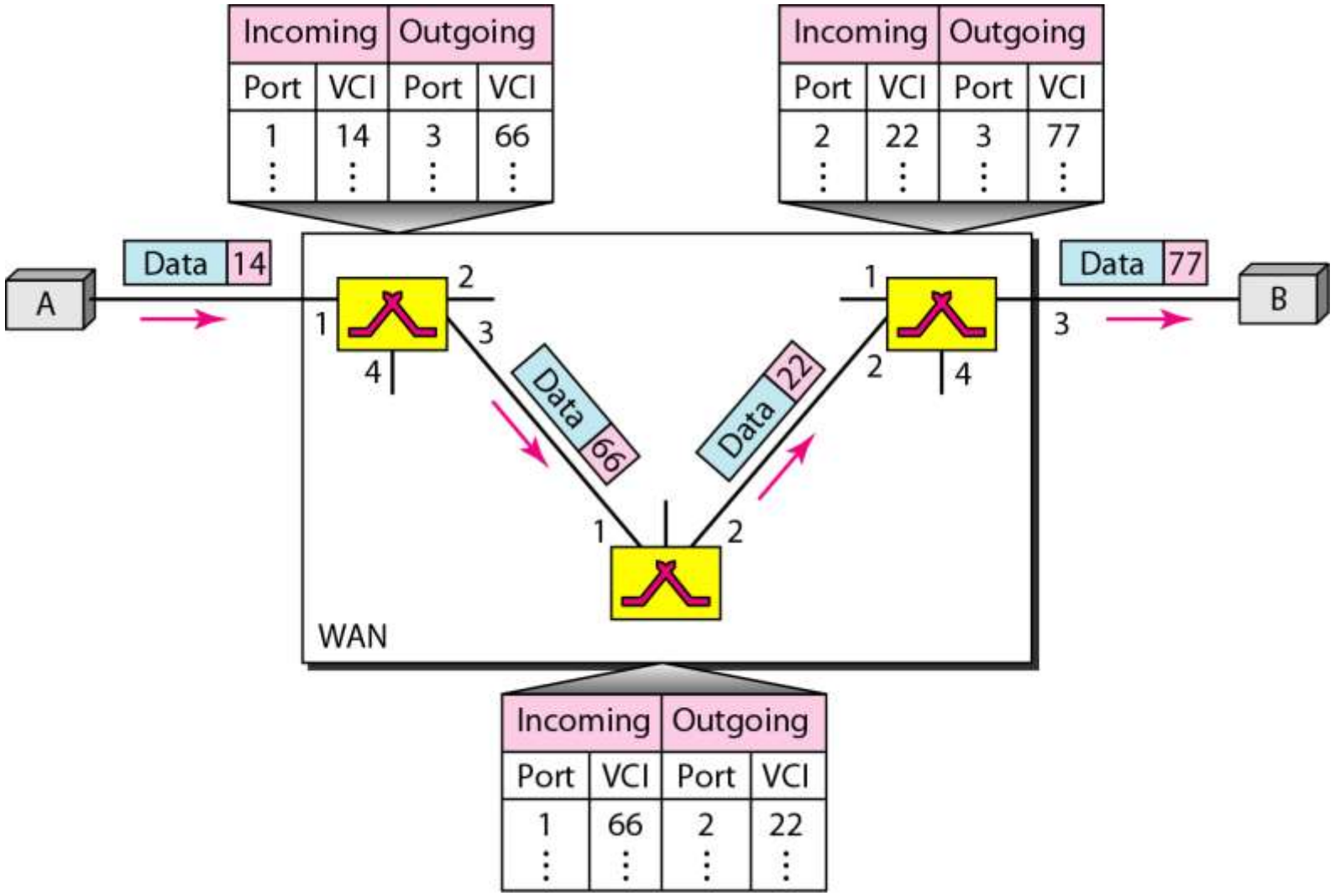


Figure 8.14 *Setup request in a virtual-circuit network*

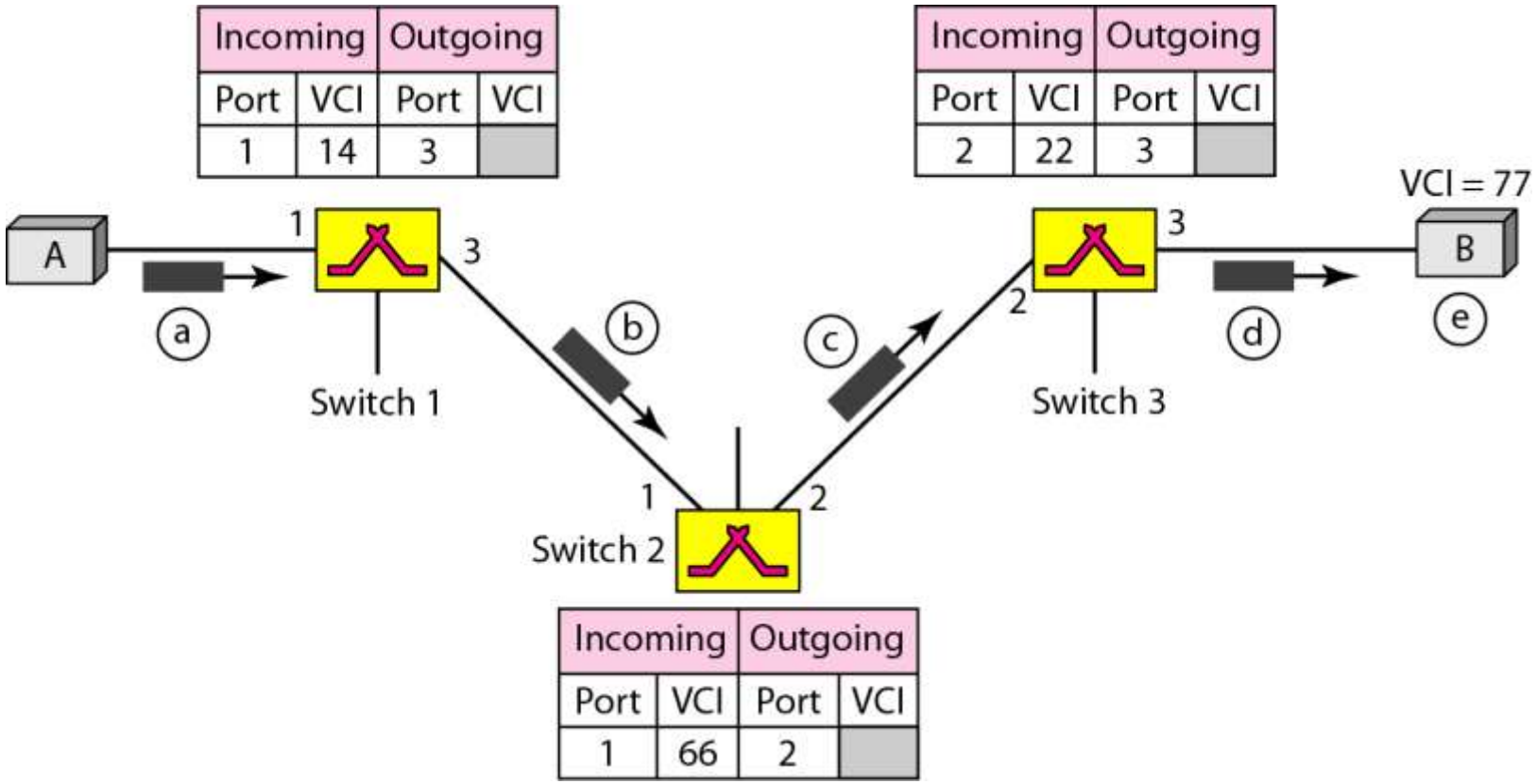
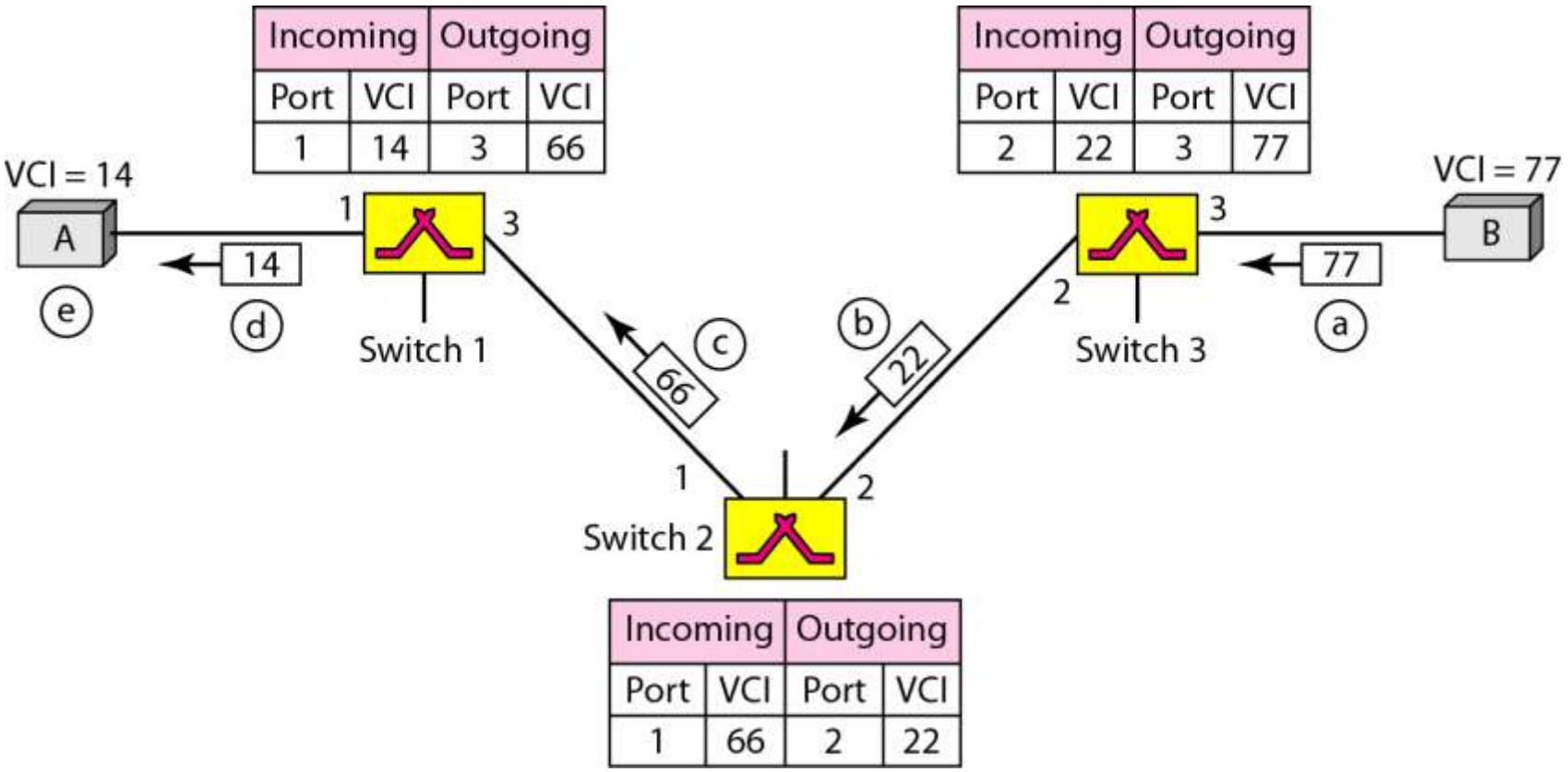


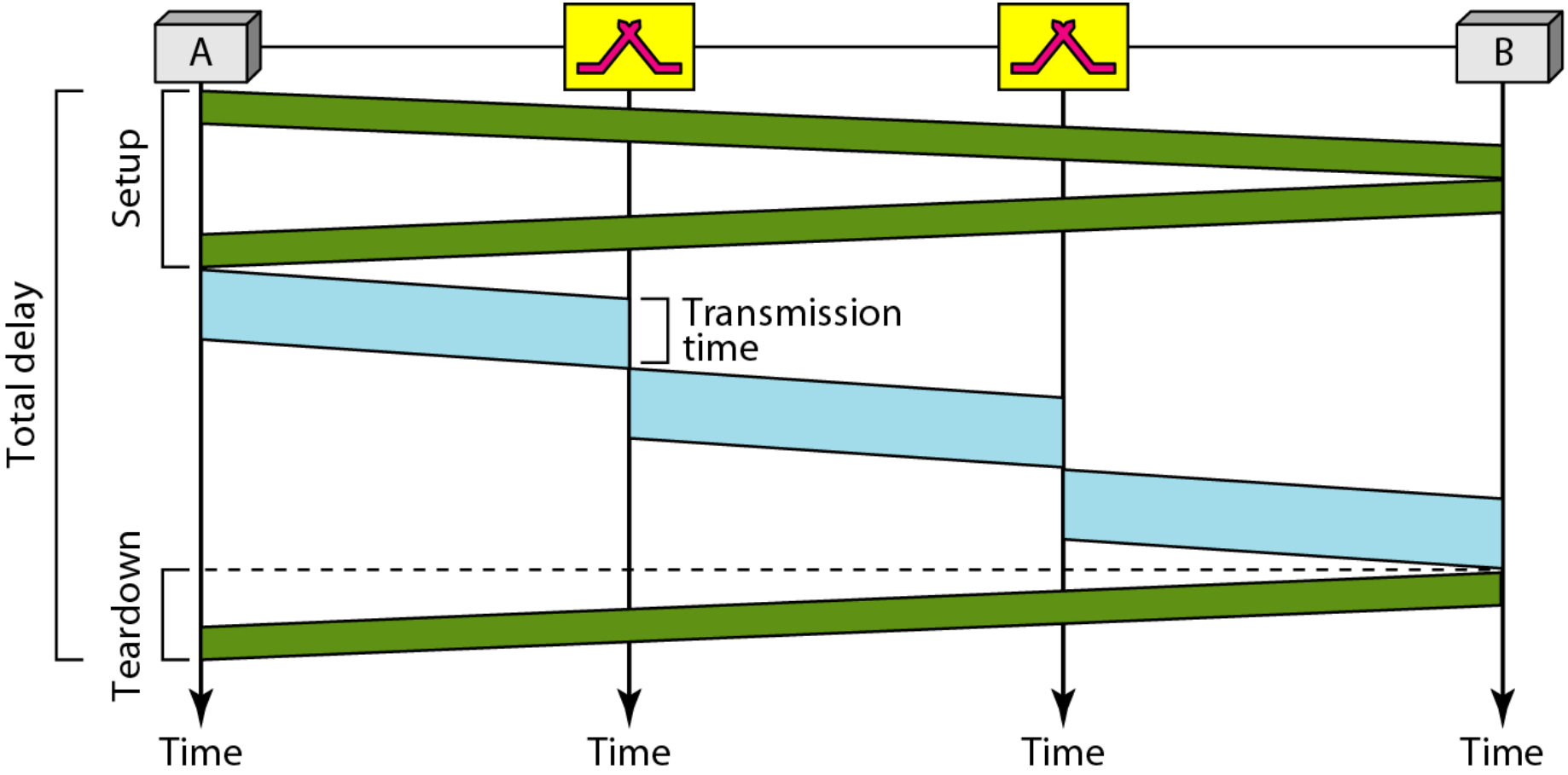
Figure 8.15 *Setup acknowledgment in a virtual-circuit network*



Note

In virtual-circuit switching, all packets belonging to the same source and destination travel the same path; but the packets may arrive at the destination with different delays if resource allocation is on demand.

Figure 8.16 *Delay in a virtual-circuit network*



Note

Switching at the data link layer in a switched WAN is normally implemented by using virtual-circuit techniques.

8-4 STRUCTURE OF A SWITCH

We use switches in circuit-switched and packet-switched networks. In this section, we discuss the structures of the switches used in each type of network.

Topics discussed in this section:

Structure of Circuit Switches
Structure of Packet Switches

Figure 8.17 *Crossbar switch with three inputs and four outputs*

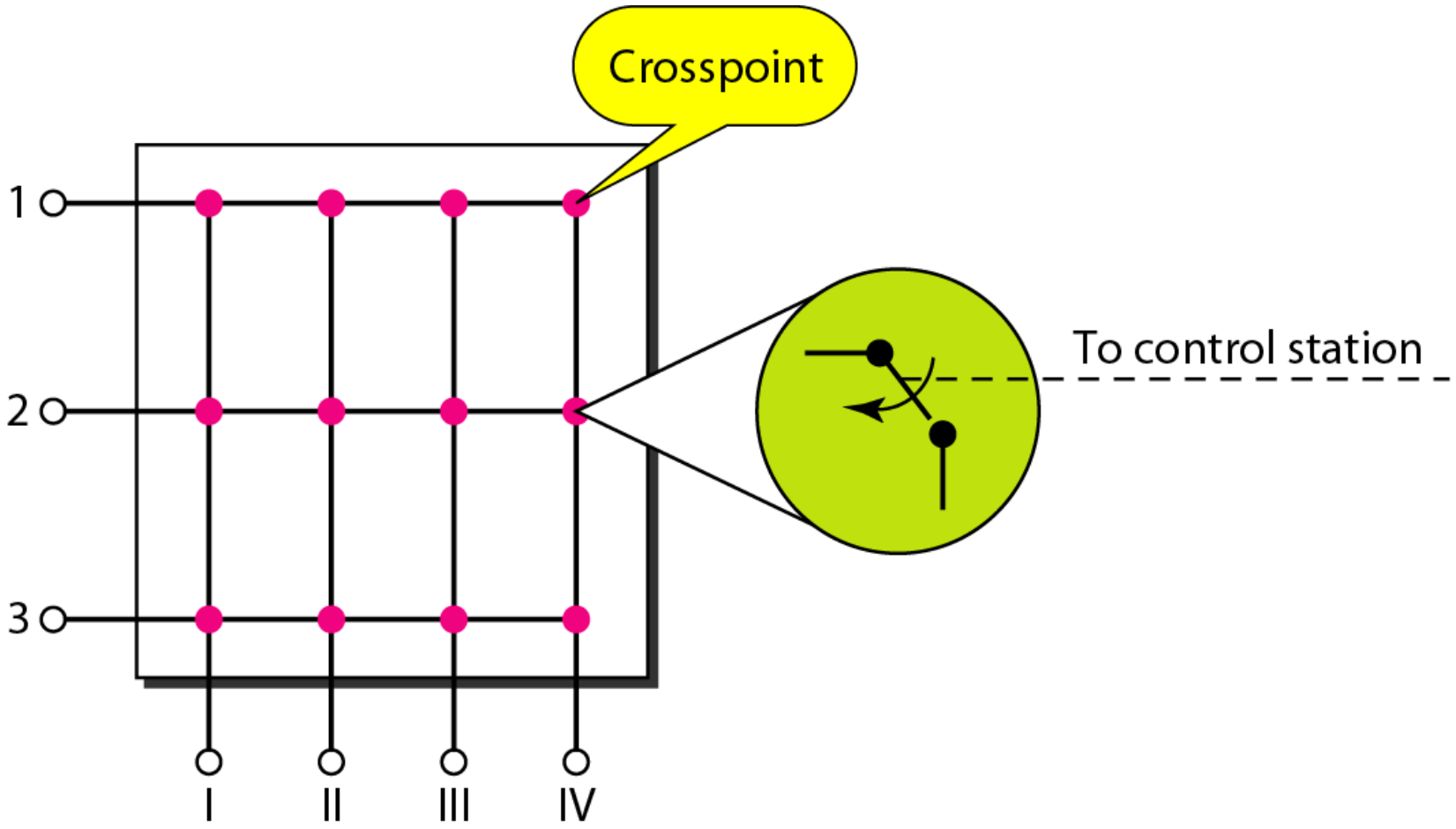
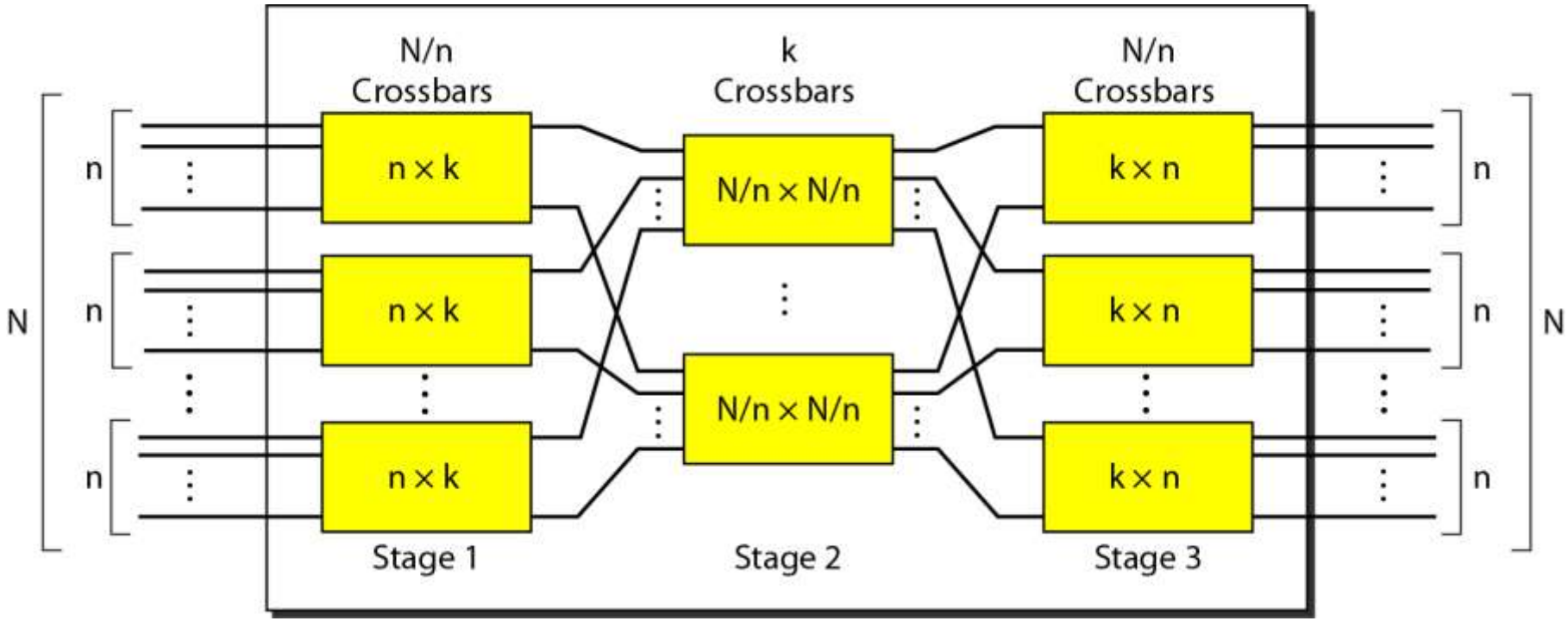


Figure 8.18 *Multistage switch*



Note

In a three-stage switch, the total number of crosspoints is $2kN + k(N/n)^2$ which is much smaller than the number of crosspoints in a single-stage switch (N^2).

Example 8.3

Design a three-stage, 200 × 200 switch (N = 200) with k = 4 and n = 20.

Solution

In the first stage we have N/n or 10 crossbars, each of size 20 × 4. In the second stage, we have 4 crossbars, each of size 10 × 10. In the third stage, we have 10 crossbars, each of size 4 × 20. The total number of crosspoints is $2kN + k(N/n)^2$, or 2000 crosspoints. This is 5 percent of the number of crosspoints in a single-stage switch (200 × 200 = 40,000).

Note

According to the Clos criterion:

$n = (N/2)^{1/2}$

$k > 2n - 1$

Crosspoints $\geq 4N\,[(2N)^{1/2} - 1]$

Example 8.4

Redesign the previous three-stage, 200 × 200 switch, using the Clos criteria with a minimum number of crosspoints.

Solution

We let $n = (200/2)^{1/2}$, or $n = 10$. We calculate $k = 2n - 1 = 19$. In the first stage, we have 200/10, or 20, crossbars, each with 10 × 19 crosspoints. In the second stage, we have 19 crossbars, each with 10 × 10 crosspoints. In the third stage, we have 20 crossbars each with 19 × 10 crosspoints. The total number of crosspoints is 20(10 × 19) + 19(10 × 10) + 20(19 ×10) = 9500.

Figure 8.19 *Time-slot interchange*

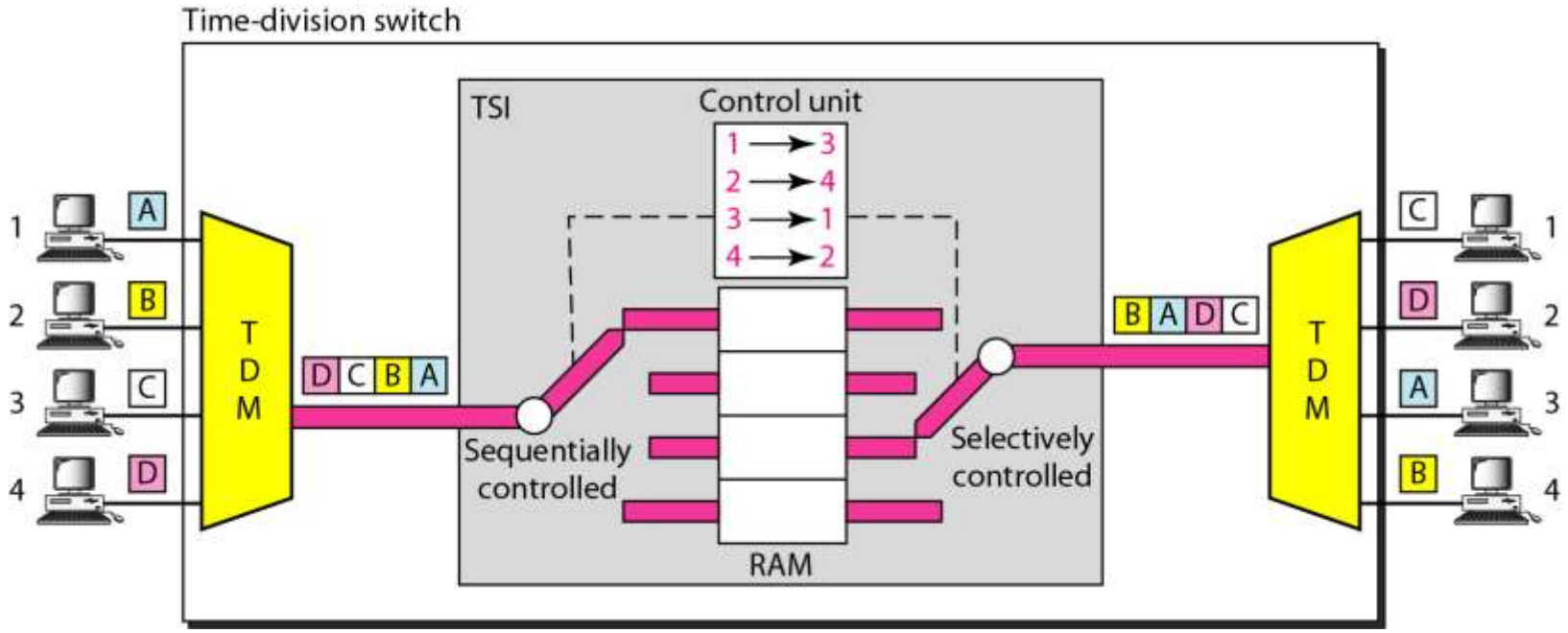


Figure 8.20 *Time-space-time switch*

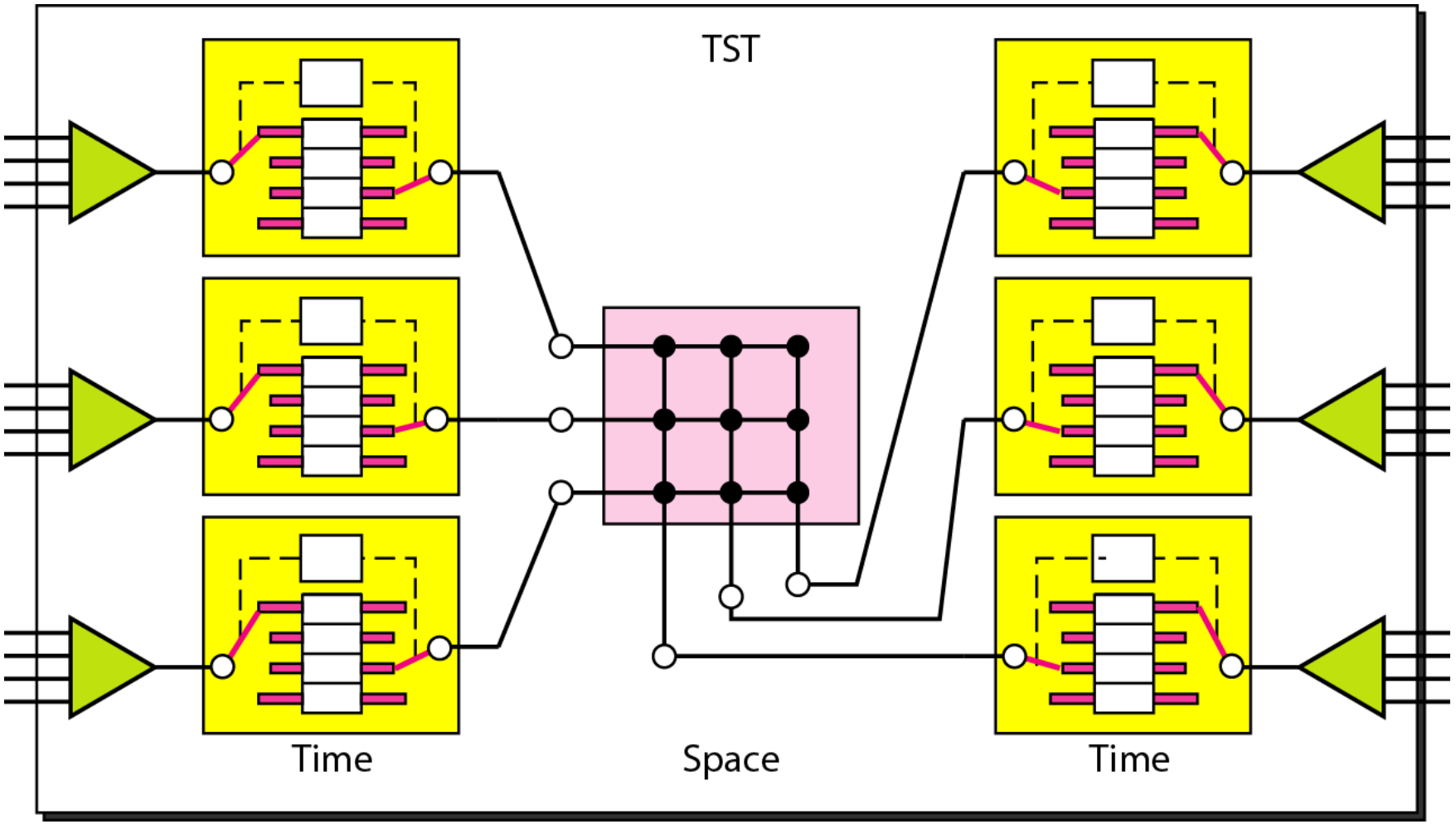


Figure 8.21 *Packet switch components*

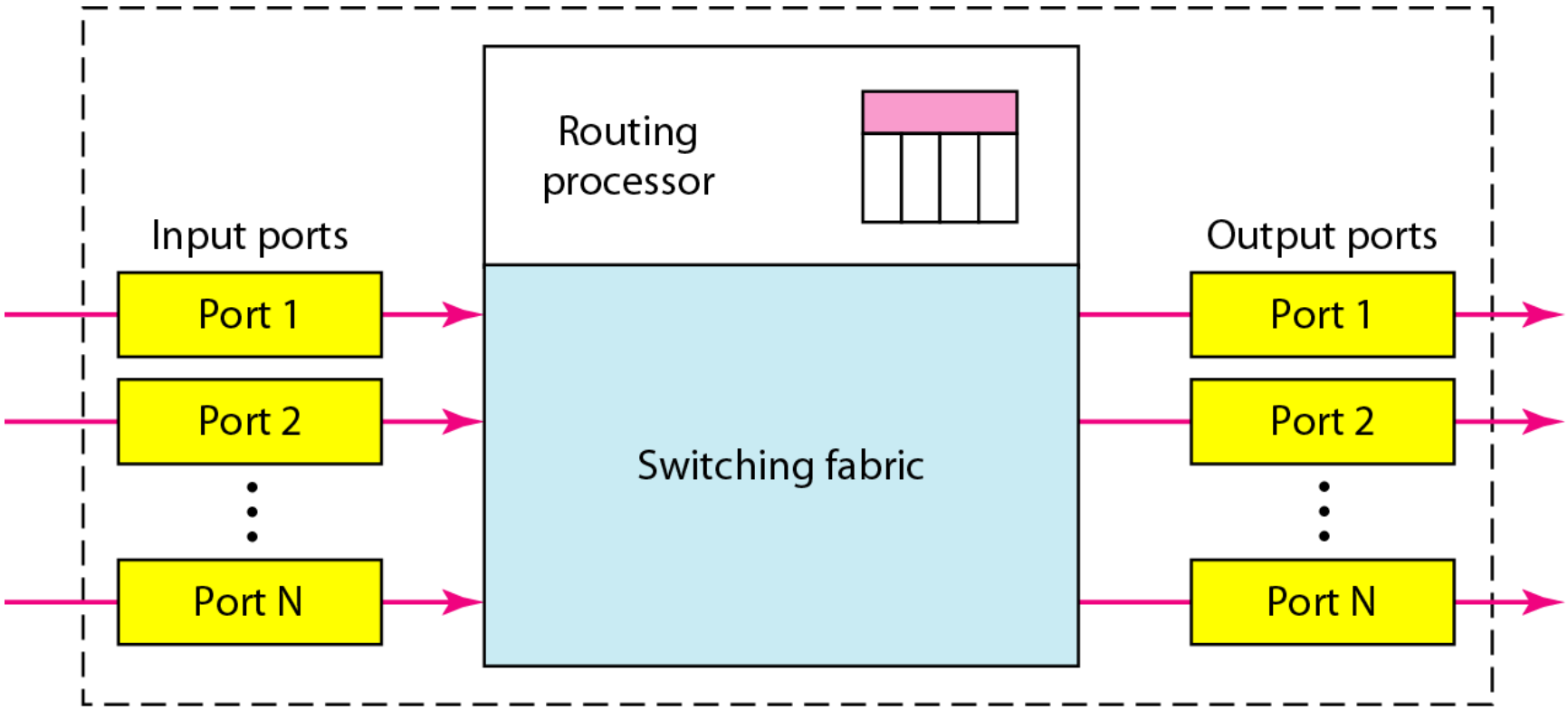


Figure 8.22 *Input port*

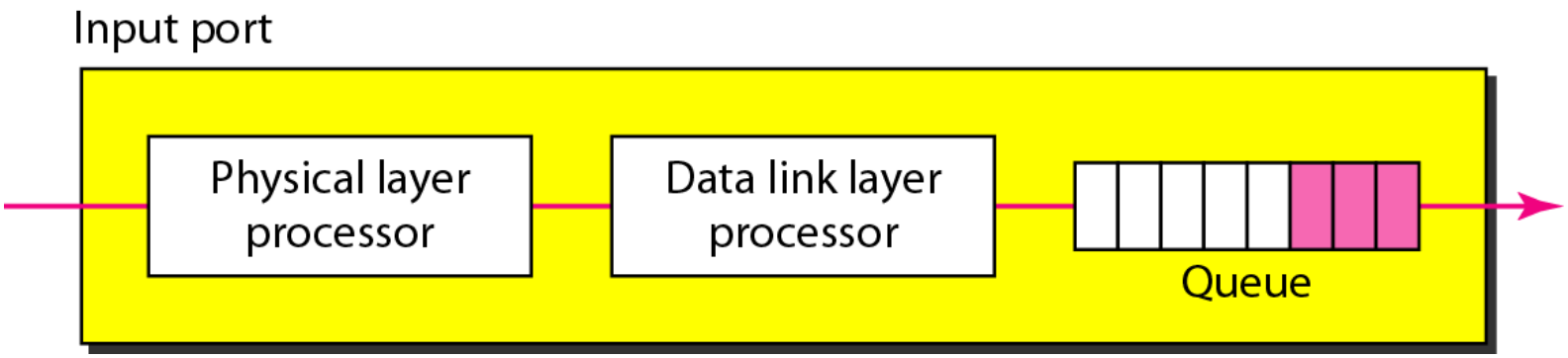


Figure 8.23 *Output port*

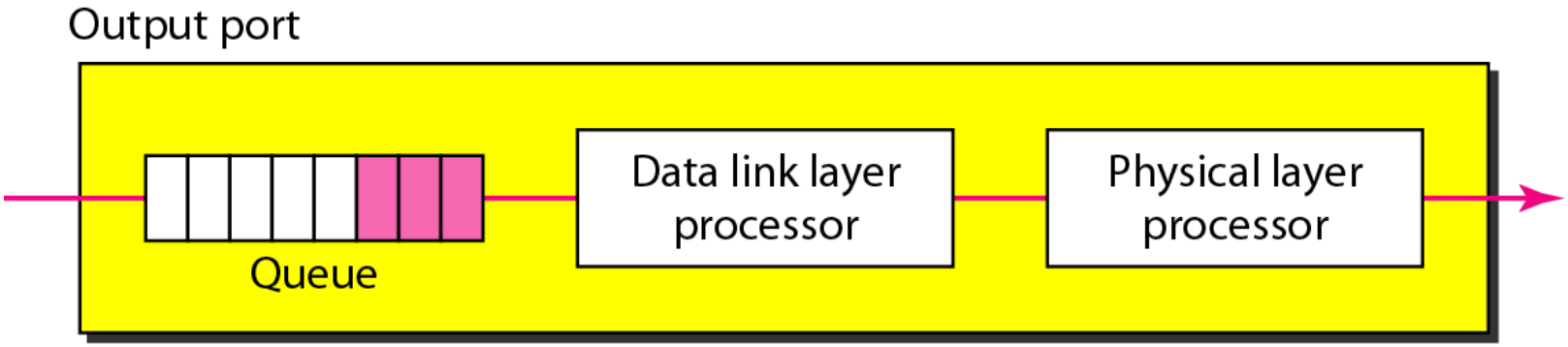


Figure 8.24 *A banyan switch*

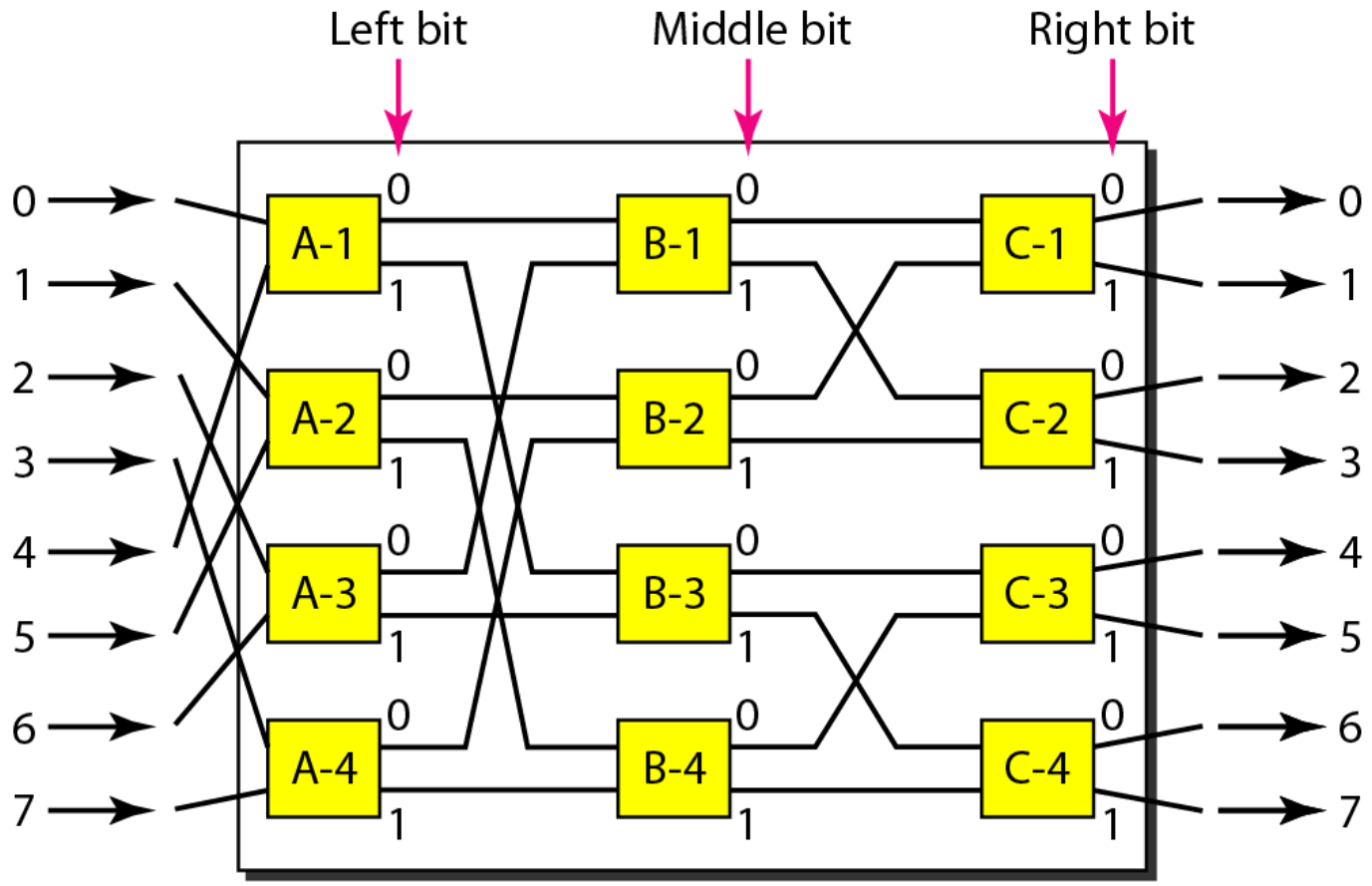
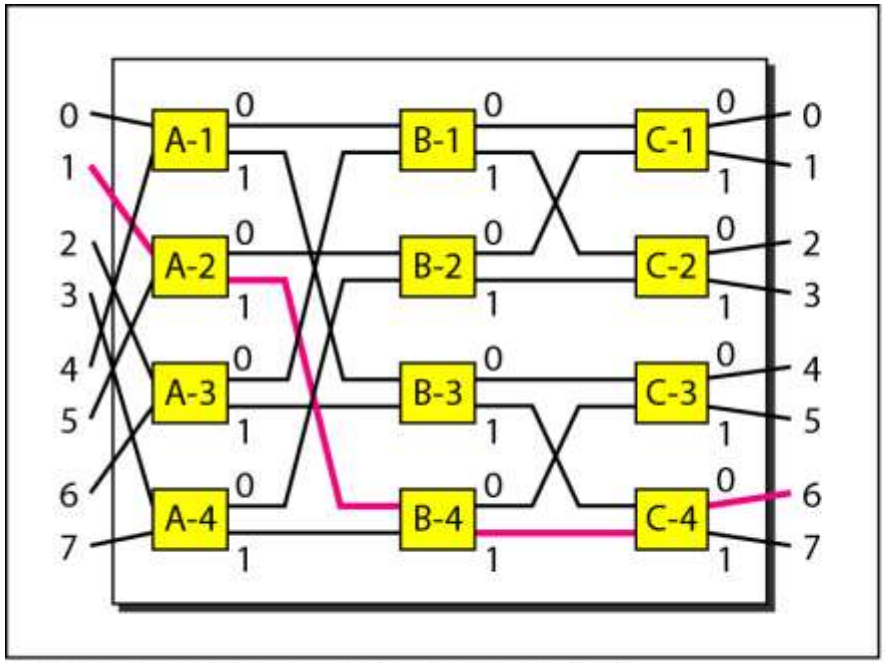
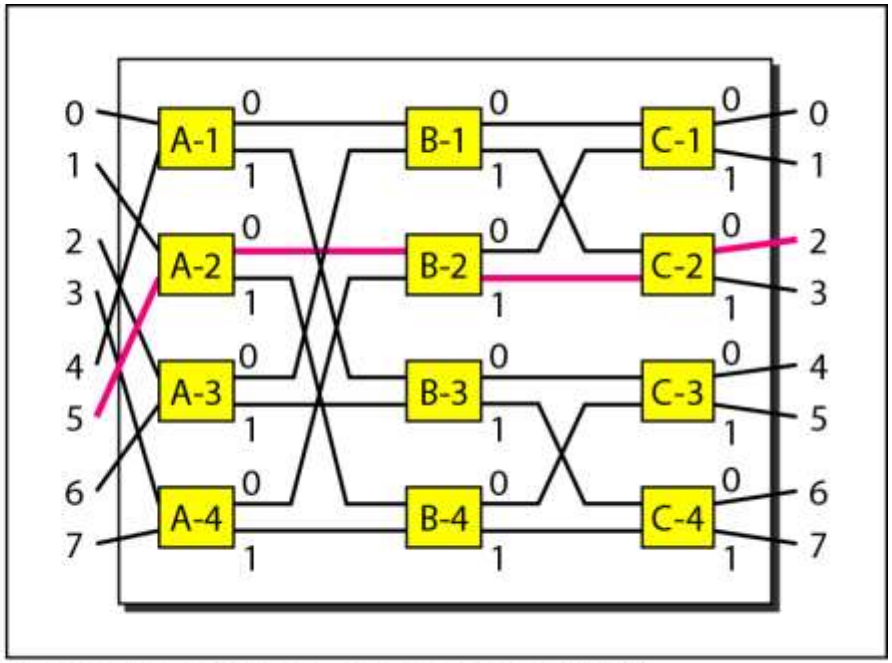


Figure 8.25 *Examples of routing in a banyan switch*



a. Input 1 sending a cell to output 6 (110)



b. Input 5 sending a cell to output 2 (010)

Figure 8.26 *Batcher-banyan switch*

